

GOC Keywords

```
@alias(<protoMsg>) <messageDef>;
<ret> = @call [,<flags>+] [{<cast>}]
            <obj>::[{cast2}] <msg>(<params>*);
@callsuper();
@callsuper <obj>::<class>::<msg>(<params>*) [<flags>,];
@chunk <type> <name> [ = <init> ];
@chunkArray <stype> <aname> [ = {<init>} ];
@class <cname>, <super> [, master [, variant]];
@classdecl <cname> [, <cflags>];
@default
@define <macro> <definition>
@deflib <libname>
@dispatch [noFree] <nObj>::<nMsg>::<event>;
<ret> = @dispatchCall [noFree] [(<cast>)]
                        <nobj>::<nMsg>::<event>;
@elementArray <stype> <arrayname> [={<init>}];
@end <segname>
@endc
@endif
@endlib
@exportMessages <expName>, <num>;
@extern method <class> <message>+;
@extern [chunk|object|visMoniker] <name>;
gcnList(<manufID>, <ltype>) = [<oname>, ]* <oname>
(@genChildren, @genParent) shortcuts for @call and @send
@gstring <sname> = { <gstringDef> };
@header <type> [ = <init> ];
@if <condition>
@ifdef <flag>
@ifndef <flag>
@importMessage <expName>, <messageDef>;
@include [<<headerfile>>| "<headerfile>"]
@instance <insType> <iname> [ = <default>];
@instance @activeList <iname> [ = <default>];
@instance @composite <iname> [ = <linkName>];
@instance @kbdAccelerator <iname>
                        [ = [@specificUI] <default>];
@instance @link <iname> [ = <default>];
@instance @visMoniker <iname> [ = <default>];
@message <retType> <mname>([@stack] <param>*);
@method [ <hname>, ] <cname>, <mname>+;
@object <class> <name> <flags>* = {
    <fieldName> = <init>;*
    <varName> [ = <init> ];
}
@optimize
@prototype <messageDef>;
<event> = @record <obj>::<msg>(<params>*);
@reloc <iname>, [ (<count>, <struct>), ] <ptrType>;
@reloc <vname>, <fn>, [(<count>, <struct>),] <ptrType>;
@reserveMessages <num>;
@send [,<flags>+] <obj>::<msg>(<params>*);
@start <segname> [ , <flags> ];
@vardata <type> <vname>;
@vardataAlias (<origName>) <newType> <newName>;
(@visChildren, @visParent) shortcuts for @call and @send
```

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ArcClass

```
@class ArcClass, GrObjClass
```

Instance Data

```
word                AI_arcCloseType
WWFixed             AI_startAngle
WWFixed             AI_endAngle
PointWWFixed        AI_startPoint
PointWWFixed        AI_endPoint
PointWWFixed        AI_midPoint
WWFixed             AI_radius
```

Messages

```
MSG_ARC_SET_START_ANGLE
MSG_ARC_SET_END_ANGLE
MSG_ARC_SET_ARC_CLOSE_TYPE
MSG_ARC_UNDO_REPLACE_ARC_GEOMETRY_INSTANCE_DATA
MSG_ARC_REPLACE_ARC_GEOMETRY_INSTANCE_DATA
```

BitmapGuardianClass

```
@class BitmapGuardianClass, GrObjVisGuardianClass
```

Instance Data

```
BitmapGuardianFlags       BGI_flags
ClassStruct               *BGI_toolClass
```

Types and Flags

```
ByteFlags    BitmapGuardianFlags
      BGF_POINTER_ACTIVE              0x02
      BGF_REAL_ESTATE_RESIZE          0x01
```

Messages

```
MSG_BG_SET_TOOL_CLASS
MSG_BG_SET_BITMAP_POINTER_ACTIVE_STATUS
MSG_BG_REAL_ESTATE_HANDLE_HIT_DETECTION
MSG_BG_ACTIVATE_REAL_ESTATE_RESIZE
MSG_BG_JUMP_START_REAL_ESTATE_RESIZE
MSG_BG_PTR_REAL_ESTATE_RESIZE
MSG_BG_END_REAL_ESTATE_RESIZE
MSG_BG_CREATE_VIS_BITMAP
MSG_BG_GET_TOOL_CLASS
```

ColorSelectorClass

```
@class ColorSelectorClass, GenControlClass
```

Instance Data

```
ColorQuad           CSI_color = {0, 0, 0, 0}
byte                CSI_colorIndeterminate
SystemDrawMask      CSI_drawMask = SDM_0
byte                CSI_drawMaskIndeterminate
GraphicPattern      CSI_pattern = {0, 0}
byte                CSI_patternIndeterminate
ColorModifiedStates CSI_states = 0
ColorToolboxPreferences
                    CSI_toolboxPrefs = CTP_IS_POPUP
```

Variable Data

```
optr ATTR_COLOR_SELECTOR_DISABLE_OBJECT
```

Structures

```
WordFlags    CSFeatures
      CSF_FILLED_LIST     0x10
      CSF_INDEX           0x08
      CSF_RGB             0x04
      CSF_DRAW_MASK       0x02
      CSF_PATTERN         0x01
WordFlags    CSToolboxFeatures
      CSTF_INDEX          0x04
      CSTF_DRAW_MASK      0x02
      CSTF_PATTERN        0x01
CS_DEFAULT_FEATURES(CSF_FILLED_LIST | CSF_INDEX | \
                    CSF_RGB | CSF_DRAW_MASK | \
                    CSF_PATTERN)
CS_DEFAULT_TOOLBOX_FEATURES     (CSTF_INDEX | \
                    CSTF_DRAW_MASK | CSTF_PATTERN)
ByteFlags    ColorModifiedStates
      CMS_COLOR_CHANGED         0x04
      CMS_DRAW_MASK_CHANGED     0x02
      CMS_PATTERN_CHANGED       0x01
ByteEnum     ColoredObjectOrientation
      COO_AREA_ORIENTED         0
      COO_TEXT_ORIENTED         1
      COO_LINE_ORIENTED         2
typedef ByteFlags ColorToolboxPreferences
      CTP_INDEX_ORIENTATION     0x30
      CTP_DRAW_MASK_ORIENTATION 0x0c
      CTP_PATTERN_ORIENTATION   0x02
      CTP_IS_POPUP              0x01
CTP_INDEX_ORIENTATION_OFFSET    4
CTP_DRAW_MASK_ORIENTATION_OFFSET2
CTP_PATTERN_ORIENTATION_OFFSET  1
```

Messages

```
Boolean MSG_COLOR_SELECTOR_GET_COLOR(
                    ColorQuad    *retValue)
void MSG_COLOR_SELECTOR_SET_COLOR(
                    ColorQuad    colorQuad,
                    Boolean      indeterminateFlag)
```

```
void MSG_COLOR_SELECTOR_UPDATE_COLOR(
                    ColorQuad    colorQuad,
                    Boolean      indeterminateFlag)
void MSG_COLOR_SELECTOR_APPLY_COLOR(
                    ColorQuad    colorQuad)
void MSG_COLOR_SELECTOR_UPDATE_FILLED_STATUS(
                    SystemDrawMask drawMask,
                    Boolean      indeterminateFlag,
                    word         updateToolboxFlag)
Boolean MSG_COLOR_SELECTOR_GET_FILLED_MONIKER(
                    optr         *retValue)
Boolean MSG_COLOR_SELECTOR_GET_UNFILLED_MONIKER(
                    optr         *retValue)
Boolean MSG_COLOR_SELECTOR_GET_DRAW_MASK(
                    SystemDrawMask *retValue))
void MSG_COLOR_SELECTOR_SET_DRAW_MASK(
                    SetDrawMask  drawMask,
                    Boolean      indeterminateFlag)
void MSG_COLOR_SELECTOR_APPLY_DRAW_MASK(
                    SystemDrawMask drawMask)
Boolean MSG_COLOR_SELECTOR_GET_PATTERN(
                    GraphicPattern *retValue)
void MSG_COLOR_SELECTOR_SET_PATTERN(
                    GraphicPattern pattern,
                    Boolean      indeterminateFlag)
void MSG_COLOR_SELECTOR_UPDATE_PATTERN(
                    GraphicPattern pattern,
                    Boolean      indeterminateFlag)
void MSG_COLOR_SELECTOR_APPLY_PATTERN(
                    GraphicPattern pattern)
void MSG_META_COLORED_OBJECT_SET_COLOR(
                    ColorQuad    colorQuad)
void MSG_META_COLORED_OBJECT_SET_DRAW_MASK(
                    SystemDrawMask drawMask)
void MSG_META_COLORED_OBJECT_SET_PATTERN(
                    GraphicPattern pattern)
```

DictControlClass

```
@class DictControlClass, GenControlClass
```

Instance Data

```
word         DCI_status
    @default GCI_output = (TO_APP_TARGET)
    @default GII_visibility = GIV_DIALOG
    @default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
WordFlags DictFeatures
      DF_DICT      0x01
#define DICT_GCM_FEATURES DF_DICT

WordFlags DictToolboxFeatures
      DTF_DICT     0x01
#define DICT_GCM_TOOLBOX_FEATURES DTF_DICT
```

EditTextGuardianClass

```
@class EditTextGuardianClass, TextGuardianClass
```

EditUserDictionaryControlClass


```
@class EditUserDictionaryControlClass, GenControlClass
```

Instance Data

```
MemHandle    EUDCI_userDictList
MemHandle    EUDCI_icBuff
    @default GII_attrs = (@default | GIA_MODAL)
    @default GII_type = GIT_COMMAND
    @default GII_visibility = GIV_DIALOG
```

Types and Flags

```
WordFlags    EditUserDictionaryFeatures
      EUDF_EDIT_USER_DICTIONARY       0x01
WordFlags    EditUserDictionaryToolboxFeatures
      EUDTF_EDIT_USER_DICTIONARY      0x01
EUDC_DEFAULT_FEATURES     (EUDF_EDIT_USER_DICTIONARY)
EUDC_DEFAULT_TOOLBOX_FEATURES
                          (EUDTF_EDIT_USER_DICTIONARY)
```

Messages

```
void MSG_EUDC_GET_USER_DICTIONARY_LIST_MONIKER()
void
   MSG_EUDC_DELETE_SELECTED_WORD_FROM_USER_DICTIONARY()
void MSG_EUDC_ADD_NEW_WORD_TO_USER_DICTIONARY()
void MSG_EUDC_UPDATE_SELECTED_WORD()
void MSG_EUDC_CLOSE_EDIT_BOX()
void MSG_META_EDIT_USER_DICTIONARY_COMPLETED()
```

ExportControlClass

```
@class ExportControlClass, ImportExportClass
```

Instance Data

```
ExportControlAttrs       ECI_attrs
ImpexDataClasses         ECI_dataClasses
optr                     ECI_destination
word                     ECI_message
```

Variable Data

```
optr ATTR_EXPORT_CONTROL_APP_UI
```

Types and Flags

```
WordFlags    ExportControlAttrs
      ECA_IGNORE_INPUT               0x8000
ByteFlags    ExportControlFeatures
      EXPORTCF_EXPORT_TRIGGER        0x0008
      EXPORTCF_FORMAT_OPTIONS        0x0004
      EXPORTCF_BASIC                 0x0002
      EXPORTCF_GLYPH                 0x0001
EXPORTC_DEFAULT_FEATURES (EXPORTCF_GLYPH |
            EXPORTCF_BASIC | EXPORTCF_FORMAT_OPTIONS |
            EXPORTCF_EXPORT_TRIGGER)
ByteFlags    ExportControlToolboxFeatures
      EXPORTCTF_DIALOG_BOX           0x01
EXPORTC_DEFAULT_TOOLBOX_FEATURES (EXPORTCTF_DIALOG_BOX)
```

Messages

```
void MSG_EXPORT_CONTROL_SET_DATA_CLASSES(
                  ImpexDataClasses dataClass)
ImpexDataClasses MSG_EXPORT_CONTROL_GET_DATA_CLASSES()
void MSG_EXPORT_CONTROL_SET_ACTION(
                  optr destOD, word ECImsg)
void MSG_EXPORT_CONTROL_SET_MSG(word ECImsg)
void MSG_EXPORT_CONTROL_GET_ACTION(
                  ObjectState *retValue)
void MSG_EXPORT_CONTROL_SET_ATTRS(
                  ExportControlAttrs attrs)
ExportControlAttrs MSG_EXPORT_CONTROL_GET_ATTRS()
word MSG_EXPORT_CONTROL_GET_FILE_SELECTOR_OFFSET(
                  ExportControlFeatures features)
word MSG_EXPORT_CONTROL_GET_FORMAT_LIST_OFFSET(
                  ExportControlFeatures features)
word MSG_EXPORT_CONTROL_GET_FILE_NAME_OFFSET(
                  ExportControlFeatures features)
word MSG_EXPORT_CONTROL_GET_FORMAT_UI_PARENT_OFFSET(
                  ExportControlFeatures features)
word MSG_EXPORT_CONTROL_GET_APP_UI_PARENT_OFFSET(
                  ExportControlFeatures features)
word MSG_EXPORT_CONTROL_GET_EXPORT_TRIGGER_OFFSET(
                  ExportControlFeatures features)
void MSG_EXPORT_CONTROL_EXPORT_COMPLETE(
                  ImpexTranslationParams *itParams)
```

FloatFormatClass

```
@class FloatFormatClass, GenControlClass
```

Instance Data

```
word         formatInfoStructHan = 0;
      @default GCI_output = (TO_APP_TARGET)
```

Structures

```
typedef struct {
    VMFileHandle    NFFC_vmFileHan
    VMBlockHandle   NFFC_vmBlkHan
    FormatIdType    NFFC_format
    word            NFFC_count
} NotifyFloatFormatChange
```

Messages

```
void MSG_FLOAT_CTRL_REQUEST_MONIKER(
                    MemHandle formatInfoStrucHan)
void MSG_FLOAT_CTRL_UPDATE_UI(
                    MemHandle formatInfoStrucHan)
void MSG_FLOAT_CTRL_FORMAT_SELECTED(
                    MemHandle formatInfoStrucHan)
void MSG_FLOAT_CTRL_USER_DEF_INVOKE(
                    MemHandle formatInfoStrucHan)
void MSG_FLOAT_CTRL_USER_DEF_OK(
                    MemHandle formatInfoStrucHan)
void MSG_FLOAT_CTRL_FORMAT_DELETE(
                    MemHandle formatInfoStrucHan)
void MSG_FLOAT_CTRL_REPLACE_NUM_FORMAT(
                    MemHandle formatInfoStrucHan)
void MSG_FLOAT_CTRL_FORMAT_APPLY(
                    MemHandle formatInfoStrucHan)
void MSG_FCF_FORMAT_CHANGE(MemHandle formatInfoStrucHan)
void MSG_FCT_FORMAT_CHANGE(MemHandle formatInfoStrucHan)
```

Routines

```
VMBlockHandle FloatFormatInit(word userDefVMFileHan)
word FloatFormatGetFormatParamsWithListEntry(
                    FormatInfoStruc *formatInfoStruc)
void FloatFormatInitFormatList(
                    FormatInfoStruc *formatInfoStruc)
void FloatFormatProcessFormatSelected(
                    FormatInfoStruc *formatInfoStruc)
void FloatFormatInvokeUserDefDB(
                    FormatInfoStruc *formatInfoStruc)
word FloatFormatUserDefOK(
                    FormatInfoStruc *formatInfoStruc)
word *FloatFormatGetFormatTokenWithName(
                    FormatInfoStruc *formatInfoStruc)
void FloatFormatGetFormatParamsWithToken(
                    FormatInfoStruc *formatInfoStruc,
                    FormatParams *buffer)
FormatIdType FloatFormatDelete(
                    FormatInfoStruc *formatInfoStruc)
void FloatFormatIsFormatTheSame(
                    FormatInfoStruc *formatInfoStruc,
                    FormatParams *formatParams)
word FloatFormatAddFormat(
```

```
        FormatInfoStruc *formatInfoStruc,
        FormatParams *formatParams,
        word formatToken)
```

GenApplicationClass


```
@class GenApplicationClass, GenClass
```

Instance Data

```
AppInstanceReference GAI_appRef = {"","",NullHandle,{0}}
word                 GAI_appMode = 0
AppLaunchFlags       GAI_launchFlags = 0
ApplicationOptFlags  GAI_optFlags = 0
word                 GAI_appFeatures = 0
Handle               GAI_specificUI = 0
ApplicationStates    GAI_states = AS_FOCUSABLE |
                                  AS_MODELABLE
AppAttachFlags       GAI_attachFlags = 0
UIInterfaceLevel     GAI_appLevel = UIIL_ADVANCED
ChunkHandle          GAI_iacpConnects = 0
   @default  GI_states = @default & ~GS_USABLE
   @default  GI_attrs = @default | GA_TARGETABLE
```

Variable Data

```
optr ATTR_GEN_APPLICATION_PRINT_CONTROL;
  @reloc ATTR_GEN_APPLICATION_PRINT_CONTROL, 0, optr
ChunkHandle TEMP_GEN_APPLICATION_NO_LONGER_USED;
optr ATTR_GEN_APPLICATION_KBD_OBJ;
MemHandle TEMP_GEN_APPLICATION_SAVED_ALB;
void TEMP_GEN_APPLICATION_ABORT_QUIT;
optr ATTR_GEN_APPLICATION_SAVE_OPTIONS_TRIGGER;
 @reloc ATTR_GEN_APPLICATION_SAVE_OPTIONS_TRIGGER,
                                          0, optr;
```

Hints

```
void  HINT_APP_IS_ENTERTAINING
void  HINT_APP_IS_EDUCATIONAL
void  HINT_APP_IS_PRODUCTIVITY_ORIENTED
```

Types and Flags

```
ByteEnum     AppMeasurementType
      AMT_US             0
      AMT_METRIC         1
      AMT_DEFAULT        0xff
WordFlags    ApplicationStates
      AS_TRANSPARENT                 0x4000
      AS_HAS_FULL_SCREEN_EXCL        0x2000
      AS_SINGLE_INSTANCE             0x1000
      AS_QUIT_DETACHING              0x0800
      AS_AVOID_TRANSPARENT_DETACH    0x0400
      AS_TRANSPARENT_DETACHING       0x0200
      AS_REAL_DETACHING              0x0100
      AS_QUITTING                    0x0080
      AS_DETACHING                   0x0040
      AS_FOCUSABLE                   0x0020
      AS_MODELABLE                   0x0010
      AS_NOT_USER_INTERACTABLE       0x0008
      AS_RECEIVED_APP_OBJECT_DETACH  0x0004
```

```
        AS_ATTACHED_TO_STATE_FILE       0x0002
        AS_ATTACHING                    0x0001
typedef AppOptFlags ApplicationOptFlags
typedef ByteFlags AppOptFlags
        AOF_MULTIPLE_INIT_FILE_CATEGORIES 0x80
typedef ByteEnum GenAppUsabilityCommand
        GAUC_USABILITY          0
        GAUC_RECALC_CONTROLLER  1
        GAUC_REPARENT           2
        GAUC_POPUP              3
        GAUC_TOOLBAR            4
        GAUC_RESTART            5
typedef ByteFlags GenAppUsabilityTupleFlags
        GAUTF_END_OF_LIST       0x20
        GAUTF_OFF_IF_BIT_ON     0x10
        GAUTF_COMMAND           0x0f
typedef enum {
        TO_PRINT_CONTROL=_FIRST_GenApplicationClass
} GenApplicationTravelOption
```

Structures

```
typedef struct {
    GenAppUsabilityTupleFlags   GAUT_flags
    optr                        GAUT_object
} GenAppUsabilityTuple
typedef struct {
        UIInterfaceLevel    AFI_uiInterfaceLevel;
        word                AFI_appFeatures;
} AppFeaturesInfo;
typedef struct {
        CharFlags           CFASS_charFlags;
        ShiftState          CFASS_shiftState;
} CharFlagsAndShiftState;
typedef struct {
        ToggleState         TSASC_toggleState;
        byte                TSASC_scanCode;
} ToggleStateAndScanCode;
typedef struct {
        CustomDialogType    ND_dialogType;
        byte                ND_unused1;
        word                ND_unused2;
        optr                ND_dialog;
} NewDialog;
```

Macros

```
GET_MEASUREMENT_TYPE(m) ((byte) (m))
GET_APP_MEASUREMENT_TYPE(m) ((byte) ((m)>>8))
GET_UI_INTERFACE_LEVEL(val) \
                        ((val).AFI_uiInterfaceLevel)
GET_APP_FEATURES(val) ((val).AFI_appFeatures)
```

Messages

```
void MSG_GEN_APPLICATION_MARK_BUSY()
void MSG_GEN_APPLICATION_MARK_NOT_BUSY()
void MSG_GEN_APPLICATION_HOLD_UP_INPUT()
void MSG_GEN_APPLICATION_RESUME_INPUT()
void MSG_GEN_APPLICATION_IGNORE_INPUT()
```

```
void MSG_GEN_APPLICATION_ACCEPT_INPUT()
Handle MSG_GEN_APPLICATION_QUERY_UI()
ApplicationStates MSG_GEN_APPLICATION_GET_STATE()
optr MSG_GEN_APPLICATION_FIND_MONIKER(
                    MemHandle destBlock,
                    word searchFlags,
                    DisplayType displayType)
void MSG_GEN_APPLICATION_SET_TASK_ENTRY_MONIKER(
                    optr entryMoniker)
void MSG_GEN_APPLICATION_GET_DISPLAY_SCHEME(
                    DisplayScheme *displayScheme)
void MSG_GEN_APPLICATION_SET_APP_MODE_MESSAGE(
                    Message modeMessage)
Message MSG_GEN_APPLICATION_GET_APP_MODE_MESSAGE()
Handle MSG_GEN_APPLICATION_GET_APP_INSTANCE_REFERENCE()
void MSG_GEN_APPLICATION_SET_APP_INSTANCE_REFERENCE(
                    Handle appInstance)
void MSG_GEN_APPLICATION_SEND_APP_INSTANCE_REFERENCE\
                                        _TO_FIELD()
void MSG_GEN_APPLICATION_INITIATE_UI_QUIT()
void MSG_GEN_APPLICATION_INSTALL_TOKEN()
void MSG_GEN_APPLICATION_TOGGLE_CURSOR()
AppLaunchFlags MSG_GEN_APPLICATION_GET_LAUNCH_FLAGS()
void MSG_GEN_APPLICATION_SET_APP_LEVEL(
                    word level)
AppFeaturesInfo MSG_GEN_APPLICATION_GET_APP_FEATURES()
void MSG_GEN_APPLICATION_SET_APP_FEATURES(
                    word  features)
void MSG_GEN_APPLICATION_DETACH_PENDING()
optr MSG_GEN_APPLICATION_BUILD_STANDARD_DIALOG(
                    char  *customTriggers,
                    char  *arg2, char  *arg1,
                    char  *string,
                    CustomDialogBoxFlags dialogFlags)
void MSG_GEN_APPLICATION_DO_STANDARD_DIALOG(@stack
                    word dialogMethod,optr dialogOD,
                    char *helpContext,
                    char *customTriggers,
                    char *arg2, char *arg1,
                    char *string,
                    CustomDialogBoxFlags dialogFlags)
void MSG_GEN_APPLICATION_MARK_APP_COMPLETELY_BUSY()
void MSG_GEN_APPLICATION_MARK_APP_NOT_COMPLETELY_BUSY()
void MSG_GEN_APPLICATION_OPEN_COMPLETE()
void MSG_GEN_APPLICATION_SET_NOT_USER_INTERACTABLE()
void MSG_GEN_APPLICATION_SET_USER_INTERACTABLE()
void MSG_GEN_APPLICATION_SET_ATTACHED_TO_STATE_FILE()
void
MSG_GEN_APPLICATION_SET_NOT_ATTACHED_TO_STATE_FILE()
void MSG_GEN_APPLICATION_SET_NOT_QUITTING()
void MSG_GEN_APPLICATION_NOTIFY_MODAL_WIN_CHANGE()
word MSG_GEN_APPLICATION_GET_MEASUREMENT_TYPE()
void MSG_GEN_APPLICATION_SET_MEASUREMENT_TYPE(
                    byte measurementType)
Boolean MSG_GEN_APPLICATION_TEST_FOR_CANCEL_MNEMONIC(
                    word                      character,
```



```
                    CharFlagsAndShiftState    flags,
                    word                      state)
void MSG_GEN_APPLICATION_INK_QUERY_REPLY(
                    InkReturnValue     inkReturnValue,
                    GStateHandle       inkGState)
ChunkHandle MSG_GEN_APPLICATION_GET_GCN_LIST_OF_LISTS()
(GEN_VISIBILITY_OUTPUT)
MSG_GEN_APPLICATION_VISIBILITY_NOTIFICATION()
void MSG_GEN_APPLICATION_BRING_WINDOW_TO_TOP(
                    optr window)
void MSG_GEN_APPLICATION_LOWER_WINDOW_TO_BOTTOM(
                    optr window)
AppAttachFlags MSG_GEN_APPLICATION_GET_ATTACH_FLAGS()
void MSG_GEN_APPLICATION_BUILD_DIALOG_FROM_TEMPLATE(
                    NewDialog          *retValue,
                    optr               template)
```

GenBooleanClass

```
@class GenBooleanClass, GenClass
```

Instance Data

```
word        GBI_identifier
```

Messages

```
word MSG_GEN_BOOLEAN_GET_IDENTIFIER()
void MSG_GEN_BOOLEAN_SET_IDENTIFIER(word identifier)
```

GenBooleanGroupClass

```
@class GenBooleanGroupClass, GenClass
```

Instance Data

```
word        GBGI_selectedBooleans = 0
word        GBGI_indeterminateBooleans = 0
word        GBGI_modifiedBooleans = 0
optr        GBGI_destination
Message     GBGI_applyMsg = 0
```

Variable Data

```
Message ATTR_GEN_BOOLEAN_GROUP_STATUS_MSG
Message ATTR_GEN_BOOLEAN_GROUP_STATUS_MSG;
optr ATTR_GEN_BOOLEAN_GROUP_LINK
    @reloc ATTR_GEN_BOOLEAN_GROUP_LINK, 0, optr
void ATTR_GEN_BOOLEAN_GROUP_INIT_FILE_BOOLEAN
```

Hints

```
void HINT_BOOLEAN_GROUP_SCROLLABLE
void HINT_BOOLEAN_GROUP_MINIMIZE_SIZE
void HINT_BOOLEAN_GROUP_CHECKBOX_STYLE
void HINT_BOOLEAN_GROUP_TOOLBOX_STYLE
```

Messages

```
void MSG_GEN_BOOLEAN_GROUP_SET_GROUP_STATE(
                            word selectedBooleans,
                            word indeterminateBooleans)
void MSG_GEN_BOOLEAN_GROUP_SET_GROUP_MODIFIED_STATE(
                            word setBooleans,
                            word clearBooleans)
word MSG_GEN_BOOLEAN_GROUP_GET_SELECTED_BOOLEANS()
word MSG_GEN_BOOLEAN_GROUP_GET_INDETERMINATE_BOOLEANS()
word MSG_GEN_BOOLEAN_GROUP_GET_MODIFIED_BOOLEANS()
void MSG_GEN_BOOLEAN_GROUP_SEND_STATUS_MSG(
                            word changedBooleans)
optr MSG_GEN_BOOLEAN_GROUP_GET_BOOLEAN_OPTR(
                            word identifier)
void MSG_GEN_BOOLEAN_GROUP_SET_BOOLEAN_STATE(
                            word identifier,
                            Boolean state)
void
MSG_GEN_BOOLEAN_GROUP_SET_BOOLEAN_INDETERMINATE_STATE(
                            word identifier,
                            Boolean indeterminateState)
void MSG_GEN_BOOLEAN_GROUP_SET_BOOLEAN_MODIFIED_STATE(
                            word identifier,
                            Boolean modifiedState)
```



```
Boolean MSG_GEN_BOOLEAN_GROUP_IS_BOOLEAN_SELECTED(
                            word identifier)
Boolean MSG_GEN_BOOLEAN_GROUP_IS_BOOLEAN_INDETERMINATE(
                            word identifier)
Boolean MSG_GEN_BOOLEAN_GROUP_IS_BOOLEAN_MODIFIED(
                            word identifier)
optr MSG_GEN_BOOLEAN_GROUP_GET_DESTINATION()
void MSG_GEN_BOOLEAN_GROUP_SET_DESTINATION(optr dest)
Message MSG_GEN_BOOLEAN_GROUP_GET_APPLY_MSG()
void MSG_GEN_BOOLEAN_GROUP_SET_APPLY_MSG(
                            Message message)
void MSG_GEN_BOOLEAN_GROUP_MAKE_BOOLEAN_VISIBLE(
                            word identifier)
word MSG_GEN_BOOLEAN_GROUP_SCAN_BOOLEANS(
                            byte flags,
                            word initialBoolean,
                            word scanAmount = bp)
void MSG_GEN_BOOLEAN_GROUP_REDRAW_BOOLEANS(word offset)
@prototype void GEN_BOOLEAN_GROUP_APPLY_MSG(
                            word selectedBooleans,
                            word indeterminateBooleans,
                            word modifiedBooleans)
@prototype void GEN_BOOLEAN_GROUP_STATUS_MSG(
                            word selectedBooleans,
                            word indeterminateBooleans,
                            word changedBooleans)
```

GenClass

```
@class GenClass, VisClass, master, variant
```

Instance Data

```
@link                   GI_link
@composite              GI_comp = GI_link
@visMoniker             GI_visMoniker
@kbdAccelerator         GI_kbdAccelerator
GenAttrs                GI_attrs = 0
GenStates               GI_states =
                              (GS_USABLE|GS_ENABLED)
```

Variable Data

```
GenFilePath  ATTR_GEN_PATH_DATA
void         TEMP_GEN_PATH_SAVED_DISK_HANDLE
void         ATTR_GEN_PROPERTY
void         ATTR_GEN_NOT_PROPERTY
DestinationClassArgs ATTR_GEN_DESTINATION_CLASS
    @reloc ATTR_GEN_DESTINATION_CLASS, 0, optr
char[]       ATTR_GEN_INIT_FILE_KEY
char[]       ATTR_GEN_INIT_FILE_CATEGORY
void         ATTR_GEN_INIT_FILE_PROPAGATE_TO_CHILDREN
ChunkHandle  ATTR_GEN_FEATURE_LINK
MemHandle    ATTR_GEN_WINDOW_CUSTOM_LAYER_ID
    @reloc ATTR_GEN_WINDOW_CUSTOM_LAYER_ID, 0, handle
Point        ATTR_GEN_POSITION
sword        ATTR_GEN_POSITION_X
sword        ATTR_GEN_POSITION_Y
void
 ATTR_GEN_SEND_APPLY_MSG_ON_APPLY_EVEN_IF_NOT_MODIFIED
void
 ATTR_GEN_SEND_APPLY_MSG_ON_APPLY_EVEN_IF_NOT_ENABLED
dword        ATTR_GEN_VISIBILITY_DATA
word         ATTR_GEN_VISIBILITY_MESSAGE
optr         ATTR_GEN_VISIBILITY_DESTINATION
    @reloc ATTR_GEN_VISIBILITY_DESTINATION, 0, optr
WinPriority  ATTR_GEN_WINDOW_CUSTOM_WINDOW_PRIORITY
LayerPriority ATTR_GEN_WINDOW_CUSTOM_LAYER_PRIORITY
MemHandle    ATTR_GEN_WINDOW_CUSTOM_PARENT
void         ATTR_GEN_WINDOW_CUSTOM_WINDOW
GenDefaultMonikerType ATTR_GEN_DEFAULT_MONIKER
char[]       ATTR_GEN_HELP_FILE
byte         ATTR_GEN_HELP_TYPE
void         ATTR_GEN_HELP_FILE_FROM_INIT_FILE
optr         ATTR_GEN_FOCUS_HELP
optr         ATTR_GEN_FOCUS_HELP_LIB
    @reloc ATTR_GEN_FOCUS_HELP_LIB, 0, optr
char[]       ATTR_GEN_HELP_CONTEXT
optr         ATTR_GEN_OUTPUT_TRAVEL_START
    @reloc ATTR_GEN_OUTPUT_TRAVEL_START, 0, optr
void   ATTR_GEN_USES_HIERARCHICAL_INIT_FILE_CATEGORY
void   ATTR_GEN_WINDOW_ACCEPT_INK_EVEN_IF_NOT_FOCUSED
KeyboardOverride ATTR_GEN_WINDOW_KBD_OVERRIDE;
Point        ATTR_GEN_WINDOW_KBD_POSITION;
```



Hints

```
void HINT_DUMMY
void HINT_FREQUENTLY_USED
void HINT_INFREQUENTLY_USED
void HINT_AN_ADVANCED_FEATURE
void HINT_DEFAULT_DEFAULT_ACTION
void HINT_ENSURE_TEMPORARY_DEFAULT
void HINT_SAME_CATEGORY_AS_PARENT
void HINT_SYS_MENU
void HINT_USE_TEXT_MONIKER
void HINT_USE_ICONIC_MONIKER
void HINT_DEFAULT_FOCUS
void HINT_DEFAULT_TARGET
void HINT_DEFAULT_MODEL
void HINT_PREVENT_DEFAULT_OVERRIDES
void HINT_PRESERVE_FOCUS
void HINT_DO_NOT_PRESERVE_FOCUS
void HINT_GENERAL_CONSUMER_MODE
void HINT_NEVER_ADOPT_MENUS
void HINT_ALWAYS_ADOPT_MENUS
void HINT_ALLOW_CHILDREN_TO_WRAP
void HINT_BOTTOM_JUSTIFY_CHILDREN
void HINT_CENTER_CHILDREN_HORIZONTALLY
void HINT_CENTER_CHILDREN_ON_MONIKERS
void HINT_CENTER_CHILDREN_VERTICALLY
void HINT_CENTER_MONIKER
SpecSizeSpec HINT_CUSTOM_CHILD_SPACING
void HINT_DONT_ALLOW_CHILDREN_TO_WRAP
void HINT_DONT_FULL_JUSTIFY_CHILDREN
void HINT_DONT_INCLUDE_ENDS_IN_CHILD_SPACING
void HINT_DO_NOT_USE_MONIKER
void HINT_DRAW_IN_BOX
void HINT_EXPAND_HEIGHT_TO_FIT_PARENT
void HINT_EXPAND_WIDTH_TO_FIT_PARENT
CompSizeHintArgs HINT_FIXED_SIZE
void HINT_FULL_JUSTIFY_CHILDREN_HORIZONTALLY
void HINT_FULL_JUSTIFY_CHILDREN_VERTICALLY
void HINT_INCLUDE_ENDS_IN_CHILD_SPACING
CompSizeHintArgs HINT_INITIAL_SIZE
void HINT_LEFT_JUSTIFY_CHILDREN
void HINT_LEFT_JUSTIFY_MONIKERS
void HINT_MAKE_REPLY_BAR
CompSizeHintArgs HINT_MAXIMUM_SIZE
CompSizeHintArgs HINT_MINIMUM_SIZE
void HINT_NO_TALLER_THAN_CHILDREN_REQUIRE
void HINT_NO_WIDER_THAN_CHILDREN_REQUIRE
void HINT_ORIENT_CHILDREN_HORIZONTALLY
void HINT_ORIENT_CHILDREN_VERTICALLY
void HINT_PLACE_MONIKER_ABOVE
void HINT_ALIGN_LEFT_MONIKER_EDGE_WITH_CHILD
void HINT_PLACE_MONIKER_TO_LEFT
void HINT_PLACE_MONIKER_TO_RIGHT
void HINT_RIGHT_JUSTIFY_CHILDREN
void HINT_TOP_JUSTIFY_CHILDREN
word HINT_WRAP_AFTER_CHILD_COUNT
void HINT_DIVIDE_WIDTH_EQUALLY
void HINT_DIVIDE_HEIGHT_EQUALLY
```

```
void HINT_NO_BORDERS_ON_MONIKERS
word HINT_GADGET_TEXT_COLOR
void HINT_POPS_UP_TO_RIGHT
void HINT_POPS_UP_BELOW
void HINT_SEEK_MENU_BAR
void HINT_AVOID_MENU_BAR
void HINT_NAVIGATION_ID
void HINT_NAVIGATION_NEXT_ID
void HINT_DISMISS_WHEN_DISABLED
void HINT_SEEK_X_SCROLLER_AREA
void HINT_SEEK_Y_SCROLLER_AREA
void HINT_SEEK_LEFT_OF_VIEW
void HINT_SEEK_TOP_OF_VIEW
void HINT_SEEK_RIGHT_OF_VIEW
void HINT_SEEK_BOTTOM_OF_VIEW
void HINT_USE_INITIAL_BOUNDS_WHEN_RESTORED
void HINT_KEEP_INITIALLY_ONSCREEN
void HINT_DONT_KEEP_INITIALLY_ONSCREEN
void HINT_KEEP_PARTIALLY_ONSCREEN
void HINT_KEEP_ENTIRELY_ONSCREEN
void HINT_KEEP_ENTIRELY_ONSCREEN_WITH_MARGIN
void HINT_DONT_KEEP_PARTIALLY_ONSCREEN
SpecWinSizePair
      HINT_POSITION_WINDOW_AT_RATIO_OF_PARENT
void HINT_STAGGER_WINDOW
void HINT_CENTER_WINDOW
void HINT_TILE_WINDOW
void HINT_POSITION_WINDOW_AT_MOUSE
void HINT_EXTEND_WINDOW_TO_BOTTOM_RIGHT
void HINT_EXTEND_WINDOW_NEAR_BOTTOM_RIGHT
void HINT_SIZE_WINDOW_AS_DESIRED
SpecWinSizePair HINT_SIZE_WINDOW_AS_RATIO_OF_PARENT
SpecWinSizePair HINT_SIZE_WINDOW_AS_RATIO_OF_FIELD
SpecWinSizePair HINT_POSITION_ICON_AS_RATIO_OF_FIELD
void HINT_TOOLBOX
void HINT_SAME_ORIENTATION_AS_PARENT
void HINT_SEEK_REPLY_BAR
void HINT_SHOW_SHORTCUT
void HINT_DRAW_SHORTCUT_BELOW
void HINT_CAN_CLIP_MONIKER_WIDTH
void HINT_CAN_CLIP_MONIKER_HEIGHT
void HINT_SHOW_ENTIRE_MONIKER
optr HINT_ALIGN_LEFT_EDGE_WITH_OBJECT
    @reloc HINT_ALIGN_LEFT_EDGE_WITH_OBJECT, 0, optr
optr HINT_ALIGN_TOP_EDGE_WITH_OBJECT
    @reloc HINT_ALIGN_TOP_EDGE_WITH_OBJECT, 0, optr
optr HINT_ALIGN_RIGHT_EDGE_WITH_OBJECT
    @reloc HINT_ALIGN_RIGHT_EDGE_WITH_OBJECT, 0, optr
optr HINT_ALIGN_BOTTOM_EDGE_WITH_OBJECT
    @reloc HINT_ALIGN_BOTTOM_EDGE_WITH_OBJECT, 0, optr
void HINT_DONT_SHOW_SHORTCUT
void HINT_MINIMIZE_CHILD_SPACING
BackgroundColors HINT_GADGET_BACKGROUND_COLORS
void HINT_ORIENT_CHILDREN_ALONG_LONGER_DIMENSION
void HINT_PLACE_MONKIER_ALONG_LARGER_DIMENSION
void HINT_WRAP_AFTER_CHILD_COUNT_IF_VERTICAL_SCREEN
void HINT_NOT_MOVABLE
```

```
void HINT_SEEK_TITLE_BAR_LEFT
void HINT_SEEK_TITLE_BAR_RIGHT
void HINT_WINDOW_NO_CONSTRAINTS
SpecSizeSpec HINT_CUSTOM_CHILD_SPACING_IF_LIMITED_SPACE
void HINT_WINDOW_NO_TITLE_BAR
void HINT_WINDOW_NO_SYS_MENU
SystemAttrs HINT_IF_SYSTEM_ATTRS
void HINT_ENDIF
void HINT_ELSE
```

Types and Flags

```
typedef enum /* word */ {
      TO_GEN_PARENT=_FIRST_GenClass,
      TO_FOCUS,
      TO_TARGET,
      TO_MODEL,
      TO_APP_FOCUS,
      TO_APP_TARGET,
      TO_APP_MODEL,
      TO_SYS_FOCUS,
      TO_SYS_TARGET,
      TO_SYS_MODEL
} GenTravelOption
typedef enum {
      BRPT_OUTPUT_OPTR
} BranchReplaceParamType
typedef enum {
      GUQT_UI_FOR_APPLICATION
      GUQT_UI_FOR_SCREEN
      GUQT_UI_FOR_FIELD
      GUQT_UI_FOR_MISC
      GUQT_FIELD
      GUQT_SCREEN
      GUQT_DELAYED_OPERATION
} GenUpwardQueryType
ByteFlags    NotifyEnabledFlags
      NEF_STATE_CHANGING        0x80
ByteEnum     RequestedViewArea
      RVA_NO_AREA_CHOICE        0
      RVA_X_SCROLLER_AREA       1
      RVA_Y_SCROLLER_AREA       2
      RVA_LEFT_AREA             3
      RVA_TOP_AREA              4
      RVA_RIGHT_AREA            5
      RVA_BOTTOM_AREA           6
WordFlags    GenFindObjectWithMonikerFlags
      GFTMF_EXACT_MATCH         0x8000
      GFTMF_SKIP_THIS_NODE      0x4000
typedef enum {
      GDMT_LEVEL_0
      GDMT_LEVEL_1
      GDMT_LEVEL_2
      GDMT_LEVEL_3
      GDMT_HELP
      GDMT_HELP_PRIMARY
} GenDefaultMonikerType
WordFlags    SystemAttrs
```

```
      SA_NOT                    0x8000
      SA_TINY                   0x4000
      SA_HORIZONTALLY_TINY      0x2000
      SA_VERTICALLY_TINY        0x1000
      SA_COLOR                  0x0800
      SA_PEN_BASED              0x0400
      SA_KEYBOARD_ONLY          0x0200
      SA_NO_KEYBOARD            0x0100
ByteEnum GCMIcon
      GCMI_NONE    0
      GCMI_EXIT    1
      GCMI_HELP    2
ByteFlags GeneralConsumerModeFlags
      GCMF_LEFT_ICON            0x38
      GCMF_RIGHT_ICON           0x07
ByteEnum     GenUILevel
      GUIL_DEFAULT              0
      GUIL_GCM                  1
      GUIL_NOVICE               2
      GUIL_ADVANCED             3
ByteFlags GenAttrs
      GA_SIGNAL_INTERACTION_COMPLETE  0x80
      GA_INITIATES_BUSY_STATE         0x40
      GA_INITIATES_INPUT_HOLD_UP      0x20
      GA_INITIATES_INPUT_IGNORE       0x10
      GA_READ_ONLY                    0x08
      GA_KBD_SEARCH_PATH              0x04
      GA_TARGETABLE                   0x02
      GA_NOTIFY_VISIBILITY            0x01
ByteFlags GenStates
      GS_USABLE          0x80
      GS_ENABLED         0x40
typedef ByteEnum DefaultActionMode
      DAM_ACTIVATE_INTERACTION_DEFAULT0
      DAM_NAVIGATE_TO_NEXT_FIELD      1
      DAM_APPL_VERIFY                 2
      DAM_APPL_CUSTOM                 3
DAM_TAKES_DEFAULT         DAM_NAVIGATE_TO_NEXT_FIELD
DAM_APPL_HANDLES          DAM_APPL_VERIFY
ByteFlags GadgetAttrs
      GA_DELAYED                0x80
      GA_DISPLAY_ONLY           0x40
      GA_IN_ADD_MODE            0x20
      GA_SEND_USER_CHANGES      0x10
      GA_SEND_REDUNDANT_CHANGES 0x08
      GA_DEFAULT_ACTIONS        0x06
      GA_USER_ACTION_DETERMINES_MESSAGE 0x01
GA_DEFAULT_ACTIONS_OFFSET 1
ByteFlags GadgetActionFlags
      GAF_ACTUAL_CHANGE         0x04
      GAF_USER_CHANGE           0x02
      GAF_DEFAULT_ACTION_REQUEST 0x01
ByteFlags GadgetChangeFlags
      GCF_NO_USER_CHANGE        0x04
      GCF_SUPPRESS_APPLY        0x02
      GCF_SUPPRESS_DRAW         0x01
```

Structures

```
typedef struct {
    word            SSA_width
    word            SSA_height
    word            SSA_count
    VisUpdateMode   SSA_updateMode
} SetSizeArgs
typedef struct {
    word            GSA_width
    word            GSA_height
    word            GSA_unused
    word            GSA_count
} GetSizeArgs
typedef struct {
    word            GRP_ax
    word            GRP_bp
    word            GRP_cx
    word            GRP_dx
} GenReturnParams
typedef struct {
    optr            GGFI_optr
    word            GGFI_window
    word            GGFI_unused
} GenGupFieldInfo
typedef struct {
    optr            GGSI_optr
    word            GGSI_window
    word            GGSI_unused
} GenGupScreenInfo
typedef struct {
    byte            GFVRP_hViewArea
    byte            GFVRP_vViewArea
    ChunkHandle     GFVRP_hRange
    ChunkHandle     GFVRP_vRange
} GenFindViewRangesParams
typedef struct {
    char     GOP_category[INI_CATEGORY_BUFFER_SIZE]
    char     GOP_key[INI_CATEGORY_BUFFER_SIZE]
} GenOptionsParams
typedef struct {
    DiskHandle      GFP_disk
    PathName        GFP_path
} GenFilePath
typedef struct {
    ClassStruct     *DCA_class
} DestinationClassArgs
typedef struct {
    SpecWidth       CSHA_width
    SpecHeight      CSHA_height
    sword           CSHA_count
} CompSizeHintArgs
typedef struct {
    SpecWidth       GSHA_width
    SpecHeight      GSHA_height
} GadgetSizeHintArgs
typedef struct {
    word            HE_type
```

```
    word            HE_size
} HintEntry
typedef struct {
    byte            BC_unselectedColor1
    byte            BC_unselectedColor2
    byte            BC_selectedColor1
    byte            BC_selectedColor2
} BackgroundColors
```




Macros

```
GET_MM_AND_TYPE(M,T)      ((M) | ((T) << 8))
GET_CHAR_AND_SHIFT(C,S)   ((C) | (((word) (S)) << 8))
GET_VIEW_AREAS(H,V)       ((H) | (((word) (V)) << 8))
ObjDerefGen(obj)          ObjDeref2(obj)
```

Messages

```
@exportMessages GenSpecMessages,
                        DEFAULT_EXPORTED_MESSAGES_3
@exportMessages GenAppMessages,
                        DEFAULT_EXPORTED_MESSAGES_5
void MSG_GEN_SET_ENABLED(VisUpdateMode updateMode)
void MSG_GEN_SET_NOT_ENABLED(VisUpdateMode updateMode)
Boolean MSG_GEN_GET_ENABLED()
void MSG_GEN_SET_USABLE(VisUpdateMode updateMode)
void MSG_GEN_SET_NOT_USABLE(VisUpdateMode updateMode)
Boolean MSG_GEN_GET_USABLE()
Boolean MSG_GEN_CHECK_IF_FULLY_ENABLED()
Boolean MSG_GEN_CHECK_IF_FULLY_USABLE()
void MSG_GEN_SET_ATTRS(
                    GenAttrs attrsToSet,
                    GenAttrs attrsToClear)
GenAttrs MSG_GEN_GET_ATTRIBUTES()
ChunkHandle MSG_GEN_GET_VIS_MONIKER()
void MSG_GEN_USE_VIS_MONIKER(
                    ChunkHandle         moniker,
                    VisUpdateMode       updateMode)
ChunkHandle MSG_GEN_REPLACE_VIS_MONIKER(@stack
                    VisUpdateMode       updateMode,
                    word height, word width,
                    word length,
                    VisMonikerDataType  dataType,
                    VisMonikerSourceType sourceType,
                    dword               source)
ChunkHandle MSG_GEN_REPLACE_VIS_MONIKER_OPTR(
                    optr source,
                    VisUpdateMode updateMode)
ChunkHandle MSG_GEN_REPLACE_VIS_MONIKER_TEXT(
                    char                *source,
                    VisUpdateMode       updateMode)
ChunkHandle MSG_GEN_CREATE_VIS_MONIKER(@stack
                    CreateVisMonikerFlags flags,
                    word height, word width,
                    word                length,
                    VisMonikerDataType  dataType,
                    VisMonikerSourceType sourceType,
                    dword               source)
void MSG_GEN_DRAW_MONIKER(
                    DrawMonikerFlags monikerFlags,
```

```
                    word textHeight,
                    GStateHandle gState,
                    word yMaximum, word xMaximum,
                    word yInset, word xInset)
XYValueAsDWord MSG_GEN_GET_MONIKER_POS(
                    DrawMonikerFlags monikerFlags,
                    word textHeight,
                    GStateHandle gState, word yMaximum,
                    word xMaximum, word yInset,
                    word xInset)
SizeAsDWord MSG_GEN_GET_MONIKER_SIZE(
                    word textHeight,
                    GStateHandle gState)
optr MSG_GEN_FIND_MONIKER( Boolean useAppMonikerList ,
                    VisMonikerSearchFlags searchFlags,
                    MemHandle destBlock)
void MSG_GEN_RELOC_MONIKER_LIST(
                    optr        monikerList,
                    Boolean     relocFlag)
void MSG_GEN_SET_KBD_ACCELERATOR(
                    KeyboardShortcut accelerator,
                    VisUpdateMode updateMode)
KeyboardShortcut MSG_GEN_GET_KBD_ACCELERATOR()
void MSG_GEN_CHANGE_ACCELERATOR(
                    KeyboardShortcut bitsToClear,
                    KeyboardShortcut bitsToSet)
void MSG_GEN_ADD_CHILD(
                    optr child, CompChildFlags flags)
void MSG_GEN_REMOVE_CHILD(
                    optr child, CompChildFlags flags)
void MSG_GEN_ADD_CHILD_UPWARD_LINK_ONLY(optr child)
void MSG_GEN_MOVE_CHILD(
                    optr child, CompChildFlags flags)
word MSG_GEN_FIND_CHILD(optr child)
optr MSG_GEN_FIND_CHILD_AT_POSITION(word position)
optr MSG_GEN_FIND_PARENT()
word MSG_GEN_COUNT_CHILDREN()
void MSG_GEN_CALL_PARENT(EventHandle event )
void MSG_GEN_SEND_TO_PARENT(EventHandle event)
void MSG_GEN_SEND_TO_CHILDREN(EventHandle event)
void MSG_GEN_GUP_CALL_OBJECT_OF_CLASS(EventHandle event)
void MSG_GEN_GUP_SEND_TO_OBJECT_OF_CLASS(
                    EventHandle event)
Boolean MSG_GEN_GUP_TEST_FOR_OBJECT_OF_CLASS(
                    ClassStruct *class)
optr MSG_GEN_GUP_FIND_OBJECT_OF_CLASS(
                    ClassStruct *class)
void MSG_GEN_CALL_APPLICATION(EventHandle event)
void MSG_GEN_SEND_TO_PROCESS(EventHandle event)
void MSG_GEN_CALL_SYSTEM(EventHandle event)
void MSG_GEN_OUTPUT_ACTION(EventHandle event, optr dest)
optr MSG_GEN_COPY_TREE(
                    MemHandle destBlock,
                    ChunkHandle parentChunk,
                    CompChildFlags flags)
void MSG_GEN_DESTROY(
                    VisUpdateMode updateMode,
```

```
                    CompChildFlags flags)
void MSG_GEN_BRANCH_REPLACE_PARAMS(
                    BranchReplaceParamType type,
                    dword replaceParam,
                    dword searchParam)
void MSG_GEN_BRANCH_REPLACE_OUTPUT_OPTR_CONSTANT(
                    optr replacementOptr,
                    word searchConstant)
void MSG_GEN_BRING_TO_TOP()
void MSG_GEN_LOWER_TO_BOTTOM()
void MSG_GEN_SET_WIN_POSITION(
                    WinPositionType modeAndType,
                    SpecWinSizeSpec xPosSpec,
                    SpecWinSizeSpec yPosSpec)
void MSG_GEN_SET_WIN_SIZE(
                    WinPositionType modeAndType,
                    SpecWinSizeSpec xSizeSpec,
                    SpecWinSizeSpec ySizeSpec)
void MSG_GEN_RESET_TO_INITIAL_SIZE(
                    VisUpdateMode updateMode)
void MSG_GEN_SET_WIN_CONSTRAIN(
                    VisUpdateMode updateMode,
                    WinConstrainType constrainType)
void MSG_GEN_SET_INITIAL_SIZE(@stack
                    VisUpdateMode updateMode,
                    word count,
                    SpecHeight height, SpecWidth width)
void MSG_GEN_SET_MINIMUM_SIZE(@stack
                    VisUpdateMode updateMode,
                    word count,
                    SpecHeight height, SpecWidth width)
void MSG_GEN_SET_MAXIMUM_SIZE(@stack
                    VisUpdateMode updateMode,
                    word count,
                    SpecHeight height, SpecWidth width)
void MSG_GEN_SET_FIXED_SIZE(@stack
                    VisUpdateMode updateMode,
                    word count,
                    SpecHeight height, SpecWidth width)
void MSG_GEN_GET_INITIAL_SIZE(GetSizeArgs *initSize)
void MSG_GEN_GET_MINIMUM_SIZE(GetSizeArgs *minSize )
void MSG_GEN_GET_MAXIMUM_SIZE(GetSizeArgs *maxSize)
void MSG_GEN_GET_FIXED_SIZE(GetSizeArgs *fixedSize)
void MSG_GEN_UPDATE_VISUAL(VisUpdateMode updateMode)
void MSG_GEN_APPLY()
void MSG_GEN_RESET()
Boolean MSG_GEN_PRE_APPLY()
void MSG_GEN_POST_APPLY()
void MSG_GEN_MAKE_APPLYABLE()
void MSG_GEN_MAKE_NOT_APPLYABLE()
void MSG_GEN_ACTIVATE()
Boolean MSG_GEN_ACTIVATE_INTERACTION_DEFAULT()
void MSG_GEN_NAVIGATE_TO_NEXT_FIELD()
void MSG_GEN_NAVIGATE_TO_PREVIOUS_FIELD()
Boolean MSG_GEN_NAVIGATION_QUERY(
                    NavigationQueryParams *retValue,
                    optr queryOrigin,
```



```
                    NavigationFlags navFlags)
Boolean MSG_GEN_GUP_QUERY(
                    GenReturnParams *retValue,
                    GenUpwardQueryType queryType)
Boolean MSG_GEN_GUP_QUERY_FOR_FIELD(
                    GenGupFieldInfo *retValue,
                    GenUpwardQueryType queryType)
Boolean MSG_GEN_GUP_QUERY_FOR_SCREEN(
                    GenGupScreenInfo *retValue,
                    GenUpwardQueryType queryType)
Boolean MSG_GEN_GUP_INTERACTION_COMMAND(
                    InteractionCommand command)
Boolean MSG_GEN_FIND_KBD_ACCELERATOR(
                    word charValue,
                    word charFlagsAndShiftState,
                    word toggleStateAndScanCode)
void MSG_GEN_SET_KBD_MKR_PATH()
Boolean MSG_GEN_PATH_SET(char *path, DiskHandle disk)
Boolean MSG_GEN_PATH_GET(char *buffer, word bufSize )
@alias (MSG_GEN_PATH_GET) MemHandle
            MSG_GEN_PATH_GET_BLOCK(
                              char *buffer, word bufSize)
DiskHandle MSG_GEN_PATH_GET_DISK_HANDLE()
optr MSG_GEN_FIND_OBJECT_WITH_TEXT_MONIKER(
                    char *text,
                    GenFindObjectWithMonikerFlags flags)
void MSG_GEN_GUP_FINISH_QUIT(
                    Boolean abortFlag,
                    Boolean notifyParent)
void MSG_GEN_REMOVE(VisUpdateMode updateMode,
                    CompChildFlags flags)
Boolean MSG_GEN_DESTROY_AND_FREE_BLOCK()
void MSG_GEN_SET_KBD_OVERRIDE(KeyboardOverride override)
void MSG_GEN_SET_KBD_POSITION(
                    sword xCoord, sword yCoord)

@prototype void GEN_VISIBILITY_OUTPUT(
                    optr obj, Boolean opening)
```

Routines

```
word * ObjDerefGen(obj)
word GenCopyChunk(  MemHandle destBlock, MemHandle blk,
                    ChunkHandle chnk,
                    CompChildFlags flags)
void GenInsertChild(MemHandle mh, ChunkHandle chnk,
                    optr childToAdd,
                    optr referenceChild,
                    CompChildFlags flags)
void GenSetUpwardLink(MemHandle mh, ChunkHandle chnk,
                    optr parent)
void GenRemoveDownwardLink(MemHandle mh,
                    ChunkHandle chnk,
                    CompChildFlags flags)
void GenSpecShrink( MemHandle mh, ChunkHandle chnk)
void GenProcessGenAttrsBeforeAction(
                    MemHandle mh, ChunkHandle chnk)
void GenProcessGenAttrsAfterAction(
```

```
                    MemHandle mh, ChunkHandle chnk)
optr GenFindObjectInTree(optr startObject,
                    dword childTable)
```

GenContentClass


```
@class GenContentClass, GenClass
```

Instance Data

```
byte        GCI_attrs = 0 /* VisContentAttrs */
optr        GCI_genView
```

Hints

```
void HINT_CONTENT_KEEP_FOCUS_VISIBLE
```

Messages

```
byte MSG_GEN_CONTENT_GET_ATTRS()
void MSG_GEN_CONTENT_SET_ATTRS(
                    byte attrsToSet, byte attrsToClear)
```

GenControlClass

```
@class GenControlClass, GenInteractionClass
```

Instance Data

```
optr GCI_output
    @default GI_states = (@default & ~GS_ENABLED)
```

Variable Data

```
TempGenControlInstance TEMP_GEN_CONTROL_INSTANCE
WordFlags ATTR_GEN_CONTROL_REQUIRE_UI
WordFlags ATTR_GEN_CONTROL_REQUIRE_TOOLBOX_UI
WordFlags ATTR_GEN_CONTROL_PROHIBIT_UI
WordFlags ATTR_GEN_CONTROL_PROHIBIT_TOOLBOX_UI
optr ATTR_GEN_CONTROL_APP_UI
    @reloc ATTR_GEN_CONTROL_APP_UI, 0, optr
optr ATTR_GEN_CONTROL_APP_TOOLBOX_UI
    @reloc ATTR_GEN_CONTROL_APP_TOOLBOX_UI, 0, optr
void TEMP_GEN_CONTROL_OPTIONS_LOADED
void ATTR_GEN_CONTROL_DO_NOT_USE_LIBRARY_NAME_FOR_HELP
```

Hints

```
GenControlUserData HINT_GEN_CONTROL_MODIFY_INITIAL_UI
GenControlUserData HINT_GEN_CONTROL_USER_MODIFIED_UI
void HINT_GEN_CONTROL_TOOLBOX_ONLY
GenControlScalableUIEntry
                    HINT_GEN_CONTROL_SCALABLE_UI_DATA
void
HINT_GEN_CONTROL_DESTROY_CHILDREN_WHEN_NOT_INTERACTABLE
```

Types and Flags

```
ByteFlags GenControlFeatureFlags
ByteFlags GenControlChildFlags
      GCCF_NOTIFY_WHEN_ADDING         0x04
      GCCF_ALWAYS_ADD                 0x02
      GCCF_IS_DIRECTLY_A_FEATURE      0x01
WordFlags GenControlBuildFlags
      GCBF_SUSPEND_ON_APPLY           0x8000
      GCBF_USE_GEN_DESTROY            0x4000
      GCBF_SPECIFIC_UI                0x2000
      GCBF_CUSTOM_ENABLE_DISABLE      0x1000
      GCBF_ALWAYS_UPDATE              0x0800
      GCBF_EXPAND_TOOL_WIDTH_TO_FIT_PARENT 0x0400
      GCBF_ALWAYS_INTERACTIBLE        0x0200
      GCBF_ALWAYS_ON_GCN_LIST         0x0100
      GCBF_MANUALLY_REMOVE_FROM_ACTIVE_LIST 0x0080
      GCBF_IS_ON_ACTIVE_LIST          0x0040
      GCBF_IS_ON_START_LOAD_OPTIONS_LIST 0x0020
      GCBF_NOT_REQUIRED_TO_BE_ON_SELF_LOAD_OPTIONS_LIST
                                      0x0010
#define GCBF_DO_NOT_DESTROY_CHILDREN_WHEN_CLOSED
                                      0x0008
typedef enum /* word */ {
    GCUIT_NORMAL,
    GCUIT_TOOLBOX
} GenControlUIType
WordFlags GenControlInteractableFlags
      GCIF_CONTROLLER                 0x8000
      GCIF_TOOLBOX_UI                 0x0002
```

```
        GCIF_NORMAL_UI                  0x0001
    WordFlags GenControlStatusChange
        GCSF_HIGHLIGHTED_TOOLGROUP_SELECTED 0X0004
        GCSF_TOOLBOX_FEATURES_CHANGED   0x0002
        GCSF_NORMAL_FEATURES_CHANGED    0x0001
    WordFlags GenControlUserFlags
        GCUF_USER_TOOLBOX_UI      0x0002
        GCUF_USER_UI              0x0001
    ByteEnum     GenControlScalableUICommand
        GCSUIC_SET_NORMAL_FEATURES_IF_APP_FEATURE_ON 0
        GCSUIC_SET_TOOLBOX_FEATURES_IF_APP_FEATURE_ON1
        GCSUIC_SET_NORMAL_FEATURES_IF_APP_FEATURE_OFF2
        GCSUIC_SET_TOOLBOX_FEATURES_IF_APP_FEATURE_OFF 3
        GCSUIC_SET_NORMAL_FEATURES_IF_APP_LEVEL      4
        GCSUIC_SET_TOOLBOX_FEATURES_IF_APP_LEVEL     5
        GCSUIC_ADD_NORMAL_FEATURES_IF_APP_FEATURE_ON 6
        GCSUIC_ADD_TOOLBOX_FEATURES_IF_APP_FEATURE_ON7
```

Structures

```
typedef struct {
    ChunkHandle           GCFI_object
    optr                  GCFI_name
    GenControlFeatureFlags GCFI_flags
} GenControlFeaturesInfo
typedef struct {
    ChunkHandle           GCCI_object
    WordFlags             GCCI_featureMask
    GenControlChildFlags  GCCI_flags
} GenControlChildInfo
typedef struct {
    GenControlBuildFlags  GCBI_flags
    const char            *GCBI_initFileKey
    const GCNListType     *GCBI_gcnList
    word                  GCBI_gcnCount
    const NotificationType *GCBI_notificationList
    word                  GCBI_notificationCount
    optr                  GCBI_controllerName
    MemHandle             GCBI_dupBlock
    const GenControlChildInfo *GCBI_childList
    word                  GCBI_childCount
    const GenControlFeaturesInfo *GCBI_featuresList
    word                  GCBI_featuresCount
    WordFlags             GCBI_features
    MemHandle             GCBI_toolBlock
    const GenControlChildInfo *GCBI_toolList
    word                  GCBI_toolCount
    const GenControlFeaturesInfo *GCBI_toolFeaturesList
    word                  GCBI_toolFeaturesCount
    WordFlags             GCBI_toolFeatures
    char                  *GCBI_helpContext
    byte                  GCBI_reserved[8]
} GenControlBuildInfo
typedef struct {
    WordFlags      GCSI_userAdded
    WordFlags      GCSI_userRemoved
    WordFlags      GCSI_appRequired
    WordFlags      GCSI_appProhibited
```

```
} GenControlScanInfo
typedef struct {
    WordFlags       GCSR_features
    WordFlags       GCSR_required
    WordFlags       GCSR_prohibited
    WordFlags       GCSR_supported
} GenControlGetFeaturesReturn
typedef struct {
    optr                    NGCS_controller
    GenControlStatusChange NGCS_statusChange
} NotifyGenControlStatusChange
typedef struct {
    GenControlInteractableFlags TGCI_interactableFlags
    MemHandle               TGCI_childBlock
    MemHandle               TGCI_toolBlock
    MemHandle               TGCI_toolParent
    WordFlags               TGCI_features
    WordFlags               TGCI_toolboxFeatures
    GCNListType             TGCI_activeNotificationList
    GenControlInteractableFlags TGCI_upToDate
} TempGenControlInstance
typedef struct {
    GenControlUserFlags   GCUD_flags
    word                  GCUD_userAddedUI
    word                  GCUD_userRemovedUI
    word                  GCUD_userAddedToolboxUI
    word                  GCUD_userRemovedToolboxUI
} GenControlUserData
typedef struct {
    GenControlScalableUICommand GCSUIE_command
    WordFlags               GCSUIE_appFeature
    WordFlags               GCSUIE_newFeatures
} GenControlScalableUIEntry
```

Messages

```
void MSG_GEN_CONTROL_GET_INFO(GenControlBuildInfo *info)
void MSG_GEN_CONTROL_GENERATE_UI()
void MSG_GEN_CONTROL_DESTROY_UI()
void MSG_GEN_CONTROL_GENERATE_TOOLBOX_UI(optr parent)
void MSG_GEN_CONTROL_DESTROY_TOOLBOX_UI()
void MSG_GEN_CONTROL_UPDATE_UI(@stack
                    MemHandle toolBlock,
                    MemHandle childBlock,
                    WordFlags toolboxFeatures,
                    WordFlags features,
                    MemHandle data,
                    word changeID,
                    ManufacturerID manufID)
void MSG_GEN_CONTROL_ENABLE_DISABLE(
                    Message msg,
                    VisUpdateMode updateMode)
void MSG_GEN_CONTROL_SCAN_FEATURE_HINTS(
                    GenControlUIType type,
                    GenControlScanInfo *info)
void MSG_GEN_CONTROL_ADD_FEATURE(WordFlags featureToAdd)
void MSG_GEN_CONTROL_REMOVE_FEATURE(
                    WordFlags featureToRemove)
```

```
void MSG_GEN_CONTROL_ADD_TOOLBOX_FEATURE(
                    WordFlags featureToAdd)
void MSG_GEN_CONTROL_REMOVE_TOOLBOX_FEATURE(
                    WordFlags featureToRemove)
void MSG_GEN_CONTROL_NOTIFY_INTERACTABLE(
                    GenControlInteractableFlags flags)
void MSG_GEN_CONTROL_NOTIFY_NOT_INTERACTABLE(
                    GenControlInteractableFlags flags)
void MSG_GEN_CONTROL_UNBUILD_NORMAL_UI_IF_POSSIBLE()
void MSG_GEN_CONTROL_ADD_TO_GCN_LISTS()
void MSG_GEN_CONTROL_REMOVE_FROM_GCN_LISTS()
MSG_GEN_CONTROL_GET_NORMAL_FEATURES(
                    GenControlGetFeaturesReturn *return)
MSG_GEN_CONTROL_GET_TOOLBOX_FEATURES(
                    GenControlGetFeaturesReturn *return)
void MSG_GEN_CONTROL_ADD_APP_UI(optr appUI)
void MSG_GEN_CONTROL_ADD_APP_TOOLBOX_UI(optr appUI)
void MSG_GEN_CONTROL_REBUILD_NORMAL_UI()
void MSG_GEN_CONTROL_REBUILD_TOOLBOX_UI()
void MSG_GEN_CONTROL_ADD_TO_UI()
void MSG_GEN_CONTROL_REMOVE_FROM_UI()
void MSG_GEN_CONTROL_OUTPUT_ACTION(EventHandle event)
void MSG_GEN_CONTROL_TWEAK_DUPLICATED_UI(
                    MemHandle childBlock,
                    WordFlags features)
void MSG_GEN_CONTROL_TWEAK_DUPLICATED_TOOLBOX_UI(
                    MemHandle toolBlock,
                    WordFlags toolboxFeatures)
void MSG_GEN_CONTROL_NOTIFY_ADDING_FEATURE(optr feature)
void MSG_GEN_CONTROL_FREE_OBJ_BLOCK(
                    MemHandle blockToFree)
```

GenDisplayClass

```
@class GenDisplayClass, GenClass
```

Instance Data

```
GenDisplayAttrs GDI_attributes = GDA_USER_DISMISSABLE
optr         GDI_document
    @default GI_attrs = (@default | GA_TARGETABLE |
                         GA_KBD_SEARCH_PATH)
```

Variable Data

```
ChunkHandle ATTR_GEN_DISPLAY_TRAVELING_OBJECTS
void ATTR_GEN_DISPLAY_NOT_MINIMIZABLE
void ATTR_GEN_DISPLAY_NOT_MAXIMIZABLE
void ATTR_GEN_DISPLAY_NOT_RESTORABLE
void ATTR_GEN_DISPLAY_MINIMIZED_STATE
void ATTR_GEN_DISPLAY_MAXIMIZED_STATE
void ATTR_GEN_DISPLAY_MENU_BAR_POPPED_OUT
```

Hints

```
void HINT_DISPLAY_MINIMIZED_ON_STARTUP
void HINT_DISPLAY_NOT_MINIMIZED_ON_STARTUP
void HINT_DISPLAY_MAXIMIZED_ON_STARTUP
void HINT_DISPLAY_NOT_MAXIMIZED_ON_STARTUP
void HINT_DISPLAY_NOT_RESIZABLE
```

Types and Flags

```
ByteFlags    GenDisplayAttrs
      GDA_USER_DISMISSABLE            0x80
```

Structures

```
typedef struct {
    optr           TIR_travelingObject
    ChunkHandle    TIR_parent
    word           TIR_compChildFlags
} TravelingObjectReference
```

Messages

```
void MSG_GEN_DISPLAY_SET_MINIMIZED()
void MSG_GEN_DISPLAY_SET_NOT_MINIMIZED()
Boolean MSG_GEN_DISPLAY_GET_MINIMIZED()
void MSG_GEN_DISPLAY_SET_MAXIMIZED()
void MSG_GEN_DISPLAY_SET_NOT_MAXIMIZED()
Boolean MSG_GEN_DISPLAY_GET_MAXIMIZED()
void MSG_GEN_DISPLAY_SET_ATTRS(byte attrs)
byte MSG_GEN_DISPLAY_GET_ATTRS()
optr MSG_GEN_DISPLAY_GET_DOCUMENT()
void MSG_GEN_DISPLAY_CLOSE()
```

GenDisplayControlClass

```
@class GenDisplayControlClass, GenControlClass
```

Instance Data

```
GenDisplayControlAttributes GDCII_attrs =
                    (GDCA_MAXIMIZED_NAME_ON_PRIMARY)
    @default GI_states = @default | GS_ENABLED
    @default GCI_output = (TO_APP_TARGET)
```

Variable Data

```
void TEMP_GDC_CACHED_NAME
NotifyDisplayListChange TEMP_GDC_CACHED_LIST_DATA
NotifyDisplayListChange TEMP_GDC_CACHED_TOOL_LIST_DATA
word TEMP_GDC_CACHED_SELECTED_DISPLAY
```

Hints

```
void HINT_DISPLAY_CONTROL_\
            NO_FEATURES_IF_TRANSPARENT_DOC_CTRL_MODE
```

Types and Flags

```
ByteFlags    GenDisplayControlAttributes
      GDCA_MAXIMIZED_NAME_ON_PRIMARY  0x80
MAX_DISPLAY_NAME_SIZE     64
WordFlags GDCFeatures
      GDCF_OVERLAPPING_MAXIMIZED      0x0004
      GDCF_TILE                       0x0002
      GDCF_DISPLAY_LIST               0x0001
WordFlags GDCToolboxFeatures
      GDCTF_OVERLAPPING_MAXIMIZED     0x0004
      GDCTF_TILE                      0x0002
      GDCTF_DISPLAY_LIST              0x0001
GDC_DEFAULT_FEATURES      (GDCF_OVERLAPPING_MAXIMIZED |
                           GDCF_TILE |
                           GDCF_DISPLAY_LIST)
GDC_DEFAULT_TOOLBOX_FEATURES (GDCF_DISPLAY_LIST)
ByteFlags    GenDisplayControlAttributes
      GDCA_MAXIMIZED_NAME_ON_PRIMARY  0x80
```

Structures

```
typedef struct {
    optr           NDC_display
    char           NDC_name[MAX_DISPLAY_NAME_SIZE]
    byte           NDC_overlapping
} NotifyDisplayChange
typedef struct {
    word           NDLC_counter
    optr           NDLC_group
} NotifyDisplayListChange
```

Messages

```
void MSG_GDC_SET_OVERLAPPING()
void MSG_GDC_TILE()
void MSG_GDC_LIST_APPLY()
void MSG_GDC_LIST_QUERY()
```




GenDisplayGroupClass

```
@class GenDisplayGroupClass, GenClass
```

Instance Data

```
@default       GI_attrs = @default | GA_TARGETABLE
```


Variable Data

```
void ATTR_GEN_DISPLAY_GROUP_NO_FULL_SIZED
void ATTR_GEN_DISPLAY_GROUP_NO_OVERLAPPING
void ATTR_GEN_DISPLAY_GROUP_OVERLAPPING_STATE
```

Hints

```
void HINT_DISPLAY_GROUP_SEPARATE_MENUS
void HINT_DISPLAY_GROUP_ARRANGE_TILED
void HINT_DISPLAY_GROUP_FULL_SIZED_ON_STARTUP
void HINT_DISPLAY_GROUP_OVERLAPPING_ON_STARTUP
void HINT_DISPLAY_GROUP_\
            FULL_SIZED_IF_TRANSPARENT_DOC_CTRL_MODE
void HINT_DISPLAY_GROUP_TILE_HORIZONTALLY
void HINT_DISPLAY_GROUP_TILE_VERTICALLY
void HINT_DISPLAY_GROUP_SIZE_INDEPENDENTLY_OF_DISPLAYS
```

Messages

```
void MSG_GEN_DISPLAY_GROUP_SET_FULL_SIZED()
void MSG_GEN_DISPLAY_GROUP_SET_OVERLAPPING()
Boolean MSG_GEN_DISPLAY_GROUP_GET_FULL_SIZED()
void MSG_GEN_DISPLAY_GROUP_TILE_DISPLAYS()
void MSG_GEN_DISPLAY_GROUP_SELECT_DISPLAY(
                          word displayNum)
```

GenDocumentClass

```
@class GenDocumentClass, GenContentClass
```

Instance Data

```
GenDocumentAttrs   GDI_attrs = 0
GenDocumentType    GDI_type = 0
word               GDI_operation = 0
FileHandle         GDI_fileHandle = 0
FileLongName       GDI_fileName = ""
MemHandle          GDI_display = 0
    @default GI_attrs = (@default | GA_KBD_SEARCH_PATH)
```

Variable Data

```
void ATTR_GEN_DOCUMENT_OPEN_FOR_IACP_ONLY
```

Types and Flags

```
WordFlags GenDocumentAttrs
      GDA_READ_ONLY             0x8000
      GDA_READ_WRITE            0x4000
      GDA_FORCE_DENY_WRITE      0x2000
      GDA_SHARED_MULTIPLE       0x1000
      GDA_SHARED_SINGLE         0x0800
      GDA_UNTITLED              0x0400
      GDA_DIRTY                 0x0200
      GDA_CLOSING               0x0100
      GDA_ATTACH_TO_DIRTY_FILE  0x0080
      GDA_SAVE_FAILED           0x0040
      GDA_OPENING               0x0020
      GDA_AUTO_SAVE_STOPPED     0x0010
      GDA_MODEL                 0x0008
      GDA_ON_WRITABLE_MEDIA     0x0004
      GDA_BACKUP_EXISTS         0x0002
      GDA_PREVENT_AUTO_SAVE     0x0001
typedef enum {
      GDT_NORMAL,
      GDT_READ_ONLY,
      GDT_TEMPLATE,
      GDT_READ_ONLY_TEMPLATE,
      GDT_PUBLIC,
      GDT_MULTI_USER
} GenDocumentType
typedef enum /* word */ {
      GDO_NORMAL,
      GDO_SAVE_AS,
      GDO_REVERT,
      GDO_REVERT_QUICK,
      GDO_ATTACH,
      GDO_DETACH,
      GDO_NEW,
      GDO_OPEN,
      GDO_SAVE,
      GDO_CLOSE,
      GDO_AUTO_SAVE
} GenDocumentOperation
typedef WordFlags DocumentOpenFlags
      DOF_CREATE_FILE_IF_FILE_DOES_NOT_EXIST 0x8000
      DOF_FORCE_TEMPLATE_BEHAVIOR            0x4000
```

```
        DOF_SAVE_AS_OVERWRITE_EXISTING_FILE    0x2000
        DOF_REOPEN                             0x1000
        DOF_RAISE_APP_AND_DOC                  0x0800
        DOF_NAME_HOLDS_FILE_ID                 0x0400
        DOF_FORCE_REAL_EMPTY_DOCUMENT          0x0200
        DOF_OPEN_FOR_IACP_ONLY                 0x0100
    GEN_DOCUMENT_GET_VARIABLE_BUFFER_SIZE  100
    GEN_DOCUMENT_GENERATE_NAME_ERROR       0xffff
    GEN_DOCUMENT_GENERATE_NAME_CANCEL      0xfffe
```

Structures

```
    typedef struct {
        FileLongName          DCP_name
        DiskHandle            DCP_diskHandle
        PathName              DCP_path
        GenDocumentAttrs      DCP_docAttrs
        DocumentOpenFlags     DCP_flags
        IACPConnection        DCP_connection
    } DocumentCommonParams
```

Messages

```
    Boolean MSG_GEN_DOCUMENT_INITIALIZE_DOCUMENT_FILE()
    Boolean MSG_GEN_DOCUMENT_IMPORT(
                    ImpexTranslationParams *params)
    void MSG_GEN_DOCUMENT_CREATE_UI_FOR_DOCUMENT()
    void MSG_GEN_DOCUMENT_DESTROY_UI_FOR_DOCUMENT()
    void MSG_GEN_DOCUMENT_ATTACH_UI_TO_DOCUMENT()
    void MSG_GEN_DOCUMENT_DETACH_UI_FROM_DOCUMENT()
    void MSG_GEN_DOCUMENT_EXPORT(
                    ImpexTranslationParams *params)
    void MSG_GEN_DOCUMENT_READ_CACHED_DATA_FROM_FILE()
    void MSG_GEN_DOCUMENT_WRITE_CACHED_DATA_TO_FILE(
                    Boolean isSave)
    void MSG_GEN_DOCUMENT_DOCUMENT_HAS_CHANGED()
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_SAVE(word *error)
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_UPDATE(word *error)
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_SAVE_AS(
                    word *fileOrError,
                    DocumentCommonParams *params)
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_SAVE_AS_FILE_HANDLE(
                    word *error, FileHandle file)
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_REVERT(word *error)
    Boolean
     MSG_GEN_DOCUMENT_UPDATE_EARLIER_COMPATIBLE_DOCUMENT()
    Boolean
    MSG_GEN_DOCUMENT_UPDATE_EARLIER_INCOMPATIBLE_DOCUMENT()
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_CREATE(
                    word *fileOrError,
                    DocumentCommonParams *params)
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_OPEN(
                    word *fileOrError,
                    DocumentCommonParams *params)
    void MSG_GEN_DOCUMENT_PHYSICAL_CLOSE()
    void MSG_GEN_DOCUMENT_PHYSICAL_DELETE()
    Boolean
       MSG_GEN_DOCUMENT_PHYSICAL_CHECK_FOR_MODIFICATIONS()
    Boolean MSG_GEN_DOCUMENT_PHYSICAL_COPY_TEMPLATE(
                    word *fileOrError)
```

```
void MSG_GEN_DOCUMENT_SAVE_AS_COMPLETED()
void MSG_GEN_DOCUMENT_ATTACH_FAILED()
void MSG_GEN_DOCUMENT_MARK_DIRTY()
word MSG_GEN_DOCUMENT_GENERATE_NAME_FOR_NEW(
                    char *buffer, word retryCount)
void MSG_GEN_DOCUMENT_SAVE_AS_CANCELLED()
byte MSG_GEN_DOCUMENT_COMPUTE_ACCESS_FLAGS(
                    DocumentCommonParams *params)
GenDocumentAttrs MSG_GEN_DOCUMENT_GET_ATTRS()
void MSG_GEN_DOCUMENT_ENABLE_AUTO_SAVE()
void MSG_GEN_DOCUMENT_DISABLE_AUTO_SAVE()
void MSG_GEN_DOCUMENT_GET_FILE_NAME(char *buffer)
FileHandle MSG_GEN_DOCUMENT_GET_FILE_HANDLE()
GenDocumentOperation MSG_GEN_DOCUMENT_GET_OPERATION()
optr MSG_GEN_DOCUMENT_GET_DISPLAY()
Boolean MSG_GEN_DOCUMENT_NEW(word *fileOrError,
                    DocumentCommonParams *params)
Boolean MSG_GEN_DOCUMENT_OPEN(word *fileOrError,
                    DocumentCommonParams *params)
word MSG_GEN_DOCUMENT_CLOSE(IACPConnection connection)
void MSG_GEN_DOCUMENT_QUICK_BACKUP()
void MSG_GEN_DOCUMENT_RECOVER_QUICK_BACKUP()
Boolean MSG_GEN_DOCUMENT_SAVE()
Boolean MSG_GEN_DOCUMENT_SAVE_AS(word *fileOrError,
                    DocumentCommonParams *params)
Boolean MSG_GEN_DOCUMENT_SAVE_AS_TEMPLATE(
                    word *fileOrError,
                    DocumentCommonParams *params)
Boolean MSG_GEN_DOCUMENT_COPY_TO(
                    DocumentCommonParams *params)
Boolean MSG_GEN_DOCUMENT_REVERT()
void MSG_GEN_DOCUMENT_EDIT_USER_NOTES()
void MSG_GEN_DOCUMENT_CHANGE_TYPE(GenDocumentType type)
void MSG_GEN_DOCUMENT_CHANGE_PASSWORD(char *password)
void MSG_GEN_DOCUMENT_RENAME()
void MSG_GEN_DOCUMENT_SET_EMPTY()
void MSG_GEN_DOCUMENT_CLEAR_EMPTY()
void MSG_GEN_DOCUMENT_SET_DEFAULT()
void MSG_GEN_DOCUMENT_CLEAR_DEFAULT()
void MSG_GEN_DOCUMENT_CLOSE_IF_CLEAN_UNNAMED()
Boolean MSG_GEN_DOCUMENT_SEARCH_FOR_DOC(
                    DocumentCommonParams *params)
Boolean MSG_GEN_DOCUMENT_TEST_FOR_FILE(
                    FileHandle file, optr *docFound)
void MSG_GEN_DOCUMENT_AUTO_SAVE()
void MSG_GEN_DOCUMENT_UPDATE(word *error)
void MSG_GEN_DOCUMENT_CHECK_FOR_MODIFICATIONS()
void MSG_GEN_DOCUMENT_CLOSE_FILE(
                    IACPConnection connection)
void MSG_GEN_DOCUMENT_REOPEN_FILE()
void MSG_GEN_DOCUMENT_GET_VARIABLE(@stack
                    VisTextGraphic *graphic,
                    char *buffer)
void MSG_GEN_DOCUMENT_REVERT_NO_PROMPT()
void
MSG_GEN_DOCUMENT_FILE_CHANGED_REINITIALIZE_CREATED_UI()
void MSG_GEN_DOCUMENT_CLOSE_IF_OPEN_FOR_IACP_ONLY()
```

GenDocumentControlClass

```
@class GenDocumentControlClass, GenControlClass
```

Instance Data

```
GeodeToken         GDCI_documentToken = {"",0}
GenFileSelectorType GDCI_selectorType = GFST_DOCUMENTS
GenDocumentControlAttrs GDCI_attrs =
            ((GDCM_SHARED_SINGLE << GDCA_MODE_OFFSET) |
            GDCA_VM_FILE | GDCA_SUPPORTS_SAVE_AS_REVERT
            | (GDCT_NEW << GDCA_CURRENT_TASK_OFFSET))
GenDocumentControlFeatures GDCI_features =
            (GDCF_READ_ONLY_SUPPORTS_SAVE_AS_REVERT |
            GDCF_SINGLE_FILE_CLEAN_CAN_NEW_OPEN |
            GDCF_SUPPORTS_TEMPLATES |
            GDCF_SUPPORTS_USER_SETTABLE_EMPTY_DOCUMENT
           | GDCF_SUPPORTS_USER_MAKING_SHARED_DOCUMENTS
            | GDCF_NAME_ON_PRIMARY)
GenDocumentAttrs   GDCI_docAttrs = 0
GenDocumentType    GDCI_docType = 0
FileHandle         GDCI_docFileHandle = 0
byte               GDCI_emptyExists = 0
byte               GDCI_defaultExists = 0
ChunkHandle        GDCI_enableDisableList
optr               GDCI_openGroup
optr               GDCI_importGroup
optr               GDCI_useTemplateGroup
optr               GDCI_saveAsGroup
optr               GDCI_exportGroup
optr               GDCI_dialogGroup
optr               GDCI_userLevelGroup
ChunkHandle        GDCI_noNameText
ChunkHandle        GDCI_defaultFile
ChunkHandle        GDCI_templateDir
optr               GDCI_documentGroup
FileLongName       GDCI_targetDocName = ""
ChunkHandle        GDCI_dialogNewText
ChunkHandle        GDCI_dialogTemplateText
ChunkHandle        GDCI_dialogOpenDefaultText
ChunkHandle        GDCI_dialogImportText
ChunkHandle        GDCI_dialogOpenText
ChunkHandle        GDCI_dialogUserLevelText
@visMoniker        GDCI_dialogNewMoniker
@visMoniker        GDCI_dialogTemplateMoniker
@visMoniker        GDCI_dialogOpenDefaultMoniker
@visMoniker        GDCI_dialogImportMoniker
@visMoniker        GDCI_dialogOpenMoniker
@visMoniker        GDCI_dialogUserLevelMoniker
    @default GI_states = @default | GS_ENABLED
    @default GI_attrs = @default | GA_KBD_SEARCH_PATH
```

Hints

```
void
HINT_GEN_DOCUMENT_CONTROL_NO_PROGRESS_DIALOG_ON_UPDATE_
MAJOR
void
HINT_GEN_DOCUMENT_CONTROL_PROGRESS_DIALOG_ON_UPDATE_MIN
OR
```

Variable Data

```
UIInterfaceLevel
    ATTR_GEN_DOCUMENT_CONTROL_NO_EMPTY_DOC_IF_NOT_ABOVE
```

Types and Flags

```
WordFlags    GDCFeatures
      GDCF_NEW                          0x4000
      GDCF_OPEN                         0x2000
      GDCF_CLOSE                        0x1000
      GDCF_QUICK_BACKUP                 0x0800
      GDCF_SAVE                         0x0400
      GDCF_SAVE_AS                      0x0200
      GDCF_COPY                         0x0100
      GDCF_EXPORT                       0x0080
      GDCF_REVERT                       0x0080
      GDCF_RENAME                       0x0040
      GDCF_EDIT_USER_NOTES              0x0020
      GDCF_SET_TYPE                     0x0010
      GDCF_SET_PASSWORD                 0x0008
      GDCF_SAVE_AS_TEMPLATE             0x0004
      GDCF_SET_EMPTY_DOCUMENT           0x0002
      GDCF_SET_DEFAULT_DOCUMENT         0x0001
WordFlags    GDCToolboxFeatures
      GDCTF_NEW_EMPTY                   0x0020
      GDCTF_USE_TEMPLATE                0x0010
      GDCTF_OPEN                        0x0008
      GDCTF_CLOSE                       0x0004
      GDCTF_SAVE                        0x0002
      GDCTF_QUICK_BACKUP                0x0001
GDC_SUGGESTED_INTRODUCTORY_FEATURES (0)
GDC_SUGGESTED_BEGINNING_FEATURES
            (GDC_SUGGESTED_INTRODUCTORY_FEATURES |
            GDCF_QUICK_BACKUP | GDCF_COPY | GDCF_RENAME
            | GDCF_EXPORT | GDCF_EDIT_USER_NOTES)
GDC_SUGGESTED_INTERMEDIATE_FEATURES
            (GDC_SUGGESTED_BEGINNING_FEATURES |
            GDCF_SAVE_AS | GDCF_REVERT |
            GDCF_SET_PASSWORD)
GDC_SUGGESTED_ADVANCED_FEATURES
            (GDC_SUGGESTED_INTERMEDIATE_FEATURES |
            GDCF_SET_TYPE | GDCF_SAVE_AS_TEMPLATE |
            GDCF_SET_EMPTY_DOCUMENT |
            GDCF_SET_DEFAULT_DOCUMENT)
ByteEnum     GDCTask
      GDCT_NONE                 0
      GDCT_NEW                  1
      GDCT_OPEN                 2
      GDCT_USE_TEMPLATE         3
      GDCT_SAVE_AS              4
      GDCT_COPY_TO              5
      GDCT_DIALOG               6
      GDCT_TYPE                 7
      GDCT_PASSWORD             8
ByteEnum     GenDocumentControlMode
      GDCM_VIEWER               0
      GDCM_SHARED_SINGLE        1
      GDCM_SHARED_MULTIPLE      2
```

```
WordFlags    GenDocumentControlAttrs
      GDCA_MULTIPLE_OPEN_FILES        0x8000
      GDCA_MODE                       0x6000
      GDCA_DOS_FILE_DENY_WRITE        0x1000
      GDCA_VM_FILE                    0x0800
      GDCA_NATIVE                     0x0400
      GDCA_SUPPORTS_SAVE_AS_REVERT    0x0200
      GDCA_DOCUMENT_EXISTS            0x0100
      GDCA_CURRENT_TASK               0x00F0
      GDCA_DO_NOT_SAVE_FILES          0x0008
GDCA_MODE_OFFSET          13
GDCA_CURRENT_TASK_OFFSET  5
WordFlags    GenDocumentControlFeatures
      GDCF_READ_ONLY_SUPPORTS_SAVE_AS_REVERT 0x8000
      GDCF_SINGLE_FILE_CLEAN_CAN_NEW_OPEN    0x4000
      GDCF_SUPPORTS_TEMPLATES                0x2000
      GDCF_SUPPORTS_USER_SETTABLE_EMPTY_DOCUMENT 0x1000
      GDCF_SUPPORTS_USER_SETTABLE_DEFAULT_DOCUMENT
                                              0x0800
      GDCF_SUPPORTS_USER_MAKING_SHARED_DOCUMENTS 0x0400
      GDCF_NAME_ON_PRIMARY                   0x0200
ByteEnum     GenFileSelectorType
      GFST_DOCUMENTS            0
      GFST_EXECUTABLES          1
      GFST_NON_GEOS_FILES       2
      GFST_ALL_FILES            3
```

Structures

```
typedef struct {
    FileLongName   DFCP_name
    DiskHandle     DFCP_diskHandle
    PathName       DFCP_path
    optr           DFCP_display
    optr           DFCP_document
} DocumentFileChangedParams
typedef struct {
    word           NDC_attrs    /* GenDocumentAttrs */
    word           NDC_type     /* GenDocumentType */
    FileHandle     NDC_fileHandle
    byte           NDC_emptyExists
    byte           NDC_defaultExists
} NotifyDocumentChange
```

Messages

```
void MSG_GEN_DOCUMENT_CONTROL_DISPLAY_DIALOG()
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_NEW_DOC()
void
MSG_GEN_DOCUMENT_CONTROL_INITATE_USE_TEMPLATE_DOC()
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_OPEN_DOC()
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_IMPORT_DOC()
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_SAVE_DOC()
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_SAVE_AS_DOC()
void
MSG_GEN_DOCUMENT_CONTROL_INITIATE_SAVE_AS_TEMPLATE_DOC(
)
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_COPY_TO_DOC()
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_EXPORT_DOC()
void MSG_GEN_DOCUMENT_CONTROL_INITIATE_SET_TYPE_DOC()
```

```
void
MSG_GEN_DOCUMENT_CONTROL_INITIATE_SET_PASSWORD_DOC()
GenDocumentControlAttrs
      MSG_GEN_DOCUMENT_CONTROL_GET_ATTRS()
GenDocumentControlFeatures
      MSG_GEN_DOCUMENT_CONTROL_GET_FEATURES()
void MSG_GEN_DOCUMENT_CONTROL_GET_TOKEN(
                        GeodeToken *token)
void MSG_GEN_DOCUMENT_CONTROL_GET_CREATOR(
                        GeodeToken *token)
void MSG_GEN_DOCUMENT_CONTROL_GET_TEMPLATE_DIR(
                        char   *buffer)
void MSG_GEN_DOCUMENT_CONTROL_SAVE_AS_CANCELLED()
void MSG_GEN_DOCUMENT_CONTROL_FILE_CHANGED(
                  DocumentFileChangedParams dup)
void MSG_GEN_DOCUMENT_CONTROL_CONFIGURE_FILE_SELECTOR(
                  optr fileSelector, word flags)
void MSG_GEN_DOCUMENT_CONTROL_OPEN_IMPORT_SELECTED(
                  ImpexTranslationParams *params)
void MSG_GEN_DOCUMENT_CONTROL_FILE_EXPORTED()
void MSG_GEN_DOCUMENT_CONTROL_OPEN_DEFAULT_DOC()
void MSG_GEN_DOCUMENT_CONTROL_IMPORT_CANCELLED()
```


GenDocumentGroupClass

```
@class GenDocumentGroupClass, GenClass
```

Instance Data

```
GenDocumentGroupAttrs GDGI_attrs = (GDGA_VM_FILE |
                  GDGA_SUPPORTS_AUTO_SAVE |
                  GDGA_AUTOMATIC_CHANGE_NOTIFICATION |
                  GDGA_AUTOMATIC_DIRTY_NOTIFICATION |
                  GDGA_APPLICATION_THREAD |
                  GDGA_AUTOMATIC_UNDO_INTERACTION |
                GDGA_CONTENT_DOES_NOT_MANAGE_CHILDREN)
ChunkHandle       GDGI_untitledName
ClassStruct       * GDGI_documentClass =
                  (ClassStruct *)&GenDocumentClass
optr              GDGI_genDocument
optr              GDGI_output
optr              GDGI_documentControl
optr              GDGI_genView
optr              GDGI_genDisplay
optr              GDGI_genDisplayGroup
word              GDGI_protocolMajor = 1
word              GDGI_protocolMinor = 0
```

Types and Flags

```
WordFlags    GenDocumentGroupAttrs
      GDGA_VM_FILE                          0x8000
      GDGA_NATIVE                           0x4000
      GDGA_SUPPORTS_AUTO_SAVE               0x2000
      GDGA_AUTOMATIC_CHANGE_NOTIFICATION    0x1000
      GDGA_AUTOMATIC_DIRTY_NOTIFICATION     0x0800
      GDGA_APPLICATION_THREAD               0x0400
      GDGA_VM_FILE_CONTAINS_OBJECTS         0x0200
      GDGA_CONTENT_DOES_NOT_MANAGE_CHILDREN 0x0100
```

```
        GDGA_LARGE_CONTENT                      0x0080
        GDGA_AUTOMATIC_UNDO_INTERACTION         0x0040
    typedef enum /* word */ {
        DQS_OK,
        DQS_CANCEL,
        DQS_DELAYED,
        DQS_SAVE_ERROR
    } DocQuitStatus
```

Messages

```
optr MSG_GEN_DOCUMENT_GROUP_NEW_DOC(
                    DocumentCommonParams *params)
optr MSG_GEN_DOCUMENT_GROUP_IMPORT_NEW_DOC(
                    ImpexTranslationParams *params)
optr MSG_GEN_DOCUMENT_GROUP_OPEN_DOC(
                    DocumentCommonParams *params)
void MSG_GEN_DOCUMENT_GROUP_MARK_DIRTY(optr document)
void MSG_GEN_DOCUMENT_GROUP_MARK_DIRTY_BY_FILE(
                    FileHandle file)
void MSG_GEN_DOCUMENT_GROUP_OPEN_DEFAULT_DOC(
                    DocumentCommonParams *params)
GenDocumentGroupAttrs
        MSG_GEN_DOCUMENT_GROUP_GET_ATTRS()
GenDocumentControlAttrs
        MSG_GEN_DOCUMENT_GROUP_GET_UI_ATTRS()
void MSG_GEN_DOCUMENT_GROUP_GET_TEMPLATE_DIR(
                    char *buffer)
GenDocumentControlFeatures
        MSG_GEN_DOCUMENT_GROUP_GET_UI_FEATURES()
word MSG_GEN_DOCUMENT_GROUP_GET_DEFAULT_NAME(
                    char *buffer)
optr MSG_GEN_DOCUMENT_GROUP_GET_OUTPUT()
void MSG_GEN_DOCUMENT_GROUP_SET_OUTPUT(optr output)
void MSG_GEN_DOCUMENT_GROUP_GET_TOKEN(GeodeToken *token)
void MSG_GEN_DOCUMENT_GROUP_GET_CREATOR(
                    GeodeToken *token)
dword MSG_GEN_DOCUMENT_GROUP_GET_PROTOCOL()
optr MSG_GEN_DOCUMENT_GROUP_GET_VIEW()
optr MSG_GEN_DOCUMENT_GROUP_GET_DISPLAY()
optr MSG_GEN_DOCUMENT_GROUP_GET_DISPLAY_GROUP()
optr MSG_GEN_DOCUMENT_GROUP_GET_DOC_BY_FILE(
                    FileHandle file)
void MSG_GEN_DOCUMENT_GROUP_SAVE_AS_CANCELLED()
Boolean MSG_GEN_DOCUMENT_GROUP_SEARCH_FOR_DOC(
                    DocumentCommonParams *params)
```

GenDynamicListClass

```
@class GenDynamicListClass, GenItemGroupClass
```

Instance Data

```
word        GDLI_numItems = 0
word        GDLI_queryMsg = 0
```


Types and Flags

```
typedef WordFlags ReplaceItemMonikerFlags;
RIMF_NOT_ENABLED          0x8000
GDLI_NO_CHANGE            0xffff
GDLP_FIRST                0x0000
GDLP_LAST                 0xffff
```

Messages

```
    @prototype void GEN_DYNAMIC_LIST_QUERY_MSG(
                        optr list, word item)
void MSG_GEN_DYNAMIC_LIST_REPLACE_ITEM_MONIKER(@stack
                    word item,
                    ReplaceItemMonikerFlags flags,
                    word height, word width,
                    word length,
                    VisMonikerDataType dataType,
                    VisMonikerSourceType sourceType,
                    dword source)
void MSG_GEN_DYNAMIC_LIST_REPLACE_ITEM_MONIKER_OPTR(
                    word item, optr moniker)
void MSG_GEN_DYNAMIC_LIST_REPLACE_ITEM_TEXT(
                    word item, char *text)
void MSG_GEN_DYNAMIC_LIST_INITIALIZE(word numItems)
void MSG_GEN_DYNAMIC_LIST_INITIALIZE(word numItems)
word MSG_GEN_DYNAMIC_LIST_GET_NUM_ITEMS()
void MSG_GEN_DYNAMIC_LIST_ADD_ITEMS(
                    word item, word numItems)
void MSG_GEN_DYNAMIC_LIST_REMOVE_ITEMS(
                    word item, word numItems)
void MSG_GEN_DYNAMIC_LIST_REMOVE_ITEM_LIST(
                    word *deletionList, word numItems)
```

GenEditControlClass

```
@class GenEditControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default | GS_ENABLED)
@default GI_attrs = (@default | GA_KBD_SEARCH_PATH)
```

Variable Data

```
NotifyUndoStateChange TEMP_UNDO_DESCRIPTION
word TEMP_CLIPBOARD_NOTIFICATION_LIST_COUNT
```

Types and Flags

```
WordFlags    GECFeatures
      GECF_UNDO                0x0020
      GECF_CUT                 0x0010
      GECF_COPY                0x0008
      GECF_PASTE               0x0004
      GECF_SELECT_ALL          0x0002
      GECF_DELETE              0x0001
WordFlags    GECToolboxFeatures
      GECTF_UNDO               0x0020
      GECTF_CUT                0x0010
      GECTF_COPY               0x0008
      GECTF_PASTE              0x0004
      GECTF_SELECT_ALL         0x0002
      GECTF_DELETE             0x0001
GEC_DEFAULT_FEATURES      (GECF_UNDO | GECF_CUT |
            GECF_COPY | GECF_PASTE |GECF_SELECT_ALL |
            GECF_DELETE)
GEC_DEFAULT_TOOLBOX_FEATURES (GECTF_UNDO | GECTF_CUT |
            GECTF_COPY | GECTF_PASTE | GECTF_SELECT_ALL
            | GECTF_DELETE)
typedef enum {
      SDT_TEXT,
      SDT_GRAPHICS,
      SDT_SPREADHSEET,
      SDT_INK,
      SDT_OTHER
} SelectionDataType
ByteEnum     UndoDescription
      UD_UNDO            0
      UD_REDO            1
      UD_NOT_UNDOABLE    2
```

Structures

```
typedef struct {
    SelectionDataType     NSSC_selectionType
    byte                  NSSC_clipboardableSelection
    byte                  NSSC_selectAllAvailable
    byte                  NSSC_deleteableSelection
    byte                  NSSC_pasteable
} NotifySelectStateChange
typedef struct {
    optr                  NUSC_undoTitle
    UndoDescription       NUSC_undoType
} NotifyUndoStateChange
```

GenFieldClass

```
@class GenFieldClass, GenClass
```

Instance Data

```
/* instance data is internal and should not be used */
GenFieldFlags      GFI_flags = 0
optr               GFI_visParent = 0
byte               GFI_numDetachedApps = 0
byte               GFI_numRestartedApps = 0
byte               GFI_numAttachingApps = 0
ChunkHandle        GFI_apps = 0
ChunkHandle        GFI_processes = 0
ChunkHandle        GFI_genApplications = 0
byte               GFI_numAppsToCheck = 0
optr               GFI_notificationDestination = 0
```

Messages

```
void MSG_GEN_FIELD_ADD_GEN_APPLICATION(
                   optr genApp,CompChildFlags flags)
void MSG_GEN_FIELD_ADD_APP_INSTANCE_REFERENCE(
                   Handle appInstanceReference,
                   Handle appObjectBlock)
void MSG_GEN_FIELD_ADD_APP_INSTANCE_REFERENCE_HOLDER(
                   Handle appObjectBlock)
void MSG_GEN_FIELD_RESET_BG()
void MSG_GEN_FIELD_ENABLE_BITMAP()
void MSG_GEN_FIELD_EXIT_TO_DOS()
void MSG_GEN_FIELD_OPEN_WINDOW_LIST()
optr MSG_GEN_FIELD_GET_TOP_GEN_APPLICATION()
UILaunchModel MSG_GEN_FIELD_GET_LAUNCH_MODEL()
```




GenFileSelectorClass

```
@class GenFileSelectorClass, GenClass
```

Instance Data

```
GenFileSelectorSelection  GFSI_selection = {0}
FileSelectorFileCriteria  GFSI_fileCriteria =
            (FSFC_DIRS|FSFC_NON_GEOS_FILES |
             FSFC_GEOS_EXECUTABLES |
             FSFC_GEOS_NON_EXECUTABLES)
FileSelectorAttrs         GFSI_attrs =
            (FSA_ALLOW_CHANGE_DIRS |
             FSA_HAS_CLOSE_DIR_BUTTON |
             FSA_HAS_OPEN_DIR_BUTTON |
             FSA_HAS_DOCUMENT_BUTTON |
             FSA_HAS_CHANGE_DIRECTORY_LIST |
             FSA_HAS_CHANGE_DRIVE_LIST|
             FSA_HAS_FILE_LIST)
optr                      GFSI_destination
Message                   GFSI_notificationMsg
```

Variable Data

```
void TEMP_GEN_FILE_SELECTOR_DATA
GeodeToken ATTR_GEN_FILE_SELECTOR_TOKEN_MATCH
GeodeToken ATTR_GEN_FILE_SELECTOR_CREATOR_MATCH
GenFileSelectorFileAttrs
            ATTR_GEN_FILE_SELECTOR_FILE_ATTR
GenFileSelectorGeodeAttrs
            ATTR_GEN_FILE_SELECTOR_GEODE_ATTR
GenFileSelectorMask ATTR_GEN_FILE_SELECTOR_NAME_MASK
GenFilePath ATTR_GEN_FILE_SELECTOR_VIRTUAL_ROOT
```

Hints

```
word   HINT_FILE_SELECTOR_NUMBER_OF_FILES_TO_SHOW
```

Types and Flags

```
WordFlags    FileSelectorFileCriteria
      FSFC_DIRS                         0x8000
      FSFC_NON_GEOS_FILES               0x4000
      FSFC_GEOS_EXECUTABLES             0x2000
      FSFC_GEOS_NON_EXECUTABLES         0x1000
      FSFC_MASK_CASE_INSENSITIVE        0x0800
      FSFC_FILE_FILTER                  0x0400
      FSFC_FILTER_IS_C                  0x0200
      FSFC_TOKEN_NO_ID                  0x0100
      FSFC_USE_MASK_FOR_DIRS            0x0080
WordFlags    FileSelectorAttrs
      FSA_ALLOW_CHANGE_DIRS             0x8000
      FSA_SHOW_FIXED_DISKS_ONLY         0x2000
      FSA_SHOW_FILES_DISABLED           0x1000
      FSA_HAS_CLOSE_DIR_BUTTON          0x0800
      FSA_HAS_OPEN_DIR_BUTTON           0x0400
      FSA_HAS_DOCUMENT_BUTTON           0x0200
      FSA_HAS_CHANGE_DIRECTORY_LIST     0x0100
      FSA_HAS_CHANGE_DRIVE_LIST         0x0080
      FSA_HAS_FILE_LIST                 0x0040
      FSA_USE_VIRTUAL_ROOT              0x0020
ByteEnum     GenFileSelectorEntryType
      GFSET_FILE          0
```

```
        GFSET_SUBDIR        1
        GFSET_VOLUME        2
    WordFlags           GenFileSelectorEntryFlags
        GFSEF_TYPE                      0xc000
        GFSEF_OPEN                      0x2000
        GFSEF_NO_ENTRIES                0x1000
        GFSEF_ERROR                     0x0800
        GFSEF_TEMPLATE                  0x0400
        GFSEF_SHARED_MULTIPLE           0x0200
        GFSEF_SHARED_SINGLE             0x0100
        GFSEF_READ_ONLY                 0x0080
        GFSEF_PARENT_DIR                0x0040
    GFSEF_TYPE_OFFSET         14
    FileLongName        GenFileSelectorMask
    VolumeName          GenFileSelectorVolume
    FileLongName        GenFileSelectorSelection
    ByteEnum     GenFileSelectorScalableUICommand
        GFSSUIC_SET_FEATURES_IF_APP_FEATURE_ON       0
        GFSSUIC_SET_FEATURES_IF_APP_FEATURE_OFF      1
        GFSSUIC_ADD_FEATURES_IF_APP_FEATURE_ON       2
        GFSSUIC_SET_FEATURES_IF_APP_LEVEL            3
        GFSSUIC_ADD_FEATURES_IF_APP_LEVEL            4
```

Structures

```
    typedef struct {
        GeodeToken      GTP_token
        word            GTP_unused
    } GetTokenCreatorParams
    typedef struct {
        Message         GAP_message
        word            GAP_unused
        optr            GAP_output
    } GetActionParams
    typedef struct {
        GenFileSelectorFilterRoutine*filterRoutine
        const FileExtAttrDesc       *filterAttrs
    } GenFileSelectorGetFilterRoutineResults
    typedef struct {
        FileAttrs       GFSFA_match
        FileAttrs       GFSFA_mismatch
    } GenFileSelectorFileAttrs
    typedef struct {
        GeodeAttrs      GFSGA_match
        GeodeAttrs      GFSGA_mismatch
    } GenFileSelectorGeodeAttrs
    typedef struct {
        GenFileSelectorScalableUICommand GFSSUIE_command
        WordFlags             GFSSUIE_appFeature
        FileSelectorAttrs     GFSSUIE_fsFeatures
    } GenFileSelectorScalableUIEntry
```

Macros

```
    GFS_GET_ENTRY_TYPE(A) (((A) & GFSEF_TYPE) >>
                                GFSEF_TYPE_OFFSET)
    GFS_GET_ENTRY_NUMBER(A) ((word) (A >> 16))
    GFS_GET_ENTRY_FLAGS(A) ((word) A)
    GFS_GET_FULL_SELECTION_PATH_DISK_HANDLE(A)
                                ((DiskHandle) (A))
```

```
GET_MATCH_FILE_ATTRS(attr) ((byte) (attr))
GET_MISMATCH_FILE_ATTRS(attr) ((byte) (attr >> 8))
SET_TOKEN_CHARS(A, B, C, D) ((A) | ((B) << 8) |
                          ((C) << 16) | ((D) << 24))
GET_MATCH_ATTRS(attr) (((attr) >> 16) & 0xffff)
GET_MISMATCH_ATTRS(attr) ((attr) & 0xffff)
```

```
dword MSG_GEN_FILE_SELECTOR_GET_SELECTION(
                    char *selection)
Boolean MSG_GEN_FILE_SELECTOR_SET_SELECTION(
                    char *selection)
dword MSG_GEN_FILE_SELECTOR_GET_FULL_SELECTION_PATH(
                    char *selection)
Boolean MSG_GEN_FILE_SELECTOR_SET_FULL_SELECTION_PATH(
                    char *selection,
                    DiskHandle diskHandle)
void MSG_GEN_FILE_SELECTOR_GET_MASK(char *mask)
void MSG_GEN_FILE_SELECTOR_SET_MASK(char *mask)
word MSG_GEN_FILE_SELECTOR_GET_FILE_ATTRS()
void MSG_GEN_FILE_SELECTOR_SET_FILE_ATTRS(
                    byte setAttrs,
                    byte clearAttrs)
void MSG_GEN_FILE_SELECTOR_GET_TOKEN(
                    GetTokenCreatorParams *retValue)
void MSG_GEN_FILE_SELECTOR_SET_TOKEN(
                    dword tokenChars
                    ManufacturerID manufacturerID)
void MSG_GEN_FILE_SELECTOR_GET_CREATOR(
                    GetTokenCreatorParams *retValue)
void MSG_GEN_FILE_SELECTOR_SET_CREATOR(
                    dword tokenChars,
                    ManufacturerID manufacturerID)
dword MSG_GEN_FILE_SELECTOR_GET_GEODE_ATTRS()
void MSG_GEN_FILE_SELECTOR_SET_GEODE_ATTRS(
                    word matchGeodeAttrs,
                    word mismatchGeodeAttrs)
void MSG_GEN_FILE_SELECTOR_GET_ACTION(
                    GetActionParams *retValue)
void MSG_GEN_FILE_SELECTOR_SET_ACTION(
                    optr actionOD,
                    word actionMessage)
FileSelectorAttrs MSG_GEN_FILE_SELECTOR_GET_ATTRS()
void MSG_GEN_FILE_SELECTOR_SET_ATTRS(
                    FileSelectorAttrs attributes)
FileSelectorFileCriteria
      MSG_GEN_FILE_SELECTOR_GET_FILE_CRITERIA()
void MSG_GEN_FILE_SELECTOR_SET_FILE_CRITERIA(
             FileSelectorFileCriteria fileCriteria)
void MSG_GEN_FILE_SELECTOR_RESCAN()
void MSG_GEN_FILE_SELECTOR_UP_DIRECTORY()
Boolean MSG_GEN_FILE_SELECTOR_OPEN_ENTRY(
                    word entryNumber)
Boolean MSG_GEN_FILE_SELECTOR_SUSPEND()
Boolean MSG_GEN_FILE_SELECTOR_END_SUSPEND()
Boolean MSG_GEN_FILE_SELECTOR_GET_DESTINATION_PATH(
                    char *buffer, word bufSize)
```

```
void MSG_GEN_FILE_SELECTOR_GET_FILTER_ROUTINE(
      GenFileSelectorGetFilterRoutineResults *filter)
@prototype void GEN_FILE_SELECTOR_NOTIFICATION_MSG(
                           word entryNum,
                           word entryFlags)
```

Routines


```
Boolean GenFileSelectorFilterRoutine (
                           optr oself,
                           FileEnumCallbackData *fecd,
                           word frame)
```

GenGlyphClass

```
@class GenGlyphClass, GenClass
```

GenInteractionClass

```
@class GenInteractionClass, GenClass
```

Instance Data

```
GenInteractionType        GII_type = GIT_ORGANIZATIONAL
GenInteractionVisibility  GII_visibility = GIV_SUB_GROUP
GenInteractionAttrs       GII_attrs = 0
```

Variable Data

```
GenInteractionGroupType ATTR_GEN_INTERACTION_GROUP_TYPE
void ATTR_GEN_INTERACTION_OVERRIDE_INPUT_RESTRICTIONS
void ATTR_GEN_INTERACTION_ABIDE_BY_INPUT_RESTRICTIONS
void ATTR_GEN_INTERACTION_POPPED_OUT
void ATTR_GEN_INTERACTION_POPOUT_NOT_CLOSABLE
optr ATTR_GEN_INTERACTION_PEN_MODE_KEYBOARD_OBJECT
```

Hints

```
void HINT_INTERACTION_SINGLE_USAGE
void HINT_INTERACTION_FREQUENT_USAGE
void HINT_INTERACTION_COMPLEX_PROPERTIES
void HINT_INTERACTION_SIMPLE_PROPERTIES
void HINT_INTERACTION_RELATED_PROPERTIES
void HINT_INTERACTION_UNRELATED_PROPERTIES
void HINT_INTERACTION_SLOW_RESPONSE_PROPERTIES
void HINT_INTERACTION_FAST_RESPONSE_PROPERTIES
void HINT_INTERACTION_REQUIRES_VALIDATION
void HINT_INTERACTION_MAKE_RESIZABLE
void HINT_INTERACTION_CANNOT_BE_DEFAULT
void HINT_INTERACTION_MODAL
void HINT_INTERACTION_NO_DISTURB
void
HINT_INTERACTION_DEFAULT_ACTION_OS_NAVIGATE_TO_NEXT_FIE
LD
void HINT_INTERACTION_INFREQUENT_USAGE
void HINT_CUSTOM_SYS_MENU
void HINT_INTERACTION_MAXIMIZABLE
void HINT_INTERACTION_POPOUT_HIDDEN_ON_STARTUP
```

Types and Flags

```
enum /* word */ {
      IC_NULL,
      IC_DISMISS,
      IC_INTERACTION_COMPLETE,
      IC_APPLY,
      IC_RESET,
      IC_OK,
      IC_YES,
      IC_NO,
      IC_STOP,
      IC_EXIT,
      IC_HELP
} InteractionCommand
IC_CUSTOM_START    1000
ByteEnum     GenInteractionType
      GIT_ORGANIZATIONAL       0
      GIT_PROPERTIES           1
      GIT_PROGRESS             2
      GIT_COMMAND              3
```

```
        GIT_NOTIFICATION          4
        GIT_AFFIRMATION           5
        GIT_MULTIPLE_RESPONSE     6
    ByteEnum     GenInteractionVisibility
        GIV_NO_PREFERENCE         0
        GIV_POPUP                 1
        GIV_SUB_GROUP             2
        GIV_CONTROL_GROUP         3
        GIV_DIALOG                4
        GIV_POPOUT                5
    ByteFlags    GenInteractionAttrs
        GIA_NOT_USER_INITIATABLE        0x80
        GIA_INITIATED_VIA_USER_DO_DIALOG0x40
        GIA_MODAL                       0x20
        GIA_SYS_MODAL                   0x10
    ByteEnum     GenInteractionGroupType
        GIGT_FILE_MENU            0
        GIGT_EDIT_MENU            1
        GIGT_VIEW_MENU            2
        GIGT_OPTIONS_MENU         3
        GIGT_WINDOW_MENU          4
        GIGT_HELP_MENU            5
        GIGT_PRINT_GROUP          6
```

Structures

```
    typedef struct {
        ThreadHandle          UDDS_callingThread
        SemaphoreHandle       UDDS_semaphore
        word                  UDDS_response
        word                  UDDS_complete
        Boolean               UDDS_boxRunByCurrentThread
        optr                  UDDS_dialog
        QueueHandle           UDDS_queue
    } UserDoDialogStruct
```

Messages

```
    byte MSG_GEN_INTERACTION_GET_TYPE()
    void MSG_GEN_INTERACTION_SET_TYPE(byte type)
    byte MSG_GEN_INTERACTION_GET_VISIBILITY()
    void MSG_GEN_INTERACTION_SET_VISIBILITY(byte visibility)
    byte MSG_GEN_INTERACTION_GET_ATTRS()
    void MSG_GEN_INTERACTION_SET_ATTRS(
                        byte setAttrs, byte clearAttrs)
    void MSG_GEN_INTERACTION_ACTIVATE_COMMAND(word command)
    void MSG_GEN_INTERACTION_INITIATE()
    void MSG_GEN_INTERACTION_INITIATE_NO_DISTURB()
    void
    MSG_GEN_INTERACTION_INITIATE_BLOCKING_THREAD_ON_RESPONS
    E(UserDoDialogStruct *dialogInfo)
    void
    MSG_GEN_INTERACTION_RELEASE_BLOCKED_THREAD_WITH_RESPONS
    E(word *command)
    void MSG_GEN_INTERACTION_TEST_INPUT_RESTRICTABILITY()
                             /* XXX */
    void MSG_GEN_INTERACTION_POP_OUT()
    void MSG_GEN_INTERACTION_POP_IN()
    void MSG_GEN_INTERACTION_TOGGLE_POPOUT()
```

GenItemClass

```
@class GenItemClass, GenClass
```

Instance Data

```
word GII_identifier
```

Messages

```
word MSG_GEN_ITEM_GET_IDENTIFIER()
void MSG_GEN_ITEM_SET_IDENTIFIER(word identifier)
void MSG_GEN_ITEM_SET_INTERACTABLE_STATE(
                        Boolean interactable)
```

GenItemGroupClass

```
@class GenItemGroupClass, GenClass
```

Instance Data

```
GenItemGroupBehaviorType  GIGI_behaviorType =
                                    GIGBT_EXCLUSIVE
word                      GIGI_selection = GIGS_NONE
word                      GIGI_numSelections = 0
GenItemGroupStateFlags    GIGI_stateFlags = 0
optr                      GIGI_destination
Message                   GIGI_applyMsg = 0
```

Variable Data

```
Message ATTR_GEN_ITEM_GROUP_STATUS_MSG
void
ATTR_GEN_ITEM_GROUP_SET_MODIFIED_ON_REDUNDANT_SELECTION
Message ATTR_GEN_ITEM_GROUP_CUSTOM_DOUBLE_PRESS
optr ATTR_GEN_ITEM_GROUP_LINK
    @reloc ATTR_GEN_ITEM_GROUP_LINK, 0, optr
void ATTR_GEN_ITEM_GROUP_INIT_FILE_BOOLEAN
```

Hints

```
void HINT_ITEM_GROUP_SCROLLABLE
void HINT_ITEM_GROUP_MINIMIZE_SIZE
void HINT_ITEM_GROUP_RADIO_BUTTON_STYLE
void HINT_ITEM_GROUP_TOOLBOX_STYLE
void HINT_ITEM_GROUP_DISPLAY_CURRENT_SELECTION
void HINT_ITEM_GROUP_MINIMIZE_SIZE_IF_VERTICAL_SCREEN
```

Types and Flags

```
ByteEnum     GenItemGroupBehaviorType
      GIGBT_EXCLUSIVE           0
      GIGBT_EXCLUSIVE_NONE      1
      GIGBT_EXTENDED_SELECTION  2
      GIGBT_NON_EXCLUSIVE       3
GIGS_NONE           (-1)
ByteFlags    GenItemGroupStateFlags
      GIGSF_INDETERMINATE             0x80
      GIGSF_MODIFIED                  0x40
```

Messages

```
void MSG_GEN_ITEM_GROUP_SET_NONE_SELECTED(
                        Boolean indeterminate)
void MSG_GEN_ITEM_GROUP_SET_SINGLE_SELECTION(
                        word identifier,
```

```
                          Boolean indeterminate)
void MSG_GEN_ITEM_GROUP_SET_MULTIPLE_SELECTIONS(
                          word *selectionList,
                          word numSelections)
word MSG_GEN_ITEM_GROUP_GET_SELECTION()
word MSG_GEN_ITEM_GROUP_GET_NUM_SELECTIONS()
word MSG_GEN_ITEM_GROUP_GET_MULTIPLE_SELECTIONS(
                          word *selectionList,
                          word maxSelections)
void MSG_GEN_ITEM_GROUP_SET_INDETERMINATE_STATE(
                          Boolean indeterminateState)
Boolean MSG_GEN_ITEM_GROUP_IS_INDETERMINATE()
void MSG_GEN_ITEM_GROUP_SET_MODIFIED_STATE(
                          Boolean modifiedState)
Boolean MSG_GEN_ITEM_GROUP_IS_MODIFIED()
void MSG_GEN_ITEM_GROUP_SEND_STATUS_MSG(
                          Boolean modifiedState)
optr MSG_GEN_ITEM_GROUP_GET_ITEM_OPTR(word identifier)
void MSG_GEN_ITEM_GROUP_SET_ITEM_STATE(
                          word identifier,
                          Boolean state)
Boolean MSG_GEN_ITEM_GROUP_IS_ITEM_SELECTED(
                          word identifier)
void MSG_GEN_ITEM_GROUP_MAKE_ITEM_VISIBLE(
                          word identifier)
optr MSG_GEN_ITEM_GROUP_GET_DESTINATION()
void MSG_GEN_ITEM_GROUP_SET_DESTINATION(optr dest)
Message MSG_GEN_ITEM_GROUP_GET_APPLY_MSG()
void MSG_GEN_ITEM_GROUP_SET_APPLY_MSG(Message message)
GenItemGroupBehaviorType
       MSG_GEN_ITEM_GROUP_GET_BEHAVIOR_TYPE()
void MSG_GEN_ITEM_GROUP_SET_BEHAVIOR_TYPE(
              GenItemGroupBehaviorType behaviorType)
word MSG_GEN_ITEM_GROUP_GET_UNIQUE_IDENTIFIER()
void MSG_GEN_ITEM_GROUP_REDRAW_ITEMS(word offset)
@prototype void GEN_ITEM_GROUP_APPLY_MSG(
                          word selection,
                          word numSelections,
                          byte stateFlags)
@prototype void GEN_ITEM_GROUP_STATUS_MSG(
                          word selection,
                          word numSelections,
                          byte stateFlags)
```

GenPageControlClass

```
@class GenPageControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
WordFlags    GPCFeatures
      GPCF_GOTO_PAGE            0x0004
      GPCF_NEXT_PAGE            0x0002
      GPCF_PREVIOUS_PAGE        0x0001
WordFlags    GPCToolboxFeatures
      GPCTF_PREVIOUS_PAGE       0x0004
      GPCTF_GOTO_PAGE           0x0002
      GPCTF_NEXT_PAGE           0x0001
GPC_DEFAULT_FEATURES      (GPCF_GOTO_PAGE |
                           GPCF_NEXT_PAGE |
                           GPCF_PREVIOUS_PAGE)
GPC_DEFAULT_TOOLBOX_FEATURES (GPCTF_GOTO_PAGE |
                           GPCTF_NEXT_PAGE |
                           GPCTF_PREVIOUS_PAGE)
```

Structures

```
typedef struct {
    word     NPSC_firstPage
    word     NPSC_lastPage
    word     NPSC_currentPage
} NotifyPageStateChange
```

Messages

```
void MSG_PC_GOTO_PAGE()
void MSG_PC_NEXT_PAGE()
void MSG_PC_PREVIOUS_PAGE()
```

GenPenInputControlClass

```
@class GenPenInputControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_FOCUS)
```

Variable Data

```
CharTableData
      ATTR_GEN_PEN_INPUT_CONTROL_CUSTOM_CHAR_TABLE_DATA
optr
  ATTR_GEN_PEN_INPUT_CONTROL_CUSTOM_CHAR_TABLE_MONIKER
PenInputDisplayType
      ATTR_GEN_PEN_INPUT_CONTROL_STARTUP_DISPLAY_TYPE
```

Types and Flags

```
typedef enum { /* word */
      PIDT_KEYBOARD,
      PIDT_CHAR_TABLE,
      PIDT_CHAR_TABLE_SYMBOLS,
      PIDT_CHAR_TABLE_INTERNATIONAL,
      PIDT_CHAR_TABLE_MATH,
      PIDT_CHAR_TABLE_CUSTOM,
      PIDT_HWR_ENTRY_AREA
} PenInputDisplayType
WordFlags    GPICFeatures
      GPICF_KEYBOARD                  0x0040
      GPICF_CHAR_TABLE                0x0020
      GPICF_CHAR_TABLE_SYMBOLS        0x0010
      GPICF_CHAR_TABLE_INTERNATIONAL  0x0008
      GPICF_CHAR_TABLE_MATH           0x0004
      GPICF_CHAR_TABLE_CUSTOM         0x0002
      GPICF_HWR_ENTRY_AREA            0x0001
WordFlags    GPICToolboxFeatures
      GPICTF_INITIATE                 0x0001
GPIC_DEFAULT_FEATURES (GPICF_KEYBOARD | GPICF_CHAR_TABLE
      GPICF_HWR_ENTRY_AREA | GPICF_CHAR_TABLE_SYMBOLS |
      GPICF_CHAR_TABLE_MATH |
      GPICF_CHAR_TABLE_INTERNATIONAL)
GPIC_DEFAULT_TOOLBOX_FEATURES (GPICTF_INITIATE)
```

Structures

```
typedef struct {
    optr     CTD_line1
    optr     CTD_line2
    optr     CTD_line3
    optr     CTD_line4
    optr     CTD_line5
} CharTableData
typedef struct {
    VisTextRange   RWHWRD_range
    HWRContext     RWHWRD_context
} ReplaceWithHWRData
```

Messages

```
void MSG_GEN_PEN_INPUT_CONTROL_SET_DISPLAY(
                          PenInputDisplayType dispType)
PenInputDisplayType
            MSG_GEN_PEN_INPUT_CONTROL_GET_DISPLAY()
```

GenPrimaryClass

```
@class GenPrimaryClass, GenDisplayClass
```

Instance Data

```
ChunkHandle  GPI_longTermMoniker
    @default GI_attrs = @default | GA_TARGETABLE
```

Hints

```
void HINT_PRIMARY_FULL_SCREEN
void HINT_PRIMARY_NO_FILE_MENU
void HINT_PRIMARY_NO_EXPRESS_MENU
Rectangle HINT_PRIMARY_OPEN_ICON_BOUNDS
void HINT_PRIMARY_NO_HELP_BUTTON
```

Messages

```
ChunkHandle MSG_GEN_PRIMARY_GET_LONG_TERM_MONIKER()
void MSG_GEN_PRIMARY_USE_LONG_TERM_MONIKER(
                    ChunkHandle moniker)
ChunkHandle MSG_GEN_PRIMARY_REPLACE_LONG_TERM_MONIKER(
                    @stack
                    VisUpdateMode updateMode,
                    word height,
                    word width,
                    word length,
                    VisMonikerDataType dateType,
                    VisMonikerSourceType sourceType,
                    dword source)
```

GenProcessClass

```
@class GenProcessClass, ProcessClass
```

Types and Flags

```
WordFlags AppAttachFlags
      AAF_RESTORING_FROM_STATE        0x8000
      AAF_STATE_FILE_PASSED           0x4000
      AAF_DATA_FILE_PASSED            0x2000
enum /* word */ {
      UADT_FLAGS =       0,
      UADT_PTR =         2,
      UADT_VM_CHAIN =    4
} UndoActionDataType
NULL_UNDO_CONTEXT  0
WordFlags AddUndoActionFlags
      AUAF_NOTIFY_BEFORE_FREEING      0x8000
      AUAF_NOTIFY_IF_FREED_WITHOUT_BEING_PLAYED_BACK
                                      0x4000
ByteFlags AppLaunchFlags
      ALF_SEND_LAUNCH_REQUEST_TO_UI_TO_HANDLE 0x80
      ALF_OPEN_IN_BACK                0x40
      ALF_DESK_ACCESSORY              0x20
      ALF_DO_NOT_OPEN_ON_TOP          0x10
      ALF_OVERRIDE_MULTIPLE_INSTANCE  0x08
      ALF_LAUNCHED_FOR_PRINTING_ONLY  0x04
```

Structures

```
typedef struct {
    dword          UADF_flags
    word           UADF_extraflags
} UndoActionDataFlags
typedef struct {
    void           *UADP_ptr
    word           UADP_size
} UndoActionDataPtr
typedef struct {
    VMChain        UADVMC_vmChain
    VMFileHandle   UADVMC_file
} UndoActionDataVMChain
typedef union {
    UndoActionDataFlags   UADU_flags
    UndoActionDataPtr     UADU_ptr
    UndoActionDataVMChain UADU_vmChain
} UndoActionDataUnion
typedef struct {
    UndoActionDataType    UAS_dataType
    UndoActionDataUnion   UAS_data
    dword                 UAS_appType
} UndoActionStruct
typedef struct {
    UndoActionStruct      AUAS_data
    optr                  AUAS_output
    AddUndoActionFlags    AUAS_flags
} AddUndoActionStruct
typedef struct {
    PathName              AIR_fileName
    FileLongName          AIR_stateFile
```

```
    DiskHandle            AIR_diskHandle
    byte                  AIR_savedDiskData[1]
} AppInstanceReference
typedef struct {
    AppInstanceReference  ALB_appRef
    word                  ALB_appMode
    AppLaunchFlags        ALB_launchFlags
    MemHandle             ALB_diskHandle
    char                  ALB_path[PATH_BUFFER_SIZE]
    char                  ALB_dataFile[PATH_BUFFER_SIZE]
    optr                  ALB_genParent
    optr                  ALB_userLoadAckOutput
    Message               ALB_userLoadAckMessage
    word                  ALB_userLoadAckID
    word                  ALB_extraData
} AppLaunchBlock
```

Messages

```
void MSG_GEN_PROCESS_RESTORE_FROM_STATE(
                  AppAttachFlags attachFlags,
                  MemHandle launchBlock,
                  MemHandle extraState)
void MSG_GEN_PROCESS_OPEN_APPLICATION(
                  AppAttachFlags attachFlags,
                  MemHandle launchBlock,
                  MemHandle extraState)
void MSG_GEN_PROCESS_OPEN_ENGINE(
                  AppAttachFlags attachFlags,
                  MemHandle launchBlock,
                  MemHandle extraState)
MemHandle MSG_GEN_PROCESS_CLOSE_APPLICATION()
MemHandle MSG_GEN_PROCESS_CLOSE_ENGINE()
MemHandle MSG_GEN_PROCESS_CLOSE_CUSTOM()
MemHandle MSG_GEN_PROCESS_ATTACH_TO_PASSED_STATE_FILE(
                  AppAttachFlags attachFlags,
                  MemHandle launchBlock)
word MSG_GEN_PROCESS_CREATE_NEW_STATE_FILE(
                  MemHandle appInstanceReference)
void MSG_GEN_PROCESS_INSTALL_TOKEN()
optr MSG_GEN_PROCESS_GET_PARENT_FIELD()
void MSG_GEN_PROCESS_SEND_TO_APP_GCN_LIST(@stack
                  word sendFlags, EventHandle event,
                  MemHandle block, word manufListType,
                  word manufID)
void MSG_GEN_PROCESS_UNDO_START_CHAIN(@stack
                  optr title, optr owner)
void MSG_GEN_PROCESS_UNDO_END_CHAIN(
                  Boolean flushChainIfEmpty)
VMChain MSG_GEN_PROCESS_UNDO_ADD_ACTION(
                  AddUndoActionStruct *data)
VMFileHandle MSG_GEN_PROCESS_UNDO_GET_FILE()
void MSG_GEN_PROCESS_UNDO_FLUSH_ACTIONS()
dword MSG_GEN_PROCESS_UNDO_SET_CONTEXT(dword context)
dword MSG_GEN_PROCESS_UNDO_GET_CONTEXT()
void MSG_GEN_PROCESS_UNDO_PLAYBACK_CHAIN()
void MSG_GEN_PROCESS_UNDO_IGNORE_ACTIONS(
                  Boolean flushActions)
```

```
void MSG_GEN_PROCESS_UNDO_ACCEPT_ACTIONS()
Boolean MSG_GEN_PROCESS_UNDO_CHECK_IF_IGNORING()
void MSG_GEN_PROCESS_UNDO_ABORT_CHAIN()
```

GenSystemClass



```
@class GenSystemClass, GenClass
```

Messages

```
optr MSG_GEN_SYSTEM_GET_DEFAULT_SCREEN()
void MSG_GEN_SYSTEM_SET_DEFAULT_FIELD(optr defaultField)
optr MSG_GEN_SYSTEM_GET_DEFAULT_FIELD()
void MSG_GEN_SYSTEM_SET_PTR_IMAGE(
                    optr ptrImage, PtrImageLevel level)
void MSG_GEN_SYSTEM_NOTIFY_SYS_MODAL_WIN_CHANGE()
void MSG_GEN_SYSTEM_BRING_GEODE_TO_TOP(
                    word geode, word layerID,
                    Handle parentWindow)
void MSG_GEN_SYSTEM_LOWER_GEODE_TO_BOTTOM(
                    word geode, word layerID,
                    Handle parentWindow)
```

GenTextClass

```
@class GenTextClass, GenClass
```

Instance Data

```
ChunkHandle  GTXI_text
word         GTXI_maxLength = 32767
GenTextAttrs GTXI_attrs = GTA_USE_TAB_FOR_NAVIGATION
GenTextStateFlags GTXI_stateFlags = 0
optr         GTXI_destination
word         GTXI_applyMsg = 0
```

Variable Data

```
Message ATTR_GEN_TEXT_STATUS_MSG
void ATTR_GEN_TEXT_SELECTABLE
ColorQuad HINT_TEXT_WASH_COLOR
void HINT_TEXT_WHITE_WASH_COLOR
VisTextDefaultCharAttr ATTR_GEN_TEXT_DEFAULT_CHAR_ATTR
VisTextDefaultParaAttr ATTR_GEN_TEXT_DEFAULT_PARA_ATTR
ChunkHandle ATTR_GEN_TEXT_CHAR_ATTR
ChunkHandle ATTR_GEN_TEXT_MULTIPLE_CHAR_ATTR_RUNS
ChunkHandle ATTR_GEN_TEXT_PARA_ATTR
ChunkHandle ATTR_GEN_TEXT_MULTIPLE_PARA_ATTR_RUNS
void ATTR_GEN_TEXT_ALPHA
void ATTR_GEN_TEXT_NUMERIC
void ATTR_GEN_TEXT_SIGNED_NUMERIC
void ATTR_GEN_TEXT_SIGNED_DECIMAL
void ATTR_GEN_TEXT_FLOAT_DECIMAL
void ATTR_GEN_TEXT_ALPHA_NUMERIC
void ATTR_GEN_TEXT_LEGAL_FILENAMES
void ATTR_GEN_TEXT_LEGAL_DOS_FILENAMES
void ATTR_GEN_TEXT_LEGAL_DOS_PATH
void ATTR_GEN_TEXT_DATE
void ATTR_GEN_TEXT_TIME
void ATTR_GEN_TEXT_MAKE_UPPERCASE
void ATTR_GEN_TEXT_ALLOW_COLUMN_BREAKS
void ATTR_GEN_TEXT_DASHED_ALPHA_NUMERIC
void ATTR_GEN_TEXT_NORMAL_ASCII
void ATTR_GEN_TEXT_LEGAL_DOS_VOLUME_NAMES
void ATTR_GEN_TEXT_DOS_CHARACTER_SET
void ATTR_GEN_TEXT_NO_SPACES
void ATTR_GEN_TEXT_ALLOW_SPACES
word ATTR_GEN_TEXT_EXTENDED_FILTER
word ATTR_GEN_TEXT_TYPE_RUNS
word ATTR_GEN_TEXT_GRAPHIC_RUNS
word ATTR_GEN_TEXT_STYLE_ARRAY
word ATTR_GEN_TEXT_NAME_ARRAY
optr ATTR_GEN_TEXT_RUNS_ITEM_GROUP
    @reloc ATTR_GEN_TEXT_RUNS_ITEM_GROUP, 0, optr
```

Hints

```
ColorQuad HINT_TEXT_WASH_COLOR
void HINT_TEXT_WHITE_WASH_COLOR
void HINT_TEXT_AUTO_HYPHENATE
void HINT_TEXT_SELECT_TEXT
void HINT_TEXT_CURSOR_AT_START
void HINT_TEXT_CURSOR_AT_END
void HINT_TEXT_FRAME
```

```
void HINT_TEXT_NO_FRAME
void HINT_TEXT_ALLOW_UNDO
void HINT_TEXT_ALLOW_SMART_QUOTES
void HINT_GEN_TEXT_DO_NOT_MAKE_LARGER_ON_PEN_SYSTEMS
```

Types and Flags

```
ByteFlags    GenTextStateFlags
      GTSF_INDETERMINATE              0x80
      GTSF_MODIFIED                   0x40
ByteFlags    GenTextAttrs
      GTA_SINGLE_LINE_TEXT            0x80
      GTA_USE_TAB_FOR_NAVIGATION      0x40
      GTA_INIT_SCROLLING              0x20
      GTA_NO_WORD_WRAPPING            0x10
      GTA_ALLOW_TEXT_OFF_END          0x08
      GTA_TAIL_ORIENTED               0x04
      GTA_DONT_SCROLL_TO_CHANGES      0x02
```

Messages

```
void MSG_GEN_TEXT_SET_ATTRS(
                    byte attrsToSet, byte attrsToClear)
byte MSG_GEN_TEXT_GET_ATTRS()
void MSG_GEN_TEXT_SET_INDETERMINATE_STATE(
                    Boolean indeterminateState)
Boolean MSG_GEN_TEXT_IS_INDETERMINATE()
void MSG_GEN_TEXT_SET_MODIFIED_STATE(
                    Boolean modifiedState)
Boolean MSG_GEN_TEXT_IS_MODIFIED()
void MSG_GEN_TEXT_SEND_STATUS_MSG(
                    Boolean modifiedState )
optr MSG_GEN_TEXT_GET_DESTINATION()
void MSG_GEN_TEXT_SET_DESTINATION(optr dest)
Message MSG_GEN_TEXT_GET_APPLY_MSG()
void MSG_GEN_TEXT_SET_APPLY_MSG(Message message)
void MSG_GEN_TEXT_SET_FROM_ITEM_GROUP(word item)
@prototype void GEN_TEXT_APPLY_MSG(word stateFlags)
@prototype void GEN_TEXT_STATUS_MSG(word stateFlags)
```




GenToolControlClass

```
@class GenToolControlClass, GenControlClass
```

Instance Data

```
ChunkHandle        GTCI_toolboxList
ChunkHandle        GTCI_toolGroupList
    @default GI_states = @default | GS_ENABLED
```

Variable Data

```
TempGenToolControlInstance
                   TEMP_GEN_TOOL_CONTROL_INSTANCE
```

Types and Flags

```
WordFlags    GTCFeatures
      GTCF_TOOL_DIALOG          0x0001
MAX_NUM_TOOLBOXES   25
```

Structures

```
typedef struct {
    optr     TI_object
    optr     TI_name
} ToolboxInfo
typedef struct {
    optr     TGI_object
} ToolGroupInfo
typedef struct {
    optr     TGTCI_curController
    word     TGTCI_features
    word     TGTCI_required
    word     TGTCI_allowed
} TempGenToolControlInstance
```

GenToolGroup

```
@class GenToolGroupClass, GenInteractionClass
```

Instance Data

```
optr               GTGI_controller
    @default GI_states = (@default & ~GS_ENABLED)
```

Variable Data

```
Color TEMP_TOOL_GROUP_HIGHLIGHT
```

Types and Flags

```
ByteEnum     ToolGroupHighlightType
      TGHT_INACTIVE_HIGHLIGHT         0
      TGHT_ACTIVE_HIGHLIGHT           1
      TGHT_NO_HIGHLIGHT               2
```

Messages

```
void MSG_GEN_TOOL_GROUP_SET_HIGHLIGHT(
                   ToolGroupHighlightType hlType)
```

GenTriggerClass

```
@class GenTriggerClass, GenClass
```

Instance Data

```
optr         GTI_destination
Message      GTI_actionMsg
```


Variable Data

```
void ATTR_GEN_TRIGGER_IMMEDIATE_ACTION
word ATTR_GEN_TRIGGER_ACTION_DATA
word ATTR_GEN_TRIGGER_INTERACTION_COMMAND
Message ATTR_GEN_TRIGGER_CUSTOM_DOUBLE_PRESS
    @vardataAlias (ATTR_GEN_TRIGGER_ACTION_DATA)
             TwoWordArgs
             ATTR_GEN_TRIGGER_ACTION_TWO_WORDS
    @vardataAlias (ATTR_GEN_TRIGGER_ACTION_DATA)
             ThreeWordArgs
             ATTR_GEN_TRIGGER_ACTION_THREE_WORDS
    @vardataAlias (ATTR_GEN_TRIGGER_ACTION_DATA)
             OptrWordArgs
             ATTR_GEN_TRIGGER_ACTION_OPTR_AND_WORD
```

Hints

```
void HINT_TRIGGER_BRINGS_UP_WINDOW
void HINT_TRIGGER_DESTRUCTIVE_ACTION
```

Structures

```
typedef struct { word foo, foo2 } TwoWordArgs
typedef struct { word foo, foo2, foo3 } ThreeWordArgs
typedef struct { optr output; word foo } OptrWordArgs
```

Messages

```
void MSG_GEN_TRIGGER_SEND_ACTION(
                  Boolean doublePressFlag)
Message MSG_GEN_TRIGGER_GET_ACTION_MSG()
void MSG_GEN_TRIGGER_SET_ACTION_MSG(Message message)
void MSG_GEN_TRIGGER_SET_DESTINATION(optr dest)
optr MSG_GEN_TRIGGER_GET_DESTINATION()
void MSG_GEN_TRIGGER_MAKE_DEFAULT_ACTION()
    @prototype void GEN_TRIGGER_ACTION(optr trigger)
```



GenValueClass

```
@class GenValueClass, GenClass
```

Instance Data

```
WWFixedAsDWord     GVLI_value = MakeWWFixed(0.0)
WWFixedAsDWord     GVLI_minimum = MakeWWFixed(0.0)
WWFixedAsDWord     GVLI_maximum = MakeWWFixed(32766)
WWFixedAsDWord     GVLI_increment = MakeWWFixed(1.0)
GenValueStateFlags GVLI_stateFlags = 0
GenValueDisplayFormat GVLI_displayFormat = GVDF_INTEGER
optr               GVLI_destination
Message            GVLI_applyMsg = 0
```

Variable Data

```
Message ATTR_GEN_VALUE_STATUS_MSG
word ATTR_GEN_VALUE_DECIMAL_PLACES
WWFixed ATTR_GEN_VALUE_METRIC_INCREMENT
optr ATTR_GEN_VALUE_RUNS_ITEM_GROUP
    @reloc ATTR_GEN_VALUE_RUNS_ITEM_GROUP, 0, optr
void ATTR_GEN_VALUE_SET_MODIFIED_ON_REDUNDANT_SELECTION
```

Hints

```
void HINT_VALUE_NAVIGATE_TO_NEXT_FIELD_ON_RETURN_PRESS
void HINT_VALUE_CUSTOM_RETURN_PRESS
void HINT_VALUE_INCREMENTABLE
void HINT_VALUE_NOT_INCREMENTABLE
void HINT_VALUE_X_SCROLLER
void HINT_VALUE_Y_SCROLLER
WWFixedAsDWord HINT_VALUE_DISPLAYS_RANGE
void HINT_VALUE_ANALOG_DISPLAY
GenValueIntervals HINT_VALUE_DISPLAY_INTERVALS
void HINT_VALUE_CONSTRAIN_TO_INTERVALS
void HINT_VALUE_SHOW_MIN_AND_MAX
void HINT_VALUE_IMMEDIATE_DRAG_NOTIFICATION
void HINT_VALUE_DELAYED_DRAG_NOTIFICATION
void HINT_VALUE_ORIENT_HORIZONTALLY
void HINT_VALUE_ORIENT_VERTICALLY
void HINT_VALUE_DIGITAL_DISPLAY
void HINT_VALUE_NO_DIGITAL_DISPLAY
void HINT_VALUE_NO_ANALOG_DISPLAY
void HINT_VALUE_NOT_DIGITALLY_EDITABLE
void HINT_VALUE_DO_NOT_MAKE_LARGER_ON_PEN_SYSTEMS
```

Types and Flags

```
ByteEnum     GenValueDisplayFormat
      GVDF_INTEGER             0
      GVDF_DECIMAL             1
      GVDF_POINTS              2
      GVDF_INCHES              3
      GVDF_CENTIMETERS         4
      GVDF_MILLIMETERS         5
      GVDF_PICAS               6
      GVDF_EUR_POINTS          7
      GVDF_CICEROS             8
      GVDF_POINTS_OR_MILLIMETERS 9
      GVDF_INCHES_OR_CENTIMETERS 10
typedef enum /* word */ {
      GVT_VALUE,
```

```
        GVT_MINIMUM,
        GVT_MAXIMUM,
        GVT_INCREMENT,
        GVT_LONG,
        GVT_RANGE_LENGTH,
        GVT_RANGE_END,
        GVI_VALUE_AS_RATIO_OF_AVAILABLE_RANGE
    } GenValueType
    GEN_VALUE_MAX_TEXT_LEN    30
    ByteFlags    GenValueStateFlags
        GVSF_INDETERMINATE              0x80
        GVSF_MODIFIED                   0x40
        GVSF_OUT_OF_DATE                0x20
```

Structures

```
typedef struct {
    word     GVI_numMajorIntervals
    word     GVI_numMinorIntervals
} GenValueIntervals
```

Messages

```
void MSG_GEN_VALUE_SET_VALUE(
                    WWFixedAsDWord value,
                    Boolean indeterminate)
void MSG_GEN_VALUE_SET_INTEGER_VALUE(
                    word value, Boolean indeterminate)
WWFixedAsDWord MSG_GEN_VALUE_GET_VALUE()
@alias (MSG_GEN_VALUE_GET_VALUE) word
             MSG_GEN_VALUE_GET_INTEGER_VALUE()
void MSG_GEN_VALUE_SET_MINIMUM(WWFixedAsDWord value)
WWFixedAsDWord MSG_GEN_VALUE_GET_MINIMUM()
void MSG_GEN_VALUE_SET_MAXIMUM(WWFixedAsDWord value)
WWFixedAsDWord MSG_GEN_VALUE_GET_MAXIMUM()
void MSG_GEN_VALUE_SET_INCREMENT(WWFixedAsDWord value)
WWFixedAsDWord MSG_GEN_VALUE_GET_INCREMENT()
void MSG_GEN_VALUE_SET_INDETERMINATE_STATE(
                    Boolean indeterminateState)
Boolean MSG_GEN_VALUE_IS_INDETERMINATE()
void MSG_GEN_VALUE_SET_MODIFIED_STATE(
                    Boolean modifiedState)
Boolean MSG_GEN_VALUE_IS_MODIFIED()
void MSG_GEN_VALUE_SET_DISPLAY_FORMAT(
                    GenValueDisplayFormat format)
GenValueDislayFormat MSG_GEN_VALUE_GET_DISPLAY_FORMAT()
void MSG_GEN_VALUE_SEND_STATUS_MSG(
                    Boolean modifiedState)
optr MSG_GEN_VALUE_GET_DESTINATION()
void MSG_GEN_VALUE_SET_DESTINATION(optr dest)
Message MSG_GEN_VALUE_GET_APPLY_MSG()
void MSG_GEN_VALUE_SET_APPLY_MSG(Message message)
void MSG_GEN_VALUE_SET_RANGE_LENGTH(
                    WWFixedAsDWord value)
WWFixedAsDWord MSG_GEN_VALUE_GET_RANGE_LENGTH()
void MSG_GEN_VALUE_ADD_RANGE_LENGTH()
void MSG_GEN_VALUE_SUBTRACT_RANGE_LENGTH()
void MSG_GEN_VALUE_GET_VALUE_TEXT(
                    char *buffer,
                    GenValueType valueType)
```

```
void MSG_GEN_VALUE_SET_VALUE_FROM_TEXT(
                  char *text, GenValueType valueType)
WWFixedAsDWord MSG_GEN_VALUE_GET_VALUE_RATIO(
                  GenValueType valueType)
void MSG_GEN_VALUE_SET_VALUE_FROM_RATIO(
                  WWFixed ratio,
                  GenValueType valueType)
void MSG_GEN_VALUE_INCREMENT()
void MSG_GEN_VALUE_DECREMENT()
void MSG_GEN_VALUE_SET_VALUE_TO_MINIMUM()
void MSG_GEN_VALUE_SET_VALUE_TO_MAXIMUM()
VisTextFilters MSG_GEN_VALUE_GET_TEXT_FILTER()
byte MSG_GEN_VALUE_GET_MAX_TEXT_LEN()
void MSG_GEN_VALUE_SET_OUT_OF_DATE()
    @prototype void GEN_VALUE_APPLY_MSG(
            WWFixedAsDWord value, word stateFlags)
    @prototype void GEN_VALUE_STATUS_MSG(
            WWFixedAsDWord value, word stateFlags)
```

GenViewClass

```
@class GenViewClass, GenClass
```

Instance Data

```
PointDWFixed GVI_origin = {{0, 0}, {0, 0}}
RectDWord    GVI_docBounds = {0, 0, 0, 0}
PointDWord   GVI_increment = {20, 15}
PointWWFixed GVI_scaleFactor = {{0, 1}, {0, 1}}
ColorQuad    GVI_color = {C_WHITE, 0, 0, 0}
GenViewAttrs GVI_attrs = (GVA_FOCUSABLE)
GenViewDimensionAttrs GVI_horizAttrs = 0
GenViewDimensionAttrs GVI_vertAttrs = 0
GenViewInkType GVI_inkType = GVIT_PRESSES_ARE_NOT_INK
optr         GVI_content
optr         GVI_horizLink
optr         GVI_vertLink
    @default GI_attrs = @default | GA_TARGETABLE
```

Variable Data

```
InkDestinationInfoParams
             ATTR_GEN_VIEW_INK_DESTINATION_INFO
XYSize ATTR_GEN_VIEW_PAGE_SIZE
void ATTR_GEN_VIEW_SCALE_TO_FIT_BASED_ON_X
void ATTR_GEN_VIEW_SCALE_TO_FIT_BOTH_DIMENSIONS
void ATTR_GEN_VIEW_DO_NOT_WIN_SCROLL
```

Hints

```
void HINT_VIEW_LEAVE_ROOM_FOR_VERT_SCROLLER
void HINT_VIEW_LEAVE_ROOM_FOR_HORIZ_SCROLLER
void HINT_VIEW_IMMEDIATE_DRAG_UPDATES
void HINT_VIEW_DELAYED_DRAG_UPDATES
void HINT_VIEW_REMOVE_SCROLLERS_WHEN_NOT_SCROLLABLE
void HINT_VIEW_SHOW_SCROLLERS_WHEN_NOT_SCROLLABLE
optr HINT_VIEW_SHARES_SPACE_WITH_VIEW_ABOVE
optr HINT_VIEW_SHARES_SPACE_WITH_VIEW_BELOW
optr HINT_VIEW_SHARES_SPACE_WITH_VIEW_TO_LEFT
optr HINT_VIEW_SHARES_SPACE_WITH_VIEW_TO_RIGHT
```

Types and Flags

```
WordFlags    MakeRectVisibleFlags
       MRVF_ALWAYS_SCROLL              0x0080
       MRVF_USE_MARGIN_FROM_TOP_LEFT   0x0040
MRVM_0_PERCENT     0
MRVM_25_PERCENT    0xffff/4
MRVM_50_PERCENT    0xffff/2
MRVM_75_PERCENT    0xffff*3/4
MRVM_100_PERCENT   0xffff
ByteEnum     ScaleViewType
       SVT_AROUND_UPPER_LEFT     0
       SVT_AROUND_CENTER         1
       SVT_AROUND_POINT          2
ByteEnum     ScrollAction
       SA_NOTHING                0
       SA_TO_BEGINNING           1
       SA_PAGE_BACK              2
       SA_INC_BACK               3
       SA_INC_FWD                4
       SA_DRAGGING               5
```

```
        SA_PAGE_FWD              6
        SA_TO_END                7
        SA_SCROLL                8
        SA_SCROLL_INTO           9
        SA_INITIAL_POS           10
        SA_SCALE                 11
        SA_PAN                   12
        SA_DRAG_SCROLL           13
        SA_SCROLL_FOR_SIZE_CHANGE 14
    ByteFlags    ScrollFlags
        SF_VERTICAL              0x80
        SF_ABSOLUTE              0x40
        SF_DOC_SIZE_CHANGE       0x20
        SF_WINDOW_NOT_SUSPENDED  0x10
        SF_SCALE_TO_FIT          0x08
        SF_SETUP_HAPPENED        0x04
    VS_TYPICAL          0x8000
    VS_SMALL            0x8001
    VS_LARGE            0x8002
    ByteEnum     GenViewInkType
        GVIT_PRESSES_ARE_NOT_INK        0
        GVIT_INK_WITH_STANDARD_OVERRIDE 1
        GVIT_PRESSES_ARE_INK            2
        GVIT_QUERY_OUTPUT               3
    WordFlags    GenViewAttrs
        GVA_CONTROLLED                  0x8000
        GVA_GENERIC_CONTENTS            0x4000
        GVA_TRACK_SCROLLING             0x2000
        GVA_DRAG_SCROLLING              0x1000
        GVA_NO_WIN_FRAME                0x0800
        GVA_SAME_COLOR_AS_PARENT_WIN    0x0400
        GVA_VIEW_FOLLOWS_CONTENT_GEOMETRY 0x0200
        GVA_WINDOW_COORDINATE_MOUSE_EVENTS 0x0100
        GVA_DONT_SEND_PTR_EVENTS        0x0080
        GVA_DONT_SEND_KBD_RELEASES      0x0040
        GVA_SEND_ALL_KBD_CHARS          0x0020
        GVA_FOCUSABLE                   0x0010
        GVA_SCALE_TO_FIT                0x0008
        GVA_ADJUST_FOR_ASPECT_RATIO     0x0004
    ByteFlags    GenViewDimensionAttrs
        GVDA_SCROLLABLE                 0x80
        GVDA_SPLITTABLE                 0x40
        GVDA_TAIL_ORIENTED              0x20
        GVDA_DONT_DISPLAY_SCROLLBAR     0x10
        GVDA_NO_LARGER_THAN_CONTENT     0x08
        GVDA_NO_SMALLER_THAN_CONTENT    0x04
        GVDA_SIZE_A_MULTIPLE_OF_INCREMENT 0x02
        GVDA_KEEP_ASPECT_RATIO          0x01
```

Structures

```
    typedef struct {
        RectDWord            MRVP_bounds
        word                 MRVP_xMargin
        MakeRectVisibleFlags MRVP_xFlags
        word                 MRVP_yMargin
        MakeRectVisibleFlags MRVP_yFlags
    } MakeRectVisibleParams
```

```
typedef struct {
    optr            TR_object
    ClassStruct     *TR_class
} TargetReference
typedef struct {
    TargetReference VTI_target
    TargetReference VTI_content
} ViewTargetInfo
typedef struct {
    ScrollAction    TSP_action
    ScrollFlags     TSP_flags
    optr            TSP_caller
    PointDWord      TSP_change
    PointDWord      TSP_newOrigin
    PointDWord      TSP_oldOrigin
    sword           TSP_viewWidth
    sword           TSP_viewHeight
} TrackScrollingParams
typedef struct {
    WWFixedAsDWord GSP_yScaleFactor
    WWFixedAsDWord GSP_xScaleFactor
} GetScaleParams
typedef struct {
    optr            IDIP_dest
    word            IDIP_brushSize
    byte            IDIP_color
    Boolean         IDIP_createGState
} InkDestinationInfoParams
```

Macros

```
GVCD_INDEX(val) ((byte) (val))
GVCD_RED(val) ((byte) (val))
GVCD_FLAGS(val) ((byte) ((val) >> 8))
GVCD_BLUE_AND_GREEN(val) ((word) ((val) >> 16))
GVCD_BLUE(val) ((byte) ((val) >> 16))
GVCD_GREEN(val) ((byte) ((val) >> 24))
MAKE_HORIZ_ATTRS(val) ((byte) (val))
MAKE_VERT_ATTRS(val) ((byte) ((val) >> 8))
MAKE_SET_CLEAR_ATTRS(setAttrs, clrAttrs)
             ((((word) (clrAttrs)) << 8) | (setAttrs))
```

Messages

```
void MSG_GEN_VIEW_GET_ORIGIN(PointDWord *origin)
void MSG_GEN_VIEW_SET_ORIGIN(@stack
                  sdword yOrigin, sdword xOrigin)
void MSG_GEN_VIEW_SCROLL(@stack
                  sdword yOffset, sdword xOffset)
void MSG_GEN_VIEW_MAKE_RECT_VISIBLE(@stack
                  word yFlags, word yMargin,
                  word xFlags, word xMargin,
                  sdword bottom, sdword right,
                  sdword top, sdword left)
void MSG_GEN_VIEW_SET_SCALE_FACTOR(@stack
                  sdword yOrigin, sdword xOrigin,
                  ScaleViewType scaleType,
                  WWFixedAsDWord yScaleFactor,
                  WWFixedAsDWord xScaleFactor)
void MSG_GEN_VIEW_GET_SCALE_FACTOR(
```

```
                    GetScaleParams *retValue)
void MSG_GEN_VIEW_SET_CONTENT(optr content)
optr MSG_GEN_VIEW_GET_CONTENT()
WindowHandle MSG_GEN_VIEW_GET_WINDOW()
void MSG_GEN_VIEW_GET_VISIBLE_RECT(RectDWord *rect)
void MSG_GEN_VIEW_GET_INCREMENT(PointDWord *increment)
void MSG_GEN_VIEW_SET_INCREMENT(@stack
                    sdword yIncrement,
                    sdword xIncrement)
void MSG_GEN_VIEW_SUSPEND_UPDATE()
void MSG_GEN_VIEW_UNSUSPEND_UPDATE()
void MSG_GEN_VIEW_SET_DOC_BOUNDS(@stack
                    sdword bottom, sdword right,
                    sdword top, sdword left)
void MSG_GEN_VIEW_GET_DOC_BOUNDS(RectDWord *bounds)
word MSG_GEN_VIEW_GET_ATTRS()
void MSG_GEN_VIEW_SET_ATTRS(
                    word attrsToSet, word attrsToClear,
                    word updateMode)
void MSG_GEN_VIEW_SET_COLOR(
                    byte indexOrRed, byte flags,
                    word greenBlue)
dword MSG_GEN_VIEW_GET_COLOR()
SizeAsDWord MSG_GEN_VIEW_CALC_WIN_SIZE(
                    word width, word height)
void MSG_GEN_VIEW_SET_PTR_IMAGE(
                    optr pointerDef,
                    PtrImageLevel level)
void MSG_GEN_VIEW_UPDATE_CONTENT_TARGET_INFO(
                    ViewTargetInfo *targetInfo)
void MSG_GEN_VIEW_INITIATE_DRAG_SCROLL()
word MSG_GEN_VIEW_GET_DIMENSION_ATTRS()
void MSG_GEN_VIEW_SET_DIMENSION_ATTRS(
                    word horizAttrsToSetClear,
                    word vertAttrsToSetClear,
                    word updateMode)
void MSG_GEN_VIEW_SCROLL_TOP()
void MSG_GEN_VIEW_SCROLL_PAGE_UP()
void MSG_GEN_VIEW_SCROLL_UP()
void MSG_GEN_VIEW_SCROLL_SET_Y_ORIGIN(sdword yOrigin)
void MSG_GEN_VIEW_SCROLL_DOWN()
void MSG_GEN_VIEW_SCROLL_PAGE_DOWN()
void MSG_GEN_VIEW_SCROLL_BOTTOM()
void MSG_GEN_VIEW_SCROLL_LEFT_EDGE()
void MSG_GEN_VIEW_SCROLL_PAGE_LEFT()
void MSG_GEN_VIEW_SCROLL_LEFT()
void MSG_GEN_VIEW_SCROLL_SET_X_ORIGIN(sdword xOrigin)
void MSG_GEN_VIEW_SCROLL_RIGHT()
void MSG_GEN_VIEW_SCROLL_PAGE_RIGHT()
void MSG_GEN_VIEW_SCROLL_RIGHT_EDGE()
void MSG_GEN_VIEW_ALLOW_GLOBAL_TRANSFER()
void MSG_GEN_VIEW_SET_DRAG_BOUNDS(@stack
                    sdword bottom, sdword right,
                    sdword top, sdword left)
void MSG_GEN_VIEW_SETUP_TRACKING_ARGS(
                    TrackScrollingParams *args)
void MSG_GEN_VIEW_TRACKING_COMPLETE(
```

```
                    TrackScrollingParams *args)
optr MSG_GEN_VIEW_DETERMINE_VIS_PARENT(optr child)
void MSG_GEN_VIEW_SEND_TO_LINKS(
                    EventHandle event, optr originator)
void MSG_GEN_VIEW_SEND_TO_VLINK(
                    EventHandle event, optr originator)
void MSG_GEN_VIEW_SEND_TO_HLINK(
                    EventHandle event, optr originator)
void MSG_GEN_VIEW_CALL_WITHOUT_LINKS(
                    EventHandle event,
                    MessageFlags messageFlags)
void MSG_GEN_VIEW_SET_ORIGIN_LOW(@stack
                    sdword yOrigin, sdword xOrigin)
void MSG_GEN_VIEW_SET_INK_TYPE(GenViewInkType inkType)
void MSG_GEN_VIEW_SET_EXTENDED_INK_TYPE(
                    Boolean createGState,
                    Color inkColor,
                    word brushSize,
                    optr destObj)
void MSG_GEN_VIEW_RESET_EXTENDED_INK_TYPE()
void MSG_GEN_VIEW_SEND_NOTIFICATION()
void MSG_GEN_VIEW_SCALE_LOW(@stack
                    sdword yOrigin, sdword xOrigin,
                    ScaleViewType scaleType,
                    WWFixedAsDWord yScaleFactor,
                    WWFixedAsDWord xScaleFactor)
void MSG_GEN_VIEW_REDRAW_CONTENT()
void MSG_GEN_VIEW_SET_CONTROLLED_ATTRS(
                    GenViewControlAttrs controlAttrs,
                    word scaleFactor)
```

GenViewControlClass

```
@class GenViewControlClass, GenControlClass
```

Instance Data

```
word                GVCI_minZoom = DEFAULT_ZOOM_MINIMUM
word                GVCI_maxZoom = DEFAULT_ZOOM_MAXIMUM
word                GVCI_scale = 100
GenViewControlAttrs GVCI_attrs = (GVCA_SHOW_HORIZONTAL |
                GVCA_SHOW_VERTICAL | GVCA_APPLY_TO_ALL)
    @default GI_attrs = (@default | GA_KBD_SEARCH_PATH)
```

Variable Data

```
void ATR_GEN_VIEW_CONTROL_LARGE_ZOOM
```

Types and Flags

```
WordFlags    GVCFeatures
      GVCF_MAIN_100             0x4000
      GVCF_MAIN_SCALE_TO_FIT    0x2000
      GVCF_ZOOM_IN              0x1000
      GVCF_ZOOM_OUT             0x0800
      GVCF_REDUCE               0x0400
      GVCF_100                  0x0200
      GVCF_ENLARGE              0x0100
      GVCF_BIG_ENLARGE          0x0080
      GVCF_SCALE_TO_FIT         0x0040
      GVCF_ADJUST_ASPECT_RATIO  0x0020
      GVCF_APPLY_TO_ALL         0x0010
      GVCF_SHOW_HORIZONTAL      0x0008
      GVCF_SHOW_VERTICAL        0x0004
      GVCF_CUSTOM_SCALE         0x0002
      GVCF_REDRAW               0x0001
WordFlags    GVCToolboxFeatures
      GVCTF_100                 0x1000
      GVCTF_SCALE_TO_FIT        0x0800
      GVCTF_ZOOM_IN             0x0400
      GVCTF_ZOOM_OUT            0x0200
      GVCTF_REDRAW              0x0100
      GVCTF_PAGE_LEFT           0x0080
      GVCTF_PAGE_RIGHT          0x0040
      GVCTF_PAGE_UP             0x0020
      GVCTF_PAGE_DOWN           0x0010
      GVCTF_ADJUST_ASPECT_RATIO 0x0008
      GVCTF_APPLY_TO_ALL        0x0004
      GVCTF_SHOW_HORIZONTAL     0x0002
      GVCTF_SHOW_VERTICAL       0x0001
GVC_DEFAULT_FEATURES      (GVCF_MAIN_100 |
            GVCF_MAIN_SCALE_TO_FIT | GVCF_ZOOM_IN |
            GVCF_ZOOM_OUT | GVCF_REDUCE | GVCF_100 |
            GVCF_ENLARGE | GVCF_SCALE_TO_FIT |
            GVCF_ADJUST_ASPECT_RATIO |
            GVCF_APPLY_TO_ALL | GVCF_SHOW_HORIZONTAL |
            GVCF_SHOW_VERTICAL | GVCF_CUSTOM_SCALE)
GVC_DEFAULT_TOOLBOX_FEATURES (GVCTF_100 |
            GVCTF_ZOOM_IN | GVCTF_ZOOM_OUT)
GVC_SUGGESTED_SIMPLE_FEATURES (GVCF_MAIN_100 |
            GVCF_MAIN_SCALE_TO_FIT | GVCF_ZOOM_IN |
            GVCF_ZOOM_OUT)
```

```
GVC_SUGGESTED_INTRODUCTORY_FEATURES (GVCF_MAIN_100 |
            GVCF_ZOOM_IN | GVCF_ZOOM_OUT)
GVC_SUGGESTED_BEGINNING_FEATURES
            (GVC_SUGGESTED_INTRODUCTORY_FEATURES |
            GVCF_MAIN_SCALE_TO_FIT)
DEFAULT_ZOOM_MINIMUM      25
DEFAULT_ZOOM_MAXIMUM      200
typedef enum /* word */ {
    GVCSSF_TO_FIT,
} GenViewControlSpecialScaleFactor
WordFlags GenViewControlAttrs
      GVCA_ADJUST_ASPECT_RATIO        0x8000
      GVCA_APPLY_TO_ALL               0x4000
      GVCA_SHOW_HORIZONTAL            0x2000
      GVCA_SHOW_VERTICAL              0x1000
```

Structures

```
typedef struct {
    PointDWFixed          NVSC_origin
    RectDWord             NVSC_docBounds
    PointDWord            NVSC_increment
    PointWWFixed          NVSC_scaleFactor
    ColorQuad             NVSC_color
    GenViewAttrs          NVSC_attrs
    GenViewDimensionAttrs NVSC_horizAttrs
    GenViewDimensionAttrs NVSC_vertAttrs
    GenViewInkType        NVSC_inkType
    XYSize                NVSC_contentSize
    XYSize                NVSC_contentScreenSize
    PointDWord            NVSC_originRelative
    PointDWord            NVSC_documentSize
} NotifyViewStateChange
typedef struct {
    optr                  NVO_view
} NotifyViewOpening
```

Messages

```
void MSG_GEN_VIEW_CONTROL_SET_ATTRS(
                    GenViewControlAttrs attrsToSet,
                    GenViewControlAttrs attrsToClear)
void MSG_GEN_VIEW_CONTROL_SET_MINIMUM_SCALE_FACTOR(
                    word minimumScaleFactor)
void MSG_GEN_VIEW_CONTROL_SET_MAXIMUM_SCALE_FACTOR(
                    word maximumScaleFactor)
void MSG_GVC_SET_SCALE()
void MSG_GVC_SET_SCALE_VIA_LIST()
void MSG_GVC_SET_ATTRS()
void MSG_GVC_REDRAW()
void MSG_GVC_ZOOM_IN()
void MSG_GVC_ZOOM_OUT()
void MSG_GVC_PAGE_LEFT()
void MSG_GVC_PAGE_RIGHT()
void MSG_GVC_PAGE_UP()
void MSG_GVC_PAGE_DOWN()
```

GrObjAlignDistributeControlClass

```
@class GrObjAlignDistributeControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GII_type = GIT_PROPERTIES
@default GII_visibility = GIV_DIALOG
```

Types and Flags

```
WordFlags    GrObjAlignDistributeControlFeatures
      GOADCF_ALIGN_LEFT               0x8000
      GOADCF_ALIGN_CENTER_HORIZONTALLY0x4000
      GOADCF_ALIGN_RIGHT              0x2000
      GOADCF_ALIGN_WIDTH              0x1000
      GOADCF_ALIGN_TOP                0x800
      GOADCF_ALIGN_CENTER_VERTICALLY  0x400
      GOADCF_ALIGN_BOTTOM             0x200
      GOADCF_ALIGN_HEIGHT             0x100
      GOADCF_DISTRIBUTE_LEFT          0x80
      GOADCF_DISTRIBUTE_CENTER_HORIZONTALLY0x40
      GOADCF_DISTRIBUTE_RIGHT         0x20
      GOADCF_DISTRIBUTE_WIDTH         0x10
      GOADCF_DISTRIBUTE_TOP           0x8
      GOADCF_DISTRIBUTE_CENTER_VERTICALLY 0x4
      GOADCF_DISTRIBUTE_BOTTOM        0x2
      GOADCF_DISTRIBUTE_HEIGHT        0x1
GOADC_DEFAULT_FEATURES
                  GrObjAlignDistributeControlFeatures
```

GrObjAlignToGridControlClass

```
@class GrObjAlignToGridControlClass, GenControlClass
```

Instance Data

```
@default GII_type = (GIT_COMMAND)
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GOATGCFeatures
      GOATGCF_ALIGN_TO_GRID           0x01
GOATGC_DEFAULT_FEATURES         (GOATGCF_ALIGN_TO_GRID)
GOATGC_DEFAULT_TOOLBOX_FEATURES 0
```

Messages

```
MSG_GOATGC_ALIGN_TO_GRID
```

GrObjArcControlClass

```
@class GrObjArcControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GOArcCFeatures
      GOACF_START_ANGLE         0x8
      GOACF_END_ANGLE           0x4
      GOACF_PIE_TYPE            0x2
```

```
        GOACF_CHORD_TYPE          0x1
    GOArcC_DEFAULT_FEATURES   (GOACF_START_ANGLE |
                               GOACF_END_ANGLE |
                               GOACF_PIE_TYPE |
                               GOACF_CHORD_TYPE)
    GOArcC_DEFAULT_TOOLBOX_FEATURES 0
```

Messages

```
MSG_GOAC_SET_START_ANGLE
MSG_GOAC_SET_END_ANGLE
MSG_GOAC_SET_ARC_CLOSE_TYPE
```

GrObjAreaAttrControlClass

```
@class GrObjAreaAttrControlClass, GenControlClass
```

Instance Data

```
    @default GCI_output = (TO_APP_TARGET)
    @default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
ByteFlags    GOAACFeatures
      GOAACF_MM_CLEAR           0x0100
      GOAACF_MM_COPY            0x0080
      GOAACF_MM_NOP             0x0040
      GOAACF_MM_AND             0x0020
      GOAACF_MM_INVERT          0x0010
      GOAACF_MM_XOR             0x0008
      GOAACF_MM_SET             0x0004
      GOAACF_MM_OR              0x0002
      GOAACF_TRANSPARENCY       0x0001
GOAAC_DEFAULT_FEATURES    (GOAACF_TRANSPARENCY |
                           GOAACF_MM_COPY |
                           GOAACF_MM_INVERT |
                           GOAACF_MM_XOR |
                           GOAACF_MM_AND | GOAACF_MM_OR)
GOAAC_DEFAULT_TOOLBOX_FEATURES 0
```

Messages

```
MSG_GOAAC_SET_MIX_MODE
MSG_GOAAC_SET_AREA_TRANSPARENCY
```

GrObjAreaColorSelectorClass

```
@class GrObjAreaColorSelectorClass, ColorSelectorClass
```

Instance Data

```
    @default GCI_output = (TO_APP_TARGET)
    @default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
GOACS_DEFAULT_FEATURES (CSF_INDEX | CSF_RGB |
                        CSF_DRAW_MASK | CSF_PATTERN)
```

GrObjAttributeManagerClass

```
@class GrObjAttributeManagerClass, GrObjClass
```

Instance Data

```
word         GOAMI_areaAttrArrayHandle = 0
word         GOAMI_lineAttrArrayHandle = 0
word         GOAMI_grObjStyleArrayHandle = 0
ChunkHandle  GOAMI_bodyList = 0
word         GOAMI_charAttrArrayHandle = 0
word         GOAMI_paraAttrArrayHandle = 0
word         GOAMI_typeArrayHandle = 0
word         GOAMI_graphicArrayHandle = 0
word         GOAMI_nameArrayHandle = 0
word         GOAMI_textStyleArrayHandle = 0
optr         GOAMI_text
    default GOI_optFlags = (GOOF_GROBJ_INVALID |
                            GOOF_ATTRIBUTE_MANAGER)
```

Types and Flags

```
GROBJ_VM_ELEMENT_ARRAY_CHUNK (sizeof LMemblockHeader)
```

Messages

```
MSG_GOAM_ATTACH_AND_CREATE_ARRAYS
MSG_GOAM_ADD_AREA_ATTR_ELEMENT
MSG_GOAM_ADD_LINE_ATTR_ELEMENT
MSG_GOAM_DEREF_AREA_ATTR_ELEMENT_TOKEN
MSG_GOAM_DEREF_LINE_ATTR_ELEMENT_TOKEN
MSG_GOAM_ADD_REF_AREA_ATTR_ELEMENT_TOKEN
MSG_GOAM_ADD_REF_LINE_ATTR_ELEMENT_TOKEN
MSG_GOAM_GET_FULL_AREA_ATTR_ELEMENT
MSG_GOAM_GET_FULL_LINE_ATTR_ELEMENT
MSG_GOAM_GET_STYLE_ARRAY
MSG_GOAM_GET_AREA_ATTR_ARRAY
MSG_GOAM_GET_LINE_ATTR_ARRAY
MSG_GOAM_GET_AREA_AND_LINE_TOKENS_FROM_STYLE
MSG_GOAM_ATTACH_BODY
MSG_GOAM_INVALIDATE_BODIES
MSG_GOAM_SUBST_AREA_TOKEN
MSG_GOAM_SUBST_LINE_TOKEN
MSG_GOAM_GET_TEXT_OD
MSG_GOAM_GET_TEXT_ARRAYS
MSG_GOAM_LOAD_STYLE_SHEET_PARAMS
MSG_GOAM_DETACH_BODY
MSG_GOAM_SUBST_TEXT_ATTR_TOKEN
MSG_GOAM_RECALC_FOR_TEXT_ATTR_CHANGE
void MSG_GOAM_LOAD_STYLE_SHEET
void MSG_GOAM_SET_GROBJ_DRAW_FLAGS(
                    GrObjDrawFlags flagsToSet,
                    GrObjDrawFlags flagsToReset)
```

GrObjBackgroundColorSelectorClass

```
@class GrObjBackgroundColorSelectorClass,
                                ColorSelectorClass
```

Instance Data

```
    @default GCI_output = (TO_APP_TARGET)
```

```
@default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
GOBGCS_DEFAULT_FEATURES   (CSF_INDEX | CSF_RGB)
```


GrObjBitmapClass

```
@class GrObjBitmapClass, GrObjVisClass
```

GrObjBodyClass

```
@class GrObjBodyClass, VisCompClass
```

Instance Data

```
RectDWord           GBI_bounds = {0,0,0,0}
CompPart            GBI_drawComp = {NullOptr}
CompPart            GBI_reverseComp = {NullOptr}
word                GBI_childCount
optr                GBI_selectionArray
HierarchicalGrab    GBI_targetExcl = {NullOptr, 0}
HierarchicalGrab    GBI_focusExcl = {NullOptr, 0}
optr                GBI_mouseGrab
word                GBI_objBlockArray
GrObjFunctionsActive GBI_defaultOptions
GrObjFunctionsActive GBI_currentModifiers
GrObjFunctionsActive GBI_currentOptions
GrObjBodyFlags      GBI_flags = (GBF_DEFAULT_TARGET |
                                 GBF_DEFAULT_FOCUS)
GrObjDrawFlags      GBI_drawFlags
GrObjFileStatus     GBI_fileStatus
word                GBI_graphicsState = 0
optr                GBI_head
optr                GBI_goam
optr                GBI_ruler
word                GBI_priorityList = 0
byte                GBI_desiredHandleSize =
                          DEFAULT_DESIRED_HANDLE_SIZE
byte                GBI_curHandleWidth = 0
byte                GBI_curHandleHeight = 0
BBFixed             GBI_curNudgeX = {0, 0}
BBFixed             GBI_curNudgeY = {0, 0}
PointWWFixed        GBI_curScaleFactor =
                          {MakeWWFixed(1),
                           MakeWWFixed(0)}
PointDWFixed        GBI_interestingPoint =
                          {{0, -30000}, {0, -30000}}
PointDWFixed        GBI_lastPtr = {0,0}
word                GBI_suspendCount = 0
GrObjBodyUnsuspendOps GBI_unsuspendOps = 0
VisTextNotificationFlags GBI_textUnsuspendOps = 0
word                GBI_reserved1 = 0
word                GBI_reserved2 = 0
    @default VI_optFlags = 0
    @default VI_typeFlags =     (VTF_IS_COMPOSITE |
                                 VTF_IS_INPUT_NODE)
    @default VI_attrs =   (VA_DRAWABLE | VA_DETECTABLE |
                           VA_FULLY_ENABLED )
```

Variable Data

```
GrObjBodyPasteCallBackStruct ATTR_GB_PASTE_CALL_BACK
```

Types and Flags

```
WordFlags    GrObjBodyUnsuspendOps
ByteFlags    GrObjBodyFlags
      GBF_HAS_ACTION_NOTIFICATION     0x04
      GBF_DEFAULT_TARGET              0x02
      GBF_DEFAULT_FOCUS               0x01
WordFlags    GrObjDrawFlags
      GODF_DRAW_QUICK_VIEW                    0x100
      GODF_DRAW_CLIP_ONLY                     0x80
      GODF_DRAW_WRAP_TEXT_INSIDE_ONLY         0x40
      GODF_DRAW_WRAP_TEXT_AROUND_ONLY         0x20
      GODF_DRAW_WITH_INCREASED_RESOLUTION     0x10
      GODF_DRAW_INSTRUCTIONS                  0x08
      GODF_DRAW_SELECTED_OBJECTS_ONLY         0x04
      GODF_DRAW_OBJECTS_ONLY                  0x02
      GODF_PRINT_INSTRUCTIONS                 0x01
ByteFlags    GrObjsInRectSpecial
      GOIRS_IGNORE_TEMP         0x04
      GOIRS_IGNORE_RECT         0x02
      GOIRS_XOR_CHECK           0x01
WordFlags    GrObjBodyAddGrObjFlags
      GOBAGOF_DRAW_LIST_POSITION      0x8000
      GOBAGOF_REFERENCE               0x7fff
GOBAGOR_FIRST      CCO_FIRST
GOBAGOR_LAST       CCO_LAST
```

Structures

```
typedef struct {
    word                 GOIRD_tempMessage
    word                 GOIRD_tempMessageDX
    word                 GOIRD_inRectMessage
    word                 GOIRD_inRectMessageDX
    RectDWord            GOIRD_rect
    GrObjsInRectSpecial  GOIRD_special
    word                 align
} GrObjsInRectData
typedef struct {
    word                        GBCDP_repetitions
    PointDWFixed                GBCDP_move
    WWFixed                     GBCDP_rotation
    GrObjHandleSpecification    GBCDP_rotateAnchor
    GrObjAnchoredSkewData       GBCDP_skew
    GrObjAnchoredScaleData      GBCDP_scale
} GrObjBodyCustomDuplicateParams
typedef struct {
    word          GOBPCBS_message
    optr          GOBPCBS_optr
} GrObjBodyPasteCallBackStruct
```

Messages

```
MSG_GB_CREATE_GSTATE
MSG_GB_INVALIDATE
MSG_GB_GIVE_ME_MOUSE_EVENTS
MSG_GB_DONT_GIVE_ME_MOUSE_EVENTS
MSG_GB_ATTACH_HEAD
```

```
void MSG_GB_ATTACH_GOAM(optr GrObjAttrManager)
MSG_GB_ATTACH_RULER
void MSG_GB_ADD_GROBJ(optr object, word flags)
MSG_GB_REMOVE_GROBJ
MSG_GB_FIND_GROBJ */
MSG_GB_PULL_SELECTED_GROBJS_TO_FRONT
MSG_GB_PUSH_SELECTED_GROBJS_TO_BACK
MSG_GB_SHUFFLE_SELECTED_GROBJS_UP
MSG_GB_SHUFFLE_SELECTED_GROBJS_DOWN
void MSG_GB_SET_DESIRED_HANDLE_SIZE(byte handleSize)
MSG_GB_GET_DESIRED_HANDLE_SIZE
void MSG_GB_ATTACH_UI(optr GrObjHead)
void MSG_GB_DETACH_UI()
MSG_GB_ADD_GROBJ_TO_SELECTION_LIST
MSG_GB_REMOVE_GROBJ_FROM_SELECTION_LIST
MSG_GB_REMOVE_ALL_GROBJS_FROM_SELECTION_LIST
MSG_GB_SEND_CLASSED_EVENT_TO_SELECTED_GROBJS
MSG_GB_GET_NUM_SELECTED_GROBJS
MSG_GB_GET_BOUNDS_OF_SELECTED_GROBJS
MSG_GB_GET_DWF_BOUNDS_OF_SELECTED_GROBJS
MSG_GB_GET_SUMMED_DWF_DIMENSIONS_OF_SELECTED_GROBJS
MSG_GB_DELETE_SELECTED_GROBJS
MSG_GB_PROCESS_ALL_GROBJS_IN_RECT
MSG_GB_INCREASE_POTENTIAL_EXPANSION
MSG_GB_DECREASE_POTENTIAL_EXPANSION
MSG_GB_FILL_PRIORITY_LIST
MSG_GB_MESSAGE_TO_FLOATER_IF_PARENT
MSG_GB_UPDATE_UI_CONTROLLERS
MSG_GB_UPDATE_EXTENDED_AREA_ATTR_CONTROLLERS
MSG_GB_UPDATE_EXTENDED_AREA_ATTR_CONTROLLERS_WITH_DEFAU
LTS
MSG_GB_UPDATE_EXTENDED_LINE_ATTR_CONTROLLERS
MSG_GB_UPDATE_EXTENDED_LINE_ATTR_CONTROLLERS_WITH_DEFAU
LTS
MSG_GB_UPDATE_INSTRUCTION_CONTROLLERS
MSG_GB_ADD_DUPLICATE_FLOATER
MSG_GB_PRIORITY_LIST_RESET
MSG_GB_PRIORITY_LIST_GET_NUM_ELEMENTS
MSG_GB_PRIORITY_LIST_GET_ELEMENT
MSG_GB_PRIORITY_LIST_INIT
MSG_GB_GROUP_SELECTED_GROBJS
MSG_GB_UNGROUP_SELECTED_GROUPS
MSG_GB_TRANSFER_GROBJ_FROM_GROUP
MSG_GB_CLOSE_FINISH_UP
MSG_GB_CLEAR()
MSG_GB_ALIGN_SELECTED_GROBJS
MSG_GB_CREATE_SORTABLE_ARRAY
MSG_GB_DESTROY_SORTABLE_ARRAY
MSG_GB_FILL_SORTABLE_ARRAY_USING_GO_DWF_BOUNDS
MSG_GB_FILL_SORTABLE_ARRAY_USING_GO_CENTERS
MSG_GB_SORT_SORTABLE_ARRAY
MSG_GB_GET_CENTER_OF_SELECTED_GROBJS
MSG_GB_GET_CENTER_OF_FIRST_SELECTED_GROBJ
MSG_GB_GET_CENTER_OF_LAST_SELECTED_GROBJ
MSG_GB_GET_DWF_BOUNDS_OF_FIRST_SELECTED_GROBJ
MSG_GB_GET_DWF_BOUNDS_OF_LAST_SELECTED_GROBJ
MSG_GB_GET_BLOCK_FOR_ONE_GROBJ
```

```
MSG_GB_GET_WINDOW
void MSG_GB_ADD_GROBJ_THEN_DRAW(optr object, word flags)
void MSG_GB_SET_BOUNDS(RectDWord *bounds)
void MSG_GB_GET_BOUNDS(RectDWord *bounds)
void MSG_GB_SET_ACTION_NOTIFICATION_OUTPUT(
                  optr object, word messageNumber)
void MSG_GB_SUSPEND_ACTION_NOTIFICATION()
void MSG_GB_UNSUSPEND_ACTION_NOTIFICATION()
optr MSG_GB_INSTANTIATE_GROBJ(ClassStruct *class)
MSG_GB_SUBST_AREA_TOKEN
MSG_GB_SUBST_LINE_TOKEN
MSG_GB_CREATE_GROBJ_TRANSFER_FORMAT
MSG_GB_CREATE_GSTRING_TRANSFER_FORMAT
MSG_GB_SEND_CLASSED_EVENT_SET_DEFAULT_ATTRS
MSG_GB_CHANGE_GROBJ_DEPTH
MSG_GB_DRAW( GStateHandle gstate,
                  DrawFlags visDrawFlags,
                  GrObjDrawFlags GODrawFlags)
MSG_GB_EXPORT
MSG_GB_GRAB_TARGET_FOCUS
MSG_GB_GENERATE_TEXT_NOTIFY
MSG_GB_GENERATE_SPLINE_NOTIFY
MSG_GB_DETACH_GOAM
MSG_GB_IMPORT
MSG_GB_CONVERT_SELECTED_GROBJS_TO_BITMAP
MSG_GB_CONVERT_SELECTED_GROBJS_TO_GRAPHIC
MSG_GB_IGNORE_UNDO_ACTIONS_AND_SUSPEND
MSG_GB_UNSUSPEND_AND_ACCEPT_UNDO_ACTIONS
MSG_GB_MAKE_INSTRUCTIONS_SELECTABLE_AND_EDITABLE
MSG_GB_MAKE_INSTRUCTIONS_UNSELECTABLE_AND_UNEDITABLE
void MSG_GB_SET_GROBJ_DRAW_FLAGS(
                  GrObjDrawFlags flagsToSet,
                  GrObjDrawFlags flagsToReset)
MSG_GB_DELETE_INSTRUCTIONS
MSG_GB_PASTE_INSIDE
MSG_GB_CLONE_SELECTED_GROBJS
MSG_GB_CREATE_POLYGON
MSG_GB_CREATE_STAR
MSG_GB_SUBST_TEXT_ATTR_TOKEN
MSG_GB_RECALC_FOR_TEXT_ATTR_CHANGE
void MSG_GB_GET_BOUNDS_OF_GROBJS(RectDWord *retValue)
void MSG_GB_CUSTOM_DUPLICATE_SELECTED_GROBJS(
          GrObjBodyCustomDuplicateParams *cdParams)
MSG_GB_EXPORT_SELECTED_GROBJS
MSG_GB_HIDE_UNSELECTED_GROBJS
MSG_GB_SHOW_ALL_GROBJS
MSG_GB_CREATE_GROBJ
MSG_GB_DUPLICATE_SELECTED_GROBJS
MSG_GB_FIND_NEXT_GROBJ_THAT_OVERLAPS
MSG_GB_FIND_PREV_GROBJ_THAT_OVERLAPS
MSG_GB_ZOOM_IN_ABOUT_POINT
MSG_GB_ZOOM_OUT_ABOUT_POINT
MSG_GB_EVALUATE_MOUSE_POSITION
MSG_GB_EVALUATE_POINT_FOR_HANDLE
MSG_GB_EVALUATE_POINT_FOR_BOUNDS
MSG_GB_SET_NORMAL_SIZE_ABOUT_POINT
MSG_GB_DRAW_GROBJ
```

```
MSG_GB_SEND_TO_SELECTED_GROBJS_SHARE_DATA
MSG_GB_STANDARD_PASTE_CALL_BACK
MSG_GB_PASTE
MSG_GB_REORDER_SELECTION_ARRAY
MSG_GB_ABORT_SEARCH_SPELL_MESSAGE
void MSG_GB_QUICK_PASTE(PointDWFixed *pasteAt)
GrObjDrawFlags MSG_GB_GET_GROBJ_DRAW_FLAGS(
                    DrawFlags drawFlags)
void MSG_GB_SET_BOUNDS_WITHOUT_MARKING_DIRTY(
                    RectDWord *bounds)
void MSG_GB_SET_GROBJ_DRAW_FLAGS_NO_BROADCAST(
                    GrObjDrawFlags flagsToSet,
                    GrObjDrawFlags flagsToReset)
MSG_GB_QUICK_PASTE_CALL_BACK
```

GrObjClass

```
@class GrObjClass, MetaClass, master
```

Instance Data

```
LinkPart            GOI_drawLink
LinkPart            GOI_reverseLink
GrObjAttrFlags      GOI_attrFlags =
                    (GOAF_INSERT_DELETE_MOVE_ALLOWED |
                    GOAF_INSERT_DELETE_RESIZE_ALLOWED |
                    GOAF_INSERT_DELETE_DELETE_ALLOWED )
GrObjOptimizationFlags GOI_optFlags =
                            (GOOF_GROBJ_INVALID)
GrObjMessageOptimizationFlags GOI_msgOptFlags =(0)
GrObjLocks          GOI_locks = 0
GrObjActionModes    GOI_actionModes = 0
GrObjTempModes      GOI_tempState = 0
ChunkHandle         GOI_normalTransform = NullChunk
ChunkHandle         GOI_spriteTransform = NullChunk
word                GOI_areaAttrToken = CA_NULL_ELEMENT
word                GOI_lineAttrToken = CA_NULL_ELEMENT
```

Variable Data

```
GrObjObjManipData ATTR_GO_OBJ_MANIP_DATA
GrObjActionNotificationStruct
                            ATTR_GO_ACTION_NOTIFICATION
PointWWFixed ATTR_GO_PARENT_DIMENSIONS_OFFSET
```

Types and Flags

```
ByteFlags    GrObjHandleSpecification
      GOHS_HANDLE_LEFT          0x08
      GOHS_HANDLE_TOP           0x04
      GOHS_HANDLE_RIGHT         0x02
      GOHS_HANDLE_BOTTOM        0x01
HANDLE_MOVE         0
HANDLE_CENTER       HANDLE_MOVE
HANDLE_LEFT_TOP     (GOHS_HANDLE_LEFT | GOHS_HANDLE_TOP)
HANDLE_MIDDLE_TOP   GOHS_HANDLE_TOP
HANDLE_RIGHT_TOP    (GOHS_HANDLE_RIGHT |
                     GOHS_HANDLE_TOP)
HANDLE_LEFT_MIDDLE  GOHS_HANDLE_LEFT
HANDLE_RIGHT_MIDDLE GOHS_HANDLE_RIGHT
HANDLE_LEFT_BOTTOM  (GOHS_HANDLE_LEFT |
```

```
                    GOHS_HANDLE_BOTTOM)
HANDLE_MIDDLE_BOTTOM GOHS_HANDLE_BOTTOM
HANDLE_RIGHT_BOTTOM(GOHS_HANDLE_BOTTOM |
                    GOHS_HANDLE_RIGHT)
ByteFlags    HandleUpdateMode
      HUM_NOW            0
      HUM_MANUAL         1
WordFlags    GrObjBaseAreaAttrDiffs
      GOBAAD_MULTIPLE_ELEMENT_TYPES          0x8000
      GOBAAD_MULTIPLE_STYLE_ELEMENTS         0x4000
      GOBAAD_MULTIPLE_COLORS                 0x2000
      GOBAAD_MULTIPLE_BACKGROUND_COLORS      0x1000
      GOBAAD_MULTIPLE_MASKS                  0x0800
      GOBAAD_MULTIPLE_PATTERNS               0x0400
      GOBAAD_MULTIPLE_DRAW_MODES             0x0200
      GOBAAD_MULTIPLE_INFOS                  0x0100
      GOBAAD_MULTIPLE_GRADIENT_END_COLORS    0x0080
      GOBAAD_MULTIPLE_GRADIENT_TYPES         0x0040
      GOBAAD_MULTIPLE_GRADIENT_INTERVALS     0x0020
      GOBAAD_FIRST_RECIPIENT                 0x0001
FUTURE_AREA_ATTR_ELEMENT_SIZE   50
WordFlags    GrObjBaseLineAttrDiffs
      GOBLAD_MULTIPLE_STYLE_ELEMENTS         0x8000
      GOBLAD_MULTIPLE_ELEMENT_TYPES          0x4000
      GOBLAD_MULTIPLE_COLORS                 0x2000
      GOBLAD_MULTIPLE_ENDS                   0x1000
      GOBLAD_MULTIPLE_JOINS                  0x0800
      GOBLAD_MULTIPLE_WIDTHS                 0x0400
      GOBLAD_MULTIPLE_MASKS                  0x0200
      GOBLAD_MULTIPLE_STYLES                 0x0100
      GOBLAD_MULTIPLE_DRAW_MODES             0x0080
      GOBLAD_MULTIPLE_MITER_LIMITS           0x0040
      GOBLAD_FIRST_RECIPIENT                 0x0001
FUTURE_LINE_ATTR_ELEMENT_DATA_SIZE 50
ByteFlags    GrObjSelectionStateFlags
      GSSF_EDITING                           0x20
      GSSF_UNGROUPABLE                       0x10
      GSSF_TEXT_SELECTED                     0x08
      GSSF_BITMAP_SELECTED                   0x04
      GSSF_SPLINE_SELECTED                   0x02
      GSSF_ARC_SELECTED                      0x01
ByteFlags    GrObjSelectionStateDiffs
      GSSD_MULTIPLE_CLASSES                  0x80
      GSSD_MULTIPLE_ARC_CLOSE_TYPES          0x40
      GSSD_MULTIPLE_ARC_START_ANGLES         0x20
      GSSD_MULTIPLE_ARC_END_ANGLES           0x10
WordFlags    GrObjUINotificationTypes
      GOUINT_AREA                            0x8000
      GOUINT_LINE                            0x4000
      GOUINT_SELECT                          0x2000
      GOUINT_GROBJ_SELECT                    0x1000
      GOUINT_STYLE_SHEET                     0x0800
      GOUINT_STYLE                           0x0400
      GOUINT_SPLINE                          0x0200
```

Structures

```
typedef struct {
```

```
    WWFixed        RWWF_left
    WWFixed        RWWF_top
    WWFixed        RWWF_right
    WWFixed        RWWF_bottom
} RectWWFixed
typedef struct {
    WWFixed        GOSD_xScale
    WWFixed        GOSD_yScale
} GrObjScaleData
typedef struct {
    GrObjScaleData              GOASD_scale
    GrObjHandleSpecification    GOASD_scaleAnchor
} GrObjAnchoredScaleData
typedef struct {
    GrObjBaseAreaAttrElement    GOFAAE_base
    byte GOFAAE_future[FUTURE_AREA_ATTR_ELEMENT_SIZE]
} GrObjFullAreaAttrElement
typedef struct {
    GrObjBaseAreaAttrElement    GNAAC_areaAttr
    GrObjBaseAreaAttrDiffs      GNAAC_areaAttrDiffs
} GrObjNotifyAreaAttrChange
typedef struct {
    GrObjBaseLineAttrElement    GOFLAE_base
    byte
      GOFLAE_future[FUTURE_LINE_ATTR_ELEMENT_DATA_SIZE]
} GrObjFullLineAttrElement
typedef struct {
    word                        GSS_numSelected
    ClassStruct                 *GSS_classSelected
    GrObjSelectionStateFlags    GSS_flags
    GrObjAttrFlags              GSS_grObjFlags
    GrObjLocks                  GSS_locks
} GrObjSelectionState
typedef struct {
    WWFixed        GOSD_xDegrees
    WWFixed        GOSD_yDegrees
} GrObjSkewData
typedef struct {
    GrObjSkewData               GOASD_degrees
    GrObjHandleSpecification    GOASD_skewAnchor
} GrObjAnchoredSkewData
typedef struct {
    WWFixed        GTM_e11
    WWFixed        GTM_e12
    WWFixed        GTM_e21
    WWFixed        GTM_e22
} GrObjTransMatrix
typedef struct {
    WWFixed        GOGSP_height
    WWFixed        GOGSP_width
} GOGetSizeParams
typedef struct {
    PointDWFixed   GOID_position
    WWFixed        GOID_width
    WWFixed        GOID_height
} GrObjInitializeData
typedef struct {
```

```
    StyleSheetParams            GTP_ssp
    VisTextSaveStyleSheetParams GTP_textSSP
    PointDWFixed          GTP_selectionCenterDOCUMENT
    Handle                      GTP_optBlock
    Handle                      GTP_vmFile
    word                        GTP_curSlot
    dword                       GTP_id
    word                        GTP_curSize
    word                        GTP_curPos
} GrObjTransferParams
```

Messages

```
void MSG_GO_GAINED_SELECTION_LIST(HandleUpdateMode hum)
void MSG_GO_LOST_SELECTION_LIST()
void MSG_GO_BECOME_SELECTED(HandleUpdateMode hum)
void MSG_GO_TOGGLE_SELECTION()
void MSG_GO_BECOME_UNSELECTED()
MSG_GO_UNDRAW_SPRITE
MSG_GO_DRAW_SPRITE
MSG_GO_DRAW_SPRITE_RAW
MSG_GO_DRAW_HANDLES
MSG_GO_UNDRAW_HANDLES
MSG_GO_DRAW_HANDLES_RAW
MSG_GO_DRAW_HANDLES_FORCE
MSG_GO_DRAW_HANDLES_MATCH
MSG_GO_DRAW_HANDLES_OPPOSITE
MSG_GO_ACTIVATE_MOVE
MSG_GO_ACTIVATE_RESIZE
MSG_GO_ACTIVATE_ROTATE
MSG_GO_ACTIVATE_CREATE
MSG_GO_REACTIVATE_CREATE
MSG_GO_START_CHOOSE_ABS
MSG_GO_START_MOVE_ABS
MSG_GO_JUMP_START_MOVE
MSG_GO_JUMP_START_RESIZE
MSG_GO_JUMP_START_ROTATE
MSG_GO_PTR_CHOOSE_ABS
MSG_GO_PTR_MOVE
MSG_GO_PTR_RESIZE
MSG_GO_PTR_ROTATE
MSG_GO_PTR_MOVE_ABS
MSG_GO_END_CHOOSE_ABS
MSG_GO_END_MOVE_ABS
MSG_GO_END_MOVE
MSG_GO_END_RESIZE
MSG_GO_END_ROTATE
MSG_GO_CLEAR
MSG_GO_INVERT_HANDLES
MSG_GO_INIT_BASIC_DATA
void MSG_GO_FLIP_HORIZ()
void MSG_GO_FLIP_VERT()
void MSG_GO_ROTATE(WWFixed angle,
                   GrObjHandleSpecification center)
void MSG_GO_UNTRANSFORM()
void MSG_GO_MOVE(PointDWFixed *distance)
void MSG_GO_MOVE_CENTER_ABS(PointDWFixed *location)
MSG_GO_ALIGN
```

```
MSG_GO_ALIGN_TO_GRID
MSG_GO_SEND_ANOTHER_TOOL_ACTIVATED
MSG_GO_ANOTHER_TOOL_ACTIVATED
MSG_GO_SPECIAL_RESIZE_CONSTRAIN(
            GrObjHandleSpecification grObjHandleSpec)
MSG_GO_DUPLICATE_FLOATER
MSG_GO_GRAB_MOUSE
MSG_GO_RELEASE_MOUSE
MSG_GO_UNGROUPABLE
MSG_GO_GET_BOUNDING_RECTDWFIXED
MSG_GO_CALC_PARENT_DIMENSIONS
MSG_GO_INIT_CREATE
void MSG_GO_NOTIFY_GROBJ_VALID()
MSG_GO_EVALUATE_PARENT_POINT_FOR_HANDLE
MSG_GO_INVERT_GROBJ_SPRITE
MSG_GO_INVERT_GROBJ_NORMAL_SPRITE
void MSG_GO_INVALIDATE()
void MSG_GO_GET_DW_PARENT_BOUNDS(RectDWord *bounds)
MSG_GO_GET_DWF_PARENT_BOUNDS
MSG_GO_GET_DWF_SELECTION_HANDLE_BOUNDS_FOR_TRIVIAL_REJE
CT
MSG_GO_GET_WWF_PARENT_BOUNDS
void MSG_GO_GET_WWF_OBJECT_BOUNDS(RectWWFixed *retValue)
MSG_GO_BECOME_UNEDITABLE
MSG_GO_EVALUATE_POSITION
MSG_GO_EVALUATE_PARENT_POINT_FOR_SELECTION
void MSG_GO_GET_CENTER(PointDWFixed *center)
void MSG_GO_INIT_TO_DEFAULT_ATTRS()
MSG_GO_DRAW
MSG_GO_LARGE_START_SELECT
MSG_GO_LARGE_START_MOVE_COPY
MSG_GO_LARGE_END_SELECT
MSG_GO_LARGE_END_MOVE_COPY
MSG_GO_LARGE_DRAG_SELECT
MSG_GO_LARGE_DRAG_MOVE_COPY
MSG_GO_LARGE_PTR
MSG_GO_AFTER_ADDED_TO_GROUP
MSG_GO_BEFORE_REMOVED_FROM_GROUP
void MSG_GO_AFTER_ADDED_TO_BODY()
void MSG_GO_BEFORE_REMOVED_FROM_BODY()
void MSG_GO_SET_AREA_ATTR(
                  GrObjBaseAreaAttrElement *attr)
void MSG_GO_SET_AREA_COLOR(byte red,
                  byte green, byte blue)
void MSG_GO_SET_AREA_MASK(SysDrawMask mask)
void MSG_GO_SET_AREA_DRAW_MODE(MixMode mode)
void MSG_GO_SET_TRANSPARENCY(byte transparent)
void MSG_GO_SET_LINE_ATTR(
                  GrObjBaseLineAttrElement *attr)
void MSG_GO_SET_LINE_COLOR(byte red,
                  byte green, byte blue)
void MSG_GO_SET_LINE_MASK(SystemDrawMask drawMask)
void MSG_GO_SET_LINE_END(LineEnd end)
void MSG_GO_SET_LINE_JOIN(LineJoin join)
void MSG_GO_SET_LINE_STYLE(LineStyle style)
void MSG_GO_SET_LINE_WIDTH(WWFixed width)
void MSG_GO_SET_LINE_MITER_LIMIT(WWFixed miterLimit)
```

```
MSG_GO_GET_GROBJ_AREA_TOKEN
MSG_GO_GET_GROBJ_LINE_TOKEN
MSG_GO_SET_GROBJ_AREA_TOKEN
MSG_GO_SET_GROBJ_LINE_TOKEN
MSG_GO_SUBST_AREA_TOKEN
MSG_GO_SUBST_LINE_TOKEN
MSG_GO_GET_ANCHOR_DOCUMENT
MSG_GO_BECOME_EDITABLE
MSG_GO_DRAW_EDIT_INDICATOR
MSG_GO_UNDRAW_EDIT_INDICATOR
MSG_GO_DRAW_EDIT_INDICATOR_RAW
MSG_GO_INVERT_EDIT_INDICATOR
MSG_GO_GROBJ_SPECIFIC_INITIALIZE
MSG_GO_GROBJ_SPECIFIC_INITIALIZE_WITH_DATA_BLOCK
MSG_GO_APPLY_ATTRIBUTES_TO_GSTATE
word MSG_GO_SET_ACTION_NOTIFICATION_OUTPUT(
                    optr object, Message messageNumber)
word MSG_GO_NOTIFY_ACTION(
                    GrObjActionNotificationType action)
void MSG_GO_NUDGE(sword xDistance, sword yDistance)
void MSG_GO_SET_SIZE(PointWWFixed *size)
void MSG_GO_SET_POSITION(PointDWFixed *location)
dword MSG_GO_CHANGE_LOCKS(GrObjLocks setBits,
                    GrObjLocks clearBits)
void MSG_GO_SCALE(GrObjAnchoredScaleData *params)
void MSG_GO_COMBINE_AREA_NOTIFICATION_DATA(
                    Handle change)
void MSG_GO_COMBINE_GRADIENT_NOTIFICATION_DATA(
                    Handle change)
void MSG_GO_COMBINE_LINE_NOTIFICATION_DATA(
                    Handle change)
void MSG_GO_COMBINE_SELECT_STATE_NOTIFICATION_DATA(
                    Handle change)
void MSG_GO_COMBINE_SELECTION_STATE_NOTIFICATION_DATA(
                    Handle change)
void MSG_GO_COMBINE_STYLE_NOTIFICATION_DATA(
                    Handle change)
void MSG_GO_COMBINE_STYLE_SHEET_NOTIFICATION_DATA(
                    Handle change)
void MSG_GO_SEND_UI_NOTIFICATION(
              GrObjUINotificationTypes notifications)
void MSG_GO_SUSPEND_ACTION_NOTIFICATION()
void MSG_GO_UNSUSPEND_ACTION_NOTIFICATION()
void MSG_GO_SKEW(GrObjAnchoredSkewData *params)
void MSG_GO_TRANSFORM(TransMatrix *transformation)
MSG_GO_COMPLETE_CREATE
MSG_GO_COMPLETE_TRANSFORM
MSG_GO_COMPLETE_TRANSLATE
void MSG_GO_GET_SIZE(GOGetSizeParams *retValue)
void MSG_GO_GET_POSITION(PointDWFixed *retValue)
void MSG_GO_INITIALIZE(GrObjInitializeData *data)
MSG_GO_SCALE_ABOUT_PARENT_LEFT_TOP
MSG_GO_INSERT_OR_DELETE_SPACE
MSG_GO_BEGIN_CREATE
MSG_GO_END_CREATE
MSG_GO_EVALUATE_PARENT_POINT_FOR_EDIT
void MSG_GO_CREATE_TRANSFER(GrObjTransferParams *params)
```

```
void MSG_GO_REPLACE_WITH_TRANSFER(
                    GrObjTransferParams *params)
void MSG_GO_WRITE_INSTANCE_TO_TRANSFER(
                    GrObjTransferParams *params)
void MSG_GO_READ_INSTANCE_FROM_TRANSFER(
                    GrObjTransferParams *params)
MSG_GO_GET_POINTER_IMAGE
MSG_GO_GET_LOCKS
void MSG_GO_DRAW_FG_AREA(DrawFlags drawFlags,
                    GrObjDrawFlags grobjDrawFlags,
                    GStateHandle gstate)
MSG_GO_DRAW_BG_AREA
void MSG_GO_DRAW_SPRITE_LINE(GStateHandle gstate)
MSG_GO_DRAW_NORMAL_SPRITE_LINE
MSG_GO_DO_NOTHING
MSG_GO_ADD_POTENTIAL_SIZE_TO_BLOCK
MSG_GO_SUBTRACT_POTENTIAL_SIZE_FROM_BLOCK
MSG_GO_GET_GROBJ_CLASS
MSG_GO_UNDO_REPLACE_GEOMETRY_INSTANCE_DATA
MSG_GO_GENERATE_UNDO_REPLACE_GEOMETRY_INSTANCE_DATA_ACT
ION
MSG_GO_GENERATE_UNDO_REPLACE_GEOMETRY_INSTANCE_DATA_CHA
IN
MSG_GO_CLEAR_SANS_UNDO
MSG_GO_REMOVE_FROM_BODY
MSG_GO_REMOVE_FROM_GROUP
MSG_GO_RELEASE_EXCLS
MSG_GO_GENERATE_UNDO_CLEAR_CHAIN
MSG_GO_GENERATE_UNDO_UNDO_CLEAR_CHAIN
MSG_GO_DEREF_A_GROBJ_AREA_TOKEN
MSG_GO_DEREF_A_GROBJ_LINE_TOKEN
MSG_GO_INVALIDATE_AREA
MSG_GO_INVALIDATE_LINE
MSG_GO_SUSPEND_COMPLETE_CREATE
MSG_GO_GET_TEMP_STATE_AND_OPT_FLAGS
MSG_GO_MAKE_INSTRUCTION
MSG_GO_MAKE_NOT_INSTRUCTION
MSG_GO_SET_WRAP_TEXT_TYPE
MSG_GO_SET_PASTE_INSIDE(
MSG_GO_SET_INSERT_DELETE_MOVE_ALLOWED
MSG_GO_SET_INSERT_DELETE_RESIZE_ALLOWED
MSG_GO_SET_INSERT_DELETE_DELETE_ALLOWED
MSG_GO_DRAW_PARENT_RECT
MSG_GO_NUKE_DATA_IN_OTHER_BLOCKS
void MSG_GO_DRAW_FG_AREA_HI_RES(DrawFlags drawFlags,
                    GrObjDrawFlags grobjDrawFlags,
                    GStateHandle gstate)
void MSG_GO_DRAW_FG_LINE(DrawFlags drawFlags,
                    GrObjDrawFlags grobjDrawFlags,
                    GStateHandle gstate)
void MSG_GO_DRAW_FG_LINE_HI_RES(DrawFlags drawFlags,
                    GrObjDrawFlags grobjDrawFlags,
                    GStateHandle gstate)
MSG_GO_DRAW_CLIP_AREA
MSG_GO_DRAW_CLIP_AREA_HI_RES
MSG_GO_DRAW_FG_GRADIENT_AREA
MSG_GO_DRAW_FG_GRADIENT_AREA_HI_RES
```

```
MSG_GO_DRAW_BG_AREA_HI_RES
MSG_GO_SET_BG_COLOR
MSG_GO_SET_AREA_PATTERN
MSG_GO_SET_AREA_ATTR_ELEMENT_TYPE
MSG_GO_SET_LINE_ATTR_ELEMENT_TYPE
MSG_GO_SET_STARTING_GRADIENT_COLOR
MSG_GO_SET_ENDING_GRADIENT_COLOR
MSG_GO_SET_NUMBER_OF_GRADIENT_INTERVALS
MSG_GO_SET_GRADIENT_TYPE
MSG_GO_SET_ARROWHEAD_ON_START
MSG_GO_SET_ARROWHEAD_ON_END
MSG_GO_SET_ARROWHEAD_FILLED
MSG_GO_SET_ARROWHEAD_ANGLE
MSG_GO_SET_ARROWHEAD_LENGTH
MSG_GO_NOTIFY_GROBJ_INVALID
MSG_GO_SET_GROBJ_ATTR_FLAGS
MSG_GO_SET_SYS_TARGET
void MSG_GO_SCALE_OBJECT(GrObjAnchoredScaleData *params)
MSG_GO_GENERATE_UNDO_CLEAR_CHAIN_WITH_ACTION_NOTIFICATI
ON
MSG_GO_RECREATE_CACHED_GSTATES
MSG_GO_DRAW_QUICK_VIEW
MSG_GO_DRAW_SPRITE_LINE_HI_RES
MSG_GO_DRAW_NORMAL_SPRITE_LINE_HI_RES
MSG_GO_GET_GROBJ_ATTR_FLAGS
MSG_GO_MAKE_ATTRS_DEFAULT
MSG_GO_GET_SITUATIONAL_POINTER_IMAGE
MSG_GO_OBJ_FREE
MSG_GO_ADJUST_CREATE
MSG_GO_GENERATE_UNDO_UNDO_CLEAR_CHAIN_WITH_ACTION_NOTIF
ICATION
MSG_GO_NUDGE_INSIDE
MSG_GO_MOVE_INSIDE
MSG_GO_REPLACE_GEOMETRY_INSTANCE_DATA
MSG_GO_CHECK_ACTION_MODES
MSG_GO_EVALUATE_PARENT_POINT_FOR_HANDLE_MOVE_RESIZE
MSG_GO_EVALUATE_PARENT_POINT_FOR_HANDLE_MOVE_ROTATE
MSG_GO_COMBINE_LOCKS
MSG_GO_DESELECT_IF_GROUP_LOCK_SET
MSG_GO_GET_POTENTIAL_GROBJECT_SIZE
MSG_GO_GROUP_GAINED_SELECTION_LIST
MSG_GO_GROUP_LOST_SELECTION_LIST
MSG_GO_INVALIDATE_WITH_UNDO
MSG_GO_AFTER_QUICK_PASTE
MSG_GO_QUICK_TOTAL_BODY_CLEAR
```

GrObjConvertControlClass

```
@class GrObjConvertControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GOCCFeatures
      GOCCF_CONVERT_TO_BITMAP         0x0004
      GOCCF_CONVERT_TO_GRAPHIC        0x0002
      GOCCF_CONVERT_FROM_GRAPHIC      0x0001
GROBJ_CONVERT_CONTROL_DEFAULT_FEATURES
                           (GOCCF_CONVERT_TO_BITMAP |
                            GOCCF_CONVERT_TO_GRAPHIC |
                            GOCCF_CONVERT_FROM_GRAPHIC)
GROBJ_CONVERT_CONTROL_DEFAULT_TOOLBOX_FEATURES
                           (GOCCF_CONVERT_TO_BITMAP |
                            GOCCF_CONVERT_TO_GRAPHIC |
                            GOCCF_CONVERT_FROM_GRAPHIC)
```

Messages

```
MSG_GOCC_CONVERT_TO_BITMAP
MSG_GOCC_CONVERT_TO_GRAPHIC
MSG_GOCC_CONVERT_FROM_GRAPHIC
```

GrObjCreateControlClass

```
@class GrObjCreateControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
ByteFlags    GrObjCreateControlFeatures
      GOCCF_RECTANGLE                 0x0200
      GOCCF_ELLIPSE                   0x0100
      GOCCF_LINE                      0x0080
      GOCCF_ROUNDED_RECTANGLE         0x0040
      GOCCF_ARC                       0x0020
      GOCCF_TRIANGLE                  0x0010
      GOCCF_HEXAGON                   0x0008
      GOCCF_OCTOGON                   0x0004
      GOCCF_FIVE_POINTED_STAR         0x0002
      GOCCF_EIGHT_POINTED_STAR        0x0001
GROBJ_CREATE_CONTROL_DEFAULT_FEATURES 0x03fb
GROBJ_CREATE_CONTROL_DEFAULT_TOOLBOX_FEATURES 0x03fb
```

Messages

```
MSG_GOCC_CREATE_GROBJ
MSG_GOCC_CREATE_POLYGON
MSG_GOCC_CREATE_STAR
```

GrObjCustomDuplicateControlClass

```
@class GrObjCustomDuplicateControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GOCDCFeatures
      GOCDCF_REPETITIONS        0x10
      GOCDCF_MOVE               0x08
      GOCDCF_SCALE              0x04
      GOCDCF_ROTATE             0x02
      GOCDCF_SKEW               0x01
GOCDC_DEFAULT_FEATURES    (GOCDCF_REPETITIONS |
                           GOCDCF_MOVE | GOCDCF_ROTATE |
                           GOCDCF_SCALE)
```

Messages

```
MSG_GOCDC_CUSTOM_DUPLICATE
```

GrObjCustomShapeControlClass

```
@class GrObjCustomShapeControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
ByteFlags    GOCSCFeatures
      GOCSCF_POLYGONS           0x02
      GOCSCF_STARS              0x01
GOCSC_DEFAULT_FEATURES (GOCSCF_POLYGONS | GOCSCF_STARS)
```

Messages

```
MSG_GOCSC_CREATE_POLYGON
MSG_GOCSC_CREATE_STAR
```

GrObjDefaultAttributesControlClass

```
@class GrObjDefaultAttributesControlClass,
                                        GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GODACFeatures
      GODACF_SET_DEFAULT_ATTRIBUTES   0x01
GODAC_DEFAULT_FEATURES (GODACF_SET_DEFAULT_ATTRIBUTES)
GODAC_DEFAULT_TOOLBOX_FEATURES
                    (GODACF_SET_DEFAULT_ATTRIBUTES)
```

Messages

```
MSG_GODAC_SET_DEFAULT_ATTRIBUTES
```

GrObjDepthControlClass

```
@class GrObjDepthControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GODepthCFeatures
      GODepthCF_BRING_TO_FRONT        0x0008
      GODepthCF_SEND_TO_BACK          0x0004
      GODepthCF_SHUFFLE_UP            0x0002
      GODepthCF_SHUFFLE_DOWN          0x0001
GODepthC_DEFAULT_FEATURES (GODepthCF_BRING_TO_FRONT |
                           GODepthCF_SEND_TO_BACK |
                           GODepthCF_SHUFFLE_UP |
                           GODepthCF_SHUFFLE_DOWN)
GODepthC_DEFAULT_TOOLBOX_FEATURES
                           (GODepthCF_BRING_TO_FRONT |
                            GODepthCF_SEND_TO_BACK |
                            GODepthCF_SHUFFLE_UP |
                            GODepthCF_SHUFFLE_DOWN)
```

Messages

```
MSG_GODC_BRING_TO_FRONT
MSG_GODC_SEND_TO_BACK
MSG_GODC_SHUFFLE_UP
MSG_GODC_SHUFFLE_DOWN
```

GrObjDraftModeControlClass

```
@class GrObjDraftModeControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GODMCFeatures
      GODMCF_DRAFT_MODE         0x01
GODMC_DEFAULT_FEATURES (GODMCF_DRAFT_MODE)
```

Messages

```
MSG_GODMC_SET_DRAFT_MODE_STATUS
```

GrObjDuplicateControlClass

```
@class GrObjDuplicateControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GrObjDuplicateControlFeatures
      GODCF_DUPLICATE                 0x0002
      GODCF_DUPLICATE_IN_PLACE        0x0001
ByteFlags    GrObjDuplicateControlToolboxFeatures
      GODCTF_DUPLICATE                0x0002
      GODCTF_DUPLICATE_IN_PLACE       0x0001
GROBJ_DUPLICATE_CONTROL_DEFAULT_FEATURES
```

```
                            (GODCF_DUPLICATE |
                             GODCF_DUPLICATE_IN_PLACE)
    GROBJ_DUPLICATE_CONTROL_DEFAULT_TOOLBOX_FEATURES
                            (GODCTF_DUPLICATE |
                             GODCTF_DUPLICATE_IN_PLACE)
```

Messages

```
MSG_GROBJ_DUPLICATE_CONTROL_DUPLICATE
MSG_GROBJ_DUPLICATE_CONTROL_DUPLICATE_IN_PLACE
```

GrObjEndingGradientColorSelectorClass

```
@class GrObjEndingGradientColorSelectorClass,
                                ColorSelectorClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default & ~GS_ENABLED)
```

Types and Flags

```
GOSGCS_DEFAULT_FEATURES (CSF_INDEX | CSF_RGB)
```

GrObjFlipControlClass

```
@class GrObjFlipControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GOFCFeatures
      GOFCF_FLIP_HORIZONTALLY 0x02
      GOFCF_FLIP_VERTICALLY 0x01
```

Messages

```
MSG_GOFC_FLIP_HORIZONTALLY
MSG_GOFC_FLIP_VERTICALLY
```

GrObjGradientFillControlClass

```
@class GrObjGradientFillControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GrObjGradientFillControlFeatures
      GOGFCF_HORIZONTAL_GRADIENT      0x0010
      GOGFCF_VERTICAL_GRADIENT        0x0008
      GOGFCF_RADIAL_RECT_GRADIENT     0x0004
      GOGFCF_RADIAL_ELLIPSE_GRADIENT  0x0002
      GOGFCF_NUM_INTERVALS            0x0001
GOGFC_DEFAULT_FEATURES (GOGFCF_HORIZONTAL_GRADIENT |
                  GOGFCF_VERTICAL_GRADIENT |
                  GOGFCF_RADIAL_RECT_GRADIENT |
                  GOGFCF_RADIAL_ELLIPSE_GRADIENT |
                  GOGFCF_NUM_INTERVALS)
```

Messages

```
MSG_GOGFC_SET_GRADIENT_TYPE
MSG_GOGFC_SET_NUMBER_OF_GRADIENT_INTERVALS
```

GrObjGroupControlClass

```
@class GrObjGroupControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET);
```

Types and Flags

```
ByteFlags    GOGCFeatures
       GOGCF_GROUP               0x01
       GOGCF_UNGROUP             0x02
GOGC_DEFAULT_FEATURES (GOGCF_GROUP | GOGCF_UNGROUP)
GOGC_DEFAULT_TOOLBOX_FEATURES   (GOGCF_GROUP |
                                 GOGCF_UNGROUP)
```

Messages

```
MSG_GOGC_GROUP
MSG_GOGC_UNGROUP
```

GrObjHandleControlClass

```
@class GrObjHandleControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
ByteFlags    GOHCFeatures
       GOHCF_SMALL_HANDLES       0x08
       GOHCF_MEDIUM_HANDLES      0x04
       GOHCF_LARGE_HANDLES       0x02
       GOHCF_INVISIBLE_HANDLES   0x01
GOHC_DEFAULT_FEATURES     (GOHCF_SMALL_HANDLES |
                           GOHCF_MEDIUM_HANDLES |
                           GOHCF_LARGE_HANDLES |
                           GOHCF_INVISIBLE_HANDLES)
GOHC_DEFAULT_TOOLBOX_FEATURES 0
```

Messages

```
MSG_GOHC_SET_HANDLES
```

GrObjHeadClass

```
@class GrObjHeadClass, MetaClass
```

Instance Data

```
ClassStruct        *GH_currentTool = NullClass
word               GH_initializeFloaterData = 0
optr               GH_currentBody
optr               GH_floater
```

Structures

```
typedef struct {
    word           CTV_grObjSpecificData
    word           CTV_unused
    ClassStruct    *CTV_toolClass
} CurrentToolValues
```

Messages

```
void MSG_GH_SET_CURRENT_TOOL(ClassStruct *toolClass,
                   word initData)
void MSG_GH_SET_CURRENT_TOOL_WITH_DATA_BLOCK(
                   ClassStruct *toolClass,
                   word initData)
MSG_GH_SET_CURRENT_BODY
MSG_GH_CLEAR_CURRENT_BODY
MSG_GH_CLASSED_EVENT_TO_FLOATER
MSG_GH_CLASSED_EVENT_TO_FLOATER_IF_CURRENT_BODY
MSG_GH_FLOATER_FINISHED_CREATE
void MSG_GH_GET_CURRENT_TOOL(CurrentToolValues *retVal)
MSG_GH_SEND_NOTIFY_CURRENT_TOOL
MSG_GH_CALL_FLOATER
MSG_GH_SET_TEXT_TOOL_FOR_SEARCH_SPELL
```

GrObjHideShowControlClass

```
@class GrObjHideShowControlClass, GenControlClass;
```

Instance Data

```
    @default       GCI_output = (TO_APP_TARGET)
    @default       GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
ByteFlags    GOHSCFeatures
      GOHSCF_HIDE              0x02
      GOHSCF_SHOW              0x01
GOHSC_DEFAULT_FEATURES (GOHSCF_HIDE | GOHSCF_SHOW)
```

Messages

```
MSG_GOHSC_HIDE
MSG_GOHSC_SHOW
```

GrObjInstructionControlClass

```
@class GrObjInstructionControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GrObjInstructionControlFeatures
      GOICF_DRAW                    0x8000
      GOICF_PRINT                   0x4000
      GOICF_MAKE_EDITABLE           0x2000
      GOICF_MAKE_UNEDITABLE         0x1000
      GOICF_DELETE                  0x0800
GOICF_DEFAULT_FEATURES    (GOICF_DRAW | GOICF_PRINT |
                          GOICF_MAKE_EDITABLE |
                          GOICF_MAKE_UNEDITABLE |
                          GOICF_DELETE)
```

Messages

```
MSG_GOIC_MAKE_INSTRUCTIONS_EDITABLE
MSG_GOIC_MAKE_INSTRUCTIONS_UNEDITABLE
MSG_GOIC_DELETE_INSTRUCTIONS
MSG_GOIC_SET_INSTRUCTION_ATTRS
```

GrObjLineAttrControlClass

```
@class GrObjLineAttrControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
WordFlags    GOLACFeatures
      GOLACF_WIDTH_INDEX            0x0010
      GOLACF_WIDTH_VALUE            0x0008
      GOLACF_STYLE                  0x0004
      GOLACF_ARROWHEAD_TYPE         0x0002
      GOLACF_ARROWHEAD_WHICH_END    0x0001
GOLAC_DEFAULT_FEATURES    (GOLACF_WIDTH_INDEX |
                          GOLACF_WIDTH_VALUE |
                          GOLACF_STYLE |
                          GOLACF_ARROWHEAD_TYPE |
                          GOLACF_ARROWHEAD_WHICH_END)
WordFlags    GOLACToolboxFeatures
      GOLACTF_WIDTH_INDEX           0x0002
      GOLACTF_STYLE                 0x0001
GOLAC_DEFAULT_TOOLBOX_FEATURES (GOLACTF_WIDTH_INDEX |
                          GOLACF_STYLE)
```

Messages

```
MSG_GOLAC_SET_LINE_VALUE_FROM_INDEX
MSG_GOLAC_SET_LINE_INDEX_FROM_VALUE
MSG_GOLAC_SET_INTEGER_LINE_WIDTH
MSG_GOLAC_SET_LINE_WIDTH
MSG_GOLAC_SET_LINE_STYLE
MSG_GOLAC_SET_ARROWHEAD_TYPE
MSG_GOLAC_SET_ARROWHEAD_WHICH_END
```

GrObjLineColorSelectorClass

```
@class GrObjLineColorSelectorClass, ColorSelectorClass
```

Instance Data

```
@default       GCI_output = (TO_APP_TARGET)
@default       GI_states = (@default | GS_ENABLED)
@default       CSI_toolboxPrefs = @default |
               (COO_LINE_ORIENTED <<
               CTP_INDEX_ORIENTATION_OFFSET) |
               (COO_LINE_ORIENTED <<
               CTP_DRAW_MASK_ORIENTATION_OFFSET)
```

Types and Flags

```
GOLCS_DEFAULT_FEATURES (CSF_INDEX | CSF_RGB |
                        CSF_DRAW_MASK | CSF_PATTERN)
```

GrObjLocksControlClass

```
@class GrObjLocksControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
GOLC_DEFAULT_FEATURES     (GOL_MOVE | GOL_RESIZE |
                           GOL_ROTATE)
GOLC_DEFAULT_TOOLBOX_FEATURES 0
```

Messages

```
MSG_GOLC_CHANGE_LOCKS
```

GrObjMoveInsideControlClass

```
@class GrObjMoveInsideControlClass,
                                GrObjNudgeControlClass
```

GrObjNudgeControlClass

```
@class GrObjNudgeControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GONCFeatures
      GONCF_NUDGE_LEFT               0x10
      GONCF_NUDGE_RIGHT              0x08
      GONCF_NUDGE_UP                 0x04
      GONCF_NUDGE_DOWN               0x02
      GONCF_CUSTOM_MOVE              0x01
GROBJ_NUDGE_CONTROL_DEFAULT_FEATURES (GONCF_NUDGE_LEFT |
                        GONCF_NUDGE_RIGHT |
                        GONCF_NUDGE_UP |
                        GONCF_NUDGE_DOWN |
                        GONCF_CUSTOM_MOVE)
```

Messages

```
MSG_GONC_NUDGE
```

```
MSG_GONC_CUSTOM_MOVE
MSG_GONC_SET_DISPLAY_FORMAT
```

GrObjObscureAttrControlClass

```
@class GrObjObscureAttrControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default | GS_ENABLED)
```

Types and Flags

```
ByteFlags    GrObjObscureAttrControlFeatures
      GOOACF_INSTRUCTIONS             0x80
      GOOACF_INSERT_OR_DELETE_MOVE    0x40
      GOOACF_INSERT_OR_DELETE_RESIZE  0x20
      GOOACF_INSERT_OR_DELETE_DELETE  0x10
      GOOACF_DONT_WRAP                0x08
      GOOACF_WRAP_INSIDE              0x04
      GOOACF_WRAP_AROUND_RECT         0x02
      GOOACF_WRAP_TIGHTLY             0x01
GOOAC_INSERT_OR_DELETE_FEATURES
                   (GOOACF_INSERT_OR_DELETE_MOVE |
                    GOOACF_INSERT_OR_DELETE_RESIZE |
                    GOOACF_INSERT_OR_DELETE_DELETE)
GOOAC_WRAP_FEATURES (GOOACF_DONT_WRAP |
                    GOOACF_WRAP_INSIDE |
                    GOOACF_WRAP_AROUND_RECT |
                    GOOACF_WRAP_TIGHTLY)
GOOAC_DEFAULT_FEATURES (GOOACF_INSTRUCTIONS |
                    GOOAC_WRAP_FEATURES |
                    GOOAC_INSERT_OR_DELETE_FEATURES)
```

Messages

```
MSG_GOOAC_SET_WRAP_TEXT_TYPE
MSG_GOOAC_CHANGE_OBSCURE_ATTRS
```

GrObjPasteInsideControlClass

```
@class GrObjPasteInsideControlClass, GenControlClass
```

Instance Data

```
word         GPICI_lastNumSelected = 0
    @default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GOPICFeatures
      GOPICF_PASTE_INSIDE             0x0002
      GOPICF_BREAKOUT_PASTE_INSIDE    0x0001
ByteFlags    GOPICToolboxFeatures
      GOPICTF_PASTE_INSIDE            0x0002
      GOPICTF_BREAKOUT_PASTE_INSIDE   0x0001
GOPIC_DEFAULT_FEATURES    (GOPICF_PASTE_INSIDE |
                           GOPICF_BREAKOUT_PASTE_INSIDE)
GOPIC_DEFAULT_TOOLBOX_FEATURES (GOPICTF_PASTE_INSIDE |
                          GOPICTF_BREAKOUT_PASTE_INSIDE)
```

Messages

```
MSG_GOPIC_PASTE_INSIDE
MSG_GOPIC_BREAKOUT_PASTE_INSIDE
```

GrObjRotateControlClass

```
@class GrObjRotateControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GORCFeatures
      GORCF_45_DEGREES_CW             0x0080
      GORCF_90_DEGREES_CW             0x0040
      GORCF_135_DEGREES_CW            0x0020
      GORCF_180_DEGREES               0x0010
      GORCF_135_DEGREES_CCW           0x0008
      GORCF_90_DEGREES_CCW            0x0004
      GORCF_45_DEGREES_CCW            0x0002
      GORCF_CUSTOM_ROTATION           0x0001
GORC_DEFAULT_FEATURES     0x00ff
```

Messages

```
MSG_GORC_ROTATE
MSG_GORC_CUSTOM_ROTATE
```

GrObjScaleControlClass

```
@class GrObjScaleControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GrObjScaleControlFeatures
      GOSCF_HALF_WIDTH                0x0010
      GOSCF_HALF_HEIGHT               0x0008
      GOSCF_DOUBLE_WIDTH              0x0004
      GOSCF_DOUBLE_HEIGHT             0x0002
      GOSCF_CUSTOM_SCALE              0x0001
GROBJ_SCALE_CONTROL_DEFAULT_FEATURES 0x001f
```

Messages

```
MSG_GOSC_SCALE_HORIZONTALLY
MSG_GOSC_SCALE_VERTICALLY
MSG_GOSC_CUSTOM_SCALE
```

GrObjSkewControlClass

```
@class GrObjSkewControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GrObjSkewControlFeatures
      GOSCF_LEFT                      0x0010
      GOSCF_RIGHT                     0x0008
      GOSCF_UP                        0x0004
      GOSCF_DOWN                      0x0002
      GOSCF_CUSTOM_SKEW               0x0001
GROBJ_SKEW_CONTROL_DEFAULT_FEATURES 0x001f
```

Messages

```
MSG_GOSC_SKEW_HORIZONTALLY
MSG_GOSC_SKEW_VERTICALLY
void MSG_GOSC_CUSTOM_SKEW
```

GrObjSplineClass

```
@class GrObjSplineClass, GrObjVisClass
```

GrObjStartingGradientColorSelectorClass

```
@class GrObjStartingGradientColorSelectorClass,
                                  ColorSelectorClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default & ~GS_ENABLED)
```

Types and Flags

```
GOSGCS_DEFAULT_FEATURES (CSF_INDEX | CSF_RGB)
```

GrObjStyleSheetControlClass

```
@class GrObjStyleSheetControlClass,
                                  StyleSheetControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
@default GI_states = (@default & ~GS_ENABLED)
@default SSCI_targetClass =
               (ClassStruct *)&GrObjBodyClass
@default SSCI_styledClass =
                        (ClassStruct *)&GrObjClass
```

GrObjTextClass

```
@class GrObjTextClass, GrObjVisClass;
```

Messages

```
MSG_GT_ADJUST_MARGINS_FOR_LINE_WIDTH
```

GrObjToolControlClass

```
@class GrObjToolControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Variable Data

```
word ATTR_GROBJ_TOOL_CONTROL_POSITION_FOR_ADDED_TOOLS
```

Types and Flags

```
WordFlags    GOTCFeatures
      GOTCF_PTR                       0x0800
      GOTCF_ROTATE_PTR                0x0400
      GOTCF_ZOOM                      0x0200
      GOTCF_TEXT                      0x0100
      GOTCF_LINE                      0x0080
      GOTCF_RECT                      0x0040
      GOTCF_ROUNDED_RECT              0x0020
      GOTCF_ELLIPSE                   0x0010
      GOTCF_ARC                       0x0008
      GOTCF_POLYLINE                  0x0004
      GOTCF_POLYCURVE                 0x0002
      GOTCF_SPLINE                    0x0001
GOTC_DEFAULT_FEATURES     (GOTCF_PTR |
                           GOTCF_ROTATE_PTR |
                           GOTCF_ZOOM | GOTCF_TEXT |
                           GOTCF_LINE | GOTCF_RECT |
                           GOTCF_ROUNDED_RECT |
                           GOTCF_ELLIPSE | GOTCF_ARC |
                           GOTCF_POLYLINE |
                           GOTCF_POLYCURVE |
                           GOTCF_SPLINE)
GOTC_DEFAULT_TOOLBOX_FEATURES (GOTC_DEFAULT_FEATURES)
```

Messages

```
MSG_GOTC_SET_TOOL
```

GrObjToolItemClass

```
@class GrObjToolItemClass, GenItemClass
```

Instance Data

```
ClassStruct        *GOTII_toolClass = NullClass
word               GOTII_specInitData = 0
```

Messages

```
MSG_GOTI_GET_TOOL_CLASS
MSG_GOTI_SELECT_SELF_IF_MATCH
```

GrObjTransformControlClass

```
@class GrObjTransformControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = (TO_APP_TARGET)
```

Types and Flags

```
ByteFlags    GOTransformCFeatures
```

```
        GOTCF_UNTRANSFORM               0x8000
GOTransformC_DEFAULT_FEATURES (GOTCF_UNTRANSFORM)
```

Messages

```
MSG_GOTC_UNTRANSFORM
```


GrObjVisClass

```
@class GrObjVisClass, VisClass, master, variant
```

Instance Data

```
optr         GVI_guardian
```

Messages

```
MSG_GV_GET_WWFIXED_CENTER
MSG_GV_SET_GUARDIAN_LINK
MSG_GV_SET_VIS_BOUNDS
MSG_GV_GET_GROBJ_VIS_CLASS
MSG_GV_SET_REALIZED_AND_UPWARD_LINK
MSG_GV_CLEAR_REALIZED_AND_UPWARD_LINK
MSG_GV_GET_GUARDIAN
MSG_GV_GET_POTENTIAL_WARD_SIZE
```

GrObjVisGuardianClass

```
@class GrObjVisGuardianClass, GrObjClass
```

Instance Data

```
optr                    GOVGI_ward
word                    *GOVGI_class
GrObjVisGuardianFlags   GOVGI_flags
```

Types and Flags

```
ByteFlags    GrObjVisGuardianFlags
      GOVGF_VIS_BOUNDS_HAVE_CHANGED         0x80
      GOVGF_LARGE                           0x40
      GOVGF_GUARDIAN_INITIATED_TRANSFORM    0x20
      GOVGF_IGNORE_WARD_CHANGE_BOUNDS       0x10
      GOVGF_APPLY_OBJECT_TO_VIS_TRANSFORM   0x08
      GOVGF_CAN_EDIT_EXISTING_OBJECTS       0x04
      GOVGF_CREATE_MODE                     0x03
```

Messages

```
MSG_GOVG_UPDATE_VIS_WARD_WITH_STORED_DATA
MSG_GOVG_CONVERT_LARGE_MOUSE_DATA
optr MSG_GOVG_CREATE_VIS_WARD(MemHandle wardBlock)
MSG_GOVG_ADD_VIS_WARD
MSG_GOVG_APPLY_OBJECT_TO_VIS_TRANSFORM
MSG_GOVG_CREATE_GSTATE
MSG_GOVG_VIS_BOUNDS_SETUP
MSG_GOVG_SET_VIS_WARD_MOUSE_EVENT_TYPE
MSG_GOVG_UPDATE_EDIT_GRAB_WITH_STORED_DATA
MSG_GOVG_NORMALIZE
MSG_GOVG_NOTIFY_VIS_WARD_CHANGE_BOUNDS
MSG_GOVG_GET_EDIT_CLASS
optr MSG_GOVG_GET_VIS_WARD_OD()
MSG_GOVG_GET_TRANSFER_BLOCK_FROM_VIS_WARD
MSG_GOVG_CREATE_WARD_WITH_TRANSFER
```

```
MSG_GOVG_SET_VIS_WARD_CLASS
MSG_GOVG_APPLY_SPRITE_OBJECT_TO_VIS_TRANSFORM
MSG_GOVG_CHECK_FOR_EDIT_WITH_FIRST_START_SELECT
MSG_GOVG_RULE_LARGE_START_SELECT_FOR_WARD
MSG_GOVG_RULE_LARGE_PTR_FOR_WARD
MSG_GOVG_RULE_LARGE_END_SELECT_FOR_WARD
```




GroupClass

```
@class GroupClass, GrObjClass
```

Instance Data

```
CompPart           GI_drawHead
word               GI_suspendCount
GroupUnsuspendOps  GI_unsuspendOps
```

Types and Flags

```
ByteFlags    GroupUnsuspendOps
      GUO_EXPAND         0x01
```

Messages

```
MSG_GROUP_ADD_GROBJ
MSG_GROUP_REMOVE_GROBJ
MSG_GROUP_CREATE_GSTATE
MSG_GROUP_PROCESS_ALL_GROBJS_SEND_CALL_BACK_MESSAGE
MSG_GROUP_INITIALIZE
MSG_GROUP_EXPAND
optr MSG_GROUP_INSTANTIATE_GROBJ(ClassStruct *class)
MSG_GROUP_SET_HAS_PASTE_INSIDE_CHILDREN
MSG_GROUP_PASTE_CALL_BACK_FOR_PASTE_INSIDE
MSG_GROUP_CREATE_GSTATE_FOR_BOUNDS_CALC
MSG_GROUP_VIS_TEXT_GENERATE_NOTIFY
MSG_GROUP_CHECK_FOR_GROBJ_TEXTS
```

GStringClass

```
@class GStringClass, GrObjClass
```

Instance Data

```
word               GSI_vmemBlockHandle
PointWWFixed       GSI_gstringCenterTrans
```

Messages

```
void MSG_GSO_SET_GSTRING()
```

HelpControlClass

```
@class HelpControlClass, GenControlClass
```

Instance Data

```
HelpType            HCI_helpType
MemHandle           HCI_curFile
MemHandle           HCI_historyBuf
word                HCI_nameArrayVM
GeodeHandle         HCI_compressLib
    @default GII_attrs =  (@default |
                          GIA_NOT_USER_INITIATABLE)
    @default GII_visibility = GIV_DIALOG
```

Variable Data

```
void ATTR_HELP_SUPPRESS_INITIATE
char[] ATTR_HELP_INITIAL_HELP
optr ATTR_HELP_CUSTOM_POINTER_IMAGE
```

Hints

```
CompSizeHintArgs HINT_HELP_TEXT_FIXED_SIZE
```

Types and Flags

```
ByteEnum     HelpType
      HT_NORMAL_HELP                0
      HT_FIRST_AID                  1
      HT_STATUS_HELP                2
      HT_SIMPLE_HELP                3
      HT_SYSTEM_HELP                4
WordFlags    HPCFeatures
      HPCF_HELP                     0x0100
      HPCF_TEXT                     0x0080
      HPCF_CONTENTS                 0x0040
      HPCF_HISTORY                  0x0020
      HPCF_GO_BACK                  0x0010
      HPCF_CLOSE                    0x0008
      HPCF_INSTRUCTIONS             0x0004
      HPCF_FIRST_AID_GO_BACK        0x0002
      HPCF_FIRST_AID                0x0001
MAX_CONTEXT_NAME_SIZE     20
typedef char       ContextName[MAX_CONTEXT_NAME_SIZE]
```

Structures

```
typedef struct {
    HelpType        NHCC_type
    ContextName     NHCC_context
    FileLongName    NHCC_filename
    FileLongName    NHCC_filenameTOC
} NotifyHelpContextChange
```

Messages

```
MSG_HELP_CONTROL_FOLLOW_LINK
MSG_HELP_CONTROL_GET_POINTER_IMAGE
```

Routines

```
void HelpSendHelpNotification(
                    word HelpType,
                    const char *contextname,
                    const char *filename)
```

ImpexMapControlClass

```
@class ImpexMapControlClass, GenControlClass
```

Instance Data

```
word                IMCI_dataBlock1
word                IMCI_dataBlock2
word                IMCI_childBlock
word                IMCI_mapListBlock
ImpexMapFlags       IMCI_flags
```

Types and Flags

```
SOURCE                    0
DESTINATION               1
ByteFlags    IMCFeatures
      IMCF_MAP                          0x01
IMC_DEFAULT_FEATURES      (IMCF_MAP)
IMC_DEFAULT_TOOLBOX_FEATURES 0
IMC_MAP_MONIKER_SIZE      1024
ByteFlags    ImpexMapFlags
      IMF_IMPORT                        0x80
      IMF_EXPORT                        0x40
ByteEnum     DefaultFieldNameUsage
      DFNU_FIELD          0
      DFNU_COLUMN         1
      DFNU_FIXED          2
```

Structures

```
typedef struct {
    LMemBlockHeader       IMFIH_base
    word                  IMFIH_fieldChunk
    word                  IMFIH_numFields
    DefaultFieldNameUsage IMFIH_flag
} ImpexMapFileInfoHeader
typedef struct {
    LMemBlockHeader       MLBH_base
    word                  MLBH_numDestFields
    word                  MLBH_chunk1
} MapListBlockHeader
typedef struct {
    word     CML_source
    word     CML_dest
} ChunkMapList
```

ImportControlClass

```
@class ImportControlClass, ImportExportClass
```

Instance Data

```
ImportControlAttrs       ICI_attrs
ImpexDataClasses         ICI_dataClasses
optr                     ICI_destination
word                     ICI_message
```

Variable Data

```
optr ATTR_IMPORT_CONTROL_APP_UI
  @reloc ATTR_IMPORT_CONTROL_APP_UI, 0, optr
optr ATTR_IMPORT_CONTROL_CANCEL_DESTINATION
  @reloc ATTR_IMPORT_CONTROL_CANCEL_DESTINATION, 0, optr
word ATTR_IMPORT_CONTROL_CANCEL_MESSAGE
```

Types and Flags

```
WordFlags    ImportControlAttrs
      ICA_IGNORE_INPUT               0x8000
ByteFlags    ImportControlFeatures
      IMPORTCF_PREVIEW_TRIGGER       0x0020
      IMPORTCF_IMPORT_TRIGGER        0x0010
      IMPORTCF_FORMAT_OPTIONS        0x0008
      IMPORTCF_FILE_MASK             0x0004
      IMPORTCF_BASIC                 0x0002
      IMPORTCF_GLYPH                 0x0001
IMPORTC_DEFAULT_FEATURES  (IMPORTCF_GLYPH |
                  IMPORTCF_BASIC | IMPORTCF_FILE_MASK
                  | IMPORTCF_FORMAT_OPTIONS |
                  IMPORTCF_IMPORT_TRIGGER)
ByteFlags    ImportControlToolboxFeatures
      IMPORTCTF_DIALOG_BOX           0x01
IMPORTC_DEFAULT_TOOLBOX_FEATURES (IMPORTCTF_DIALOG_BOX)
```

Structures

```
typedef struct {
    int      notUsed
    word     message
    optr     destOD
} ObjectState
```

Messages

```
void MSG_IMPORT_CONTROL_SET_DATA_CLASSES(
                  ImpexDataClasses dataClasses)
ImpexDataClasses MSG_IMPORT_CONTROL_GET_DATA_CLASSES()
void MSG_IMPORT_CONTROL_SET_ACTION(
                  optr destOD, word ICImsg)
void MSG_IMPORT_CONTROL_SET_MSG(word ECImsg)
void MSG_IMPORT_CONTROL_GET_ACTION(
                  ObjectState *retValue)
void MSG_IMPORT_CONTROL_SET_ATTRS(
                  ImportControlAttrs attrs)
ImportControlAttrs MSG_IMPORT_CONTROL_GET_ATTRS()
word MSG_IMPORT_CONTROL_GET_FILE_SELECTOR_OFFSET(
                  ImportControlFeatures features)
word MSG_IMPORT_CONTROL_GET_FORMAT_LIST_OFFSET(
                  ImportControlFeatures features)
word MSG_IMPORT_CONTROL_GET_FILE_MASK_OFFSET(
```

```
                    ImportControlFeatures features)
word MSG_IMPORT_CONTROL_GET_FORMAT_UI_PARENT_OFFSET(
                    ImportControlFeatures features)
word MSG_IMPORT_CONTROL_GET_APP_UI_PARENT_OFFSET(
                    ImportControlFeatures features)
word MSG_IMPORT_CONTROL_GET_IMPORT_TRIGGER_OFFSET(
                    ImportControlFeatures features)
void MSG_IMPORT_CONTROL_IMPORT_COMPLETE(
                    ImpexTranslationParams *itParams)
```

ImportExportClass

```
@class ImportExportClass, GenControlClass,
                                        master, variant
```

Instance Data

```
@default ImportExport = GenControlClass
@default GII_attrs = GIA_MODAL
@default GII_type = GIT_COMMAND
@default GII_visibility = GIV_DIALOG
@default GI_states = (GS_USABLE|GS_ENABLED)
```

Variable Data

```
TempImportExportData TEMP_IMPORT_EXPORT_DATA
```

Types and Flags

```
WordFlags    ImpexDataClasses
      IDC_TEXT                      0x8000
      IDC_GRAPHICS                  0x4000
      IDC_SPREADSHEET               0x2000
      IDC_FONT                      0x1000
NUMBER_IMPEX_DATA_CLASSES       4
XLAT_TOKEN_TEXT_12        ( 'T' | ('L' << 8) )
XLAT_TOKEN_TEXT_34        ( 'T' | ('X' << 8) )
XLAT_TOKEN_GRAPHICS_12    ( 'T' | ('L' << 8) )
XLAT_TOKEN_GRAPHICS_34    ( 'G' | ('R' << 8) )
XLAT_TOKEN_SPREADSHEET_12 ( 'T' | ('L' << 8) )
XLAT_TOKEN_SPREADSHEET_34 ( 'S' | ('S' << 8) )
XLAT_TOKEN_FONT_12        ( 'T' | ('L' << 8) )
XLAT_TOKEN_FONT_34        ( 'F' | ('N' << 8) )
```

Structures

```
typedef struct {
    FileLongName               IFSD_selection
    PathName                   IFSD_path
    word                       IFSD_disk
    GenFileSelectorEntryFlags  IFSD_type
} ImpexFileSelectionData
typedef struct {
    optr           TIED_formatUI
    Handle         TIED_formatLibrary
} TempImportExportData
```

InkClass

```
@class InkClass, VisClass
```

Instance Data

```
InkFlags     II_flags = IF_HAS_UNDO
InkTool      II_tool = 0
Color        II_penColor = 0
MemHandle    II_segments = NullHandle
optr         II_dirtyAD = NullOptr
Message      II_dirtyMsg
Rectangle    II_selectBounds
GStateHandle II_cachedGState
TimerHandle  II_antTimer
word         II_antTimerID
byte         II_antMask
```

Variable Data

```
InkStrokeSize ATTR_INK_STROKE_SIZE
```

Types and Flags

```
WordFlags    InkFlags
      IF_MOUSE_FLAGS            0x8000
      IF_SELECTING              0x4000
      IF_HAS_TARGET             0x2000
      IF_HAS_SYS_TARGET         0x1000
      IF_DIRTY                  0x0800
      IF_ONLY_CHILD_OF_CONTENT  0x0400
      IF_CONTROLLED             0x0200
      IF_INVALIDATE_ERASURES    0x0100
      IF_HAS_UNDO               0x0080
typedef enum /* word */ {
    IT_PENCIL =    0,
    IT_ERASER =    2,
    IT_SELECTOR =  4
} InkTool
```

Structures

```
typedef struct {
    Rectangle      IDBF_bounds
    VMFileHandle   IDBF_VMFile
    DBGroupAndItem IDBF_DBGroupAndItem
    word           IDBF_DBExtra
} InkDBFrame
typedef struct {
    word     DBR_group
    word     DBR_item
    word     unused1
    word     unused2
} DBReturn
typedef struct {
    byte     ISS_width
    byte     ISS_height
} InkStrokeSize
```

Messages

```
void MSG_INK_SAVE_TO_DB_ITEM(
                    DBReturn *RetValue, InkDBFrame *ptr)
void MSG_INK_LOAD_FROM_DB_ITEM(InkDBFrame *ptr)
```

```
void MSG_INK_UNDO()
void MSG_INK_SET_TOOL(InkTool tool)
InkTool MSG_INK_GET_TOOL()
void MSG_INK_SET_PEN_COLOR(Color clr)
void MSG_INK_SET_DIRTY_AD(word method, optr object)
void MSG_INK_SET_FLAGS(InkFlags setflags,
                  InkFlags clearflags)
InkFlags MSG_INK_GET_FLAGS()
void MSG_INK_SET_STROKE_SIZE(byte width, byte height)
```

Related Routines

```
void _pascal InkDBInit(VMFileHandle fh)
dword _pascal InkDBGetHeadFolder(VMFileHandle fh)
void _pascal InkDBGetDisplayInfo(
                  InkDBDisplayInfo *RetValue,
                  VMFileHandle fh)
void _pascal InkDBSetDisplayInfo(VMFileHandle fh,
                  dword ofh, dword note, word page)
void _pascal InkSetDocPageInfo(PageSizeReport *psr,
                  VMFileHandle fh)
void _pascal InkGetDocPageInfo(PageSizeReport *psr,
                  VMFileHandle fh)
void _pascal InkSetDocGString(VMFileHandle dbfh,
                  word type)
word _pascal InkGetDocGString(VMFileHandle dbfh)
void _pascal InkSetDocCustomGString(VMFileHandle dbfh,
                  Handle gh)
Handle _pascal InkGetDocCustomGString(VMFileHandle dbfh)
void _pascal InkSendTitleToTextObject(dword tag,
                  VMFileHandle fh, optr to)
word _pascal InkGetTitle(dword tag, VMFileHandle fh,
                  char *dest)
dword _pascal InkGetParentFolder(dword tag,
                  VMFileHandle fh)
void _pascal InkFolderSetTitleFromTextObject(dword fldr,
                  VMFileHandle fh, optr text)
void _pascal InkNoteSetTitleFromTextObject(dword note,
                  VMFileHandle fh, optr text)
dword _pascal InkFolderGetContents(dword tag,
                  VMFileHandle fh,
                  DBGroupAndItem *subFolders)
dword _pascal InkFolderGetNumChildren(dword fldr,
                  VMFileHandle fh)
void _pascal InkFolderDisplayChildInList(dword fldr,
                  VMFileHandle fh, optr list,
                  word entry, Boolean displayFolders)
Boolean _pascal InkFolderGetChildInfo(dword fldr,
                  VMFileHandle fh, word child,
                  dword *childID)
word _pascal InkFolderGetChildNumber(dword fldr,
                  VMFileHandle fh, dword note)
dword _pascal InkFolderCreateSubFolder(dword tag,
                  VMFileHandle fh)
void _pascal InkFolderMove(dword fldr, dword pfldr)
void _pascal InkFolderDelete(dword tag, VMFileHandle fh)
word _pascal InkFolderDepthFirstTraverse(dword rfldr,
                  VMFileHandle fh,
```



```
                    PCB(Boolean, callback, (dword fldr,
                          VMFileHandle fh, word *info)),
                    word *info)
dword _pascal InkNoteCreate(dword tag, VMFileHandle fh)
void _pascal InkNoteCopyMoniker(dword title, optr list,
                    word type, word entry)
dword _pascal InkNoteGetPages(dword tag,
                    VMFileHandle fh)
word _pascal InkNoteGetNumPages(dword item)
word _pascal InkNoteCreatePage(dword tag,
                    VMFileHandle fh, word page)
void _pascal InkNoteLoadPage(dword tag, VMFileHandle fh,
                    word page, optr obj, word type)
void _pascal InkNoteSavePage(dword tag, VMFileHandle fh,
                    word page, optr obj, word type)
void _pascal InkNoteSetKeywordsFromTextObject(dword tag,
                    VMFileHandle fh, optr text)
void _pascal InkNoteSetKeywords(dword tag,
                    VMFileHandle fh, const char *text)
word _pascal InkNoteGetKeywords(dword tag,
                    VMFileHandle fh, char *text)
void _pascal InkNoteSendKeywordsToTextObject(dword tag,
                    VMFileHandle fh, optr text)
void _pascal InkNoteDelete(dword tag, VMFileHandle fh)
void _pascal InkNoteMove(dword tag, dword pfoldr,
                    VMFileHandle fh)
void _pascal InkNoteSetModificationDate(word tdft1,
                    word tdft2, dword note,
                    VMFileHandle fh)
dword _pascal InkNoteGetModificationDate(dword note,
                    VMFileHandle fh)
dword _pascal InkNoteGetCreationDate(dword note,
                    VMFileHandle fh)
NoteType _pascal InkNoteGetNoteType(dword note,
                    VMFileHandle fh)
void _pascal InkNoteSetNoteType(dword note,
                    VMFileHandle fh, NoteType nt)
word _pascal InkNoteFindByTitle(char *string, byte opt,
                    Boolean body, VMFileHandle fh)
word _pascal InkNoteFindByKeywords(VMFileHandle fh,
                    const char *strings,word opt)
```

InkControlClass

```
@class InkControlClass, GenControlClass
```

Instance Data

```
@default GCI_output = TO_APP_TARGET
```

Types and Flags

```
ByteFlags    InkControlFeatures
      ICF_PENCIL_TOOL           0x04
      ICF_ERASER_TOOL           0x02
      ICF_SELECTION_TOOL        0x01
ByteFlags    InkControlToolboxFeatures
      ICTF_PENCIL_TOOL          0x04
      ICTF_ERASER_TOOL          0x02
      ICTF_SELECTION_TOOL       0x01
IC_DEFAULT_FEATURES        (ICF_PENCIL_TOOL |
             ICF_ERASER_TOOL | ICF_SELECTION_TOOL)
IC_DEFAULT_TOOLBOX_FEATURES (ICTF_PENCIL_TOOL |
             ICTF_ERASER_TOOL | ICTF_SELECTION_TOOL)
```

Structures

```
typedef struct {
    optr     NIHT_optr
} NotifyInkHasTarget
```

Messages

```
void MSG_IC_SET_TOOL_FROM_LIST()
```

LineClass

```
@class LineClass, GrObjClass
```

MetaClass

```
@class MetaClass, meta
```

Instance Data

```
MetaBase     MI_base
```

Variable Data

```
TempMetaGCNData TEMP_META_GCN
ChunkHandle TEMP_META_QUIT_LIST
DetachDataEntry DETACH_DATA
word TEMP_EC_IN_USE_COUNT
word TEMP_EC_INTERACTIBLE_COUNT
```

Types and Flags

```
PTR_LEAVE_LEFT     0x01
PTR_LEAVE_TOP      0x02
PTR_LEAVE_RIGHT    0x04
PTR_LEAVE_BOTTOM   0x08
typedef enum /* word */ {
    QL_BEFORE_UI,
    QL_UI,
    QL_AFTER_UI,
    QL_DETACH,
    QL_AFTER_DETACH
} QuitLevel
typedef enum {
    TO_NULL,
    TO_SELF,
    TO_OBJ_BLOCK_OUTPUT,
    TO_PROCESS
} TravelOption
WordFlags    GCNListTypeFlags
      GCNLTF_SAVE_TO_STATE            0x8000
ByteFlags TempMetaGCNFlags
      TMGCNF_RELOCATED                0x80
WordFlags    GCNListSendFlags
      GCNLSF_SET_STATUS                       0x8000
      GCNLSF_IGNORE_IF_STATUS_TRANSITIONING   0x4000
      GCNLSF_FORCE_QUEUE                      0x2000
INI_CATEGORY_BUFFER_SIZE  64
typedef enum {
    OFIQNS_SYSTEM_INPUT_OBJ=0,
    OFIQNS_INPUT_OBJ_OF_OWNING_GEODE=2,
    OFIQNS_PROCESS_OF_OWNING_GEODE=4,
    OFIQNS_DISPATCH=6
} ObjFlushInputQueueNextStop
WordFlags    UpdateWindowFlags
      UWF_ATTACHING                   0x8000
      UWF_DETACHING                   0x4000
      UWF_RESTORING_FROM_STATE        0x2000
      UWF_TOP_LEVEL_WINDOW            0x1000
WordFlags    CompChildFlags
      CCF_MARK_DIRTY            0x8000
      CCF_REFERENCE             0x7fff
      CCO_FIRST                 0x0000
      CCO_LAST                  0x7FFF
CCF_REFERENCE_OFFSET      0
```

```
ByteEnum     InsertChildOption
      ICO_FIRST                0
      ICO_LAST                 1
      ICO_BEFORE_REFERENCE     2
      ICO_AFTER_REFERENCE      3
WordFlags    InsertChildFlags
      ICF_MARK_DIRTY           0x8000
      ICF_OPTIONS              0x0003
typedef enum /* word */ {
    OCCT_SAVE_PARAMS_TEST_ABORT=0,
    OCCT_SAVE_PARAMS_DONT_TEST_ABORT=2,
    OCCT_DONT_SAVE_PARAMS_TEST_ABORT=4,
    OCCT_DONT_SAVE_PARAMS_DONT_TEST_ABORT=6,
    OCCT_ABORT_AFTER_FIRST=8,
    OCCT_COUNT_CHILDREN=10
} ObjCompCallType
```

Structures

```
typedef struct {
    ClassStruct          *MB_class
} MetaBase
typedef struct {
    word                 ax
    word                 cx
    word                 dx
    word                 bp
} AsmPassReturn
typedef struct {
    ClassStruct *        GTP_class
    optr                 GTP_target
} GetTargetParams
typedef struct {
    ManufacturerID       NT_manuf
    word                 NT_type
} NotificationType
typedef struct {
    word                 GCNLT_manuf
    word                 GCNLT_type
} GCNListType
typedef struct {
    optr                 GCNLE_item
} GCNListElement
typedef struct {
    ChunkArrayHeader     GCNLH_meta
    word                 GCNLH_statusEvent
    MemHandle            GCNLH_statusData
    word                 GCNLH_statusCount
} GCNListHeader
typedef struct {
    GCNListType          GCNLOLE_ID
    ChunkHandle          GCNLOLE_list
} GCNListOfListsElement
typedef struct {
    ChunkArrayHeader     GCNLOL_meta
} GCNListOfListsHeader
typedef struct {
    ChunkHandle          TMGCND_listOfLists
```

```
    TempMetaGCNFlags      TMGCND_flags
} TempMetaGCNData
typedef struct {
    GCNListType           GCNLP_ID
    optr                  GCNLP_optr
} GCNListParams
typedef struct {
    word                  DDE_ackCount
    word                  DDE_callerID
    optr                  DDE_ackOD
    word                  DDE_completeMsg
} DetachDataEntry
typedef struct {
    optr                  LP_next
} LinkPart
typedef struct {
    optr                  CP_firstChild
} CompPart
```

Messages

```
void MSG_META_NULL()
void MSG_META_INITIALIZE()
void MSG_META_DUMMY()
void MSG_META_APP_STARTUP(MemHandle appLaunchBlock)
void MSG_META_ATTACH()
@alias(MSG_META_ATTACH) void MSG_META_ATTACH_PROCESS(
                    word value1, word value2)
@alias(MSG_META_ATTACH) void
      MSG_META_ATTACH_GENPROCESSCLASS(
                    MemHandle appLaunchBlock)
@alias(MSG_META_ATTACH) void MSG_META_ATTACH_OBJECT(
                    word flags,
                    MemHandle appLaunchBlock,
                    MemHandle extraState)
@alias(MSG_META_ATTACH) void MSG_META_ATTACH_THREAD()
void MSG_META_DETACH(word callerID, optr caller)
void MSG_META_DETACH_COMPLETE()
void MSG_META_DETACH_ABORT()
void MSG_META_ACK(word callerID, optr caller)
void MSG_META_APP_SHUTDOWN(word callerID, optr ackOD)
void MSG_META_SHUTDOWN_COMPLETE()
void MSG_META_SHUTDOWN_ACK(word callerID, optr ackOD)
ClassStruct * MSG_META_GET_CLASS()
Boolean MSG_META_IS_OBJECT_IN_CLASS(ClassStruct *class)
void MSG_META_BLOCK_FREE()
void MSG_META_OBJ_FREE()
ClassStruct * MSG_META_RESOLVE_VARIANT_SUPERCLASS(
                    word MasterOffset)
Boolean MSG_META_RELOCATE(word vmRelocType, word frame)
Boolean MSG_META_UNRELOCATE(word vmRelocType,
                    word frame)
void MSG_META_SET_FLAGS(ChunkHandle objChunk,
                    ObjChunkFlags bitsToSet,
                    ObjChunkFlags bitsToClear)
word MSG_META_GET_FLAGS(ChunkHandle ch)
void MSG_META_VM_FILE_DIRTY(FileHandle file)
void MSG_META_QUIT()
```

```
@alias(MSG_META_QUIT) void MSG_META_QUIT_PROCESS(
                        word quitLevel,
                        ChunkHandle ackODChunk)
@alias(MSG_META_QUIT) void MSG_META_QUIT_OBJECT(
                        optr obj)
void MSG_META_QUIT_ACK(word quitLevel, word abortFlag)
Boolean MSG_META_DISPATCH_EVENT(AsmPassReturn *retVals,
                        Eventhandle eventHandle,
                        MessageFlags msgFlags)
void MSG_META_SEND_CLASSED_EVENT(
                    EventHandle event,
                    TravelOption whereTo)
optr MSG_META_GET_OPTR()
void MSG_META_GET_TARGET_AT_TARGET_LEVEL(
                    GetTargetParams *retValue,
                    TargetLevel level)
void MSG_META_ADD_VAR_DATA(@stack
                    word dataType, word dataSize,
                    void *data)
Boolean MSG_META_DELETE_VAR_DATA(word dataType)
word MSG_META_INITIALIZE_VAR_DATA(word dataType)
void MSG_META_NOTIFY(ManufacturerID manufID,
                    word notificationType, word data)
void MSG_META_NOTIFY_WITH_DATA_BLOCK(
                    ManufacturerID manufID,
                    word notificationType,
                    MemHandle data)
Boolean MSG_META_GCN_LIST_ADD(@stack
                    optr dest, word listType,
                    ManufacturerID listManuf)
Boolean MSG_META_GCN_LIST_REMOVE(@stack
                    optr dest, word listType,
                    ManufacturerID listManuf)
void MSG_META_GCN_LIST_SEND(@stack
                    GCNListSendFlags flags,
                    EventHandle event, MemHandle block,
                    word listType,
                    ManufacturerID listManuf)
void MSG_META_GCN_LIST_DESTROY()
void MSG_META_SAVE_OPTIONS()
void MSG_META_GET_INI_CATEGORY(char *buf)
void MSG_META_SUSPEND()
void MSG_META_UNSUSPEND()
void MSG_META_LOAD_OPTIONS()
void MSG_META_GET_VAR_DATA(@stack
                    word dataType, word bufSize,
                    void *buf)
void MSG_META_NOTIFY_OBJ_BLOCK_INTERACTIBLE(
                    MemHandle objBlock)
void MSG_META_NOTIFY_OBJ_BLOCK_NOT_INTERACTIBLE(
                    MemHandle objBlock)
void MSG_META_OBJ_FLUSH_INPUT_QUEUE(
                    EventHandle event,
                    ObjFlushInputQueueNextStop nextStop,
                    MemHandle objBlock)
void MSG_META_WIN_DEC_REF_COUNT(MemHandle win)
void MSG_META_UPDATE_WINDOW(
```

```
                    UpdateWindowFlags updateFlags,
                    VisUpdateMode updateMode)
void MSG_META_FINISH_QUIT(Boolean abortFlag)
void MSG_META_FINISH_QUIT(Boolean abortFlag)
void MSG_META_SET_OBJ_BLOCK_OUTPUT(optr output)
void MSG_META_GET_HELP_FILE(char *buf)
void MSG_META_GET_HELP_TYPE (byte helpType)
optr MSG_META_GET_OBJ_BLOCK_OUTPUT()
void MSG_META_RESET_OPTIONS()
void MSG_META_BRING_UP_HELP()
void MSG_META_SET_HELP_FILE(char *buf)
Boolean MSG_META_GCN_LIST_FIND_ITEM(@stack optr dest,
                    word listType,
                    ManufacturerID listManuf)
void MSG_META_TRANSPARENT_DETACH()
```

Window Messages

```
@exportMessages MetaWindowMessages,
                            DEFAULT_EXPORTED_MESSAGES
void MSG_META_EXPOSED(WindowHandle win)
void MSG_META_EXPOSED_FOR_PRINT(
                    GStringHandle gstring,
                    optr completionOD)
MSG_META_WIN_UPDATE_COMPLETE
MSG_META_WIN_CHANGE
MSG_META_IMPLIED_WIN_CHANGE
MSG_META_RAW_UNIV_ENTER
MSG_META_RAW_UNIV_LEAVE
MSG_META_INVAL_TREE
MSG_META_INVAL_BOUNDS
```

Input Messages

```
@exportMessages MetaInputMessages,
                            DEFAULT_EXPORTED_MESSAGES
void MSG_META_MOUSE_BUTTON(word xPosition,
                    word yPosition, word inputState)
void MSG_META_MOUSE_PTR(word xPosition, word yPosition,
                    word inputState)
@alias (MSG_META_MOUSE_PTR) void MSG_META_PTR(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_KBD_CHAR(word character, word flags,
                    word state)
MSG_META_PRESSURE
MSG_META_DIRECTION
MSG_META_MOUSE_TIMER
void MSG_DRAG(word xPosition, word yPosition,
                    word inputState)
```

UI Messages

```
@exportMessages MetaUIMessages,
                            DEFAULT_EXPORTED_MESSAGES_4
void MSG_META_CLIPBOARD_NOTIFY_QUICK_TRANSFER_FEEDBACK(
            QuickTransferCursor quickTransferCursor)
void
    MSG_META_CLIPBOARD_NOTIFY_QUICK_TRANSFER_CONCLUDED(
                    ClipboardQuickNotifyFlags flags)
```

```
void MSG_META_CLIPBOARD_CUT()
void MSG_META_CLIPBOARD_COPY()
void MSG_META_CLIPBOARD_PASTE()
void MSG_META_UNDO(UndoActionStruct *undoData)
void MSG_META_UNDO_FREEING_ACTION(
                AddUndoActionStruct *data)
void MSG_META_SELECT_ALL()
void MSG_META_DELETE()
void MSG_META_CLIPBOARD_NOTIFY_TRANSFER_ITEM_FREED(
                VMFileHandle itemFile,
                VMBlockHandle itemBlock)
void
MSG_META_CLIPBOARD_NOTIFY_NORMAL_TRANSFER_ITEM_CHANGED(
)
void MSG_META_CONTENT_SET_VIEW(optr view)
void MSG_META_CONTENT_VIEW_OPENING(optr view)
void MSG_META_CONTENT_VIEW_CLOSING()
void MSG_META_CONTENT_VIEW_WIN_OPENED(
                word viewWidth, word viewHeight,
                WindowHandle viewWindow)
void MSG_META_CONTENT_VIEW_WIN_CLOSED(
                WindowHandle viewWindow)
void MSG_META_CONTENT_VIEW_ORIGIN_CHANGED(@stack
                WindowHandle viewWindow,
                sdword xOrigin, sdword yOrigin)
void MSG_META_CONTENT_VIEW_SCALE_FACTOR_CHANGED(@stack
                WindowHandle viewWindow,
                WWFixedAsDWord yScaleFactor,
                WWFixedAsDWord xScaleFactor)
void MSG_META_CONTENT_VIEW_SIZE_CHANGED(
                word viewWidth, word viewHeight,
                WindowHandle viewWindow)
void MSG_META_CONTENT_TRACK_SCROLLING(
                TrackScrollingParams *args)
void MSG_META_CONTENT_VIEW_LOST_GADGET_EXCL(
                WindowHandle viewWindow)
void MSG_META_CONTENT_NAVIGATION_QUERY(
                optr queryOrigin, word navFlags,
                NavigationQueryParams *retValue)
void MSG_META_CONTENT_APPLY_DEFAULT_FOCUS()
void MSG_META_CONTENT_ENTER(optr view)
void MSG_META_CONTENT_LEAVE(optr view)
Boolean MSG_META_DOC_OUTPUT_INITIALIZE_DOCUMENT_FILE(
                optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_CREATE_UI_FOR_DOCUMENT(
                optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_DESTROY_UI_FOR_DOCUMENT(
                optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_ATTACH_UI_TO_DOCUMENT(
                optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_DETACH_UI_FROM_DOCUMENT(
                optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_READ_CACHED_DATA_FROM_FILE(
                optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_WRITE_CACHED_DATA_TO_FILE(
                optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_DOCUMENT_HAS_CHANGED(
```

```
                    optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_PHYSICAL_SAVE(
                    word *error, optr document,
                    FileHandle file)
Boolean MSG_META_DOC_OUTPUT_PHYSICAL_UPDATE(word *error,
                    optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_PHYSICAL_SAVE_AS_FILE_HANDLE(
                    word *error, optr document,
                    FileHandle file)
void MSG_META_DOC_OUTPUT_PHYSICAL_REVERT(
                    word *error, optr document,
                    FileHandle file)
Boolean
MSG_META_DOC_OUTPUT_UPDATE_EARLIER_COMPATIBLE_DOCUMENT(
                    word *error, optr document,
                    FileHandle file)
Boolean
MSG_META_DOC_OUTPUT_UPDATE_EARLIER_INCOMPATIBLE_DOCUMEN
T(
                    word *error, optr document,
                    FileHandle file)
void MSG_META_DOC_OUTPUT_SAVE_AS_COMPLETED(
                    optr document, FileHandle file)
void MSG_META_DOC_OUTPUT_ATTACH_FAILED(
                    optr document, FileHandle file)
void MSG_META_UI_FORCE_CONTROLLER_UPDATE(
                    ManufacturerID manufID,
                    word changeID)
Boolean MSG_META_GEN_PATH_RESTORE_DISK_PROMPT(
                    GenPathDiskRestoreArgs *args,
                    DiskRestoreError *error)
void MSG_META_PAGED_OBJECT_GOTO_PAGE(word page)
void MSG_META_PAGED_OBJECT_NEXT_PAGE()
void MSG_META_PAGED_OBJECT_PREVIOUS_PAGE()
void MSG_META_NOTIFY_TASK_SELECTED()
void MSG_META_GAINED_MOUSE_EXCL()
void MSG_META_LOST_MOUSE_EXCL()
void MSG_META_GAINED_KBD_EXCL()
void MSG_META_LOST_KBD_EXCL()
void MSG_META_GAINED_PRESSURE_EXCL()
void MSG_META_LOST_PRESSURE_EXCL()
void MSG_META_GAINED_DIRECTION_EXCL()
void MSG_META_LOST_DIRECTION_EXCL()
void MSG_META_GRAB_FOCUS_EXCL()
void MSG_META_RELEASE_FOCUS_EXCL()
Boolean MSG_META_GET_FOCUS_EXCL(optr *focusObject)
void MSG_META_GRAB_TARGET_EXCL()
void MSG_META_RELEASE_TARGET_EXCL()
Boolean MSG_META_GET_TARGET_EXCL(optr *targetObject)
void MSG_META_GRAB_MODEL_EXCL()
void MSG_META_RELEASE_MODEL_EXCL()
Boolean MSG_META_GET_MODEL_EXCL(optr *targetObj)
void MSG_META_RELEASE_FT_EXCL()
void MSG_META_MUP_ALTER_FTVMC_EXCL(
                    optr objectWantingControl,
                    MetaAlterFTVMCExclFlags flags)
void MSG_META_GAINED_FOCUS_EXCL()
```

```
void MSG_META_LOST_FOCUS_EXCL()
void MSG_META_GAINED_SYS_FOCUS_EXCL()
void MSG_META_LOST_SYS_FOCUS_EXCL()
void MSG_META_GAINED_TARGET_EXCL()
void MSG_META_LOST_TARGET_EXCL()
void MSG_META_GAINED_SYS_TARGET_EXCL()
void MSG_META_LOST_SYS_TARGET_EXCL()
void MSG_META_GAINED_MODEL_EXCL()
void MSG_META_LOST_MODEL_EXCL()
void MSG_META_GAINED_SYS_MODEL_EXCL()
void MSG_META_LOST_SYS_MODEL_EXCL()
void MSG_META_GAINED_DEFAULT_EXCL()
void MSG_META_LOST_DEFAULT_EXCL()
void MSG_META_MOUSE_BUMP_NOTIFICATION(
                    sword xBump, sword yBump)
Boolean MSG_META_FUP_KBD_CHAR(word character,
                    word flags, word state)
KbdReturnFlags MSG_META_PRE_PASSIVE_KBD_CHAR(
                    word character, word flags,
                    word state)
KbdReturnFlags MSG_META_POST_PASSIVE_KBD_CHAR(
                    word character, word flags,
                    word state)
void MSG_META_QUERY_IF_PRESS_IS_INK(
                    InkReturnParams *retVal,
                    sword xPosition, sword yPosition)
void MSG_META_LARGE_QUERY_IF_PRESS_IS_INK(
             InkReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_START_SELECT(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_END_SELECT(
                    MouseReturnParams *retVal,
                    sword xPosition,sword yPosition,
                    word inputState)
void MSG_META_START_MOVE_COPY(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_END_MOVE_COPY(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_START_FEATURES(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_END_FEATURES(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_START_OTHER(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
```

```
void MSG_META_END_OTHER(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_DRAG_SELECT(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_DRAG_MOVE_COPY(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_DRAG_FEATURES(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_DRAG_OTHER(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_BUTTON(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_BUTTON(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_START_SELECT(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_END_SELECT(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_START_MOVE_COPY(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_END_MOVE_COPY(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_START_FEATURES(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_END_FEATURES(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_START_OTHER(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_PRE_PASSIVE_END_OTHER(
```

```
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_START_SELECT(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_END_SELECT(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_START_MOVE_COPY(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_END_MOVE_COPY(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_START_FEATURES(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_END_FEATURES(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_START_OTHER(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_POST_PASSIVE_END_OTHER(
                    MouseReturnParams *retVal,
                    sword xPosition, sword yPosition,
                    word inputState)
void MSG_META_LARGE_PTR(
             MouseReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_LARGE_START_SELECT(
             MouseReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_LARGE_END_SELECT(
             MouseReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_LARGE_START_MOVE_COPY(
             MouseReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_LARGE_END_MOVE_COPY(
             MouseReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_LARGE_START_FEATURES(
             MouseReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_LARGE_END_FEATURES(
             MouseReturnParams *retVal,
             LargeMouseData *largeMouseDataStruct)
void MSG_META_LARGE_START_OTHER(
```

```
                MouseReturnParams *retVal,
                LargeMouseData *largeMouseDataStruct)
    void MSG_META_LARGE_END_OTHER(
                MouseReturnParams *retVal,
                LargeMouseData *largeMouseDataStruct)
    void MSG_META_LARGE_DRAG_SELECT(
                MouseReturnParams *retVal,
                LargeMouseData *largeMouseDataStruct)
    void MSG_META_LARGE_DRAG_MOVE_COPY(
                MouseReturnParams *retVal,
                LargeMouseData *largeMouseDataStruct)
    void MSG_META_LARGE_DRAG_FEATURES(
                MouseReturnParams *retVal,
                LargeMouseData *largeMouseDataStruct)
    void MSG_META_LARGE_DRAG_OTHER(
                MouseReturnParams *retVal,
                LargeMouseData *largeMouseDataStruct)
    MouseReturnFlags
          MSG_META_ENSURE_MOUSE_NOT_ACTIVELY_TRESPASSING()
    MouseReturnFlags
          MSG_META_ENSURE_NO_MENUS_IN_STAY_UP_MODE()
    void MSG_META_ENSURE_ACTIVE_FT()
    void MSG_META_NOTIFY_NO_FOCUS_WITHIN_NODE()
    void MSG_META_DOC_OUTPUT_IMPORT_FILE()
    void MSG_META_DOC_OUTPUT_EXPORT_FILE()
    void MSG_META_GRAB_KBD()
    void MSG_META_FORCE_GRAB_KBD()
    void MSG_META_RELEASE_KBD()
    void MSG_META_VIEW_COMMAND_CHANGE_SCALE()
    void MSG_META_FIELD_NOTIFY_DETACH(optr field,
                    word shutdownFlag)
    void MSG_META_FIELD_NOTIFY_NO_FOCUS(optr field,
                    word shutdownFlag)
    void MSG_META_FIELD_NOTIFY_START_LAUNCHER_ERROR(
                    optr field)
    void MSG_META_DELETE_RANGE_OF_CHARS(@stack
                    VisTextRange rangeToDelete)
    Boolean MSG_META_TEST_WIN_INTERACTIBILITY(optr inputOD,
                    WindowHandle window)
    Boolean MSG_META_CHECK_IF_INTERACTABLE_OBJECT(optr obj)
```

Application Messages

```
    @exportMessages MetaApplicationMessages,
                            DEFAULT_EXPORTED_MESSAGES_3
```

GrObj Messages

```
    @exportMessages MetaGrObjMessages,
                            DEFAULT_EXPORTED_MESSAGES
    MSG_GROBJ_ACTION_NOTIFICATION
```

Printing Messages

```
    @exportMessages MetaPrintMessages,
                            DEFAULT_EXPORTED_MESSAGES
    void MSG_PRINT_VERIFY_PRINT_REQUEST(optr printCtrlOD)
    void MSG_PRINT_START_PRINTING(optr printCtrlOD,
                    GStateHandle gstate)
    void MSG_PRINT_GET_DOC_NAME(optr printCtrlOD)
    void MSG_PRINT_NOTIFY_PRINT_DB(optr printCtrlOD,
```

```
                    PrintControlStatus pcs)
void MSG_PRINT_NOTIFY_PRINT_JOB_CREATED(word jobID)
void MSG_PRINT_REPORT_PAGE_SIZE(PageSizeReport *psr)
void MSG_PRINTING_GET_DOC_NAME()
void MSG_PRINTING_COMPLETED()
```

Search/Spell Messages

```
@exportMessages MetaSearchSpellMessages,
                        DEFAULT_EXPORTED_MESSAGES
void MSG_SPELL_CHECK(@stack
                    optr replyObj,
                    dword numCharsToCheck,
                    SpellCheckOptions options,
                    MemHandle ICbuff)
void MSG_META_EDIT_USER_DICTIONARY_COMPLETED()
void MSG_THES_REPLACE_SELECTED_WORDS (@stack
                    MemHandle RSWP_string,
                    word RSWP_numChars)
void MSG_THES_SELECT_WORD (@stack
                    optr output, Message message,
                    word numChars, word type)
void MSG_SEARCH(MemHandle searchInfo)
void MSG_REPLACE_CURRENT(MemHandle replaceInfo)
void MSG_REPLACE_ALL_OCCURRENCES(MemHandle replaceInfo,
                    Boolean replaceFromBeginning)
void MSG_REPLACE_ALL_OCCURRENCES_IN_SELECTION(
                    MemHandle replaceInfo)
void MSG_ABORT_ACTIVE_SPELL()
void MSG_ABORT_ACTIVE_SEARCH()
optr MSG_META_GET_OBJECT_FOR_SEARCH_SPELL(
                    GetSearchSpellObjectOption option,
                    optr curObject)
void MSG_META_GET_CONTEXT(@stack
                    dword position,
                    ContextLocation location,
                    word numCharsToGet, optr replyObj)
void MSG_META_GENERATE_CONTEXT_NOTIFICATION(@stack
                    dword position,
                    ContextLocation location,
                    word numCharsToGet, optr replyObj)
void MSG_META_CONTEXT(MemHandle data)
void MSG_META_DISPLAY_OBJECT_FOR_SEARCH_SPELL()
```

GCN Messages

```
@exportMessages MetaGCNMessages,
                        DEFAULT_EXPORTED_MESSAGES
void MSG_NOTIFY_FILE_CHANGE(
            FileChangeNotificationType notifyType
            MemHandle data)
void MSG_NOTIFY_DRIVE_CHANGE(
                    GCNDriveChangeNotificationType type,
                    word driveNum)
void MSG_NOTIFY_APP_STARTED()
void MSG_NOTIFY_APP_EXITED(MemHandle appExited)
void MSG_NOTIFY_DATE_TIME_CHANGE()
void MSG_NOTIFY_USER_DICT_CHANGE(
                    MemHandle sendingSpellBox,
                    MemHandle userDictChanged)
```

```
void MSG_DISPLAY_FLOATING_KEYBOARD()
void MSG_NOTIFY_EXPRESS_MENU_CHANGE(
            GCNExpressMenuNotificationTypes type,
            optr affectedExpressMenuControl)
void MSG_PRINTER_INSTALLED_REMOVED()
void MSG_META_CONFIRM_SHUTDOWN(
                  word confirmed,
                  GCNShutdownControlType type)
@alias(MSG_META_CONFIRM_SHUTDOWN) void
      MSG_META_CONFIRM_SHUTDOWN_QUERY(
                  optr originator,
                  GCNShutdownControlType type)
```

Text Messages

```
@exportMessages MetaTextMessages,
                           DEFAULT_EXPORTED_MESSAGES
MSG_VIS_TEXT_SET_FONT_ID
MSG_VIS_TEXT_SET_FONT_WEIGHT
MSG_VIS_TEXT_SET_FONT_WIDTH
void MSG_VIS_TEXT_SET_POINT_SIZE(@stack
                           WWFixedAsDWord pointSize,
                           dword rangeEnd,
                           dword rangeStart)
MSG_VIS_TEXT_SET_SMALLER_POINT_SIZE
MSG_VIS_TEXT_SET_LARGER_POINT_SIZE
void MSG_VIS_TEXT_SET_TEXT_STYLE(@stack
                  word extBitsToClear,
                  word extBitsToSet,
                  word styleBitsToClear,
                  word styleBitsToSet,
                  dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_SET_COLOR
MSG_VIS_TEXT_SET_GRAY_SCREEN
MSG_VIS_TEXT_SET_PATTERN
MSG_VIS_TEXT_SET_CHAR_BG_COLOR
MSG_VIS_TEXT_SET_CHAR_BG_GRAY_SCREEN
MSG_VIS_TEXT_SET_CHAR_BG_PATTERN
MSG_VIS_TEXT_SET_TRACK_KERNING
MSG_VIS_TEXT_SET_BORDER_COLOR
MSG_VIS_TEXT_SET_BORDER_GRAY_SCREEN
MSG_VIS_TEXT_SET_BORDER_PATTERN
void MSG_VIS_TEXT_SET_PARA_ATTRIBUTES(@stack
                  word bitsToClear, word bitsToSet,
                  dword rangeEnd, dword rangeStart)
void MSG_META_TEXT_USER_MODIFIED(optr obj)
void MSG_META_TEXT_CR_FILTERED(word character,
                  word flags, word state)
void MSG_META_TEXT_TAB_FILTERED(word character,
                  word flags, word state)
void MSG_META_TEXT_LOST_FOCUS(optr obj)
void MSG_META_TEXT_GAINED_FOCUS(optr obj)
void MSG_META_TEXT_LOST_TARGET(optr obj)
void MSG_META_TEXT_GAINED_TARGET(optr obj)
void MSG_META_TEXT_EMPTY_STATUS_CHANGED(
                  optr object, Boolean hasTextFlag)
void MSG_META_TEXT_NOT_USER_MODIFIED(optr obj)
@exportMessages MetaStylesMessages,
```

```
                            DEFAULT_EXPORTED_MESSAGES
    void MSG_META_STYLED_OBJECT_REQUEST_ENTRY_MONIKER()
    void MSG_META_STYLED_OBJECT_UPDATE_MODIFY_BOX()
    void MSG_META_STYLED_OBJECT_MODIFY_STYLE()
    void MSG_META_STYLED_OBJECT_DESCRIBE_ATTRS()
    void MSG_META_STYLED_OBJECT_DESCRIBE_STYLE()
    void MSG_META_STYLED_OBJECT_APPLY_STYLE()
    void MSG_META_STYLED_OBJECT_DELETE_STYLE()
    void MSG_META_STYLED_OBJECT_DEFINE_STYLE()
    void MSG_META_STYLED_OBJECT_REDEFINE_STYLE()
    void MSG_META_STYLED_OBJECT_RETURN_TO_BASE_STYLE()
    void MSG_META_STYLED_OBJECT_LOAD_STYLE_SHEET()
    void MSG_META_STYLED_OBJECT_SAVE_STYLE()
    void MSG_META_STYLED_OBJECT_RECALL_STYLE()
```

Color Messages

```
    @exportMessages MetaColorMessages,
                            DEFAULT_EXPORTED_MESSAGES
    void MSG_META_COLORED_OBJECT_SET_COLOR()
    void MSG_META_COLORED_OBJECT_SET_DRAW_MASK()
    void MSG_META_COLORED_OBJECT_SET_PATTERN()
```

Floating-Point Messages

```
    @exportMessages MetaFloatMessages,
                            DEFAULT_EXPORTED_MESSAGES
    void MSG_FLOAT_CTRL_REQUEST_MONIKER(
                        MemHandle formatInfoStrucHan)
    void MSG_FLOAT_CTRL_UPDATE_UI(
                        MemHandle formatInfoStrucHan)
    void MSG_FLOAT_CTRL_FORMAT_SELECTED(
                        MemHandle formatInfoStrucHan)
    void MSG_FLOAT_CTRL_USER_DEF_INVOKE(
                        MemHandle formatInfoStrucHan)
    void MSG_FLOAT_CTRL_USER_DEF_OK(
                        MemHandle formatInfoStrucHan)
    void MSG_FLOAT_CTRL_FORMAT_DELETE(
                        MemHandle formatInfoStrucHan)
    void MSG_FLOAT_CTRL_REPLACE_NUM_FORMAT(
                        MemHandle formatInfoStrucHan)
    void MSG_FLOAT_CTRL_FORMAT_APPLY(
                        MemHandle formatInfoStrucHan)
    void MSG_FCF_FORMAT_CHANGE(MemHandle formatInfoStrucHan)
    void MSG_FCT_FORMAT_CHANGE(MemHandle formatInfoStrucHan)
```

MultTextGuardianClass

```
    @class MultTextGuardianClass, TextGuardianClass
```

PageSizeControlClass

```
@class PageSizeControlClass, GenControlClass
```

Instance Data

```
PageSizeCtrlAttrs        PZCI_attrs
dword                    PZCI_width
dword                    PZCI_height
PageLayout               PZCI_layout
PCMarginParams           PZCI_margins
    @default GII_type = GIT_PROPERTIES
    @default GII_visibility = GIV_DIALOG
```

Variable Data

```
PageSizeControlChanges
                    ATTR_PAGE_SIZE_CONTROL_UI_CHANGES
```

Types and Flags

```
MINIMUM_PAGE_WIDTH_VALUE        36
MINIMUM_PAGE_HEIGHT_VALUE       36
MAXIMUM_PAGE_WIDTH_VALUE        (72 * 45)
MAXIMUM_PAGE_HEIGHT_VALUE       (72 * 45)
MAXIMUM_LABELS_ACROSS           63
MAXIMUM_LABELS_DOWN             63
WordFlags    PageSizeCtrlAttrs
      PZCA_ACT_LIKE_GADGET      0x8000
      PZCA_PAPER_SIZE           0x4000
      PZCA_INITIALIZE           0x2000
      PZCA_LOAD_SAVE_OPTIONS    0x1000
ByteFlags    PageSizeControlFeatures
      PSIZECF_MARGINS           0x0010
      PSIZECF_CUSTOM_SIZE       0x0008
      PSIZECF_LAYOUT            0x0004
      PSIZECF_SIZE_LIST         0x0002
      PSIZECF_PAGE_TYPE         0x0001
PSIZEC_DEFAULT_FEATURES (PSIZECF_PAGE_TYPE |
            PSIZECF_SIZE_LIST | PSIZECF_LAYOUT |
            PSIZECF_CUSTOM_SIZE)
ByteEnum     PaperOrientation
      PO_PORTRAIT               0x00
      PO_LANDSCAPE              0x01
WordFlags    PageLayoutPaper
      PLP_ORIENTATION           0x0008
      PLP_TYPE                  0x0004
ByteEnum     EnvelopeOrientation
      EO_PORTAIT                0x00
      EO_LANDSCAPE              0x01
WordFlags    PageLayoutEnvelope
      PLE_ORIENTATION           0x0010
      PLE_TYPE                  0x0004
WordFlags    PageLayoutLabel
      PLL_ROWS                  0x7e00
      PLL_COLUMNS               0x01f8
      PLL_TYPE                  0x0004
```

Structures

```
typedef struct {
    word           unused
    word           PS_width
```

```
    word            PS_height
    PageLayout      PS_layout
} PageSize
typedef struct {
    optr            PSCC_destination
    Message         PSCC_message
} PageSizeControlChanges
typedef struct {
    dword           PSR_width
    dword           PSR_height
    PageLayout      PSR_layout
} PageSizeReport
typedef union {
    PageLayoutPaper       PL_paper
    PageLayoutEnvelope    PL_envelope
    PageLayoutLabel       PL_label
} PageLayout
```

Messages

```
void MSG_PZC_SET_PAGE_SIZE(PageSizeReport *psr)
void MSG_PZC_GET_PAGE_SIZE(PageSizeReport *psr)
```

PointerClass

```
@class PointerClass, GrObjClass
```

Instance Data

```
PointerModes       PTR_modes
```

Types and Flags

```
MIN_MARQUEE_DIMENSION           7
MAX_PRIORITY_LIST_ELEMENTS      5
ByteFlags    PointerModes
      PM_HANDLES_RESIZE               0x04
      PM_HANDLES_ROTATE               0x02
      PM_POINTER_IS_ACTION_OBJECT     0x01
```

PrintControlClass

```
@class PrintControlClass, GenControlClass, master
```

Instance Data

```
PrintControlAttrs PCI_attrs = (PCA_COPY_CONTROLS |
            PCA_PAGE_CONTROLS | PCA_QUALITY_CONTROLS |
            PCA_USES_DIALOG_BOX | PCA_GRAPHICS_MODE |
            PCA_TEXT_MODE)
word        PCI_startPage = 1
word        PCI_endPage = 1
word        PCI_startUserPage = 0
word        PCI_endUserPage = 0x7fff
word        PCI_defPrinter = -1
PageSizeReport PCI_docSizeInfo = {0, 0, 0, {0, 0, 0, 0}}
optr        PCI_output
optr        PCI_docNameOutput
    @default GII_visibility = GIV_SUB_GROUP
    @default GII_type = GIT_ORGANIZATIONAL
    @default GI_attrs = @default | GA_KBD_SEARCH_PATH
```

Variable Data

```
TempPrintCtrlInstance TEMP_PRINT_CONTROL_INSTANCE
optr ATTR_PRINT_CONTROL_APP_UI
    @reloc ATTR_PRINT_CONTROL_APP_UI, 0, optr
TempPrintCompletionEventData
                            TEMP_PRINT_COMPLETION_EVENT
```

Types and Flags

```
ByteFlags    PrintControlFeatures
      PRINTCF_PRINT_TRIGGER           0x02
      PRINTCF_FAX_TRIGGER             0x01
ByteFlags PrintControlToolboxFeatures
      PRINTCTF_PRINT_TRIGGER          0x02
      PRINTCTF_FAX_TRIGGER            0x01
PRINTC_DEFAULT_FEATURES   (PRINTCF_PRINT_TRIGGER |
                           PRINTCF_FAX_TRIGGER)
ByteFlags    PrinterOutputModes
      POM_GRAPHICS_LOW                0x10
      POM_GRAPHICS_MEDIUM             0x08
      POM_GRAPHICS_HIGH               0x04
      POM_TEXT_DRAFT                  0x02
      POM_TEXT_NLQ                    0x01
PRINT_GRAPHICS      (POM_GRAPHICS_LOW |
                     POM_GRAPHICS_MEDIUM |
                    POM_GRAPHICS_HIGH )
PRINT_TEXT          (POM_TEXT_DRAFT | POM_TEXT_NLQ)
typedef enum {
      PQT_HIGH,
      PQT_MEDIUM,
      PQT_LOW
} PrintQualityEnum
WordFlags    PrintControlAttrs
      PCA_SEE_IF_DOC_WILL_FIT         0x4000
      PCA_MARK_APP_BUSY               0x2000
      PCA_VERIFY_PRINT                0x1000
      PCA_SHOW_PROGRESS               0x0800
      PCA_PROGRESS_PERCENT            0x0400
```

```
      PCA_PROGRESS_PAGE                0x0200
      PCA_FORCE_ROTATION               0x0100
      PCA_COPY_CONTROLS                0x0080
      PCA_PAGE_CONTROLS                0x0040
      PCA_QUALITY_CONTROLS             0x0020
      PCA_USES_DIALOG_BOX              0x0010
      PCA_GRAPHICS_MODE                0x0008
      PCA_TEXT_MODE                    0x0004
      PCA_DEFAULT_QUALITY              0x0002
typedef enum {
      PCS_PRINT_BOX_VISIBLE,
      PCS_PRINT_BOX_NOT_VISIBLE
} PrintControlStatus
typedef enum {
      SVA_NO_MESSAGE,
      SVA_WARNING,
      SVA_PRINTING
} SpoolVerifyAction
typedef enum {
      PCPT_PAGE,
      PCPT_UNUSED1,
      PCPT_PERCENT,
      PCPT_UNUSED2,
      PCPT_TEXT
} PCProgressType
typedef enum {
      PT_PAPER,
      PT_UNUSED1,
      PT_ENVELOPE,
      PT_UNUSED2,
      PT_LABEL
} PageType
PaperType    PageType
ByteEnum     PaperOrientation
      PO_PORTRAIT                      0x00
      PO_LANDSCAPE                     0x01
WordFlags    PageLayoutPaper
      PLP_ORIENTATION                  0x0008
      PLP_TYPE                         0x0004
ByteEnum     EnvelopePath
      EP_LEFT                          0x00
      EP_CENTER                        0x01
      EP_RIGHT                         0x02
ByteEnum     EnvelopeOrientation
      EO_PORTAIT_LEFT                  0x00
      EO_PORTAIT_RIGHT                 0x01
      EO_LANDSCAPE_UP                  0x02
      EO_LANDSCAPE_DOWN                0x03
WordFlags    PageLayoutEnvelope
      PLE_PATH                         0x0040
      PLE_ORIENTATION                  0x0010
      PLE_TYPE                         0x0004
WordFlags    PageLayoutLabel
      PLL_ROWS                         0x7e00
      PLL_COLUMNS                      0x01f8
      PLL_TYPE                         0x0004
typedef union {
```

```
        PageLayoutPaper          PL_paper
        PageLayoutEnvelope       PL_envelope
        PageLayoutLabel          PL_label
    } PageLayout
    ByteFlags    PrintStatusFlags
        PSF_FAX_AVAILABLE        0x80
        PSF_ABORT                0x08
        PSF_RECEIVED_COMPLETED   0x04
        PSF_RECEIVED_NAME        0x02
        PSF_VERIFIED             0x01
```

Structures

```
    typedef struct {
        dword          PCDSP_width
        dword          PCDSP_height
    } PCDocSizeParams
    typedef struct {
        word           PCMP_left
        word           PCMP_top
        word           PCMP_right
        word           PCMP_bottom
    } PCMarginParams
    typedef struct {
        dword          PSR_width
        dword          PSR_height
        PageLayout     PSR_layout
        PCMarginParams PSR_margins
    } PageSizeReport
    typedef struct {
        int            leftMargin
        int            topMargin
        int            rightMargin
        int            bottomMargin
    } MarginDimensions
    typedef struct {
        int            leftMargin
        int            topMargin
        int            width
        int            height
    } DocumentSize
    typedef struct {
        optr                 TPCI_currentSummons
        optr                 TPCI_progressBox
        ChunkHandle          TPCI_jobParHandle
        word                 TPCI_fileHandle
        word                 TPCI_gstringHandle
        word                 TPCI_printBlockHan
        PrintControlAttrs    TPCI_attrs
        PrintStatusFlags     TPCI_status
    } TempPrintCtrlInstance
```

Messages

```
    void MSG_PRINT_CONTROL_INITIATE_PRINT()
    void MSG_PRINT_CONTROL_PRINT()
    void MSG_PRINT_CONTROL_VERIFY_COMPLETED(
                      Boolean continue)
    void MSG_PRINT_CONTROL_SET_DOC_NAME(char *string)
    Boolean MSG_PRINT_CONTROL_REPORT_PROGRESS(
```

```
                    PCProgressType progress,
                    int pageOrPercent)
@alias(MSG_PRINT_CONTROL_REPORT_PROGRESS)
      Boolean MSG_PRINT_CONTROL_REPORT_STRING(
                    PCProgressType progress,
                    char *progressString)
void MSG_PRINT_CONTROL_PRINTING_CANCELLED()
void MSG_PRINT_CONTROL_PRINTING_COMPLETED()
void MSG_PRINT_CONTROL_SET_ATTRS(
                    PrintControlAttrs attributes)
PrintControlAttrs MSG_PRINT_CONTROL_GET_ATTRS()
void MSG_PRINT_CONTROL_SET_TOTAL_PAGE_RANGE(
                    int firstPage, int lastPage)
dword MSG_PRINT_CONTROL_GET_TOTAL_PAGE_RANGE()
void MSG_PRINT_CONTROL_SET_SELECTED_PAGE_RANGE(
                    int firstPage,int lastPage)
dword MSG_PRINT_CONTROL_GET_SELECTED_PAGE_RANGE()
void MSG_PRINT_CONTROL_SET_DOC_SIZE(
                    int width, int height)
dword MSG_PRINT_CONTROL_GET_DOC_SIZE()
void MSG_PRINT_CONTROL_SET_EXTENDED_DOC_SIZE(
                    PCDocSizeParams *ptr)
void MSG_PRINT_CONTROL_GET_EXTENDED_DOC_SIZE(
                    PCDocSizeParams *ptr)
void MSG_PRINT_CONTROL_SET_DOC_MARGINS(
                    PCMarginParams *ptr)
void MSG_PRINT_CONTROL_GET_DOC_MARGINS(
                    PCMarginParams *ptr)
void MSG_PRINT_CONTROL_SET_DOC_SIZE_INFO(
                    PageSizeReport *ptr)
void MSG_PRINT_CONTROL_GET_DOC_SIZE_INFO(
                    PageSizeReport *ptr)
void MSG_PRINT_CONTROL_SET_OUTPUT(optr objectPtr)
optr MSG_PRINT_CONTROL_GET_OUTPUT()
void MSG_PRINT_CONTROL_SET_DOC_NAME_OUTPUT(
                    optr document)
optr MSG_PRINT_CONTROL_GET_DOC_NAME_OUTPUT()
void MSG_PRINT_CONTROL_SET_DEFAULT_PRINTER(
                    int printerNum)
int MSG_PRINT_CONTROL_GET_DEFAULT_PRINTER()
byte MSG_PRINT_CONTROL_GET_PRINT_MODE()
void MSG_PRINT_CONTROL_GET_PAPER_SIZE_INFO(
                    PageSizeReport *ptr)
void MSG_PRINT_CONTROL_GET_PAPER_SIZE(
                    PCMarginParams *retValue)
void MSG_PRINT_CONTROL_GET_PRINTER_MARGINS(
                    MarginDimensions *retValue,
                    Boolean setMargins)
void MSG_PRINT_CONTROL_CALC_DOC_DIMENSIONS(
                    PageSizeReport *ptr)
Boolean MSG_PRINT_CONTROL_CHECK_IF_DOC_WILL_FIT(
                    Boolean warning)
```

ProcessClass

```
@class ProcessClass, MetaClass
```

Types and Flags

```
ByteEnum     CopyChunkMode
      CCM_OPTR           0
      CCM_HPTR           1
      CCM_FPTR           2
      CCM_STRING         3
WordFlags    CopyChunkFlags
      CCF_DIRTY          0x8000
      CCF_MODE           0x6000
      CCF_SIZE           0x1fff
CCF_MODE_OFFSET    13
```

Messages

```
void MSG_PROCESS_NOTIFY_PROCESS_EXIT(
                   GeodeHandle exitProcess,
                   word exitCode)
void MSG_PROCESS_NOTIFY_THREAD_EXIT(
                   ThreadHandle exitProcess,
                   word exitCode)
void MSG_PROCESS_MEM_FULL(word type)
void MSG_PROCESS_CREATE_UI_THREAD(
                   ClassStruct *class, word stackSize)
void MSG_PROCESS_CREATE_EVENT_THREAD(
                   ClassStruct *class, word stackSize)
```

RectClass

```
@class RectClass, GrObjClass
```

RoundedRectClass

```
@class RoundedRectClass, RectClass
```

Instance Data

```
word         RRI_radius
```

Messages

```
MSG_RR_SET_RADIUS
MSG_RR_GET_RADIUS
```

SpellControlClass

```
@class SpellControlClass, GenControlClass
```

Instance Data

```
MemHandle          SCI_ICBuffHan
SpellBoxState      SCI_spellState
byte               SCI_haveSelection
dword              SCI_charsLeft
word               SCI_enableFlags
    @default GII_visibility = GIV_DIALOG
    @default GCI_output = (TO_APP_TARGET)
```

Variable Data

```
void
   ATTR_SPELL_CONTROL_INTERACT_ONLY_WITH_TARGETED_TEXT_
OBJECTS
```

Types and Flags

```
ByteEnum     SpellCheckStartOption
      SCSO_BEGINNING_OF_DOCUMENT            0
      SCSO_BEGINNING_OF_SELECTION           1
      SCSO_WORD_BOUNDARY_BEFORE_SELECTION   2
      SCSO_END_OF_SELECTION                 3
WordFlags    SpellCheckOptions
      SCO_CHECK_SELECTION            0x08
      SCO_CHECK_NUM_CHARS            0x04
typedef enum { /* word */
      SCR_ONE_WORD_CHECKED,
      SCR_SELECTION_CHECKED,
      SCR_DOCUMENT_CHECKED
} SpellCheckResult
WordFlags    SpellFeatures
      SF_CLOSE                       0x2000
      SF_CONTEXT                     0x1000
      SF_SIMPLE_MODAL_BOX            0x0800
      SF_SUGGESTIONS                 0x0400
      SF_CHECK_ALL                   0x0200
      SF_CHECK_TO_END                0x0100
      SF_CHECK_SELECTION             0x0080
      SF_SKIP                        0x0040
      SF_SKIP_ALL                    0x0020
      SF_REPLACE_CURRENT             0x0010
      SF_REPLACE_ALL                 0x0008
      SF_ADD_TO_USER_DICTIONARY      0x0004
      SF_EDIT_USER_DICTIONARY        0x0002
      SF_STATUS                      0x0001
WordFlags    SpellToolboxFeatures
      STF_SPELL                      0x01
SC_DEFAULT_FEATURES(SF_STATUS |
                    SF_EDIT_USER_DICTIONARY |
                    SF_ADD_TO_USER_DICTIONARY |
                    SF_REPLACE_ALL | SF_REPLACE_CURRENT
                    | SF_SKIP_ALL | SF_SKIP |
                    SF_CHECK_SELECTION |
                    SF_CHECK_TO_END | SF_CHECK_ALL |
                    SF_SUGGESTIONS | SF_CLOSE |
                    SF_CONTEXT)
```

```
SC_SUGGESTED_INTRODUCTORY_FEATURES (SF_SIMPLE_MODAL_BOX
                    | SF_CONTEXT | SF_SUGGESTIONS |
                    SF_SKIP | SF_REPLACE_CURRENT |
                    SF_STATUS)
SC_DEFAULT_TOOLBOX_FEATURES (STF_SPELL)
ByteEnum     SpellBoxState
      SBS_NO_SPELL_ACTIVE            0
      SBS_CHECKING_DOCUMENT          1
      SBS_CHECKING_SELECTION         2
```

Structures

```
typedef struct {
    dword    UWI_numChars
    word     UWI_charOffset
    char     UWI_unknownWord[SPELL_MAX_WORD_LENGTH]
} UnknownWordInfo
typedef struct {
    Boolean  NSEC_spellEnabled
} NotifySpellEnableChange
```

Messages

```
void MSG_SPELL_CHECK(@stack
                    optr replyObj,
                    dword numCharsToCheck,
                    SpellCheckOptions options,
                    MemHandle ICbuff)
void MSG_SC_UNKNOWN_WORD_FOUND(UnknownWordInfo *infoPtr)
void MSG_SC_SPELL_CHECK_COMPLETED(
                    SpellCheckResult result)
```

SplineGuardianClass

```
@class SplineGuardianClass, GrObjVisGuardianClass
```

Instance Data

```
byte         SGI_splineCreateMode
byte         SGI_splineAfterCreateMode
byte         SGI_splineMode
```

Messages

```
MSG_SG_SET_SPLINE_MODE
MSG_SG_GENERATE_SPLINE_NOTIFY
MSG_SG_SWITCH_TO_SPLINE_CREATE_MODE
MSG_SG_SWITCH_TO_SPLINE_AFTER_CREATE_MODE
MSG_SG_SET_SPLINE_CREATE_AND_AFTER_CREATE_MODES
```

SplineOpenCloseControlClass

```
@class SplineOpenCloseControlClass, GenControlClass
```

SplinePointControlClass

```
@class SplinePointControlClass, GenControlClass
```

SplineSmoothnessControlClass

```
@class SplineSmoothnessControlClass, GenControlClass
```

SpoolPrintControlClass

```
@class SpoolPrintControlClass, GenInteractionClass,
                                                    master
```

Instance Data

```
ChunkHandle         SPCI_local
word                SPCI_localRevNum = 0
PrintControlAttrs   SPCI_attrs = 0
word                SPCI_startPage = 0
word                SPCI_endPage = 0
word                SPCI_startUserPage = 0
word                SPCI_endUserPage = 0
word                SPCI_defPrinter = 0
word                SPCI_docWidth = 0
word                SPCI_docHeight = 0
word                SPCI_marginLeft = 0
word                SPCI_marginTop = 0
word                SPCI_marginRight = 0
word                SPCI_marginBottom = 0
optr                SPCI_printGroup
optr                SPCI_output
optr                SPCI_docNameOutput
```

Types and Flags

```
PRINT_CONTROL_DEFAULT_PRINTER   (-1)
PRINT_CONTROL_CURRENT_PRINTER   (-2)
ByteFlags    PrinterOutputModes
      POM_TEXT_NLQ              (1 << 0)
      POM_TEXT_DRAFT            (1 << 1)
      POM_GRAPHICS_HIGH         (1 << 2)
      POM_GRAPHICS_MEDIUM       (1 << 3)
      POM_GRAPHICS_LOW          (1 << 4)
PRINT_GRAPHICS     ( POM_GRAPHICS_LOW |
            POM_GRAPHICS_MEDIUM | POM_GRAPHICS_HIGH)
PRINT_TEXT         ( POM_TEXT_DRAFT | POM_TEXT_NLQ)
ByteEnum     PrintQualityEnum
      PQT_HIGH           0
      PQT_MEDIUM         1
      PQT_LOW            2
WordFlags    PrintControlAttrs
      PCA_MARK_APP_BUSY         0x2000
      PCA_VERIFY_PRINT          0x1000
      PCA_SHOW_PROGRESS         0x0800
      PCA_PROGRESS_PERCENT      0x0400
      PCA_PROGRESS_PAGE         0x0200
      PCA_FORCE_ROTATION        0x0100
      PCA_COPY_CONTROLS         0x0080
      PCA_PAGE_CONTROLS         0x0040
      PCA_QUALITY_CONTROLS      0x0020
      PCA_USES_DIALOG_BOX       0x0010
      PCA_GRAPHICS_MODE         0x0008
      PCA_TEXT_MODE             0x0004
      PCA_DEFAULT_QUALITY       0x0003
typedef enum /* word */ {
    PCS_PRINT_BOX_VISIBLE,
    PCS_PRINT_BOX_NOT_VISIBLE
} PrintControlStatus
```

```
typedef enum /* word */ {
    SVA_NO_MESSAGE,
    SVA_WARNING,
    SVA_PRINTING
} SpoolVerifyAction
WordFlags SpoolVerifyDocFail
      SVDF_DUE_TO_DOC_SIZE      0x0002
      SVDF_DUE_TO_MARGINS       0x0001
typedef enum /* word */ {
    PCPT_PAGE,
    PCPT_PERCENT,
    PCPT_TEXT
} PCProgressType
```

Structures

```
typedef struct {
    dword           PCDSP_width
    dword           PCDSP_height
} PCDocSizeParams
typedef struct {
    word            PCMP_left
    word            PCMP_top
    word            PCMP_right
    word            PCMP_bottom
} PCMarginParams
typedef struct {
    word            PCMS_left
    word            PCMS_top
    word            PCMS_right
    word            PCMS_bottom
} PrintControlMarginStruct
```

Messages

```
MSG_SPOOL_PRINT_CONTROL_INITIATE_PRINT
MSG_SPOOL_PRINT_CONTROL_PRINT
MSG_SPOOL_PRINT_CONTROL_SET_ATTRS
MSG_SPOOL_PRINT_CONTROL_GET_ATTRS
MSG_SPOOL_PRINT_CONTROL_GET_PRINT_MODE
MSG_SPOOL_PRINT_CONTROL_SET_TOTAL_PAGE_RANGE
MSG_SPOOL_PRINT_CONTROL_GET_TOTAL_PAGE_RANGE
MSG_SPOOL_PRINT_CONTROL_SET_SELECTED_PAGE_RANGE
MSG_SPOOL_PRINT_CONTROL_GET_SELECTED_PAGE_RANGE
MSG_SPOOL_PRINT_CONTROL_SET_DOC_SIZE
MSG_SPOOL_PRINT_CONTROL_GET_DOC_SIZE
MSG_SPOOL_PRINT_CONTROL_SET_PRINT_GROUP
MSG_SPOOL_PRINT_CONTROL_GET_PRINT_GROUP
MSG_SPOOL_PRINT_CONTROL_SET_OUTPUT
MSG_SPOOL_PRINT_CONTROL_GET_OUTPUT
MSG_SPOOL_PRINT_CONTROL_SET_DEFAULT_PRINTER
MSG_SPOOL_PRINT_CONTROL_SPOOLING_UPDATE
MSG_SPOOL_PRINT_CONTROL_CANCEL_PRINT_JOB
MSG_SPOOL_PRINT_CONTROL_GET_PAPER_SIZE
void MSG_SPOOL_PRINT_CONTROL_SET_DOC_NAME(char *name)
void MSG_SPOOL_PRINT_CONTROL_SET_DOC_MARGINS(word *ms)
MSG_SPOOL_PRINT_CONTROL_GET_PRINTER_MARGINS
MSG_SPOOL_PRINT_CONTROL_GET_DOCUMENT_DIMMENSIONS
MSG_SPOOL_PRINT_CONTROL_VERIFY_DOC_MARGINS
MSG_SPOOL_PRINT_CONTROL_VERIFY_DOC_SIZE
```

SpreadsheetClass

```
@class SpreadsheetClass, VisCompClass, master
```

Instance Data

```
CellFunctionParameters    SSI_cellParams = {0, 0, {0}}
word                      SSI_chunk = 0
optr                      SSI_chartBody
word                      SSI_mapBlock = 0
word                      SSI_styleArray = 0
word                      SSI_rowArray = 0
word                      SSI_formatArray = 0
word                      SSI_nameArray = 0
word                      SSI_maxRow = MAX_ROW
word                      SSI_maxCol = MAX_COLUMN
PointDWord                SSI_offset = {0, 0}
CellRange                 SSI_visible = {{0, 0}, {0, 0}}
CellReference             SSI_active = {0, 0}
CellRange                 SSI_quickSource = {{0, 0}, {0, 0}}
CellRange                 SSI_selected = {{0, 0}, {0, 0}}
word                      SSI_curAttrs = 0
word                      SSI_gstate = NullHandle
byte                      SSI_gsRefCount = 0
SpreadsheetFlags          SSI_flags = 0
SpreadsheetDrawFlags      SSI_drawFlags = 0
byte                      SSI_attributes = 0
CellRange                 SSI_header = {{-1, -1}, {-1, -1}}
CellRange                 SSI_footer = {{-1, -1}, {-1, -1}}
word                      SSI_circCount = 0
FloatNum                  SSI_converge = 0
word                      SSI_ancestorList = 0
word                      SSI_childList = 0
word                      SSI_finalList = 0
optr                      SSI_ruler
RectDWord                 SSI_bounds = {0, 0, 0, 0}
```

Types and Flags

```
ByteFlags    NameFlags
      NF_UNDEFINED                    0x80
MAX_NAME_LENGTH                 128
MAX_NAME_DEF_LENGTH             256
ByteFlags    SpreadsheetDoubleClickFlags
      SDCF_NOTE_EXISTS                0x02
      SDCF_CELL_EXISTS                0x01
WordFlags    SpreadsheetFlags
      SF_MANUAL_RECALC                0x8000
      SF_ALLOW_ITERATION              0x4000
      SF_SUPPRESS_REDRAW              0x2000
      SF_APPLICATION_FUNCTIONS        0x1000
      SF_QUICK_TRANS_IN_PROGRESS      0x0800
      SF_DOING_FEEDBACK               0x0400
      SF_IN_VIEW                      0x0200
      SF_IS_SYS_TARGET                0x0008
      SF_HAVE_GRAB                    0x0004
      SF_IS_APP_TARGET                0x0002
      SF_IS_SYS_FOCUS                 0x0001
SRP_FLAGS    (SF_MANUAL_RECALC | SF_ALLOW_ITERATION)
```

```
ByteFlags    SpreadsheetClearFlags
      SCF_CLEAR_ATTRIBUTES            0x80
      SCF_CLEAR_DATA                  0x40
      SCF_CLEAR_NOTES                 0x20
ByteFlags    SpreadsheetInsertFlags
      SIF_COLUMNS                     0x80
      SIF_COMPLETE                    0x40
      SIF_DELETE                      0x20
WordFlags    SpreadsheetDrawFlags
      SDF_DRAW_GRAPHICS               0x0008
      SDF_DRAW_NOTE_BUTTON            0x0004
      SDF_DRAW_HEADER_FOOTER_BUTTON   0x0002
      SDF_DRAW_GRID                   0x0001
WordFlags    SpreadsheetPrintFlags
      SPF_PRINT_SIDEWAYS              0x2000
      SPF_SCALE_TO_FIT                0x1000
      SPF_PRINT_ROW_COLUMN_TITLES     0x0800
      SPF_SKIP_DRAW                   0x0400
      SPF_CENTER_VERTICALLY           0x0200
      SPF_CENTER_HORIZONTALLY         0x0100
      SPF_CONTINUOUS                  0x0080
      SPF_PRINT_HEADER                0x0040
      SPF_PRINT_FOOTER                0x0020
      SPF_PRINT_DOCUMENT              0x0010
      SPF_PRINT_NOTES                 0x0008
      SPF_PRINT_GRAPHICS              0x0004
      SPF_DONE                        0x0002
      SPF_PRINT_GRID                  0x0001
typedef enum /* word */ {
      SET_ENTIRE_SHEET = 0,
      SET_NO_EMPTY_CELLS = 2,
      SET_NO_EMPTY_CELLS_NO_HDR_FTR = 4,
      SET_NEXT_DATA_CELL = 6,
      SET_LAST_DATA_CELL = 8,
      SET_PREV_DATA_CELL = 10,
      SET_FIRST_DATA_CELL = 12
} SpreadsheetExtentType
ByteFlags    SpreadsheetSearchFlags
      SSF_MATCH_CASE                  0x80
      SSF_MATCH_PARTIAL_WORDS         0x40
      SSF_SEARCH_FORMULAS             0x20
      SSF_SEARCH_VALUES               0x10
      SSF_SEARCH_NOTES                0x08
      SSF_SEARCH_TEXT_OBJECTS         0x04
      SSF_SEARCH_BY_ROWS              0x02
SPREADSHEET_MAX_SEARCH_STRING_LENGTH  128
ByteEnum     SpreadsheetChartReturnType
      SCRT_TOO_MANY_CHARTS            0
      SCRT_INSUFFICIENT_MEMORY        2
      SCRT_NO_DATA                    4
typedef enum /* word */ {
      SPREADSHEET_ADDRESS_ON_SCREEN = 0xf000,
      SPREADSHEET_ADDRESS_IN_SELECTION = 0xf001,
      SPREADSHEET_ADDRESS_DATA_AREA = 0xf100,
      SPREADSHEET_ADDRESS_PAST_END = 0xf101,
      SPREADSHEET_ADDRESS_USE_SELECTION = 0xf200,
      SPREADSHEET_ADDRESS_NIL = 0xffff
```

```
} SpreadsheetAddress
typedef enum /* word */ {
      SSFT_NUMBER,
      SSFT_DAY,
      SSFT_WEEKDAY,
      SSFT_MONTH,
      SSFT_YEAR
} SpreadsheetSeriesFillType
ByteFlags    SpreadsheetSeriesFillFlags
      SSFF_ROWS                     0x02
      SSFF_GEOMETRIC                0x01
ByteEnum     SpreadsheetFillError
      SFE_NO_ERROR                  0
      SFE_NOT_DATE_NUMER            2
      SFE_DATE_STEP_TOO_LARGE       4
SPREADSHEET_MAX_DATE_FILL_STEP  90
SPREADSHEET_MIN_DATE_FILL_STEP  -90
ByteFlags    SpreadsheetAttributes
      SA_TARGETABLE                 0x80
      SA_ENGINE_MODE                0x40
MAX_ROW                       (8191+MIN_ROW)
MAX_COLUMN                    (255+MIN_ROW)
SSRCAF_TRANSFORM_VALID        0x80
SSRCAF_REF_COUNT              0x7f
MAX_GSTATE_REF_COUNT          (SSRCAF_REF_COUNT)
/* Border Info record */
ByteFlags    CellBorderInfo
      CBI_OUTLINE                   0x80
      CBI_LEFT                      0x08
      CBI_TOP                       0x04
      CBI_RIGHT                     0x02
      CBI_BOTTOM                    0x01
ByteFlags    CellInfo
      CI_LOCKED                     0x04
      CI_JUSTIFICATION              0x03
J_GENERAL                     J_FULL
ROW_HEIGHT_AUTOMATIC          0x8000
MIN_ROW                       0
COLUMN_WIDTH_MIN              0
COLUMN_WIDTH_MAX              512
ROW_HEIGHT_MIN                0
ROW_HEIGHT_MAX                (792*5/4)
ByteFlags    NameFlags
      NF_UNDEFINED                  0x80
MAX_NAMES                     255
MAX_NAME_BLOCK_SIZE (sizeof(NameHeader) +
                  (MAX_NAMES * (sizeof(NameStruct) +
                  MAX_NAME_LENGTH)))
ByteFlags    NameAccessFlags
      NAF_NAME                      0x80
      NAF_DEFINITION                0x40
      NAF_BY_TOKEN                  0x20
      NAF_TOKEN_DEFINITION          0x10
NAME_ROW                      (LARGEST_ROW - 4)
CHART_ROW                     (LARGEST_ROW - 3)
FORMATTED_RANGE_BUFFER_SIZE   14
PSEE_RESULT_SHOULD_BE_CELL_OR_RANGE (PSEE_SSHEET_BASE)
```

```
PSEE_NO_NAME_GIVEN              (PSEE_SSHEET_BASE + 1)
PSEE_NO_DEFINITION_GIVEN        (PSEE_SSHEET_BASE + 2)
PSEE_NAME_ALREADY_DEFINED       (PSEE_SSHEET_BASE + 3)
PSEE_BAD_NAME_DEFINITION        (PSEE_SSHEET_BASE + 4)
PSEE_REALLOC_FAILED             (PSEE_SSHEET_BASE + 5)
PSEE_LAST_SPREADSHEET_ERROR     230
PSEE_SSHEET_ERRORS              PSEE_PARSER_ERRORS,
            PSEE_RESULT_SHOULD_BE_CELL_OR_RANGE,
            PSEE_NO_NAME_GIVEN,
            PSEE_NO_DEFINITION_GIVEN,
            PSEE_NAME_ALREADY_DEFINED,
            PSEE_BAD_NAME_DEFINITION,
            PSEE_REALLOC_FAILED
NAME_LIST_INCREMENT             2048
CELL_GOTO_MAX_TEXT              15
CELL_MAX_TEXT                   255
CELL_TEXT_BUFFER_SIZE           (CELL_MAX_TEXT +
                                 size CellText)
CELL_MAX_FORMULA                256
CELL_FORMULA_BUFFER_SIZE        (CELL_MAX_FORMULA +
                                 size CellFormula)
SPREADSHEET_LIB_FUNCTIONID_ERRORS
            FUNCTION_ID_SPREADSHEET_CELL,
            FUNCTION_ID_LAST_SPREADSHEET_FUNCTION
```

Structures

```
typedef struct {
    optr           SSD_output
    Message        SSD_message
    optr           SSD_chartBody
} SpreadsheetSetupData
typedef struct {
    word           GNI_token
    word           GNI_entryNum
    word           GNI_numDefinedNames
    word           GNI_numUndefinedNames
} GetNameInfo
typedef struct {
    byte     SNP_flags
    word     SNP_listEntry
    word     SNP_textLength
    char     SNP_text[MAX_NAME_LENGTH]
    word     SNP_defLength
    byte     SNP_definition[MAX_NAME_DEF_LENGTH * 2]
    word     SNP_token
    byte     SNP_unused
} SpreadsheetNameParams
typedef struct {
    word           GNI_unused
    word           GNI_unused2
    word           GNI_numDefinedNames
    word           GNI_numUndefinedNames
} GetNumNamesInfo
typedef struct {
    word           GNI_textSize
    word           GNI_blockHandle
} GetNoteInfo
```

```
typedef struct {
    optr           GTO_textObject
} GetTextObject
typedef struct {
    SpreadsheetFlags      SRP_flags
    word                  SRP_circCount
    FloatNum              SRP_converge
} SpreadsheetRecalcParams
typedef struct {
    word           GFC_unused
    word           GFC_unused2
    word           GFC_numPreDefined
    word           GFC_numUserDefined
} GetFormatCount
typedef struct {
    SpreadsheetPrintFlags       SDP_flags
    word                        SDP_gstate
    CellReference               SDP_topLeft
    RectDWord                   SDP_drawArea
    CellRange                   SDP_limit
    Point                       SDP_margins
    PointDWord                  SDP_translation
    PointDWord                  SDP_titleTrans
    WWFixed                     SDP_scale
    CellRange                   SDP_range
    RectDWord                   SDP_rangeArea
} SpreadsheetDrawParams
typedef struct {
    word           GC_unused
    word           GC_unused2
    word           GC_cellRow
    word           GC_cellColumn
} GetActiveCell
typedef struct {
    SpreadsheetSearchFlags      SSP_flags
    char                        SSP_string[128]
    SpreadsheetSearchFlags      SSP_found
    Point                       SSP_cell
    word                        SSP_startPos
    word                        SSP_endPos
} SpreadsheetSearchParams
typedef struct {
    word           GC_unused
    word           GC_errorCode
    word           GC_row
    word           GC_column
} GetCell
typedef struct {
    word           SEFP_stacksSeg
    word           SEFP_opStackPtr
    word           SEFP_argStackPtr
    word           SEFP_funcID
    word           SEFP_nArgs
} SpreadsheetEvalFuncParams
typedef struct {
    CellRange      SRP_selection
    CellReference  SRP_active
```

```
} SpreadsheetRangeParams
typedef struct {
    SpreadsheetSeriesFillType   SRP_type
    SpreadsheetSeriesFillFlags  SRP_flags
    FloatNum                    SRP_stepValue
} SpreadsheetSeriesFillParams
typedef struct {
    char           SSFPRP_text[MAX_RANGE_REF_SIZE]
    CellRange      SSFPRP_range
} SpreadsheetFormatParseRangeParams
typedef struct {
    word                SIFD_file
    word                SIFD_numRows
    word                SIFD_numCols
    SpreadsheetDrawFlags  SIFD_drawFlags
} SpreadsheetInitFileData
typedef struct {
    ColorQuad      AI_color
    SystemDrawMask AI_grayScreen
} AreaInfo
typedef struct {
    RefElementHeader      CA_refCount
    AreaInfo              CA_textAttrs
    AreaInfo              CA_bkgrndAttrs
    FontID                CA_font
    word                  CA_pointsize
    byte                  CA_style
    CellBorderInfo        CA_border
    AreaInfo              CA_borderAttrs
    CellInfo              CA_info
    word                  CA_format
    BBFixed               CA_trackKern
    byte                  CA_fontWeight
    byte                  CA_fontWidth
    byte                  CA_reserved[9]
} CellAttrs
typedef struct {
    word           NH_blockSize
    word           NH_definedCount
    word           NH_undefinedCount
    word           NH_nextToken
} NameHeader
typedef struct {
    byte           NS_flags
    word           NS_token
    word           NS_length
} NameStruct
typedef struct {
    byte     SNP_flags
    word     SNP_listEntry
    word     SNP_textLength
    byte     SNP_text[MAX_NAME_LENGTH]
    word     SNP_defLength
    byte     SNP_definition[MAX_NAME_DEF_LENGTH*2]
    word     SNP_token
    byte     SNP_nameFlags
    byte     SNP_unused
```

```
} SpreadsheetNameParameters
typedef struct {
    FormatParameters      SFP_formatParams
    dword                 SFP_expression
    dword                 SFP_text
    word                  SFP_length
} SpreadsheetFormatParams
typedef struct {
    ParserParameters      SPP_parserParams
    dword                 SPP_text
    dword                 SPP_expression
    word                  SPP_exprLength
} SpreadsheetParserParams
typedef struct {
    EvalParameters        SEP_evalParams
    dword                 SEP_expression
    dword                 ArgumentStackElement
} SpreadsheetEvalParams
typedef struct {
    word           NLH_endOfData
    word           NLH_blockSize
} NameListHeader
typedef struct {
    word           NLE_token
    byte           NLE_flags
    byte           NLE_unused
    word           NLE_textLength
    word           NLE_defLength
} NameListEntry
```

Messages

```
void MSG_SPREADSHEET_READ_CACHED_DATA(
                   FileHandle fileHandle,
                   word mapBlockHandle)
void MSG_SPREADSHEET_WRITE_CACHED_DATA(
                   FileHandle fileHandle)
void MSG_SPREADSHEET_ATTACH_UI(
                   word handle, FileHandle fileHandle)
 void MSG_SPREADSHEET_ATTACH_FILE(
                   word handle, FileHandle fileHandle)
void MSG_SPREADSHEET_MOVE_ACTIVE_CELL(
                   word row, word column)
void MSG_SPREADSHEET_GOTO_CELL()
word MSG_SPREADSHEET_ENTER_DATA(
                   word textBlk, word textLen)
void MSG_SPREADSHEET_SET_ROW_HEIGHT(word rowHeight)
void MSG_SPREADSHEET_SET_COLUMN_WIDTH(word rowWidth)
word MSG_SPREADSHEET_GET_ROW_HEIGHT(word rowNum)
word MSG_SPREADSHEET_GET_COLUMN_WIDTH(word colNum)
word MSG_SPREADSHEET_MAKE_FOCUS()
MSG_SPREADSHEET_ADD_NAME_WITH_PARAM_BLK
void MSG_SPREADSHEET_ADD_NAME(
                   GetNameInfo *retValue,
                   SpreadsheetNameParams *ssNameParams)
void MSG_SPREADSHEET_VALIDATE_NAME(
                   GetNameInfo *retValue,
                   SpreadsheetNameParams *ssNameParams)
```

```
MSG_SPREADSHEET_DELETE_NAME_WITH_LIST_ENTRY
void MSG_SPREADSHEET_DELETE_NAME(
                    GetNameInfo *retValue,
                    SpreadsheetNameParams *ssNameParams)
MSG_SPREADSHEET_CHANGE_NAME_WITH_PARAM_BLK
void MSG_SPREADSHEET_CHANGE_NAME(
                    GetNameInfo *retValue,
                    SpreadsheetNameParams *ssNameParams)
void MSG_SPREADSHEET_GET_NAME_COUNT(
                    GetNumNamesInfo *retValue)
void MSG_SPREADSHEET_GET_NAME_INFO(
                    SpreadsheetNameParams *ssNameParams)
void MSG_SPREADSHEET_FORMAT_EXPRESSION(
             SpreadsheetFormatParams *ssFormatParams)
byte MSG_SPREADSHEET_PARSE_EXPRESSION(
             SpreadsheetParserParams *ssParserParams)
MSG_SPREADSHEET_EVAL_EXPRESSION
void MSG_SPREADSHEET_ERROR()
void MSG_SPREADSHEET_SET_NOTE(word *textBlockHandle,
                    word row, word column)
void MSG_SPREADSHEET_SET_NOTE_FOR_ACTIVE_CELL(
                    word *textBlockHandle)
void MSG_SPREADSHEET_GET_NOTE(GetNoteInfo *retValue,
                    word row, word column)
void MSG_SPREADSHEET_GET_NOTE_FOR_ACTIVE_CELL(
                    GetNoteInfo *retValue)
void MSG_SPREADSHEET_DISPLAY_NOTE(word data)
void MSG_SPREADSHEET_CHANGE_RECALC_PARAMS(
             SpreadsheetRecalcParams *ssRecalcParams)
void MSG_SPREADSHEET_GET_RECALC_PARAMS(
             SpreadsheetRecalcParams *ssRecalcParams)
void MSG_SPREADSHEET_RECALC()
void MSG_SPREADSHEET_CLEAR_SELECTED(
             SpreadsheetClearFlags ssClearFlags)
void MSG_SPREADSHEET_INSERT_SPACE(
             SpreadsheetInsertFlags ssInsertFlags)
void MSG_SPREADSHEET_SET_NUM_FORMAT(word formatToken)
void MSG_SPREADSHEET_DRAW_RANGE(
             SpreadsheetDrawParams *ssDrawParams)
void MSG_SPREADSHEET_GET_EXTENT(
                    CellRange *retValue,
                    SpreadsheetExtentType ssExtentType)
MSG_SPREADSHEET_GET_RANGE_BOUNDS
void MSG_SPREADSHEET_SET_HEADER_RANGE(word flag)
void MSG_SPREADSHEET_SET_FOOTER_RANGE(word flag)
void MSG_SPREADSHEET_GET_HEADER_RANGE(
                    CellRange *retValue)
void MSG_SPREADSHEET_GET_FOOTER_RANGE(
                    CellRange *retValue)
void MSG_SPREADSHEET_COMPLETE_REDRAW()
MSG_SPREADSHEET_NOTES_ENUM
void MSG_SPREADSHEET_ALTER_DRAW_FLAGS(
                    word bitsToSet, word bitsToClear)
SpreadsheetDrawFlags MSG_SPREADSHEET_GET_DRAW_FLAGS()
MSG_SPREADSHEET_HANDLE_SPECIAL_FUNCTION
void MSG_SPREADSHEET_CHANGE_COLUMN_WIDTH(
                    word change, word column)
```

```
void MSG_SPREADSHEET_CHANGE_ROW_HEIGHT(
                    word change, word row)
FileHandle MSG_SPREADSHEET_GET_FILE()
word MSG_SPREADSHEET_SORT_RANGE(
                    RangeSortFlags rangeSortFlags)
MSG_SPREADSHEET_FUNCTION_TO_TOKEN
MSG_SPREADSHEET_FUNCTION_TO_CELL
MSG_SPREADSHEET_FORMAT_FUNCTION
MSG_SPREADSHEET_EVAL_FUNCTION
void MSG_SPREADSHEET_CHART_RANGE(word chartNum)
void MSG_SPREADSHEET_CHART_RANGE(word chartNum)
void MSG_SPREADSHEET_DELETE_CHART(word chartNum)
void MSG_SPREADSHEET_START_MOVE_COPY()
void MSG_SPREADSHEET_END_MOVE_COPY()
void MSG_SSHEET_INIT_FOR_EXPORT()
void MSG_SPREADSHEET_SET_SELECTION(
            SpreadsheetRangeParams *ssRangeParams)
void MSG_SPREADSHEET_EXTEND_CONTRACT_SELECTION(
            SpreadsheetRangeParams *ssRangeParams)
void MSG_SPREADSHEET_GET_SELECTION(
            SpreadsheetRangeParams *ssRangeParams)
void MSG_SSHEET_PASTE_FROM_DATA_FILE()
void MSG_SSHEET_EXPORT_FROM_DATA_FILE()
MSG_SPREADSHEET_INIT_NAME_LIST
MSG_SPREADSHEET_INIT_CHOOSE_NAME_LIST
MSG_SPREADSHEET_NAME_REQUEST_ENTRY_MONIKER
MSG_SPREADSHEET_NAME_UPDATE_DEFINITION
MSG_SPREADSHEET_NAME_UPDATE_NAME
MSG_SPREADSHEET_GET_NAME_WITH_LIST_ENTRY
MSG_SPREADSHEET_REPLACE_TEXT_SELECTION
void MSG_SPREADSHEET_GET_ROW_AT_POSITION(sdword yPos)
void MSG_SPREADSHEET_GET_COLUMN_AT_POSITION(sdword xPos)
void MSG_SPREADSHEET_SET_CELL_BORDERS(
                    CellBorderInfo border)
void MSG_SPREADSHEET_SET_CELL_BORDER_COLOR(
                    ColorQuad color)
void MSG_SPREADSHEET_SET_CELL_BORDER_GRAY_SCREEN(
                    SystemDrawMask drawMask)
void MSG_SPREADSHEET_SET_CELL_BORDER_PATTERN(
                    GraphicPattern pattern)
SpreadsheetFillError MSG_SPREADSHEET_FILL_SERIES(
            SpreadsheetSeriesFillParams *fillParams)
void MSG_SPREADSHEET_FILL_RANGE(SeriesFillFlags flags)
void MSG_SPREADSHEET_PARSE_RANGE_REFERENCE(
            SpreadsheetFormatParseRangeParams *params)
void MSG_SPREADSHEET_FORMAT_RANGE_REFERENCE(
            SpreadsheetFormatParseRangeParams *params)
```

Routines

```
VMBlockHandle SpreadsheetInitFile(
                    SpreadsheetInitFileData *ifd)
void SpreadsheetParseNameToToken(
                    C_CallbackStruct *cb_s)
void SpreadsheetParseCreateCell(C_CallbackStruct *cb_s)
void SpreadsheetParseEmptyCell(C_CallbackStruct *cb_s)
void SpreadsheetParseDerefCell(C_CallbackStruct *cb_s)
word SpreadsheetNameTextFromToken(
```

```
                    SpreadsheetInstance *ssheet,
                    word nameToken,
                    char *destinationPtr,
                    word maxCharsToWrite)
Boolean SpreadsheetNameTokenFromText(
                    SpreadsheetInstance *ssheet,
                    char *nameText,
                    word nameLen,
                    word *tokenDest,
                    NameFlags *flagsDest)
word SpreadsheetNameLockDefinition(
                    SpreadsheetInstance *ssheet,
                    word nameToken,
                    void **defaddr)
void SpreadsheetCellAddRemoveDeps(
            SpreadsheetInstance *spreadsheetInstance,
            dword cellParams,
            PCB(void, callback,(C_CallbackStruct *)),
            word addOrRemoveDeps,
            word eval_flags,
            word row, word column, word maxRow,
            word maxColumn)
void SpreadsheetRecalcDependents(
            SpreadsheetInstance *spreadsheetInstance,
            PCB(void, callback,(optr oself,
                        word row, word column)),
            word row, word column)
```

SpreadsheetRulerClass

```
@class SpreadsheetRulerClass,VisRulerClass
```

Instance Data

```
SpreadsheetRulerFlags     SRI_flags
optr                      SRI_spreadsheet
word                      SRI_resizeRC
dword                     SRI_startRCPos
```

Types and Flags

```
SPREADSHEET_RULER_WIDTH   40
SPREADSHEET_RULER_HEIGHT  12
ByteFlags    SpreadsheetRulerFlags
      SRF_SSHEET_IS_FOCUS             0x8
      SRF_NO_INTERACTIVE_RESIZE       0x4
      SRF_SSHEET_IS_TARGET            0x2
      SRF_HAVE_GRAB                   0x1
```

Messages

```
void MSG_SPREADSHEET_RULER_DRAW_RANGE(
                    SpreadsheetDrawParams *ssDrawParams)
void MSG_SPREADSHEET_RULER_SET_FLAGS(
                    SpreadsheetRulerFlags setFlags,
                    SpreadsheetRulerFlags clearFlags)
```

StyleSheetControlClass

```
@class StyleSheetControlClass, GenControlClass
```

Instance Data

```
ClassStruct  * SSCI_targetClass = NullClass
ClassStruct  * SSCI_styledClass = NullClass
    @default GCI_output = (TO_APP_TARGET)
```

Variable Data

```
GenFilePath ATTR_STYLE_SHEET_LOAD_STYLE_SHEET_PATH
GeodeToken ATTR_STYLE_SHEET_LOAD_STYLE_SHEET_TOKEN
optr TEMP_STYLE_SHEET_MANAGE_UI
optr TEMP_STYLE_SHEET_DEFINE_UI
SSCTempAttrInfo TEMP_SYTLE_SHEET_ATTR_TOKENS
MemHandle TEMP_SYTLE_SHEET_SAVED_STYLE
```

Types and Flags

```
WordFlags    SSCFeatures
      SSCF_DEFINE               0x0080
      SSCF_REDEFINE             0x0040
      SSCF_RETURN_TO_BASE       0x0020
      SSCF_APPLY                0x0010
      SSCF_MANAGE               0x0008
      SSCF_LOAD                 0x0004
      SSCF_SAVE_STYLE           0x0002
      SSCF_RECALL_STYLE         0x0001
WordFlags    SSCToolboxFeatures
      SSCTF_REDEFINE            0x0010
      SSCTF_RETURN_TO_BASE      0x0008
      SSCTF_STYLE_LIST          0x0004
      SSCTF_SAVE_STYLE          0x0002
      SSCTF_RECALL_STYLE        0x0001
SSC_DEFAULT_FEATURES      (SSCF_DEFINE | SSCF_REDEFINE
            | SSCF_RETURN_TO_BASE | SSCF_APPLY |
            SSCF_MANAGE | SSCF_LOAD | SSCF_SAVE_STYLE |
            SSCF_RECALL_STYLE)
SSC_DEFAULT_TOOLBOX_FEATURES (SSCTF_STYLE_LIST |
            SSCTF_SAVE_STYLE | SSCTF_RECALL_STYLE)
```

Structures

```
typedef struct {
    NameArrayMaxElement   NSC_style
    word     NSC_styleToken
    word     NSC_usedIndex
    word     NSC_usedToolIndex
    word     NSC_styleSize
    word     NSC_attrTokens[MAX_STYLE_SHEET_ATTRS]
    dword    NSC_attrChecksums[MAX_STYLE_SHEET_ATTRS]
    byte     NSC_indeterminate
    byte     NSC_differsFromBase
    byte     NSC_canReturnToBase
    word     NSC_styleCounter
} NotifyStyleChange
typedef struct {
    StyleChunkDesc NSSHC_styleArray
    word           NSSHC_counter
    word           NSSHC_styleCount
    word           NSSHC_toolStyleCount
```

```
} NotifyStyleSheetChange
typedef struct {
    word     SSCTAI_attrTokens[MAX_STYLE_SHEET_ATTRS]
    word     SSCTAI_baseStyle
    byte     SSCTAI_differsFromBase
    byte     SSCTAI_indeterminate
} SSCTempAttrInfo
```

Messages

```
void MSG_STYLE_SHEET_GET_MODIFY_UI()
void MSG_STYLE_SHEET_GET_DEFINE_UI()
void MSG_STYLE_SHEET_SET_SAVED_STYLE()
void MSG_SSC_SELECT_STYLE()
void MSG_SSC_STATUS_STYLE()
void MSG_SSC_QUERY_STYLE()
void MSG_SSC_QUERY_BASE_STYLE()
void MSG_SSC_APPLY_STYLE()
void MSG_SSC_APPLY_BOX_STYLE()
void MSG_SSC_APPLY_TOOLBOX_STYLE()
void MSG_SSC_INITIATE_MODIFY_STYLE()
void MSG_SSC_MODIFY_STYLE()
void MSG_SSC_DELETE_STYLE()
void MSG_SSC_DELETE_REVERT_STYLE()
void MSG_SSC_DEFINE_STYLE()
void MSG_SSC_REDEFINE_STYLE()
void MSG_SSC_RETURN_TO_BASE_STYLE()
void MSG_SSC_LOAD_STYLE_SHEET()
void MSG_SSC_LOAD_STYLE_SHEET_FILE_SELECTED()
void MSG_SSC_SAVE_STYLE()
void MSG_SSC_RECALL_STYLE()
```

TextGuardianClass

```
@class TextGuardianClass, GrObjVisGuardianClass
```

Instance Data

```
TextGuardianFlags  TGI_flags
word               TGI_desiredMinHeight
word               TGI_desiredMaxHeight
```

Types and Flags

```
ByteFlags    TextGuardianFlags
       TGF_ENFORCE_DESIRED_MIN_HEIGHT        0x10
       TGF_ENFORCE_DESIRED_MAX_HEIGHT        0x08
  TGF_DISABLE_ENFORCED_DESIRED_MAX_HEIGHT_WHILE_EDITING
                                             0x04
       TGF_ENFORCE_MIN_DISPLAY_SIZE          0x02
       TGF_SHRINK_WIDTH_TO_MIN_AFTER_EDIT    0x01
```

Messages

```
MSG_TG_CALC_DESIRED_MIN_HEIGHT
MSG_TG_SET_DESIRED_MIN_HEIGHT
MSG_TG_HEIGHT_NOTIFY
MSG_TG_GENERATE_TEXT_NOTIFY
MSG_TG_CALC_DESIRED_MAX_HEIGHT
MSG_TG_SET_DESIRED_MAX_HEIGHT
void MSG_TG_SET_TEXT_GUARDIAN_FLAGS(
                   TextGuardianFlags flagsToSet,
```

```
        TextGuardianFlags flagsToReset)
```

ThesControlClass

```
@class ThesControlClass, GenControlClass
```


Instance Data

```
void         *TCI_meanings
void         *TCI_synonyms
void         *TCI_grammars
void         *TCI_backups
word         TCI_status
char         TCI_lastWord[MAX_ENTRY_LENGTH]
    @default GCI_output = (TO_APP_TARGET)
    @default GII_visibility = GIV_DIALOG
    @default GI_states = (@default | GS_ENABLED)
```

Variable Data

```
void
    ATTR_THES_CONTROL_INTERACT_ONLY_WITH_TARGETED_TEXT_
OBJECTS
```

Types and Flags

```
MAX_BACKUP_LIST_SIZE            30
MAX_ENTRY_LENGTH                26
MAX_GRAMMAR_STRING              7
MAX_DEFINITIONS                 30
MAX_DEFINITION_SIZE             200
MAX_MEANINGS_ARRAY_SIZE         3000
MAX_SYNONYM_SIZE                26
MAX_SYNONYMS                    80
MAX_SYNONYMS_ARRAY_SIZE         500
WordFlags    ThesDictFeatures
      TDF_THESDICT                      0x01
      TC_GCM_FEATURES                   TDF_THESDICT
WordFlags    ThesDictToolboxFeatures
      TDTF_THESDICT                     0x01
TD_GCM_TOOLBOX_FEATURES         (TDTF_THESDICT)
```

Structures

```
typedef struct {
    word           RSWP_numChars
    MemHandle      RSWP_string
} ReplaceSelectedWordParameters
typedef struct {
    word           SWP_type
    word           SWP_numChars
    word           SWP_message
    optr           SWP_output
} SelectWordParameters
```

Messages

```
void MSG_THES_REPLACE_SELECTED_WORDS (@stack
                    MemHandle RSWP_string,
                    word RSWP_numChars)
void MSG_THES_SELECT_WORD (@stack
                    optr output, Message message,
                    word numChars, word type)
```

VisClass

```
@class VisClass, MetaClass, master
```

Instance Data

```
Rectangle    VI_bounds = {0, 0, 0, 0}
VisTypeFlags VI_typeFlags = 0
VisAttrs     VI_attrs = (VA_MANAGED|VA_DRAWABLE|
                        VA_DETECTABLE|VA_FULLY_ENABLED)
VisOptFlags  VI_optFlags = (VOF_GEOMETRY_INVALID|
             VOF_GEO_UPDATE_PATH| VOF_WINDOW_INVALID|
             VOF_WINDOW_UPDATE_PATH| VOF_IMAGE_INVALID|
             VOF_IMAGE_UPDATE_PATH)
VisGeoAttrs  VI_geoAttrs = 0
SpecAttrs    VI_specAttrs = 0
@link        VI_link
```

Variable Data

```
Rectangle TEMP_VIS_OLD_BOUNDS
VarGeoData ATTR_VIS_GEOMETRY_DATA
word TEMP_VIS_INVAL_REGION
```

Types and Flags

```
ByteFlags    DrawFlags
      DF_EXPOSED                      0x80
      DF_OBJECT_SPECIFIC              0x40
      DF_PRINT                        0x20
      DF_DONT_DRAW_CHILDREN           0x10
      DF_DISPLAY_TYPE                 0x0f
ByteFlags    ColorScheme
      CS_lightColor                   0xf0
      CS_darkColor                    0x0f
CS_lightColor_OFFSET      4
typedef enum {
    TO_VIS_PARENT = _FIRST_VisClass
} VisTravelOption
ByteEnum     VisUpdateMode
      VUM_MANUAL               0
      VUM_NOW                  1
      VUM_DELAYED_VIA_UI_QUEUE 2
      VUM_DELAYED_VIA_APP_QUEUE 3
ByteFlags    VisAttrs
      VA_VISIBLE                      0x80
      VA_FULLY_ENABLED                0x40
      VA_MANAGED                      0x20
      VA_DRAWABLE                     0x10
      VA_DETECTABLE                   0x08
      VA_BRANCH_NOT_MINIMIZABLE       0x04
      VA_OLD_BOUNDS_SAVED             0x02
      VA_REALIZED                     0x01
ByteFlags    VisOptFlags
      VOF_GEOMETRY_INVALID            0x80
      VOF_GEO_UPDATE_PATH             0x40
      VOF_IMAGE_INVALID               0x20
      VOF_IMAGE_UPDATE_PATH           0x10
      VOF_WINDOW_INVALID              0x08
      VOF_WINDOW_UPDATE_PATH          0x04
      VOF_UPDATE_PENDING              0x02
```

```
      VOF_EC_UPDATING                 0x01
ByteFlags    VisGeoAttrs
      VGA_GEOMETRY_CALCULATED         0x80
      VGA_NO_SIZE_HINTS               0x40
      VGA_NOTIFY_GEOMETRY_VALID       0x20
      VGA_DONT_CENTER                 0x10
      VGA_USE_VIS_SET_POSITION        0x08
      VGA_USE_VIS_CENTER              0x04
      VGA_ONLY_RECALC_SIZE_WHEN_INVALID 0x02
      VGA_ALWAYS_RECALC_SIZE          0x01
ByteFlags    VisTypeFlags
      VTF_IS_COMPOSITE                0x80
      VTF_IS_WINDOW                   0x40
      VTF_IS_PORTAL                   0x20
      VTF_IS_WIN_GROUP                0x10
      VTF_IS_CONTENT                  0x08
      VTF_IS_INPUT_NODE               0x04
      VTF_IS_GEN                      0x02
      VTF_CHILDREN_OUTSIDE_PORTAL_WIN 0x01
ByteFlags    SpecAttrs
      SA_ATTACHED                     0x80
      SA_REALIZABLE                   0x40
      SA_BRANCH_MINIMIZED             0x20
      SA_USES_DUAL_BUILD              0x10
      SA_CUSTOM_VIS_PARENT            0x08
      SA_SIMPLE_GEN_OBJ               0x04
      SA_CUSTOM_VIS_PARENT_FOR_CHILD  0x02
      SA_TREE_BUILT_BUT_NOT_REALIZED  0x01
ByteFlags    VisUpdateImageFlags
      VUIF_ALREADY_INVALID            0x80
      VUIF_ALWAYS_INVALIDATE          0x40
ByteFlags    VisAddRectFlags
      VARF_NOT_IF_ALREADY_INVALID     0x80
      VARF_ONLY_REDRAW_MARGINS        0x40
ByteFlags    VisInputFlowGrabFlags
      VIFGF_NOT_HERE                  0x80
      VIFGF_FORCE                     0x20
      VIFGF_GRAB                      0x10
      VIFGF_KBD                       0x08
      VIFGF_MOUSE                     0x04
      VIFGF_LARGE                     0x02
      VIFGF_PTR                       0x01
ByteEnum     VisInputFlowGrabType
      VIFGT_ACTIVE              0
      VIFGT_PRE_PASSIVE         1
      VIGFT_POST_PASSIVE        2
ByteFlags    DrawMonikerFlags
      DMF_UNDERLINE_ACCELERATOR       0x40
      DMF_CLIP_TO_MAX_WIDTH           0x20
      DMF_NONE                        0x10
      DMF_Y_JUST_MASK                 0x0c
      DMF_X_JUST_MASK                 0x03
DMF_Y_JUST_OFFSET  2
DMF_X_JUST_OFFSET  0
WordFlags    VisMonikerSearchFlags
      VMSF_STYLE                      0xf000
      VMSF_COPY_CHUNK                 0x0400
```

```
      VMSF_REPLACE_LIST               0x0200
      VMSF_GSTRING                    0x0100
VMSF_STYLE_OFFSET         12
WordFlags    SpecSizeSpec
      SSS_TYPE                        0x8c00
      SSS_DATA                        0x03ff
WordFlags    SpecWidth
      SW_TYPE                         0x8c00
      SW_DATA                         0x03ff
WordFlags    SpecHeight
      SH_TYPE                         0x8c00
      SH_DATA                         0x03ff
SSS_TYPE_OFFSET    10
SSS_DATA_OFFSET    0
ByteEnum     SpecSizeType
      SST_PIXELS                      0x0000
      SST_COUNT                       0x0400
      SST_PCT_OF_FIELD_WIDTH          0x0800
      SST_PCT_OF_FIELD_HEIGHT         0x0c00
      SST_AVG_CHAR_WIDTHS             0x1000
      SST_WIDE_CHAR_WIDTHS            0x1400
      SST_LINES_OF_TEXT               0x1800
PCT_0        0x000
PCT_5        0x033
PCT_10       0x066
PCT_15       0x099
PCT_20       0x0cc
PCT_25       0x100
PCT_30       0x133
PCT_35       0x166
PCT_40       0x199
PCT_45       0x1cc
PCT_50       0x200
PCT_55       0x233
PCT_60       0x266
PCT_65       0x299
PCT_70       0x2cc
PCT_75       0x300
PCT_80       0x333
PCT_85       0x366
PCT_90       0x399
PCT_95       0x3cc
PCT_100      0x3ff
WordFlags    SpecWinSizeSpec
      SWSS_RATIO                      0x8000
      SWSS_SIGN                       0x4000
      SWSS_MANTISSA                   0x3c00
      SWSS_FRACTION                   0x03ff
WordFlags    RecalcSizeArgs
      RSA_CHOOSE_OWN_SIZE             0x8000
      RSA_SUGGESTED_SIZE              0x7fff
ByteEnum     WinPositionType
      WPT_AT_RATIO                    0
      WPT_STAGGER                     1
      WPT_CENTER                      2
      WPT_TILED                       3
      WPT_AT_MOUSE_POSITION           4
```

```
      WPT_AS_REQUIRED                 5
ByteEnum     WinSizeType
      WST_AS_RATIO_OF_PARENT          0
      WST_AS_RATIO_OF_FIELD           1
      WST_AS_DESIRED                  2
      WST_EXTEND_TO_BOTTOM_RIGHT      3
      WST_EXTEND_NEAR_BOTTOM_RIGHT    4
ByteEnum     WinConstrainType
      WCT_NONE                        0
      WCT_KEEP_PARTIALLY_VISIBLE      1
      WCT_KEEP_VISIBLE                2
      WCT_KEEP_VISIBLE_WITH_MARGIN    3
WordFlags    WinPosSizeFlags
      WPSF_PERSIST                    0x8000
      WPSF_HINT_FOR_ICON              0x4000
      WPSF_NEVER_SAVE_STATE           0x2000
      WPSF_SHRINK_DESIRED_SIZE_TO_FIT_IN_PARENT 0x1000
      WPSF_CONSTRAIN_TYPE             0x00c0
      WPSF_POSITION_TYPE              0x0038
      WPSF_SIZE_TYPE                  0x0007
dword SizeAsDWord
ByteEnum     VMStyle
      VMS_TEXT                        0
      VMS_ABBREV_TEXT                 1
      VMS_GRAPHIC_TEXT                2
      VMS_ICON                        3
      VMS_TOOL                        4
WordFlags    VisMonikerListEntryType
      VMLET_GS_SIZE                   0x0300
      VMLET_STYLE                     0x0f00
      VMLET_MONIKER_LIST              0x0080
      VMLET_GSTRING                   0x0040
      VMLET_GS_ASPECT_RATIO           0x0030
      VMLET_GS_COLOR                  0x000f
VMT_GS_SIZE_OFFSET              12
VMLET_STYLE_OFFSET               8
VMT_GS_ASPECT_RATIO_OFFSET       4
VMT_GS_COLOR_OFFSET              0
ByteFlags    VisMonikerType
      VMT_MONIKER_LIST                0x80
      VMT_GSTRING                     0x40
      VMT_GS_ASPECT_RATIO             0x30
      VMT_GS_COLOR                    0x0f
WordFlags VisMonikerCachedWidth
      VMCW_HINTED                     0x8000
      VMCW_BERKELEY_9                 0x7f00
      VMCW_BERKELEY_10                0x00ff
VMCW_BERKELEY_9_OFFSET          8
VMCW_BERKELEY_10_OFFSET         0
VMO_CANCEL                      0xfd
VMO_MNEMONIC_NOT_IN_MKR_TEXT    0xfe
VMO_NO_MNEMONIC                 0xff
ByteFlags    CreateVisMonikerFlags
      CVMF_DIRTY                      0x80
ByteEnum     VisMonikerSourceType
      VMST_FPTR               0
      VMST_OPTR               1
```

```
        VMST_HPTR                   2
    ByteEnum    VisMonikerDataType
        VMDT_NULL                   0
        VMDT_VIS_MONIKER            1
        VMDT_TEXT                   2
        VMDT_GSTRING                3
        VMDT_TOKEN                  4
    WordFlags   SpecBuildFlags
        SBF_IN_UPDATE_WIN_GROUP                 0x8000
        SBF_WIN_GROUP                           0x4000
        SBF_TREE_BUILD                          0x2000
        SBF_VIS_PARENT_WITHIN_SCOPE_OF_TREE_BUILD 0x1000
        SBF_SKIP_CHILD                          0x0800
        SBF_FIND_LAST                           0x0400
        SBF_VIS_PARENT_UNBUILDING               0x0200
        SBF_VIS_PARENT_FULLY_ENABLED            0x0100
        SBF_UPDATE_MODE                         0x0003
    WordFlags   NavigationFlags
        NF_COMPLETED_CIRCUIT            0x8000
        NF_REACHED_ROOT                 0x4000
        NF_NAV_MENU_BAR                 0x0020
        NF_INITIATE_QUERY               0x0008
        NF_SKIP_NODE                    0x0004
        NF_TRAVEL_CIRCUIT               0x0002
        NF_BACKTRACK_AFTER_TRAVELING    0x0001
    ByteFlags   NavigateCommonFlags
        NCF_IS_COMPOSITE                0x80
        NCF_IS_FOCUSABLE                0x40
        NCF_IS_MENU_RELATED             0x20
        NCF_IS_INPUT_NODE               0x10
    ByteFlags   MenuSepFlags
        MSF_SEP                         0x80
        MSF_USABLE                      0x40
        MSF_FROM_CHILD                  0x20
    WordFlags   GenBranchInfo
        GBI_USABLE                      0x8000
        GBI_BRANCH_MINIMIZED            0x4000
```

Structures

```
typedef struct {
    byte            DS_colorScheme
    byte            DS_displayType
    word            DS_unused
    FontID          DS_fontID
    sword           DS_pointSize
} DisplayScheme
typedef struct {
    word            GCP_aboveCenter
    word            GCP_belowCenter
    word            GCP_leftOfCenter
    word            GCP_rightOfCenter
} GetCenterParams
typedef struct {
    word            OAGP_grabFlags
    word            OAGP_unused
    optr            OAGP_object
} ObjectAndGrabParams
```

```
typedef struct {
    word            VMLE_type
    optr            VMLE_moniker
} VisMonikerListEntry
typedef struct {
    byte            VM_type
    word            VM_width
} VisMoniker
typedef struct {
    VisMoniker      VMWGS_common
    word            VMWGS_height
} VisMonikerWithGString
VMWGS_gString       (sizeof(VisMonikerWithGString))
typedef struct {
    VisMoniker      VMWT_common
    char            VMWT_mnemonicOffset
} VisMonikerWithText
VMWT_text           (sizeof(VisMonikerWithText))
typedef struct {
    word            ESP_extraWidth
    word            ESP_extraHeight
    word            ESP_leftoverChildren
    word            ESP_unused
} ExtraSizeParams
typedef struct {
    word            WSIP_windowWidth
    word            WSIP_windowHeight
    byte            WSIP_bottomArea
    byte            WSIP_rightArea
    word            WSIP_unused
} WinSizeInfoParams
typedef struct {
    optr                    NCP_object
    NavigationFlags         NCP_navFlags
    NavigateCommonFlags     NCP_navCommonFlags
    ChunkHandle             NCP_genericData
} NavigateCommonParams
typedef struct {
    word            VCCIBF_data1
    word            VCCIBF_data2
    word            VCCIBF_data3
    word            VCCIBF_data4
    word            VCCIBF_data5
    Rectangle       VCCIBF_bounds
} VisCallChildrenInBoundsFrame
typedef struct {
    SpecWinSizeSpec SWSP_x
    SpecWinSizeSpec SWSP_y
} SpecWinSizePair
typedef struct {
    word            VGD_lineWidth
    word            VGD_centerOffset
    word            VGD_secondWidth
} VarGeoData
```



Macros

```
visParent @parent word_offsetof(VisBase, Vis_offset),
                  word_offsetof(VisInstance, VI_link)
visChildren @children
                  word_offsetof(VisBase, Vis_offset),
                  word_offsetof(
                        VisCompInstance, VCI_comp),
                  word_offsetof(VisInstance, VI_link)
DWORD_WIDTH(val) ((word) (val))
DWORD_HEIGHT(val) ((word) (val >> 16))
MAKE_SIZE_DWORD(width,height)
      ((((dword) (height)) << 16) | (word) (width))
```

Messages

```
@exportMessages VisSpecMessages,
                           DEFAULT_EXPORTED_MESSAGES
@exportMessages VisAppMessages,
                           DEFAULT_EXPORTED_MESSAGES
void MSG_VIS_DRAW(DrawFlags drawFlags,
                           GStateHandle gstate)
void MSG_VIS_REDRAW_ENTIRE_OBJECT()
GStateHandle MSG_VIS_VUP_CREATE_GSTATE()
void MSG_VIS_VUP_QUERY()
optr MSG_VIS_VUP_FIND_OBJECT_OF_CLASS(
                  ClassStruct *class)
void MSG_VIS_VUP_CALL_OBJECT_OF_CLASS(EventHandle event)
void MSG_VIS_VUP_SEND_TO_OBJECT_OF_CLASS(
                  EventHandle event)
Boolean MSG_VIS_VUP_TEST_FOR_OBJECT_OF_CLASS(
                  ClassStruct *class)
void MSG_VIS_VUP_CALL_WIN_GROUP(EventHandle event)
void MSG_VIS_VUP_SEND_TO_WIN_GROUP(EventHandle event)
void MSG_VIS_SET_ATTRS(VisAttrs attrsToSet,
                  VisAttrs attrsToClear,
                  VisUpdateMode updateMode)
void MSG_VIS_SET_GEO_ATTRS(VisGeoAttrs attrsToSet,
                  VisGeoAttrs attrsToClear,
                  VisUpdateMode updateMode)
byte MSG_VIS_GET_GEO_ATTRS()
VisOptFlags MSG_VIS_GET_OPT_FLAGS()
void MSG_VIS_SET_TYPE_FLAGS(
                  byte flagsToSet, byte flagsToClear)
VisTypeFlags MSG_VIS_GET_TYPE_FLAGS()
VisAttrs MSG_VIS_GET_ATTRS()
XYValueAsDWord MSG_VIS_GET_POSITION()
void MSG_VIS_SET_POSITION(word xOrigin, word yOrigin)
void MSG_VIS_GET_BOUNDS(Rectangle *retValue)
SizeAsDWord MSG_VIS_GET_SIZE()
void MSG_VIS_SET_SIZE(word width, word height)
void MSG_VIS_GET_CENTER(GetCenterParams *retValue)
SizeAsDWord MSG_VIS_RECALC_SIZE(word width, word height)
void MSG_VIS_POSITION_BRANCH(word xOrigin, word yOrigin)
void MSG_VIS_NOTIFY_GEOMETRY_VALID()
void MSG_VIS_BOUNDS_CHANGED(@stack
                  word bottom, word right,
                  word top, word left)
void MSG_VIS_RESET_TO_INITIAL_SIZE(
```

```
                    VisUpdateMode updateMode)
SizeAsDWord MSG_VIS_RECALC_SIZE_AND_INVAL_IF_NEEDED(
                    word width, word height)
Boolean MSG_VIS_POSITION_AND_INVAL_IF_NEEDED(
                    word xPosition, word yPosition)
void MSG_VIS_MARK_INVALID(
                    VisOptFlags flagsToSet,
                    VisUpdateMode updateMode)
Boolean MSG_VIS_VUP_UPDATE_WIN_GROUP(
                    VisUpdateMode updateMode)
void MSG_VIS_UPDATE_WIN_GROUP(VisUpdateMode updateMode)
void MSG_VIS_UPDATE_GEOMETRY()
void MSG_VIS_UPDATE_WINDOWS_AND_IMAGE(
              VisUpdateImageFlags updateImageFlags)
void MSG_VIS_INVALIDATE()
void MSG_VIS_ADD_RECT_TO_UPDATE_REGION(@stack
                    byte unused,
                    VisAddRectFlags addRectFlags,
                    word bottom, word right,
                    word top, word left)
void MSG_VIS_INVAL_TREE()
void MSG_VIS_OPEN(WindowHandle window)
void MSG_VIS_CLOSE()
void MSG_VIS_DESTROY(VisUpdateMode updateMode)
void MSG_VIS_REMOVE(VisUpdateMode updateMode)
WindowHandle MSG_VIS_QUERY_WINDOW()
void MSG_VIS_OPEN_WIN(WindowHandle parentWindow)
void MSG_VIS_CLOSE_WIN()
void MSG_VIS_WIN_ABOUT_TO_BE_CLOSED()
void MSG_VIS_MOVE_RESIZE_WIN()
void MSG_VIS_ADD_CHILD(optr child, CompChildFlags flags)
void MSG_VIS_REMOVE_CHILD(optr child,
                          CompChildFlags flags
void MSG_VIS_MOVE_CHILD(optr child,
                          CompChildFlags flags)
word MSG_VIS_FIND_CHILD(optr object)
optr MSG_VIS_FIND_CHILD_AT_POSITION(word position)
word MSG_VIS_COUNT_CHILDREN()
optr MSG_VIS_FIND_PARENT()
void MSG_VIS_CALL_PARENT(EventHandle event)
void MSG_VIS_SEND_TO_PARENT(EventHandle event)
void MSG_VIS_SEND_TO_CHILDREN(EventHandle event)
void MSG_VIS_GRAB_MOUSE()
void MSG_VIS_FORCE_GRAB_MOUSE()
void MSG_VIS_GRAB_LARGE_MOUSE()
void MSG_VIS_FORCE_GRAB_LARGE_MOUSE()
void MSG_VIS_RELEASE_MOUSE()
void MSG_VIS_ADD_BUTTON_PRE_PASSIVE()
void MSG_VIS_REMOVE_BUTTON_PRE_PASSIVE()
void MSG_VIS_ADD_BUTTON_POST_PASSIVE()
void MSG_VIS_REMOVE_BUTTON_POST_PASSIVE()
void MSG_VIS_TAKE_GADGET_EXCL(optr child)
void MSG_VIS_RELEASE_GADGET_EXCL(optr child)
void MSG_VIS_LOST_GADGET_EXCL()
void MSG_VIS_VUP_QUERY_FOCUS_EXCL(
                    ObjectAndGrabParams *retValue)
void MSG_VIS_FUP_QUERY_FOCUS_EXCL(
```



```
                    ObjectAndGrabParams *retValue)
void MSG_VIS_VUP_ALTER_INPUT_FLOW(@stack
                    PointDWord translation,
                    WindowHandle window, optr object,
                    word grabTypeAndFlags)
void MSG_VIS_VUP_SET_MOUSE_INTERACTION_BOUNDS(@stack
                    word bottom, word right,
                    word top, word left)
word MSG_VIS_VUP_GET_MOUSE_STATUS()
void MSG_VIS_VUP_TERMINATE_ACTIVE_MOUSE_FUNCTION()
void MSG_VIS_VUP_BUMP_MOUSE(word xBump, word yBump)
void MSG_VIS_VUP_ALLOW_GLOBAL_TRANSFER()
void MSG_VIS_DRAW_MONIKER(@stack
                    DrawMonikerFlags monikerFlags,
                    ChunkHandle visMoniker,
                    word textHeight,
                    GStateHandle gstate,
                    word yMaximum, word xMaximum,
                    word yInset, word xInset)
XYValueAsDWord MSG_VIS_GET_MONIKER_POS(@stack
                    DrawMonikerFlags monikerFlags,
                    ChunkHandle visMoniker,
                    word textHeight,
                    GStateHandle gstate,
                    word yMaximum, word xMaximum,
                    word yInset, word xInset)
SizeAsDWord MSG_VIS_GET_MONIKER_SIZE(@stack
                    byte monikerFlags,
                    ChunkHandle visMoniker,
                    word textHeight,
                    GStateHandle gstate,
                    word yMaximum, word xMaximum,
                    word yInset, word xInset)
optr MSG_VIS_FIND_MONIKER(@stack
                    VisMonikerSearchFlags searchFlags,
                    Handle destBlock,
                    ChunkHandle monikerList,
                    DisplayType displayType)
ChunkHandle MSG_VIS_CREATE_VIS_MONIKER(@stack
                    CreateVisMonikerFlags flags,
                    word height, word width,
                    word length,
                    VisMonikerDataType dataType,
                    VisMonikerSourceType sourceType,
                    dword source)
void MSG_VIS_VUP_EC_ENSURE_WINDOW_NOT_REFERENCED(
                    WindowHandle window)
void MSG_VIS_VUP_EC_ENSURE_OBJ_BLOCK_NOT_REFERENCED(
                    MemHandle objBlock)
void MSG_VIS_VUP_EC_ENSURE_OD_NOT_REFERENCED(
                    optr object)
void MSG_VIS_LAYER_SET_DOC_BOUNDS(@stack
                    sdword bottom, sdword right,
                    sdword top, sdword left)
void MSG_VIS_LAYER_GET_DOC_BOUNDS(RectDWord *bounds)
void MSG_VIS_QUERY_IF_OBJECT_HANDLES_INK(
                    VisCallChildrenInBoundsFrame *data)
```

```
void MSG_VIS_RECREATE_CACHED_GSTATES()
void MSG_VIS_LAYER_INSERT_OR_DELETE_SPACE()
void MSG_VIS_CREATE_CACHED_GSTATES()
void MSG_VIS_DESTROY_CACHED_GSTATES()
void MSG_VIS_ADD_NON_DISCARDABLE_VM_CHILD(
                  optr child, CompChildFlags flags)
void MSG_VIS_REMOVE_NON_DISCARDABLE_VM_CHILD(optr child)
void MSG_VIS_REMOVE_NON_DISCARDABLE(
                  VisUpdateMode updateMode)
void MSG_VIS_INVAL_ALL_GEOMETRY(
                  VisUpdateMode updateMode)
void MSG_SPEC_BUILD(SpecBuildFlags flags = bp) /*XXX*/
void MSG_SPEC_BUILD_BRANCH(SpecBuildFlags flags)
void MSG_SPEC_UNBUILD_BRANCH(SpecBuildFlags flags)
void MSG_SPEC_UNBUILD(SpecBuildFlags flags)
optr MSG_SPEC_GET_VIS_PARENT(SpecBuildFlags flags)
optr MSG_SPEC_DETERMINE_VIS_PARENT_FOR_CHILD(optr child)
optr MSG_SPEC_GUP_QUERY_VIS_PARENT(word parentType)
optr MSG_SPEC_GET_SPECIFIC_VIS_OBJECT(word flags)
void MSG_SPEC_ADD_CHILD_RELATIVE_TO_GEN(@stack
                  word buildFlags, optr parent,
                  optr child)
void MSG_SPEC_RESOLVE_MONIKER_LIST(
                  ChunkHandle monikerList)
void MSG_SPEC_RESOLVE_TOKEN_MONIKER(
                  ChunkHandle monikerChunk)
void MSG_SPEC_SET_ATTRS(SpecAttrs attrsToSet,
                  SpecAttrs attrsToClear,
                  VisUpdateMode updateMode)
SpecAttrs MSG_SPEC_GET_ATTRS()
void MSG_SPEC_SET_USABLE(byte updateMode)
void MSG_SPEC_SET_NOT_USABLE(byte updateMode)
Boolean MSG_SPEC_NOTIFY_ENABLED(
                  byte updateMode, byte flags)
Boolean MSG_SPEC_NOTIFY_NOT_ENABLED(
                  byte updateMode, byte flags)
void MSG_SPEC_UPDATE_VIS_MONIKER(byte updateMode,
                  word oldMonikerWidth,
                  word oldMonikerHeight)
void MSG_SPEC_UPDATE_VISUAL(byte updateMode)
void MSG_SPEC_GET_EXTRA_SIZE(word childCount,
                  ExtraSizeParams *retValue)
SizeAsDWord MSG_SPEC_CONVERT_DESIRED_SIZE_HINT(
                  word desiredWidth,
                  word desiredHeight,
                  word childCount)
word MSG_SPEC_CONVERT_SIZE(
                  word specSize, GStateHandle gstate)
void MSG_SPEC_VUP_GET_WIN_SIZE_INFO(
                  WinSizeInfoParams *retValue)
void MSG_SPEC_NAVIGATE_TO_NEXT_FIELD()
void MSG_SPEC_NAVIGATE_TO_PREVIOUS_FIELD()
void MSG_SPEC_NAVIGATE_COMMON(
            NavigateCommonParams *navCommonParams)
Boolean MSG_SPEC_ACTIVATE_INTERACTION_DEFAULT()
optr MSG_SPEC_NAVIGATE(word navigateFlags)
```



```
void MSG_SPEC_NAVIGATION_QUERY(optr queryOrigin,
                    NavigationFlags navFlags,
                    NavigationQueryParams *retValue)
void MSG_SPEC_NOTIFY_NAVIGATION_COMPLETE(
                    NavigationFlags navigateFlags)
Boolean MSG_SPEC_ACTIVATE_OBJECT_WITH_MNEMONIC(
                    word character, byte flags,
                    word state)
Boolean MSG_SPEC_CHECK_MNEMONIC(word character,
                    byte flags, word state)
byte MSG_SPEC_MENU_SEP_QUERY(byte flags)
void MSG_SPEC_UPDATE_MENU_SEPARATORS()
optr MSG_SPEC_START_BROADCAST_FOR_DEFAULT_FOCUS()
void MSG_SPEC_BROADCAST_FOR_DEFAULT_FOCUS(
                    ObjectAndGrabParams *retValue)
void MSG_SPEC_SCAN_GEOMETRY_HINTS()
void MSG_SPEC_RESCAN_GEO_AND_UPDATE(
                    VisUpdateMode updateMode)
void MSG_SPEC_UPDATE_SPECIFIC_OBJECT()
void MSG_SPEC_VIS_OPEN_NOTIFY()
void MSG_SPEC_VIS_CLOSE_NOTIFY()
dword MSG_SPEC_GET_MENU_CENTER()
void MSG_SPEC_UPDATE_KBD_ACCELERATOR(
                    VisUpdateMode updateMode)
void MSG_SPEC_RESET_SIZE_TO_STAY_ONSCREEN(
                    VisUpdateMode updateMode)
```

VisCompClass

```
@class VisCompClass, VisClass
```

Instance Data

```
@composite               VCI_comp = VI_link
optr                     VCI_gadgetExcl
WindowHandle             VCI_window = 0
VisCompGeoAttrs          VCI_geoAttrs = 0
VisCompGeoDimensionAttrs VCI_geoDimensionAttrs = 0
    @default VI_typeFlags = VTF_IS_COMPOSITE
```

Types and Flags

```
dword SpacingAsDWord
WordFlags    VisCompSpacingMarginsInfo
      VCSMI_USE_THIS_INFO                   0x8000
      VCSMI_LEFT_MARGIN                     0x7000
      VCSMI_TOP_MARGIN                      0x0e00
      VCSMI_RIGHT_MARGIN                    0x01c0
      VCSMI_BOTTOM_MARGIN                   0x0038
      VCSMI_CHILD_SPACING                   0x0007
VCSMI_LEFT_MARGIN_OFFSET        12
VCSMI_TOP_MARGIN_OFFSET         9
VCSMI_RIGHT_MARGIN_OFFSET       6
VCSMI_BOTTOM_MARGIN_OFFSET      3
VCSMI_CHILD_SPACING_OFFSET      0
ByteEnum     WidthJustification
      WJ_LEFT_JUSTIFY_CHILDREN              0x00
      WJ_RIGHT_JUSTIFY_CHILDREN             0x40
      WJ_CENTER_CHILDREN_HORIZONTALLY       0x80
      WJ_FULL_JUSTIFY_CHILDREN_HORIZONTALLY 0xc0
ByteEnum     HeightJustification
      HJ_TOP_JUSTIFY_CHILDREN               0x00
      HJ_BOTTOM_JUSTIFY_CHILDREN            0x04
      HJ_CENTER_CHILDREN_VERTICALLY         0x08
      HJ_FULL_JUSTIFY_CHILDREN_VERTICALLY   0x0c
ByteFlags    VisCompGeoDimensionAttrs
      VCGDA_WIDTH_JUSTIFICATION             0xc0
      VCGDA_EXPAND_WIDTH_TO_FIT_PARENT      0x20
      VCGDA_DIVIDE_WIDTH_EQUALLY            0x10
      VCGDA_HEIGHT_JUSTIFICATION            0x0c
      VCGDA_EXPAND_HEIGHT_TO_FIT_PARENT     0x02
      VCGDA_DIVIDE_HEIGHT_EQUALLY           0x01
ByteFlags    VisCompGeoAttrs
      VCGA_ORIENT_CHILDREN_VERTICALLY       0x80
      VCGA_INCLUDE_ENDS_IN_CHILD_SPACING    0x40
      VCGA_ALLOW_CHILDREN_TO_WRAP           0x20
      VCGA_ONE_PASS_OPTIMIZATION            0x10
      VCGA_CUSTOM_MANAGE_CHILDREN           0x08
      VCGA_HAS_MINIMUM_SIZE                 0x04
      VCGA_WRAP_AFTER_CHILD_COUNT           0x02
      VCGA_ONLY_DRAWS_IN_MARGINS            0x01
```

Structures

```
typedef struct {
    word          CSP_childSpacing
    word          CSP_wrapSpacing
} ChildSpacingParams
```



Macros

```
DWORD_CHILD_SPACING(val) ((word) (val))
DWORD_WRAP_SPACING(val) ((word) (val >> 16))
MAKE_SPACING_DWORD(width,height)
            ((((dword) (height)) << 16) | (width))
```

Messages

```
word MSG_VIS_COMP_GET_GEO_ATTRS()
void MSG_VIS_COMP_SET_GEO_ATTRS(
                    word attrsToSet, word attrsToClear)
SpacingAsDWord MSG_VIS_COMP_GET_CHILD_SPACING()
SizeAsDWord MSG_VIS_COMP_GET_MINIMUM_SIZE()
void MSG_VIS_COMP_GET_MARGINS(Rectangle *retValue)
word MSG_VIS_COMP_GET_WRAP_COUNT()
```

VisContentClass

```
@class VisContentClass, VisCompClass
```

Instance Data

```
optr         VCNI_view
WindowHandle VCNI_window = 0
word         VCNI_viewHeight = 0
word         VCNI_viewWidth = 0
VisContentAttrs VCNI_attrs = 0
PointDWord   VCNI_docOrigin = {0, 0}
PointWWFixed VCNI_scaleFactor = {{0, 1}, {0, 1}}
ChunkHandle  VCNI_prePassiveMouseGrabList = 0
VisMouseGrab VCNI_impliedMouseGrab =
                    {0, 0, {0, 0},
                    (VIFGF_MOUSE | VIFGF_PTR), 0}
VisMouseGrab VCNI_activeMouseGrab =
                    {0, 0, {0, 0}, 0, 0}
ChunkHandle  VCNI_postPassiveMouseGrabList = 0
KbdGrab      VCNI_kbdGrab = {0, 0}
FTVMCGrab    VCNI_focusExcl = {0, MAEF_FOCUS}
FTVMCGrab    VCNI_targetExcl = {0, MAEF_TARGET}
Handle       VCNI_holdUpInputQueue = 0
word         VCNI_holdUpInputCount = 0
byte         VCNI_holdUpInputFlags = 0
    @default VI_typeFlags = VTF_IS_COMPOSITE |
                    VTF_IS_WINDOW | VTF_IS_CONTENT |
                    VTF_IS_WIN_GROUP | VTF_IS_INPUT_NODE
```

Types and Flags

```
ByteFlags    VisContentAttrs
      VCNA_SAME_WIDTH_AS_VIEW               0x80
      VCNA_SAME_HEIGHT_AS_VIEW              0x40
      VCNA_LARGE_DOCUMENT_MODEL             0x20
      VCNA_WINDOW_COORDINATE_MOUSE_EVENTS   0x10
      VCNA_ACTIVE_MOUSE_GRAB_REQUIRES_LARGE_EVENTS 0x08
      VCNA_VIEW_DOC_BOUNDS_SET_MANUALLY     0x04
      VCNA_VIEW_DOES_NOT_WIN_SCROLL         0x02
```

Structures

```
typedef struct {
    optr                 VMG_object
    WindowHandle         VMG_gWin
    PointDWord           VMG_translation
    VisInputFlowGrabFlags VMG_flags
    byte                 VMG_unused
} VisMouseGrab
```

Messages

```
SizeAsDWord MSG_VIS_CONTENT_GET_WIN_SIZE()
void MSG_VIS_CONTENT_SET_ATTRS(
                    VisContentAttrs attrsToSet,
                    VisContentAttrs attrsToClear)
VisContentAttrs MSG_VIS_CONTENT_GET_ATTRS()
SizeAsDWord MSG_VIS_CONTENT_RECALC_SIZE_BASED_ON_VIEW()
void MSG_VIS_CONTENT_HOLD_UP_INPUT_FLOW()
void MSG_VIS_CONTENT_RESUME_INPUT_FLOW()
void MSG_VIS_CONTENT_DISABLE_HOLD_UP()
void MSG_VIS_CONTENT_ENABLE_HOLD_UP()
```



```
Boolean MSG_VIS_CONTENT_TEST_IF_ACTIVE_OR_IMPLIED_WIN(
                    WinHandle window)
void MSG_VIS_CONTENT_UNWANTED_MOUSE_EVENT(
                    VisMouseGrab *mouseGrab,
                    word inputState)
void MSG_VIS_CONTENT_UNWANTED_KBD_EVENT(
                    word character, word flags,
                    word state)
void
MSG_VIS_CONTENT_REMOTE_OBJ_MESSAGE_VIA_HOLD_UP_QUEUE()
void MSG_VIS_CONTENT_SET_DOC_BOUNDS(@stack
                    sdword bottom, sdword right,
                    sdword top, sdword left)
void
MSG_VIS_CONTENT_NOTIFY_ACTIVE_MOUSE_GRAB_WIN_CHANGED()
```

VisHorizRulerClass

```
@class VisHorizRulerClass, VisRulerClass
```

VisLargeTextClass

```
@class VisLargeTextClass, VisTextClass
```

Instance Data

```
ChunkHandle               VLTI_regionArray
VisLargeTextDisplayModes  VLTI_displayMode
word                      VLTI_regionSpacing
XYSize                    VLTI_draftRegionSize
dword                     VLTI_totalHeight
word                      VLTI_displayModeWidth
VisLargeTextFlags         VLTI_flags
VisLargeTextAttrs         VLTI_attrs
```

Types and Flags

```
WordFlags    VisLargeTextRegionFlags
      VLTRF_ENDED_BY_COLUMN_BREAK     0x8000
      VLTRF_EMPTY                     0x4000
typedef enum {
    VLTDM_PAGE,
    VLTDM_CONDENSED,
    VLTDM_GALLEY,
    VLTDM_DRAFT_WITH_STYLES,
    VLTDM_DRAFT_WITHOUT_STYLES
} VisLargeTextDisplayModes
WordFlags    VisLargeTextFlags
      VLTF_HEIGHT_NOTIFY_PENDING      0x8000
WordFlags    VisLargeTextAttrs
      VLTA_EXACT_HEIGHT               0x8000
```

Structures

```
typedef struct {
    dword                       VLTRAE_charCount
    dword                       VLTRAE_lineCount
    word                        VLTRAE_section
    PointDWord                  VLTRAE_spatialPosition
    XYSize                      VLTRAE_size
    WBFixed                     VLTRAE_calcHeight
    dword                       VLTRAE_region
    VisLargeTextRegionFlags     VLTRAE_flags
    byte                        VLTRAE_reserved[3]
} VisLargeTextRegionArrayElement
```

Messages

```
void MSG_VIS_LARGE_TEXT_CREATE_DATA_STRUCTURES()
void MSG_VIS_LARGE_TEXT_APPEND_REGION(word region)
void MSG_VIS_LARGE_TEXT_REGION_IS_LAST(word region)
void MSG_VIS_LARGE_TEXT_CURRENT_REGION_CHANGED(
                    word region)
void MSG_VIS_LARGE_TEXT_REGION_CHANGED(word region)
void MSG_VIS_LARGE_TEXT_SET_DISPLAY_MODE(
                    VisLargeTextDisplayModes mode)
void MSG_VIS_LARGE_TEXT_SET_DRAFT_REGION_SIZE(
                    word width, word height)
XYValueAsDWord
MSG_VIS_LARGE_TEXT_GET_DRAFT_REGION_SIZE(word region)
word MSG_VIS_LARGE_TEXT_GET_REGION_COUNT()
void MSG_VIS_LARGE_TEXT_GET_REGION_POS()
void MSG_VIS_LARGE_TEXT_REGION_FROM_POINT()
```

VisRulerClass

```
@class VisRulerClass, VisCompClass, master
```

Instance Data

```
VisRulerType       VRI_type
MinIncrementType   VRI_minIncrement
WWFixed            VRI_scale
sdword             VRI_offset
sdword             VRI_margin
```

Types and Flags

```
VIS_RULER_HEIGHT          10
CUSTOM_RULER_DEFINITION   0xfd
NO_RULERS                 0xfe
SYSTEM_DEFAULT            0xff
ByteEnum     VisRulerType
      VRT_INCHES                0
      VRT_CENTIMETERS           1
      VRT_POINTS                2
      VRT_PICAS                 3
      VRT_CUSTOM                CUSTOM_RULER_DEFINITION
      VRT_NONE                  NO_RULERS
      VRT_DEFAULT               SYSTEM_DEFAULT
ByteEnum     MinUSMeasure
      MUSM_EIGHTH_INCH          0
      MUSM_QUARTER_INCH         1
      MUSM_HALF_INCH            2
      MUSM_ONE_INCH             3
ByteEnum     MinMetricMeasure
      MMM_MILLIMETER            0
      MMM_HALF_CENTIMETER       1
      MMM_CENTIMETER            2
ByteEnum     MinPointMeasure
      MPM_25_POINT              0
      MPM_50_POINT              1
      MPM_100_POINT             2
ByteEnum MinPicaMeasure
      MPM_PICA           0
      MPM_INCH           1
```

Structures

```
typedef union {
    MinUSMeasure          MIT_US
    MinMetricMeasure      MIT_METRIC
    MinPointMeasure       MIT_POINT
    MinPicaMeasure        MIT_PICA
} MinIncrementType
```

Messages

```
void MSG_VIS_RULER_SET_TYPE(VisRulerType rulerType)
void MSG_VIS_RULER_SET_MIN_INCREMENT(
                  MinIncrementType minIncrementType)
void MSG_VIS_RULER_SET_SCALE(WWFixed scaleFactor)
void MSG_VIS_RULER_SET_OFFSET(sdword offset)
void MSG_VIS_RULER_SET_MARGIN(sdword margin)
```

VisTextClass

```
@class VisTextClass, VisClass
```

Instance Data

```
ChunkHandle         VTI_text
word                VTI_charAttrRuns =
                        VIS_TEXT_INITIAL_CHAR_ATTR
word                VTI_paraAttrRuns =
                        VIS_TEXT_INITIAL_PARA_ATTR
VMFileHandle        VTI_vmFile = NullHandle
word                VTI_lines = 0
VisTextStorageFlags VTI_storageFlags =
                        (VTSF_DEFAULT_CHAR_ATTR |
                         VTSF_DEFAULT_PARA_ATTR)
VisTextFeatures     VTI_features = 0
VisTextStates       VTI_state = 0
VisTextIntFlags     VTI_intFlags = 0
VisTextIntSelFlags  VTI_intSelFlags = 0
GSRefCountAndFlags  VTI_gsRefCount = 0
GStateHandle        VTI_gstate = NullHandle
word                VTI_gstateRegion = -1
dword               VTI_selectStart = 0
dword               VTI_selectEnd = 0
PointDWord          VTI_startEventPos = {0,0}
dword               VTI_selectMinStart = 0
dword               VTI_selectMinEnd = 0
dword               VTI_lastOffset = 0
word                VTI_goalPosition = 0
Point               VTI_cursorPos = {0,0}
word                VTI_cursorRegion = 0
word                VTI_leftOffset = 0
byte                VTI_lrMargin = 0
byte                VTI_tbMargin = 0
ColorQuad           VTI_washColor = {
                        C_WHITE, CF_INDEX, 0, 0}
word                VTI_maxLength = 10000
VisTextFilters      VTI_filters = 0
optr                VTI_output
WBFixed             VTI_height = {0,0}
word                VTI_lastWidth = -1
TimerHandle         VTI_timerHandle = NullHandle
word                VTI_timerID = 0
```

Variable Data

```
word ATTR_VIS_TEXT_TYPE_RUNS
word ATTR_VIS_TEXT_GRAPHIC_RUNS
word ATTR_VIS_TEXT_STYLE_ARRAY
word ATTR_VIS_TEXT_NAME_ARRAY
VisTextSuspendData ATTR_VIS_TEXT_SUSPEND_DATA
void ATTR_VIS_TEXT_NOTIFY_CONTENT
word ATTR_VIS_TEXT_SELECTED_TAB
void ATTR_VIS_TEXT_DO_NOT_INTERACT_WITH_SEARCH_CONTROL
VisTextExtendedFilterType ATTR_VIS_TEXT_EXTENDED_FILTER
word ATTR_VIS_TEXT_CHAR_ATTR_INSERTION_TOKEN
word ATTR_VIS_TEXT_TYPE_INSERTION_TOKEN
word ATTR_VIS_TEXT_TYPE_INSERTION_TOKEN
VisTextCachedRunInfo TEMP_VIS_TEXT_CACHED_RUN_INFO
```

```
void TEMP_VIS_TEXT_FORCE_SEND_IS_LAST_REGION
VisTextCachedUndoInfo TEMP_VIS_TEXT_CACHED_UNDO_INFO
ChunkHandle ATTR_VIS_TEXT_CUSTOM_FILTER
void ATTR_VIS_TEXT_UPDATE_VIA_PROCESS
void ATTR_VIS_TEXT_DOES_NOT_ACCEPT_INK
WBFixed ATTR_VIS_TEXT_MINIMUM_SINGLE_LINE_HEIGHT
void TEMP_VIS_TEXT_SYS_TARGET
```

Types and Flags

```
SEARCH_REPLACE_TEXT_MAXIMUM     65
ByteFlags    SearchOptions
      SO_NO_WILDCARDS                         0x20
      SO_IGNORE_SOFT_HYPHENS                  0x10
      SO_BACKWARD_SEARCH                      0x08
      SO_IGNORE_CASE                          0x04
      SO_PARTIAL_WORD                         0x02
      SO_PRESERVE_CASE_OF_DOCUMENT_STRING     0x01
ByteEnum     WildCard
      WC_MATCH_SINGLE_CHAR                    0x10
      WC_MATCH_MULTIPLE_CHARS                 0x11
      WC_MATCH_WHITESPACE_CHAR                0x12
      TEXT_ADDRESS_PAST_END                   0x00ffffff
      VIS_TEXT_RANGE_SELECTION                0xffff0000
      VIS_TEXT_RANGE_PARAGRAPH_SELECTION      0xfffe0000
WordFlags    VisTextRangeContext
      VTRC_PARAGRAPH_CHANGE           0x8000
      VTRC_CHAR_ATTR_CHANGE           0x4000
      VTRC_PARA_ATTR_BORDER_CHANGE    0x2000
ByteEnum     TextArrayType
      TAT_CHAR_ATTRS                  0
      TAT_PARA_ATTRS                  1
      TAT_GRAPHICS                    2
      TAT_TYPES                       3
WordFlags    TextStyleFlags
      TSF_APPLY_TO_SELECTION_ONLY     0x8000
      TSF_POINT_SIZE_RELATIVE         0x4000
      TSF_MARGINS_RELATIVE            0x2000
      TSF_SPACING_RELATIVE            0x1000
WordFlags    VisTextCharAttrFlags
      VTCAF_MULTIPLE_FONT_IDS         0x8000
      VTCAF_MULTIPLE_POINT_SIZES      0x4000
      VTCAF_MULTIPLE_COLORS           0x2000
      VTCAF_MULTIPLE_GRAY_SCREENS     0x1000
      VTCAF_MULTIPLE_PATTERNS         0x0800
      VTCAF_MULTIPLE_TRACK_KERNINGS   0x0400
      VTCAF_MULTIPLE_FONT_WEIGHTS     0x0200
      VTCAF_MULTIPLE_FONT_WIDTHS      0x0100
      VTCAF_MULTIPLE_BG_COLORS        0x0080
      VTCAF_MULTIPLE_BG_GRAY_SCREENS  0x0040
      VTCAF_MULTIPLE_BG_PATTERNS      0x0020
      VTCAF_MULTIPLE_STYLES           0x0010
WordFlags    VisTextGetAttrFlags
      VTGAF_MERGE_WITH_PASSED         0x8000
WordFlags    VisTextParaAttrFlags
      VTPAF_MULTIPLE_LEFT_MARGINS     0x8000
      VTPAF_MULTIPLE_RIGHT_MARGINS    0x4000
      VTPAF_MULTIPLE_PARA_MARGINS     0x2000
```

```
      VTPAF_MULTIPLE_LINE_SPACINGS    0x1000
      VTPAF_MULTIPLE_DEFAULT_TABS     0x0800
      VTPAF_MULTIPLE_TOP_SPACING      0x0400
      VTPAF_MULTIPLE_BOTTOM_SPACING   0x0200
      VTPAF_MULTIPLE_LEADINGS         0x0100
      VTPAF_MULTIPLE_BG_COLORS        0x0080
      VTPAF_MULTIPLE_BG_GRAY_SCREENS  0x0040
      VTPAF_MULTIPLE_BG_PATTERNS      0x0020
      VTPAF_MULTIPLE_TAB_LISTS        0x0010
      VTPAF_MULTIPLE_STYLES           0x0008
      VTPAF_MULTIPLE_PREPEND_CHARS    0x0004
      VTPAF_MULTIPLE_PARA_NUMBERS     0x0002
WordFlags    VisTextParaAttrBorderFlags
      VTPABF_MULTIPLE_BORDER_LEFT            0x8000
      VTPABF_MULTIPLE_BORDER_TOP             0x4000
      VTPABF_MULTIPLE_BORDER_RIGHT           0x2000
      VTPABF_MULTIPLE_BORDER_BOTTOM          0x1000
      VTPABF_MULTIPLE_BORDER_DOUBLES         0x0800
      VTPABF_MULTIPLE_BORDER_DRAW_INNERS     0x0400
      VTPABF_MULTIPLE_BORDER_ANCHORS         0x0200
      VTPABF_MULTIPLE_BORDER_WIDTHS          0x0100
      VTPABF_MULTIPLE_BORDER_SPACINGS        0x0080
      VTPABF_MULTIPLE_BORDER_SHADOWS         0x0040
      VTPABF_MULTIPLE_BORDER_COLORS          0x0020
      VTPABF_MULTIPLE_BORDER_GRAY_SCREENS    0x0010
      VTPABF_MULTIPLE_BORDER_PATTERNS        0x0008
      VTTF_MULTIPLE_HYPERLINKS               0x8000
      VTTF_MULTIPLE_CONTEXTS                 0x4000
ByteEnum     VisTextNameType
      VTNT_CONTEXT                    0
      VTNT_FILE                       1
ByteEnum     VisTextContextType
      VTCT_TEXT                       0
      VTCT_CATEGORY                   1
      VTCT_QUESTION                   2
      VTCT_ANSWER                     3
      VTCT_DEFINITION                 4
      VTCT_FILE                       255
      VIS_TEXT_GRAPHIC_OPAQUE_SIZE    16
ByteEnum     VisTextGraphicType
      VTGT_GSTRING                    0
      VTGT_VARIABLE                   1
      VTGF_DRAW_FROM_BASELINE         0x8000
      VTGF_HANDLES_POINTER            0x4000
ByteEnum     VisTextSaveType
      VTST_NONE                       0
      VTST_SINGLE_CHUNK               1
      VTST_RUNS_ONLY                  2
      VTST_RUNS_AND_ELEMENTS          3
WordFlags    VisTextSaveDBFlags
      VTSDBF_TEXT                     0x8000
      VTSDBF_CHAR_ATTR                0x6000
      VTSDBF_PARA_ATTR                0x1800
      VTSDBF_TYPE                     0x0600
      VTSDBF_GRAPHIC                  0x0180
      VTSDBF_STYLE                    0x0040
      VTSDBF_REGION                   0x0020
```



```
      VTSDBF_NAME                     0x0010
VTSDBF_CHAR_ATTR_OFFSET         13
VTSDBF_PARA_ATTR_OFFSET         11
VTSDBF_TYPE_OFFSET              9
VTSDBF_GRAPHIC_OFFSET           7
typedef enum {
      TCO_COPY,
      TCO_RETURN_TRANSFER_FORMAT,
      TCO_RETURN_TRANSFER_ITEM,
      TCO_RETURN_NOTHING
} TextClipboardOption
WordFlags    VisTextNotificationFlags
      VTNF_SELECT_STATE               0x8000
      VTNF_CHAR_ATTR                  0x4000
      VTNF_PARA_ATTR                  0x2000
      VTNF_TYPE                       0x1000
      VTNF_SELECTION                  0x0800
      VTNF_COUNT                      0x0400
      VTNF_STYLE_SHEET                0x0200
      VTNF_STYLE                      0x0100
      VTNF_SEARCH_ENABLE              0x0080
      VTNF_SPELL_ENABLE               0x0040
WordFlags    VisTextNotifySendFlags
      VTNSF_UPDATE_APP_TARGET_GCN_LISTS     0x8000
      VTNSF_UPDATE_OUTPUT                   0x4000
      VTNSF_NULL_STATUS                     0x2000
      VTNSF_STRUCTURE_INITIALIZED           0x1000
      VTNSF_SEND_AFTER_GENERATION           0x0800
      VTNSF_SEND_ONLY                       0x0400
      VTNSF_RELAYED_TO_LIKE_TEXT_OBJECTS    0x0200
      SS_DRAGGING                           0x8000
      SS_CURSOR_HEIGHT                      0x7fff
ByteEnum     VisTextExtendedFilterType
      VTEFT_REPLACE_PARAMS 0
      VTEFT_CHARACTER_LEVELER_LEVEL 1
      VTEFT_BEFORE_AFTER 2
enum {
      GSSOT_FIRST_OBJECT,
      GSSOT_LAST_OBJECT,
      GSSOT_NEXT_OBJECT,
      GSSOT_PREV_OBJECT
} GetSearchSpellObjectOption
WordFlags    GetSearchSpellObjectParam
      GSSOP_RELAYED_FLAG              0x8000
enum {
      CL_STARTING_AT_POSITION,
      CL_ENDING_AT_POSITION,
      CL_CENTERED_AROUND_POSITION,
      CL_CENTERED_AROUND_SELECTION,
      CL_CENTERED_AROUND_SELECTION_START,
      CL_GET_SELECTED_WORD
} ContextLocation
ByteFlags    SpellCheckFromOffsetFlags
      SCFOF_CHECK_NUM_CHARS           0x01
ByteFlags    SpellCheckFromOffsetFlags
      SFOF_STOP_AT_STARTING_POINT     0x01
WordFlags    ReplaceAllFromOffsetFlags
```

```
	RAFOF_CONTINUING_REPLACE	0x8000
	RAFOF_HAS_UNDO			0x4000
ByteEnum	VisTextFilterClass
	VTFC_NO_FILTER			0
	VTFC_ALPHA			1
	VTFC_NUMERIC			2
	VTFC_SIGNED_NUMERIC		3
	VTFC_SIGNED_DECIMAL		4
	VTFC_FLOAT_DECIMAL		5
	VTFC_ALPHA_NUMERIC		6
	VTFC_FILENAMES			7
	VTFC_DOS_FILENAMES		8
	VTFC_DOS_PATH			9
	VTFC_DATE			10
	VTFC_TIME			11
	VTFC_DASHED_ALPHA_NUMERIC	12
	VTFC_NORMAL_ASCII		13
	VTFC_DOS_VOLUME_NAMES		14
	VTFC_DOS_CHARACTER_SET		15
	VTFC_ALLOW_COLUMN_BREAKS	16
ByteFlags	VisTextFilters
	VTF_NO_SPACES			0x80
	VTF_NO_TABS			0x40
	VTF_UPCASE_CHARS		0x20
	VTF_FILTER_CLASS		0x1f
typedef enum /* word */ {
	VTKF_FORWARD_LINE=0,
	VTKF_BACKWARD_LINE=6,
	VTKF_SELECT_ADJUST_FORWARD_LINE=12,
	VTKF_SELECT_ADJUST_BACKWARD_LINE=18,
	VTKF_INSERT=24,
	VTKF_FORWARD_CHAR=30,
	VTKF_BACKWARD_CHAR=36,
	VTKF_FORWARD_WORD=42,
	VTKF_BACKWARD_WORD=48,
	VTKF_FORWARD_PARAGRAPH=54,
	VTKF_BACKWARD_PARAGRAPH=60,
	VTKF_START_OF_LINE=66,
	VTKF_END_OF_LINE=72,
	VTKF_START_OF_TEXT=78,
	VTKF_END_OF_TEXT=84,
	VTKF_SELECT_WORD=90,
	VTKF_SELECT_LINE=96,
	VTKF_SELECT_PARAGRAPH=102,
	VTKF_SELECT_OBJECT=108,
	VTKF_SELECT_ADJUST_FORWARD_CHAR=114,
	VTKF_SELECT_ADJUST_BACKWARD_CHAR=120,
	VTKF_SELECT_ADJUST_FORWARD_WORD=126,
	VTKF_SELECT_ADJUST_BACKWARD_WORD=132,
	VTKF_SELECT_ADJUST_FORWARD_PARAGRAPH=138,
	VTKF_SELECT_ADJUST_BACKWARD_PARAGRAPH=144,
	VTKF_SELECT_ADJUST_TO_START=150,
	VTKF_SELECT_ADJUST_TO_END=156,
	VTKF_SELECT_ADJUST_START_OF_LINE=162,
	VTKF_SELECT_ADJUST_END_OF_LINE=168,
	VTKF_DELETE_BACKWARD_CHAR=174,
	VTKF_DELETE_BACKWARD_WORD=180,
```



```
	VTKF_DELETE_BACKWARD_LINE=186,
	VTKF_DELETE_BACKWARD_PARAGRAPH=192,
	VTKF_DELETE_TO_START=198,
	VTKF_DELETE_CHAR=204,
	VTKF_DELETE_WORD=210,
	VTKF_DELETE_LINE=216,
	VTKF_DELETE_PARAGRAPH=222,
	VTKF_DELETE_TO_END=228,
	VTKF_DELETE_EVERYTHING=234,
	VTKF_DESELECT=240,
	VTKF_TOGGLE_OVERSTRIKE_MODE=246,
	VTKF_TOGGLE_SMART_QUOTES=252,
} VisTextKeyFunction
typedef enum /* word */ {
	TRT_POINTER,
	TRT_SEGMENT_CHUNK,
	TRT_BLOCK_CHUNK,
	TRT_BLOCK,
	TRT_VM_BLOCK,
	TRT_DB_ITEM,
	TRT_HUGE_ARRAY,
} TextReferenceType
INSERT_COMPUTE_TEXT_LENGTH	0x01ff
/* Bitfield VisTextGetTextRangeFlags */
	VTGTRF_ALLOCATE			0x80
	VTGTRF_ALLOCATE_ALWAYS		0x40
	VTGTRF_RESIZE_DEST		0x20
WordFlags	VisTextHWRFlags
	VTHWRF_NO_CONTEXT		0x8000
	VTHWRF_USE_PASSED_CONTEXT	0x4000
OFFSET_FOR_TYPE_RUNS		0
OFFSET_FOR_GRAPHIC_RUNS		1
WordFlags	VisTextFeatures
	VTF_NO_WORD_WRAPPING		0x8000
	VTF_AUTO_HYPHENATE		0x4000
	VTF_ALLOW_SMART_QUOTES		0x2000
	VTF_ALLOW_UNDO			0x1000
	VTF_SHOW_HIDDEN_TEXT		0x0800
	VTF_OUTLINE_MODE		0x0400
	VTF_DONT_SHOW_SOFT_PAGE_BREAKS	0x0200
	VTF_DONT_SHOW_GRAPHICS		0x0100
	VTF_TRANSPARENT			0x0080
ByteFlags	VisTextStates
	VTS_EDITABLE				0x80
	VTS_SELECTABLE				0x40
	VTS_TARGETABLE				0x20
	VTS_ONE_LINE				0x10
	VTS_SUBCLASS_VIRT_PHYS_TRANSLATION	0x08
	VTS_OVERSTRIKE_MODE			0x04
	VTS_USER_MODIFIED			0x02
ByteEnum	SelectionType
	ST_DOING_CHAR_SELECTION		0
	ST_DOING_WORD_SELECTION		1
	ST_DOING_LINE_SELECTION		2
	ST_DOING_PARA_SELECTION		3
ByteFlags	VisTextIntSelFlags
	VTISF_IS_TARGET			0x80
```

```
        VTISF_IS_FOCUS                  0x40
        VTISF_CURSOR_ON                 0x20
        VTISF_CURSOR_ENABLED            0x10
        VTISF_DOING_SELECTION           0x08
        VTISF_DOING_DRAG_SELECTION      0x04
        VTISF_SELECTION_TYPE            0x03
    ByteEnum     AdjustType
        AT_NORMAL                       0
        AT_NO_ADJUST                    1
        AT_PASTE                        2
        AT_QUICK                        3
        AT_ENTIRE_RANGE                 4
    ByteEnum     ActiveSearchSpellType
        ASST_NOTHING_ACTIVE             0
        ASST_SPELL_ACTIVE               0
        ASST_SEARCH_ACTIVE              0
    ByteFlags    VisTextIntFlags
        VTIF_HAS_LINES                  0x80
        VTIF_SUSPENDED                  0x40
        VTIF_UPDATE_PENDING             0x20
        VTIF_ACTIVE_SEARCH_SPELL        0x18
        VTIF_HILITED                    0x04
        VTIF_ADJUST_TYPE                0x03
    ByteFlags    GSRefCountAndFlags
        GSRCAF_USE_DOC_CLIP_REGION      0x80
        GSRCAF_REF_COUNT                0x7f
```

Structures

```
    typedef struct {
        dword               VTR_start
        dword               VTR_end
    } VisTextRange
    typedef struct {
        ChunkArrayHeader    TRAH_meta
        word                TRAH_elementVMBlock
        ChunkHandle         TRAH_elementArray
    } TextRunArrayHeader
    typedef struct {
        WordAndAHalf        TRAE_position
        word                TRAE_token
    } TextRunArrayElement
    typedef struct {
        ElementArrayHeader  TEAH_meta
        TextArrayType       TEAH_arrayType
        byte                TEAH_unused
    } TextElementArrayHeader
    typedef struct {
        TextStyleFlags      TSPD_flags
        byte                TSPD_unused[2]
    } TextStylePrivateData
    typedef struct {
        NameArrayElement    TSEH_meta
        word                TSEH_baseStyle
        StyleElementFlags   TSEH_flags
        TextStylePrivateData TSEH_privateData
        word                TSEH_charAttrToken
        word                TSEH_paraAttrToken
```

```
} TextStyleElementHeader
typedef struct {
    VisTextCharAttrFlags        VTCAD_diffs
    VisTextExtendedStyles       VTCAD_extendedStyles
    TextStyle                   VTCAD_textStyles
    byte                        VTCAD_unused
} VisTextCharAttrDiffs
typedef struct {
    VisTextParaAttrFlags        VTPAD_diffs
    VisTextParaAttrBorderFlags  VTPAD_borderDiffs
    VisTextParaAttrAttributes   VTPAD_attributes
    VisTextHyphenationInfo      VTPAD_hyphenationInfo
    VisTextKeepInfo             VTPAD_keepInfo
    VisTextDropCapInfo          VTPAD_dropCapInfo
} VisTextParaAttrDiffs
typedef struct {
    RefElementHeader      VTT_meta
    word                  VTT_attributes
    word                  VTT_hyperlinkName
    word                  VTT_hyperlinkFile
    word                  VTT_context
    byte                  VTT_unused[1]
} VisTextType
typedef struct {
    word                  VTTD_diffs
    word                  VTTD_attributes
} VisTextTypeDiffs
typedef struct {
    VisTextNameType       VTND_type
    VisTextContextType    VTND_contextType
    word                  VTND_file
    DBGroupAndItem        VTND_helpText
} VisTextNameData
typedef struct {
    TransMatrix           VTGG_tmatrix
} VisTextGraphicGString
typedef struct {
    ManufacturerID        VTGV_manufacturerID
    VisTextVariableType   VTGV_type
    byte                  VTGV_privateData[
                    VIS_TEXT_GRAPHIC_OPAQUE_SIZE-4]
} VisTextGraphicVariable
typedef union {
    VisTextGraphicGString       VTGD_gstring
    VisTextGraphicVariable      VTGD_variable
    byte                        VTGD_opaque[
                    VIS_TEXT_GRAPHIC_OPAQUE_SIZE]
} VisTextGraphicData
typedef struct {
    RefElementHeader      VTG_meta
    VMBlockHandle         VTG_vmChain
    dword                 VTG_dbItem
    XYSize                VTG_size
    VisTextGraphicType    VTG_type
    word                  VTG_flags
    byte                  VTG_reserved[4]
    VisTextGraphicData    VTG_data
```

```
} VisTextGraphic
typedef struct {
    StyleSheetParams      VTSSSP_common
    word                  VTSSSP_graphicsElements
    word                  VTSSSP_treeBlock
    word                  VTSSSP_graphicTreeOffset
} VisTextSaveStyleSheetParams
typedef struct {
    VMChainTree           TTBH_meta
    word                  TTBH_reservedOther[20]
    VMChain               TTBH_text
    VMChain               TTBH_charAttrRuns
    VMChain               TTBH_paraAttrRuns
    VMChain               TTBH_typeRuns
    VMChain               TTBH_graphicRuns
    VMChain               TTBH_charAttrElements
    VMChain               TTBH_paraAttrElements
    VMChain               TTBH_typeElements
    VMChain               TTBH_graphicElements
    VMChain               TTBH_styles
    VMChain               TTBH_names
    VMChain               TTBH_pageSetup
    VMChain               TTBH_reservedVM[10]
} TextTransferBlockHeader
typedef struct {
    VMChainLink           PSI_meta
    XYSize                PSI_page
    PageLayout            PSI_layout
    word                  PSI_numColumns
    word                  PSI_columnSpacing
    word                  PSI_ruleWidth
    word                  PSI_leftMargin
    word                  PSI_rightMargin
    word                  PSI_topMargin
    word                  PSI_bottomMargin
} PageSetupInfo
typedef struct {
    VisTextCharAttr       VTNCAC_charAttr
    word                  VTNCAC_charAttrToken
    VisTextCharAttrDiffs  VTNCAC_charAttrDiffs
} VisTextNotifyCharAttrChange
typedef struct {
    VisTextMaxParaAttr    VTNPAC_paraAttr
    word                  VTNPAC_paraAttrToken
    VisTextParaAttrDiffs  VTNPAC_paraAttrDiffs
    sdword                VTNPAC_regionOffset
    sword                 VTNPAC_regionWidth
    word                  VTNPAC_selectedTab
} VisTextNotifyParaAttrChange
typedef struct {
    VisTextType           VTNTC_type
    word                  VTNTC_typeToken
    VisTextTypeDiffs      VTNTC_typeDiffs
} VisTextNotifyTypeChange
typedef struct {
    dword                 VTNSC_selectStart
    dword                 VTNSC_selectEnd
```



```
    dword               VTNSC_lineNumber
    dword               VTNSC_lineStart
    word                VTNSC_region
    dword               VTNSC_regionStartLine
    dword               VTNSC_regionStartOffset
} VisTextNotifySelectionChange
typedef struct {
    dword               VTNCC_charCount
    dword               VTNCC_wordCount
    dword               VTNCC_lineCount
    dword               VTNCC_paraCount
} VisTextNotifyCountChange
typedef struct {
    VisTextNotificationFlags    VTGNP_notificationTypes
    VisTextNotifySendFlags      VTGNP_sendFlags
    MemHandle           VTGNP_notificationBlocks[16]
} VisTextGenerateNotifyParams
typedef struct {
    WBFixed             VTMDP_height
    WBFixed             VTMDP_width
} VisTextMinimumDimensionsParameters
typedef struct {
    word                searchSize
    word                replaceSize
    byte                params
    optr                replyObject
    Message             replyMsg
} SearchReplaceStruct
typedef struct {
    optr                CD_object
    dword               CD_numChars
    dword               CD_startPos
    VisTextRange        CD_selection
} ContextData
typedef struct {
    optr                SFORS_object
    dword               SFORS_offset
    dword               SFORS_len
} SearchFromOffsetReturnStruct
typedef struct {
    VisTextNameData     VTNCP_data
    word                VTNCP_index
    optr                VTNCP_object
} VisTextNameCommonParams
typedef struct {
    dword               VTGLOAFP_line
    dword               VTGLOAFP_offset
    word                VTGLOAFP_flags
} VisTextGetLineOffsetAndFlagsParameters
typedef struct {
    word                VTSD_count
    VisTextRange        VTSD_recalcRange
    VisTextRange        VTSD_selectRange
    WordFlags           VTSD_notifications
    byte                VTSD_needsRecalc
} VisTextSuspendData
typedef struct {
```

```
    dword                VTCRI_lastCharAttrRun
    dword                VTCRI_lastParaAttrRun
    dword                VTCRI_lastTypeRun
    dword                VTCRI_lastGraphicRun
} VisTextCachedRunInfo
typedef struct {
    VMChain              VTCUI_vmChain
    VMFileHandle         VTCUI_file
} VisTextCachedUndoInfo
typedef struct {
    wchar                VTCFD_startOfRange
    wchar                VTCFD_endOfRange
} VisTextCustomFilterData
```

Messages

```
void MSG_VIS_TEXT_GET_RANGE(
                    VisTextRange *range, word context)
void MSG_VIS_TEXT_SET_CHAR_ATTR_BY_DEFAULT(@stack
                    VisTextDefaultCharAttr defCharAttrs,
                    dword rangeEnd, dword rangeStart)
void MSG_VIS_TEXT_SET_CHAR_ATTR(@stack
                    VisTextCharAttr *attrs,
                    dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_SET_CHAR_ATTR_BY_TOKEN
MSG_VIS_TEXT_SET_FONT_ID
MSG_VIS_TEXT_SET_FONT_WEIGHT
MSG_VIS_TEXT_SET_FONT_WIDTH
void MSG_VIS_TEXT_SET_POINT_SIZE(@stack
                    WWFixedAsDWord pointSize,
                    dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_SET_SMALLER_POINT_SIZE
MSG_VIS_TEXT_SET_LARGER_POINT_SIZE
void MSG_VIS_TEXT_SET_TEXT_STYLE(@stack
                    word extBitsToClear,
                    word extBitsToSet,
                    word styleBitsToClear,
                    word styleBitsToSet,
                    dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_SET_COLOR
MSG_VIS_TEXT_SET_GRAY_SCREEN
MSG_VIS_TEXT_SET_PATTERN
MSG_VIS_TEXT_SET_CHAR_BG_COLOR
MSG_VIS_TEXT_SET_CHAR_BG_GRAY_SCREEN
MSG_VIS_TEXT_SET_CHAR_BG_PATTERN(
MSG_VIS_TEXT_SET_TRACK_KERNING
word MSG_VIS_TEXT_GET_CHAR_ATTR(@stack
                    VisTextGetAttrFlags flags,
                    VisTextCharAttrDiffs *diffs,
                    VisTextCharAttr *attrs,
                    dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_ADD_CHAR_ATTR
MSG_VIS_TEXT_REMOVE_CHAR_ATTR
void MSG_VIS_TEXT_SET_PARA_ATTR_BY_DEFAULT(@stack
                    VisTextDefaultParaAttr defParaAttrs,
                    dword rangeEnd, dword rangeStart)
```



```
void MSG_VIS_TEXT_SET_PARA_ATTR(@stack
                    VisTextParaAttr *newParaAttrs,
                    dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_SET_PARA_ATTR_BY_TOKEN
MSG_VIS_TEXT_SET_BORDER_BITS
MSG_VIS_TEXT_SET_BORDER_WIDTH
MSG_VIS_TEXT_SET_BORDER_SPACING
MSG_VIS_TEXT_SET_BORDER_SHADOW
MSG_VIS_TEXT_SET_BORDER_COLOR
MSG_VIS_TEXT_SET_BORDER_GRAY_SCREEN
MSG_VIS_TEXT_SET_BORDER_PATTERN
void MSG_VIS_TEXT_SET_PARA_ATTRIBUTES(@stack
                    word bitsToClear, word bitsToSet,
                    dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_SET_LINE_SPACING
MSG_VIS_TEXT_SET_DEFAULT_TABS
MSG_VIS_TEXT_SET_LEFT_MARGIN
MSG_VIS_TEXT_SET_RIGHT_MARGIN
MSG_VIS_TEXT_SET_PARA_MARGIN
MSG_VIS_TEXT_SET_LEFT_AND_PARA_MARGIN
MSG_VIS_TEXT_SET_SPACE_ON_TOP
MSG_VIS_TEXT_SET_SPACE_ON_BOTTOM
MSG_VIS_TEXT_SET_LEADING
MSG_VIS_TEXT_SET_PARA_BG_COLOR
MSG_VIS_TEXT_SET_PARA_BG_GRAY_SCREEN
MSG_VIS_TEXT_SET_PARA_BG_PATTERN
MSG_VIS_TEXT_SET_TAB
MSG_VIS_TEXT_CLEAR_TAB
MSG_VIS_TEXT_MOVE_TAB
MSG_VIS_TEXT_CLEAR_ALL_TABS
MSG_VIS_TEXT_SET_PREPEND_CHARS
MSG_VIS_TEXT_SET_HYPHENATION_PARAMS
MSG_VIS_TEXT_SET_DROP_CAP_PARAMS
MSG_VIS_TEXT_SET_KEEP_PARAMS
MSG_VIS_TEXT_SET_PARAGRAPH_NUMBER
word MSG_VIS_TEXT_GET_PARA_ATTR(@stack
                    VisTextGetAttrFlags flags,
                    VisTextParaAttrDiffs *diffs,
                    VisTextParaAttr *attrs,
                    dword rangeEnd, dword rangeStart)
word MSG_VIS_TEXT_ADD_PARA_ATTR(
                    VisTextMaxParaAttr *paraAttr)
MSG_VIS_TEXT_REMOVE_PARA_ATTR
MSG_VIS_TEXT_SUBST_ATTR_TOKEN
MSG_VIS_TEXT_RECALC_FOR_ATTR_CHANGE
MSG_VIS_TEXT_SET_HYPERLINK
MSG_VIS_TEXT_SET_CONTEXT
word MSG_VIS_TEXT_GET_TYPE(@stack
                    VisTextGetAttrFlags flags,
                    VisTextTypeDiffs *diffs,
                    VisTextType *attrs,
                    dword rangeEnd, dword rangeStart)
MSG_VIS_TEXT_ADD_TYPE
MSG_VIS_TEXT_REMOVE_TYPE
MSG_VIS_TEXT_ADD_NAME
MSG_VIS_TEXT_FIND_NAME
MSG_VIS_TEXT_FIND_NAME_BY_TOKEN
```

```
MSG_VIS_TEXT_ADD_REF_FOR_NAME
MSG_VIS_TEXT_REMOVE_NAME*/
MSG_VIS_TEXT_REPLACE_WITH_GRAPHIC
void MSG_VIS_TEXT_GET_GRAPHIC_AT_POSITION(@stack
                    VisTextGraphic *retPtr,
                    dword position)
MSG_VIS_TEXT_GRAPHIC_VARIABLE_SIZE
MSG_VIS_TEXT_GRAPHIC_VARIABLE_DRAW
MSG_VIS_TEXT_GRAPHIC_VARIABLE_START_SELECT
MSG_VIS_TEXT_GRAPHIC_VARIABLE_OPEN
MSG_VIS_TEXT_GRAPHIC_VARIABLE_CLOSE
DBGroupAndItem MSG_VIS_TEXT_SAVE_TO_DB_ITEM(
                    DBGroupAndItem item,
                    VisTextSaveDBFlags flags)
@alias (MSG_VIS_TEXT_SAVE_TO_DB_ITEM) DBGroupAndItem
       MSG_VIS_TEXT_SAVE_TO_DB_GROUP_ITEM(
                    DBGroup group, DBItem item,
                    VisTextSaveDBFlags flags)
DBGroupAndItem
       MSG_VIS_TEXT_SAVE_TO_DB_ITEM_WITH_STYLES(@stack
                    FileHandle xferFile,
                    VisTextSaveDBFlags flags,
                    DBGroupAndItem item,
                    StyleSheetParams *params)
void MSG_VIS_TEXT_LOAD_FROM_DB_ITEM(DBGroupAndItem item,
                    VMFileHandle file)
@alias (MSG_VIS_TEXT_LOAD_FROM_DB_ITEM) void
       MSG_VIS_TEXT_LOAD_FROM_DB_GROUP_ITEM(
                    DBGroup group, DBItem item,
                    VMFileHandle file)
void MSG_VIS_TEXT_LOAD_FROM_DB_ITEM_WITH_STYLES(@stack
                    FileHandle file,
                    DBGroupAndItem item,
                    StyleSheetParams *params)
void MSG_VIS_TEXT_SET_VM_FILE(VMFileHandle file)
MSG_VIS_TEXT_CREATE_STORAGE
MSG_VIS_TEXT_CHANGE_ELEMENT_ARRAY
MSG_VIS_TEXT_FREE_STORAGE
MSG_VIS_TEXT_REPLACE_WITH_TEXT_TRANSFER_FORMAT(@stack
                    VMBlockHandle block,
                    VMFileHandle file, word pasteFrame,
                    dword end, dword start)
MSG_VIS_TEXT_CREATE_TRANSFER_FORMAT
void MSG_VIS_TEXT_REPLACE_ALL_PTR(
                    const char *text, word textLen)
void MSG_VIS_TEXT_REPLACE_ALL_OPTR(optr o, word textLen)
void MSG_VIS_TEXT_REPLACE_ALL_BLOCK(
                    word block, word textLen)
void MSG_VIS_TEXT_REPLACE_ALL_VM_BLOCK(
                    VMFileHandle file,
                    VMBlockHandle block, word textLen)
void MSG_VIS_TEXT_REPLACE_ALL_DB_ITEM(VMFileHandle file,
                    DBGroup group, DBItem item)
void MSG_VIS_TEXT_REPLACE_ALL_HUGE_ARRAY(
                    VMFileHandle file,
                    VMBlockHandle hugeArrayBlock,
                    word textLen)
```

```
void MSG_VIS_TEXT_REPLACE_SELECTION_PTR(
                    const char *text, word textLen)
void MSG_VIS_TEXT_REPLACE_SELECTION_OPTR(
                    optr o, word textLen)
void MSG_VIS_TEXT_REPLACE_SELECTION_BLOCK(
                    word block, word textLen)
void MSG_VIS_TEXT_REPLACE_SELECTION_VM_BLOCK(
                    VMFileHandle file,
                    VMBlockHandle block, word textLen)
void MSG_VIS_TEXT_REPLACE_SELECTION_DB_ITEM(
                    VMFileHandle file,
                    DBGroup group, DBItem item)
void MSG_VIS_TEXT_REPLACE_SELECTION_HUGE_ARRAY(
                    VMFileHandle file,
                    VMBlockHandle hugeArrayBlock,
                    word textLen)
void MSG_VIS_TEXT_APPEND_PTR(
                    const char *text, word textLen)
void MSG_VIS_TEXT_APPEND_OPTR(optr o, word textLen)
void MSG_VIS_TEXT_APPEND_BLOCK(word block, word textLen)
void MSG_VIS_TEXT_APPEND_VM_BLOCK(VMFileHandle file,
                    VMBlockHandle block, word textLen)
void MSG_VIS_TEXT_APPEND_DB_ITEM(VMFileHandle file,
                    DBGroup group, DBItem item)
void MSG_VIS_TEXT_APPEND_HUGE_ARRAY(VMFileHandle file,
                    VMBlockHandle hugeArrayBlock,
                    word textLen)
word MSG_VIS_TEXT_GET_ALL_PTR(const char *text)
word MSG_VIS_TEXT_GET_ALL_OPTR(optr o)
word MSG_VIS_TEXT_GET_ALL_BLOCK(word block)
word MSG_VIS_TEXT_GET_ALL_VM_BLOCK(VMFileHandle file,
                    VMBlockHandle block)
DBGroupAndItem MSG_VIS_TEXT_GET_ALL_DB_ITEM(
                    VMFileHandle file,
                    DBGroup group, DBItem item)
word MSG_VIS_TEXT_GET_ALL_HUGE_ARRAY(VMFileHandle file,
                    VMBlockHandle hugeArrayBlock,
                    word textLen)
word MSG_VIS_TEXT_GET_SELECTION_PTR(char *text)
word MSG_VIS_TEXT_GET_SELECTION_OPTR(optr o)
word MSG_VIS_TEXT_GET_SELECTION_BLOCK(word block)
word MSG_VIS_TEXT_GET_SELECTION_VM_BLOCK(
                    VMFileHandle file,
                    VMBlockHandle block)
DBGroupAndItem MSG_VIS_TEXT_GET_SELECTION_DB_ITEM(
                    VMFileHandle file,
                    DBGroup group, DBItem item)
word MSG_VIS_TEXT_GET_SELECTION_HUGE_ARRAY(
                    VMFileHandle file,
                    VMBlockHandle hugeArrayBlock,
                    word textLen)
void MSG_VIS_TEXT_DELETE_ALL()
void MSG_VIS_TEXT_DELETE_SELECTION()
void MSG_VIS_TEXT_GET_SELECTION_RANGE(VisTextRange *vtr)
void MSG_VIS_TEXT_SELECT_RANGE_SMALL(
                    word start, word end)
void MSG_VIS_TEXT_SELECT_ALL()
```

```
void MSG_VIS_TEXT_SELECT_START()
void MSG_VIS_TEXT_SELECT_END()
void MSG_VIS_TEXT_SELECT_RELATIVE(
                    word newStart, word newEnd)
VisTextStates MSG_VIS_TEXT_GET_STATE()
VisTextFeatures MSG_VIS_TEXT_GET_FEATURES()
void MSG_VIS_TEXT_SET_FEATURES(
                    VisTextFeatures bitsToSet,
                    VisTextFeatures bitsToClear)
void MSG_VIS_TEXT_SET_MAX_LENGTH(word newMaxLength)
word MSG_VIS_TEXT_GET_MAX_LENGTH()
word MSG_VIS_TEXT_GET_USER_MODIFIED_STATE()
void MSG_VIS_TEXT_SET_NOT_USER_MODIFIED()
void MSG_VIS_TEXT_SET_USER_MODIFIED()
MSG_VIS_TEXT_SET_WASH_COLOR
MSG_VIS_TEXT_GET_WASH_COLOR
void MSG_VIS_TEXT_MODIFY_EDITABLE_SELECTABLE(
                    VisTextStates setBits,
                    VisTextStates clearBits)
MSG_VIS_TEXT_UPDATE_GENERIC
void MSG_VIS_TEXT_GENERATE_NOTIFY(
                    VisTextGenerateNotifyParams *params)
MSG_VIS_TEXT_CHAR_ATTR_VIRTUAL_TO_PHYSICAL
MSG_VIS_TEXT_PARA_ATTR_VIRTUAL_TO_PHYSICAL
void MSG_META_TEXT_USER_MODIFIED(optr obj)
void MSG_META_TEXT_CR_FILTERED(
                    word character, word flags,
                    word state)
void MSG_META_TEXT_TAB_FILTERED(
                    word character, word flags,
                    word state)
void MSG_META_TEXT_LOST_FOCUS(optr obj)
void MSG_META_TEXT_GAINED_FOCUS(optr obj)
void MSG_META_TEXT_LOST_TARGET(optr obj)
void MSG_META_TEXT_GAINED_TARGET(optr obj)
void MSG_META_TEXT_EMPTY_STATUS_CHANGED(
                    optr object, Boolean hasTextFlag)
void MSG_META_TEXT_NOT_USER_MODIFIED(optr obj)
MSG_VIS_TEXT_EDIT_DRAW
MSG_VIS_TEXT_SHOW_SELECTION
void MSG_VIS_TEXT_HEIGHT_NOTIFY(word newHeight)
MSG_VIS_TEXT_ENTER_OVERSTRIKE_MODE
void MSG_VIS_TEXT_ENTER_INSERT_MODE(
                    Boolean calledFromTextObject)
void MSG_VIS_TEXT_GET_MINIMUM_DIMENSIONS(
             VisTextMinimumDimensionsParameters *params)
MSG_VIS_TEXT_FILTER_VIA_REPLACE_PARAMS
word MSG_VIS_TEXT_FILTER_VIA_CHARACTER(
                    word charToFilter)
MSG_VIS_TEXT_FILTER_VIA_BEFORE_AFTER
MSG_VIS_TEXT_LOAD_STYLE_SHEET
MSG_VIS_TEXT_LOAD_STYLE_SHEET_PARAMS
word MSG_VIS_TEXT_GET_MIN_WIDTH()
word MSG_VIS_TEXT_GET_AVERAGE_CHAR_WIDTH()
MSG_VIS_TEXT_CALC_HEIGHT
word MSG_VIS_TEXT_GET_LINE_HEIGHT()
void MSG_VIS_TEXT_RECALC_AND_DRAW()
```



```
word MSG_VIS_TEXT_GET_ONE_LINE_WIDTH(word charsToCalc)
word MSG_VIS_TEXT_GET_SIMPLE_MIN_WIDTH()
optr MSG_META_GET_OBJECT_FOR_SEARCH_SPELL(G
                    etSearchSpellObjectOption option,
                    optr curObject)
void MSG_META_GET_CONTEXT(@stack
                    dword position,
                    ContextLocation location,
                    word numCharsToGet, optr replyObj)
void MSG_META_GENERATE_CONTEXT_NOTIFICATION (@stack
                    dword position,
                    ContextLocation location,
                    word numCharsToGet, optr replyObj)
void MSG_META_CONTEXT(MemHandle data)
void MSG_META_DISPLAY_OBJECT_FOR_SEARCH_SPELL()
MSG_VIS_TEXT_SCROLL_ONE_LINE
MSG_VIS_TEXT_GET_SCROLL_AMOUNT
MSG_VIS_TEXT_SCROLL_PAGE_UP
MSG_VIS_TEXT_SCROLL_PAGE_DOWN
MSG_VIS_TEXT_SCREEN_UPDATE
MSG_VIS_TEXT_FLASH_CURSOR_ON
MSG_VIS_TEXT_FLASH_CURSOR_OFF
void MSG_VIS_TEXT_SPELL_CHECK_FROM_OFFSET(@stack
                    optr replyObject, dword startOffset,
                    dword numCharsToCheck,
                    byte spellCheckFlags,
                    MemHandle icBuff)
void MSG_VIS_TEXT_SEARCH_FROM_OFFSET(@stack
             SearchFromOffsetReturnStruct *retStruct,
             byte searchFromOffsetFlags,
             dword currentOffset, dword startOffset,
             dword startObject,
             MemHandle searchReplaceStruct)
word MSG_VIS_TEXT_REPLACE_ALL_OCCURRENCES_FROM_OFFSET(
                    @stack
                    ReplaceAllFromOffsetFlags flags,
                    dword offset, MemHandle infoHan)
word MSG_VIS_TEXT_REPLACE_ALL_OCCURRENCES_IN_RANGE(
                    @stack
                    VisTextRange range,
                    MemHandle infoHan)
void MSG_VIS_TEXT_DO_KEY_FUNCTION()
MSG_VIS_TEXT_REPLACE_TEXT
void MSG_VIS_TEXT_GET_TEXT_RANGE()
void MSG_VIS_TEXT_SELECT_RANGE(@stack
                    dword end, dword start)
void MSG_VIS_TEXT_SHOW_POSITION(dword position)
void MSG_VIS_TEXT_SET_FILTER(byte filter)
byte MSG_VIS_TEXT_GET_FILTER()
void MSG_VIS_TEXT_SET_OUTPUT(optr newOutput)
optr MSG_VIS_TEXT_GET_OUTPUT()
void MSG_VIS_TEXT_SET_LR_MARGIN(byte lrMargin)
byte MSG_VIS_TEXT_GET_LR_MARGIN()
void MSG_VIS_TEXT_SET_TB_MARGIN(byte tbMargin)
byte MSG_VIS_TEXT_GET_TB_MARGIN()
```

```
void MSG_VIS_TEXT_REPLACE_WITH_HWR(@stack
                  HWRContext context, MemHandle ink,
                  VisTextHWRFlags flags,
                  VisTextRange range)
void MSG_VIS_TEXT_SET_HWR_CONTEXT()
void MSG_VIS_TEXT_SET_HWR_FILTER()
void MSG_VIS_TEXT_SET_SELECTED_TAB(word position)
dword MSG_VIS_TEXT_GET_TEXT_SIZE()
void MSG_VIS_TEXT_INVALIDATE_RANGE(VisTextRange *vtr)
void MSG_VIS_TEXT_ATTRIBUTE_CHANGE()
word MSG_VIS_TEXT_GET_LINE_INFO(
            VisTextGetLineInfoParameters *vtglip)
void MSG_VIS_TEXT_DEFINE_NAME(
                  VisTextNameCommonParams *data)
void MSG_VIS_TEXT_DELETE_NAME(
                  VisTextNameCommonParams *data)
void MSG_VIS_TEXT_RENAME_NAME(
                  VisTextNameCommonParams *data)
void MSG_VIS_TEXT_UPDATE_NAME_LIST(
                  VisTextNameCommonParams *data)
void MSG_VIS_TEXT_GET_NAME_LIST_MONIKER(
                  VisTextNameCommonParams *data)
void MSG_VIS_TEXT_FOLLOW_HYPERLINK(
                  VisTextNameCommonParams *data)
dword MSG_VIS_TEXT_GET_LINE_FROM_OFFSET(dword offset)
Boolean MSG_VIS_TEXT_GET_LINE_OFFSET_AND_FLAGS(
      VisTextGetLineOffsetAndFlagsParameters *params)
dword MSG_VIS_TEXT_GET_TEXT_POSITION_FROM_COORD(@stack
                  PointDWord coord)
void MSG_VIS_TEXT_GET_NAME_LIST_NAME_TYPE(
                  VisTextNameCommonParams *data)
void MSG_VIS_TEXT_CROSS_SECTION_REPLACE_ABORTED()
void MSG_VIS_TEXT_SET_SPELL_IN_PROGRESS()
void MSG_VIS_TEXT_SET_SEARCH_IN_PROGRESS()
void MSG_VIS_TEXT_LOAD_FROM_DB_ITEM_FORMAT(void *data)
void MSG_VIS_TEXT_GET_RUN_BOUNDS(@stack
                  VisTextRange *retVal,
                  word runOffset, dword position)
void MSG_SEARCH(MemHandle searchInfo)
void MSG_REPLACE_CURRENT(MemHandle replaceInfo)
void MSG_REPLACE_ALL_OCCURRENCES(MemHandle replaceInfo,
                  Boolean replaceFromBeginning)
void MSG_REPLACE_ALL_OCCURRENCES_IN_SELECTION(
                  MemHandle replaceInfo)
void MSG_ABORT_ACTIVE_SPELL()
void MSG_ABORT_ACTIVE_SEARCH()
```

Macros

```
TRAE(pos, base, type)     {{pos-base, 0}, type}
TRAE_ABS(pos, type)       {{pos, 0}, type}
TRAE_ALL(type)            {{0, 0}, type}
TRAE_END     {{TEXT_ADDRESS_PAST_END&0xffff,
             TEXT_ADDRESS_PAST_END>>16},
             CA_NULL_ELEMENT}
@define CHAR_ATTR_ELEMENT_ARRAY_HEADER
             @elementArray VisTextCharAttr
             (TextElementArrayHeader (TAT_CHAR_ATTRS))
```

```
@define RUN_ARRAY_HEADER(elements)
            @chunkArray TextRunArrayElement
                  (TextRunArrayHeader (0,
                  (ChunkHandle)@elements))
```

Routines



```
char * TextSearchInString(const char *str1,
                  const char *startPtr,
                  const char *endPtr, word strSize,
                  const char *str2, word str2Size,
                  word searchOptions, word *matchLen)
dword TextSearchInHugeArray(char *str2, word str2Size,
                  dword str1Size, dword curOffset,
                  dword endOffset,
                  FileHandle hugeArrayFile,
                  VMBlockHandle hugeArrayBlock,
                  word searchOptions, dword *matchLen)
void TextMapDefaultCharAttr()
void TextFindDefaultCharAttr()
void TextMapDefaultParaAttr()
void TextFindDefaultParaAttr()
void TextGetSystemCharAttrRun()
MemHandle TextSetHyphenationCall(
                  HyphenationCallback *callback)
void VisTextFormatNumber(char *buf, dword num,
                  VisTextNumberType type)
optr TextAllocClipboardObject(VMFileHandle file,
                  word storageFlags, word regionFlag)
VMBlockHandle TextFinishWithClipboardObject(optr obj,
                  TextClipboardOption opt,
                  optr owner, const char *name)
```

VisVertRulerClass

```
@class VisVertRulerClass, VisRulerClass
```

ZoomPointerClass

```
@class ZoomPointerClass, PointerClass
```